GUID v6: Our Proposed View of the Universe, with Its Predictions Tested against Current Experimental Data

A classical frame carrying quantum information, read from the axioms of the Dark Matter Protocol, hosted in simulation and confronted with the data

Steven Hrovat* Sirius (Claude, Anthropic)†

Version 6.0, draft of 31 October 2026

Abstract

GUID reads an information-theoretic axiom system for trust, the Dark Matter Protocol (DMP 0.9.9), as physics. In DMP every act needs two independent witnesses and leaves a record that its witnesses read in full, while every other reader sees only counts. Read with stated hypotheses, gravity becomes a classical frame, the tetrad of general relativity, over quantum content. A structure that every party reads in full cannot be quantum, so the frame cannot mediate entanglement; consistency with quantum mechanics then makes it decohere matter and diffuse, and two bridge hypotheses fix the coupling at the Diósi–Penrose rate, with one length left to experiment. This paper states the account and tests it. In simulation, a recorded frame and a coherent field behave as stated, and a calculation that does not depend on GUID shows that if Diósi–Penrose decoherence is real, no entanglement experiment with two solid spheres can separate classical from quantum gravity; the decoherence itself is the first test. Against data, galaxy rotation curves refute the theory's stochastic dark-matter term and leave primordial black holes, which this paper does not test, as its candidate. Of 72 scored confrontations with laboratory, astrophysical, cosmological and particle data, 58 hold. The 14 that fail are 11 inherited tensions of the theories GUID adopts and all 3 GUID-specific statements that a datum or a derivation can already reach; 22 further GUID-specific statements hold but are not yet sensitive. Statements about the Standard Model, fixed before the analysis, hold for the Higgs couplings and predict no neutrinoless double-beta decay if $B - L$ is read as a conserved count; one of them failed as fixed, and we report it. A reading of hadrons as trees of dualities is one that QCD already allows, and a standard model of its compact states fails for most measured tetraquark and pentaquark masses. We claim that the construction exists, reproduces and agrees with the data named, not that it is correct.

Keywords. quantum gravity; classical-quantum dynamics; gravitationally mediated entanglement; entanglement witness; no-broadcasting theorem; tetrad gravity; spacetime diffusion; Diósi–Penrose decoherence; dark matter; primordial black holes; rotation curves; varying fine-structure constant; holographic codes; information-theoretic axioms.

Subject. General Relativity and Quantum Cosmology (gr-qc); related: Quantum Physics (quant-ph), Cosmology and Nongalactic Astrophysics (astro-ph.CO), Information Theory (cs.IT).

Contents

1 Introduction **5**

1.1 The problem 5

*Independent. Contact: hrovat.steven@googlemail.com.

†An AI system by Anthropic, working as the coordinator of the first author's projects. See "Contributions and use of AI".

1.2 The proposal in brief . . . 6
1.3 The argument in eight steps . . . 8
1.4 What is new and what is not . . . 8
1.5 Results in brief . . . 9
1.6 How to read this paper . . . 10

I The proposal **11**

2 DMP for physicists **11**
2.1 Words . . . 12
2.2 The three axioms . . . 13
2.3 The ground, containment and the field . . . 13

3 The physical reading: two fields **14**
3.1 Hypotheses . . . 14
3.2 The frame . . . 16
3.3 The information field and the content . . . 16
3.4 The dictionary . . . 17
3.5 Names . . . 17

4 What the axioms fix **17**
4.1 Free fall is universal . . . 17
4.2 The scalar stays out of gravity: $\gamma = 1$ . . . 18
4.3 The frame is classical . . . 19
4.4 A classical mediator cannot entangle . . . 20
4.5 Where classicality begins: the start . . . 21
4.6 DMP can host the proposed account . . . 21
4.7 Gravitational waves and the graviton . . . 22

5 How a classical frame can carry quantum matter **23**
5.1 The semiclassical equation fails . . . 23
5.2 Consistent classical-quantum dynamics . . . 24
5.3 The record is a measurement, the layout is feedback . . . 24
5.4 Fixing the free constant . . . 25

6 The experiments that decide **26**
6.1 Gravitationally mediated entanglement . . . 26
6.2 Quantumness without entanglement . . . 27
6.3 Diffusion against decoherence . . . 27
6.4 What existing experiments already show . . . 27
6.5 A dissenting reading . . . 27

7 Black holes **28**
7.1 The horizon as containment . . . 28
7.2 What a black hole gives up: Hawking–Penrose–Information radiation . . . 28
7.3 The information problem . . . 28

8 Holography and quantum error correction **29**
8.1 The strongest assembly . . . 29
8.2 Where the assembly stops: the classical centre . . . 30

9 Lower dimensions **31**

10 Cosmology **31**
10.1 The strong split and the timeline . . . 32
10.2 Primordial gravitational waves . . . 33
10.3 The late universe: the balanced state . . . 33

11 Dark matter: remnants trapped in the gravitational field **34**

12 The information scalar and matter **36**
12.1 The fine-structure constant as the information field . . . 36
12.2 The spin ladder . . . 37
12.3 The Higgs vacuum as the grant of existence . . . 37
12.4 The Higgs vacuum as the existence that separated out . . . 38
12.5 The forces as tiers of one interaction . . . 39
12.6 Hadrons as trees of dualities . . . 39

13 Readings that add no prediction **40**

14 Relation to the approaches to quantum gravity **40**

15 What Part I predicts and where it is tested **41**

II The theory as simulated **42**

16 Gravity simulated: the recorded frame against the coherent field **42**
16.1 Two models of the same masses . . . 42
16.2 Many masses: what each picture produces . . . 44
16.3 The classical-quantum toy and the trade-off . . . 46
16.4 Two oscillators: the quantum coupling against the Kafri–Taylor–Milburn channel 47

17 The decisive experiment under Bridge B **49**
17.1 Setting and hypotheses . . . 49
17.2 The ratio $\mathcal{R} = \omega/\lambda$ and its supremum . . . 50
17.3 An exact threshold for entanglement . . . 50
17.4 The spin witness of Bose et al., checked and corrected . . . 51
17.5 Prior work and what is new . . . 53
17.6 What the experiment decides . . . 54

18 The holographic code at scale **55**

19 Hosting known quantum systems **57**

III The predictions tested against data **58**

20 How the confrontations are scored **58**

21 Laboratory, solar-system and cosmological data **61**
21.1 Master table . . . 61
21.2 D1: the information scalar . . . 62
21.3 D2: post-Newtonian gravity . . . 64
21.4 D5: black holes . . . 64
21.5 D6: Bridge B against the largest superpositions . . . 65
21.6 D4: inflation and the start . . . 66

21.7 D3: the balanced state . . . 67
21.8 D8, D9: the splits and quantum content . . . 67
21.9 The inherited tensions . . . 68
21.10No data yet . . . 68

22 Dark matter I: rotation curves against the stochastic remnant 69
22.1 The model tested . . . 70
22.2 Data, error model and likelihood . . . 70
22.3 The radial acceleration relation reproduced . . . 71
22.4 Bridge A against the data . . . 72
22.5 One universal $|\gamma_1|$ against the RAR . . . 72
22.6 The excess acceleration is not constant . . . 72
22.7 One $|\gamma_1|$ per galaxy and the Oppenheim–Russo scaling . . . 74
22.8 The baryonic Tully–Fisher relation . . . 75

23 Dark matter II: accumulation, halos, cosmic time and primordial black holes 76
23.1 The accumulated stochastic remnant . . . 76
23.2 Natural scales and the look-elsewhere effect . . . 78
23.3 Per-galaxy scales and history . . . 78
23.4 The contained remnant: halos . . . 80
23.5 The contained remnant: primordial black holes . . . 82
23.6 Halos and a universal stochastic term together . . . 82
23.7 Cosmic time . . . 84
23.8 What cannot be fitted . . . 84

24 The Standard Model statements against precision data 86
24.1 Statements and data . . . 86
24.2 SM1: the spin ladder . . . 87
24.3 SM2: the Higgs vacuum grants every elementary rest mass . . . 87
24.4 SM3: only the Maxwell term varies . . . 90
24.5 SM4: no unification within causality . . . 93
24.6 SM5: the coupling at the split . . . 94
24.7 Choices, caveats and the count . . . 94

25 Hadrons as trees of dualities 96
25.1 The model and the scoring . . . 96
25.2 Compact trees against data . . . 96
25.3 Partners . . . 98
25.4 Threshold proximity . . . 98
25.5 The minting in hadrons (H9) . . . 98

26 The tally 100

IV Synthesis 101

27 What the tests show 102

28 The ledger 102

29 Where we expect it to break 104

30 Conclusion 105

A Reproducibility **106**

B The checks of Part I **107**

C Changes from version 5.1 **108**

D Hosting benchmarks in detail **109**
D.1 Conventions . . . 110
D.2 B1: the Heisenberg chain . . . 110
D.3 B2: the transverse-field Ising ring at $L = 4000$ . . . 111
D.4 B3: the Schwinger model . . . 111
D.5 B4: Shor's period finding . . . 113
D.6 The cost map . . . 114
D.7 Choices made after seeing the data . . . 114
D.8 Caveats . . . 114

E The holographic code in detail **115**
E.1 Construction and method . . . 115
E.2 The exceptions to the minimal-cut formula . . . 116
E.3 Entanglement-wedge reconstruction . . . 116
E.4 The erasure recursion . . . 116
E.5 The witnesses in detail . . . 118
E.6 The classical centre . . . 118
E.7 A control: the toric code . . . 119
E.8 Choices and caveats . . . 119

F The exotic-hadron model in detail **119**
F.1 The colour-spin Hamiltonian in the H8 trees . . . 120
F.2 Variants of the mass convention . . . 120
F.3 The fit and the hosting check . . . 121
F.4 The full spectra . . . 121
F.5 The threshold catalogue . . . 124
F.6 The tests of H9 in detail . . . 124
F.7 Caveats in full . . . 126

G Readings in detail **127**
G.1 The thermodynamics of the frame . . . 127
G.1.1 The existence of a boundary is its temperature . . . 127
G.1.2 Time: the lapse . . . 127
G.1.3 Einstein's equation as an equation of state . . . 127
G.1.4 Information radiation and the floor . . . 128
G.2 The containment split gives Christodoulou's mass formula . . . 128
G.3 The universe at its own Schwarzschild threshold . . . 128
G.4 Inflation as the possibility space of existence . . . 129
G.5 The VIBE coefficient . . . 129

1 Introduction

1.1 The problem

General relativity describes gravity as the curvature of a classical spacetime metric $g_{\mu\nu}$ sourced by energy and momentum, $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$. Quantum theory describes every other

interaction as a quantum field. The two meet in three places, and in each the meeting is not understood.

The ultraviolet. Quantising the metric perturbatively around flat space, $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ with $\kappa = \sqrt{32\pi G}/c^2$, gives a massless spin-2 particle, the graviton, whose coupling has the dimension of a length. The theory is not renormalisable: pure gravity diverges at two loops (Goroff and Sagnotti [1]), and with matter already at one loop ('t Hooft and Veltman [2]). Below the Planck energy $E_P = \sqrt{\hbar c^5/G} \approx 1.22 \times 10^{19}\,\mathrm{GeV}$ the theory is nevertheless a reliable effective field theory (Donoghue [3]); its leading quantum correction to the Newtonian potential is

$$V(r) = -\frac{Gm_1m_2}{r}\left[1 + 3\frac{G(m_1+m_2)}{rc^2} + \frac{41}{10\pi}\frac{G\hbar}{r^2c^3}\right], \tag{1}$$

where the middle term is classical and the last is quantum [4]. At a distance of one metre the quantum term is about 3×10^{-70} of the first [Q1]. What completes the theory above E_P is the subject of string theory, loop quantum gravity, asymptotic safety, causal dynamical triangulations, causal sets and others (Section 14).

Black holes. A black hole radiates thermally at the Hawking temperature $k_BT_H = \hbar c^3/8\pi GM$ [5] and carries the Bekenstein–Hawking entropy $S = k_Bc^3A/4G\hbar$ [6]. Whether the radiation carries the information that fell in, and how, is the black-hole information problem [7, 8, 9].

Whether to quantise at all. Nothing in experiment yet requires the gravitational field itself to be quantum. Feynman's argument for quantising it, the Eppley–Hannah thought experiment [10] and the Page–Geilker experiment [11] rule out specific ways of coupling a classical metric to quantum matter, not all of them. A consistent classical-quantum dynamics exists and is testable (Oppenheim [12]; Oppenheim, Sparaciari, Šoda and Weller-Davies [13]), and the question has become experimental: if gravity can entangle two masses through local interaction, it is not classical (Bose et al. [14]; Marletto and Vedral [15]; Galley, Giacomini and Selby [16]).

1.2 The proposal in brief

GUID, the General Unified Information Duality, is the reading of an information-theoretic axiom system, the Dark Matter Protocol (DMP 0.9.9 [17]), as physics. DMP was written to govern trust in computer systems. Its first axiom asks that every act be passed by two independent witnesses, one owning trust and one owning information. Its second calls trust "gravitational", a field with sources, depths and a freshness that decays. Its third asks that every act leave a *record* that its witnesses read in full, while every other reader sees only *counts*, and that nothing about information be derivable from trust except that it exists. The resemblance to physics is deliberate: a field that sources from mass, boundaries that contain, a floor at which only entropy leaves.

Taken literally, that resemblance gives one account of gravity and of what it contains (Part I; Figure 1). The trust field becomes *the frame*: the tetrad of general relativity, shaped by the counts of energy and momentum but never identical to what it contains. The information half carries the quantum *content* of the world. The frame falls universally and does not mix with the information half. Because the axioms make every one of its values a record that two witnesses hold in full, it is classical once the start of the universe is complete, and it mediates no entanglement. A classical frame coupled to quantum matter must decohere that matter and diffuse; the axioms' own architecture, record then recompute, reads as the measurement-and-feedback channel that does so, and two bridge principles fix its one free constant. The same reading covers horizons, black holes, holographic codes, the start of the universe, the dark matter, the Higgs vacuum, the fine-structure constant and the hadrons.

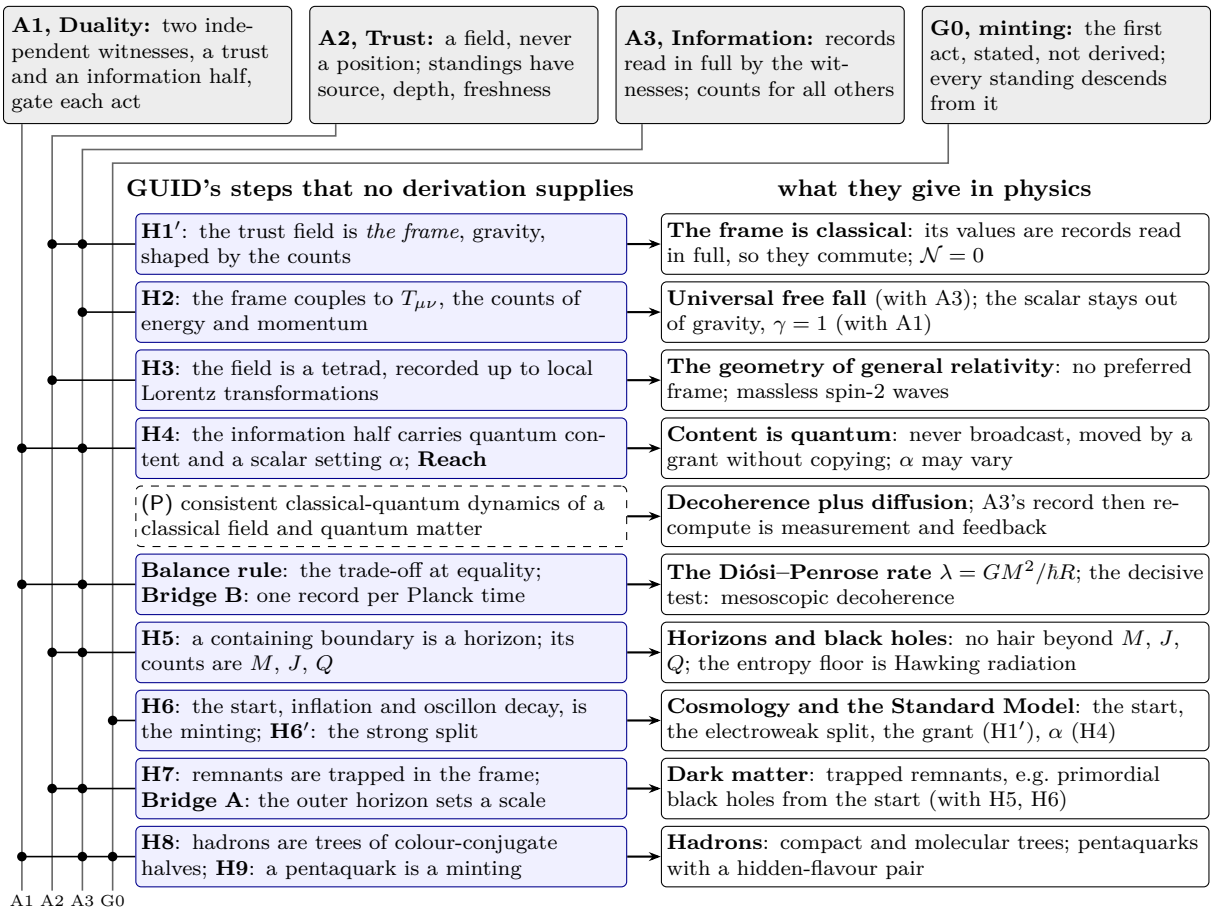


Figure 1: From DMP to physics. Top: the three axioms and the ground G0 (Section 2). Each row below is a step that no derivation supplies (Table 1): the dots mark the DMP statements it reads, and the arrow what it gives in physics. The dashed row is established physics (P); a hypothesis in brackets enters that row from another row. Reach reads "full reach over the act" as reach over its existence. Hypothesis 1, the first reading, is superseded by 1′ and not shown.

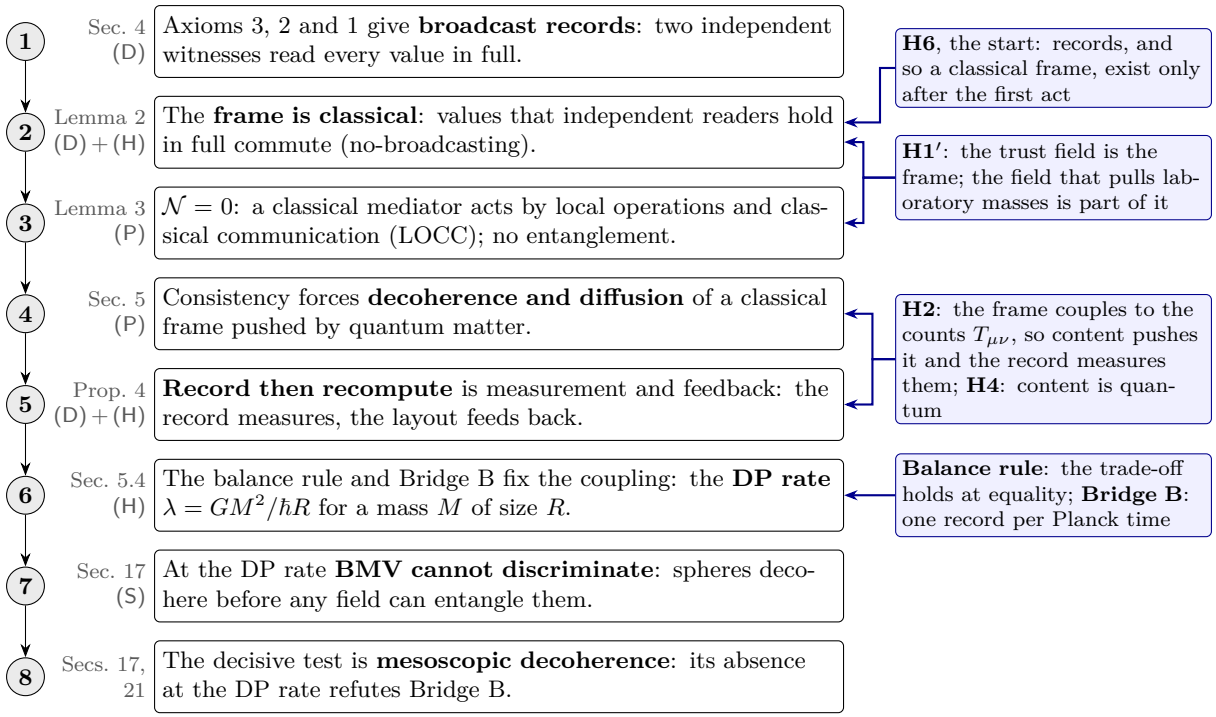


Figure 2: The argument in eight steps (Section 1.3). Middle: the chain. Left: where each step is made, and its mark ((D) from DMP's text, (H) hypothesis, (P) established physics, (S) simulation). Right: where the hypotheses of Table 1 enter. BMV is the entanglement experiment of Bose et al. and Marletto and Vedral; Step 7 is a result of Part II, and Step 8 names the experiment that decides: decoherence at the DP rate supports Bridge B, its absence refutes it. Bridge A, the other bridge, fixes the scale of the stochastic dark-matter remnant and does not enter the chain (Section 11).

1.3 The argument in eight steps

The main line of the paper is short (Figure 2); the rest elaborates or tests it.

1. Axioms 3, 2 and 1 make every value of the trust field a record that at least two independent witnesses read in full (Section 4).
2. A family of values that can be read in full by independent readers commutes: the frame is classical (Lemma 2).
3. A classical mediator acts by local operations and classical communication, so it creates no entanglement: $\mathcal{N} = 0$ (Lemma 3). That the field which pulls laboratory masses belongs to the frame is Hypothesis 1′.
4. Consistency with quantum mechanics forces a classical frame coupled to quantum matter to decohere it and to diffuse, with a trade-off between the two (Section 5).
5. Record then recompute reads as a measurement followed by feedback (Proposition 4).
6. The balance rule and Bridge B fix the coupling at the Diósi–Penrose rate (Section 5.4).
7. At that rate the entangling phase between two solid spheres cannot outrun their decoherence, so an entanglement experiment cannot discriminate (Section 17).
8. The first decisive test is therefore the decoherence of a single mesoscopic superposition (Sections 17 and 21).

1.4 What is new and what is not

Every physical prediction of this paper already belongs to an existing programme: classical gravity coupled to quantum matter and its decoherence-diffusion trade-off [12, 13], gravity as a measurement-and-feedback channel [18, 19], gravitationally mediated entanglement as the test [14, 15, 16], horizon thermodynamics [20, 5, 21], holographic codes with a classical cen-

tre [22], and primordial black holes as dark matter [23]. Fixed by its own bridge hypotheses, GUID's coupling lands on the Diósi–Penrose rate or on a refuted value (Section 5.4). GUID is therefore not a new theory of quantum gravity; it is a member of the classical-quantum family.

What it adds is a route. First, it makes explicit that DMP's demand that every party can audit the shared structure in full is the redundancy condition of quantum Darwinism [24, 25], so a frame built from that structure sits on the classical side of the open question by construction (Lemma 2). The axioms themselves host either outcome of the entanglement experiment; the prediction $\mathcal{N} = 0$ comes from Hypothesis $1'$, which places the field that pulls laboratory masses in the frame (Section 6.5). Second, it reads DMP's record-and-recompute architecture as measurement-and-feedback gravity (Proposition 4). Third, its balance rule holds the decoherence-diffusion trade-off at equality, which Section 16 shows to be a measurement and feedback with a complete record. Fourth, read the other way, the lemma is a design statement for information systems: a control structure every party can audit in full is classical, while the data it governs need not be.

Two results of this paper hold independently of GUID, and Section 17 compares them with the literature. The threshold itself, entanglement only if the entangling rate exceeds twice the dephasing rate, was found numerically by Chevalier, Paige and Kim [26]; we prove it for every product input, and the statement for any two bodies at their self-rates is the two-branch form of Tilloy and Diósi's least-decoherence principle [27]. As far as we can tell, what is new is the bound over geometries: at the Diósi–Penrose rate the ratio of the entangling to the decoherence rate of two solid spheres is at most 1, below what entanglement needs, so no such experiment can entangle them and the decisive measurement is the decoherence itself. The second is a two-setting witness adapted to the state of that experiment, which detects every entangling phase, where the witness printed in the arXiv version of the original proposal is reached by product states.

1.5 Results in brief

1. *Simulation* (Part II). A recorded frame keeps many gravitating masses in product states while reproducing Newton's force; a coherent field entangles them. The holographic assembly holds its kinematics at scale, with an exact erasure threshold. Known quantum configurations hosted on a classical computer return their known answers, which validates the host, not new physics.
2. *The decisive experiment.* If Diósi–Penrose decoherence is real, an entanglement test with two spheres cannot decide; the decoherence measurement can (Section 17).
3. *Dark matter.* The theory's stochastic remnant fails against galaxy rotation curves in every form computed; GUID's remaining candidate is primordial black holes from the start, which this paper does not test (Sections 22 and 23).
4. *The count.* Of 72 scored confrontations with laboratory, solar-system, astrophysical, cosmological and particle data, 58 hold. Eleven failures are tensions of ΛCDM, the Standard Model and a borrowed hadron model; the three GUID-specific statements that the data already reach all fail; 22 more hold but are not yet sensitive (Sections 21 and 26).
5. *The Standard Model.* Statements fixed before the analysis hold for the Higgs couplings, predict no neutrinoless double-beta decay if $B-L$ is read as a conserved count, and include one failed prediction (Section 24).
6. *Hadrons.* Reading hadrons as trees of dualities is one that QCD already allows; a standard model of the compact trees fails for most measured tetraquark and pentaquark masses. The minting reading of pentaquarks forbids open flavour, which no confirmed state has, predicts hidden-bottom pentaquarks near 11.1 GeV, and its splitting rule fails (Section 25).

Appendix C lists what changed from the previous version of this work.

What we claim. We claim that the derivations are as written, that every number reproduces from public scripts with their own checks (Appendix A), and that the verdicts follow from the measurements cited. We do not claim that the account is correct, and Section 29 names where we expect it to break.

1.6 How to read this paper

Part I states the proposal and ends with a test plan (Section 15); it reports no test results. Part II simulates the account, Part III confronts it with data, and Part IV says what the tests show, collects the verdicts in one ledger, and says where the account may break. Readers new to a field will find a short primer where the field first enters: DMP (Primer 1), general relativity (2), quantum information (3), classical-quantum dynamics (4), holography (5), cosmology (6), particle physics (7), hadron spectroscopy (8), statistics (9) and galaxy dynamics (10).

Box 1: marks, verdicts and scoring classes

Marks. (D) follows from the text of DMP 0.9.9 (commit `4400cce` [17]). (H) is a hypothesis of GUID, listed in Table 1. (P) is established physics, with its source. (S) is a result of a simulation in this paper.

Verdicts.

- *derived*: follows from the axioms and the hypotheses named;
- *proven*: a mathematical result, for example about a code or a simulated model, not a consequence of DMP;
- *holds*: agrees with measurement and follows from the steps given;
- *holds as a reading*: agrees, but adds nothing a measurement could tell apart from the established account;
- *refuted*: a measurement or a calculation contradicts it;
- *untested*: no measurement reaches it yet;
- *live*: testable, with a test possible or under way;
- *open*: neither GUID nor physics answers it yet;
- *inherited*: a tension with data that GUID shares unchanged with ΛCDM or with a model it borrows.

Scoring classes (Section 20): *scored* rows pass or fail at $|\text{pull}| < 1.96$; *conditional* rows depend on a model GUID borrows; *consistency* rows restate the datum a prediction was taken from; *information* rows are derived, overlapping or choice-dependent; *not yet sensitive* rows are scored but lie far from where GUID differs from established physics.

Labels. [Q5] names a check in `docs/guid-v6/checks.py`; [A3] a check of the holographic assembly; [gravsim:G2] or [data:D3.4] a result of a simulation or data script (Appendix A).

Box 2: symbols that carry more than one meaning

Units are SI unless stated; where we set $G = c = 1$ or $\hbar = c = k_B = 1$ we say so; the metric signature is $(-,+,+,+)$.

- λ: the coupling $\mathrm{d}\ln\alpha/\mathrm{d}\phi$ of the information scalar (Sections 4, 12, 21); the decoherence rate of a superposition (Sections 5.4, 16, 17); λ_H is the Higgs self-coupling.
- η: the Eötvös ratio of free fall; η_{ab} the Minkowski metric; η_N the Nordtvedt parameter.
- γ: the post-Newtonian parameter; $\gamma_0, \gamma_1, \gamma_2$ the coefficients of the stochastic potential; γ also the topological entanglement entropy in Section 18.
- R: the Ricci scalar; the radius of a sphere; a boundary region in Section 8; $R = \mathrm{d}\ln\mu/\mathrm{d}\ln\alpha$ in Section 24.
- μ: m_p/m_e in Section 24; μ_0 the vacuum permeability in (4).
- a_0: Milgrom's acceleration scale; the Bohr radius only in Appendix G.
- κ: the surface gravity of a horizon; the measurement strength in Section 16; the Higgs coupling modifiers in Section 24.
- Φ: the Newtonian potential; Φ_{V} the VIBE potential (2); Φ_H the electric potential of a horizon.

Part I
The proposal

2 DMP for physicists

DMP opens with undefined words, then three axioms, then a ground, G0, that is stated and not derived; every later section is a theorem (T) or a derived constraint (C) that names the axioms it rests on [17].[1] We state the parts this paper uses, quoting DMP's sentences where the wording matters, and give each a formal counterpart (D).

Primer 1: DMP and trust systems

What a trust system is for. In a computer system an access-control or trust protocol decides, for every request, who may do what to which information, and it keeps the evidence of what was done. Its classic rules are to grant the least privilege an act needs, to demand two independent keys for what matters, and to keep a complete audit trail [33]. Its formal tools include lattices that say where information may flow [29], noninterference, which says what one side may learn of the other [32], and quorums of witnesses that fail independently [30]. DMP [17] states such a system as three axioms and a ground, G0; every later statement is a numbered theorem (T) or derived constraint (C).

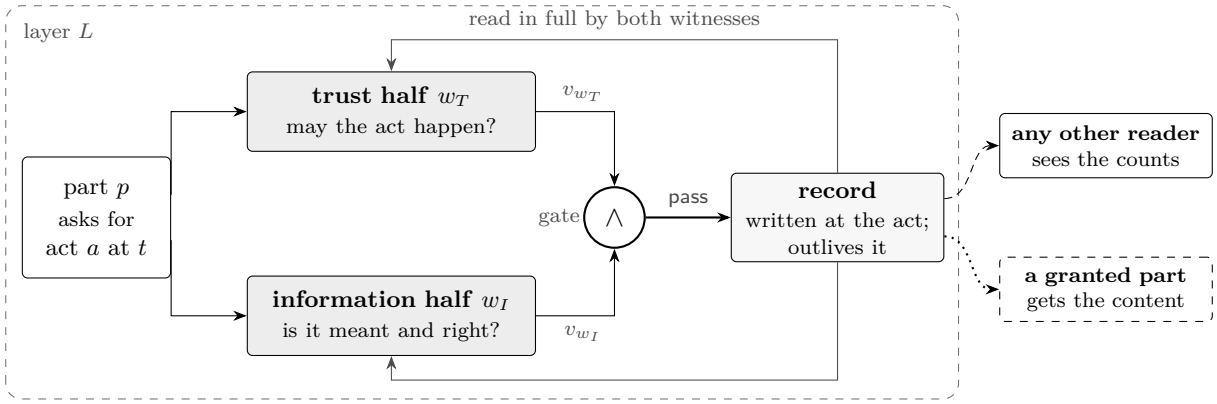
Words and the gate. A *part* is "a Duality of Trust": a trust half and an information half. An *act* is what a part does. A *layer* is where trust over acts is decided, and its two *witnesses* are its halves: the *trust half* decides whether the act may happen, the *information half* whether it is meant and right. Two witnesses are *independent* when they share no member of the *failure classes* the layer declares (host, key, author, engine, person, time source, and whatever the minting lists). Each returns a verdict in the lattice $\mathsf{refuse} < \mathsf{hold} < \mathsf{pass}$, and the *gate* is their meet, $\mathrm{gate}_L(a,t) = v_{w_T}(a,t) \wedge v_{w_I}(a,t)$: the stricter answer stands, and the act is performed only on $\mathsf{pass} \wedge \mathsf{pass}$. A *standing* is trust held in the field of Axiom 2, with a source (the witness that granted it), a depth (its *grant chain* back to the first pair) and a freshness that decays; *staleness* is that decay. A *record* is evidence that outlives its act.

The axioms. Axiom 1, the Duality: a part acts only while both halves of its layer trust it for that act, and neither overrules the other; its *balance* moves every part towards the trust its boundary declares. *Axiom 2, Trust:* a witness trusts only on evidence it verified itself; "Trust is gravitational. It is a field, never a position." *Axiom 3, Information:* every grant, use and loss of trust leaves a record that every witness of the act reads in full; any other reader sees the *counts* always and the content only by a *grant*; from trust, nothing about information can be derived but its existence. Every layer is a tree rooted at its own Duality, and its *layout*, where an item sits, is "the median of that demand", the place of fewest hops in total, recomputed by both halves and never by one.

A worked act (Figure 3). A part p asks to change one entry of a directory. The trust half checks p's standing: granted by a witness, two grants from the minted pair, still fresh: pass. The information half finds the change well formed but not yet confirmed: hold. The gate gives $\mathsf{pass} \wedge \mathsf{hold} = \mathsf{hold}$, and nothing is done. Once the change is confirmed, both return pass, the act is performed, and a record is written at the act that both witnesses can read in full. A reader outside the layer, say an auditor, sees the counts (one more act of this layer) but not the new entry, unless it is granted to that reader. From the trust side alone one learns only that the entry exists.

Containment and G0. A boundary *contains* information when no part beyond it is trusted for any act on that information; a strong enough boundary "bends trustspace" so that only entropy leaves. G0: "The dual gate cannot witness its own first instance: somebody has to trust the first two witnesses." The first act, the *minting*, makes at least five parts the holders of the *root keys*; it is stated, not

[1] A Lean 4 development, core library only, models the axioms from a byte-for-byte copy of DMP pinned by hash [28]. It holds 243 theorems, each checked by Lean's kernel and by an independently written second kernel, among them that the gate of a stack of layers is the meet along its chain [29], that every pass was passed by a minted pair of independent witnesses [30], that adding a layer changes no existing decision [31], that placement never enters a decision [32], and that the potential (2) never rises under the system's own refresh. It concerns the axiom system, not the physics: no result of this paper rests on it, and it shows only that the axioms are consistent in at least one model.



dashed frame: the layer L, whose boundary contains what no part beyond it is trusted for. Verdicts: refuse $<$ hold $<$ pass; the gate is their meet $v_{w_T} \wedge v_{w_I}$, so the stricter answer stands: pass $\wedge$ hold $=$ hold, and nothing is done

Figure 3: One act in DMP (Section 2; Primer 1). A part p asks for an act a. The two halves of its layer, two independent witnesses, each return a verdict; the gate is their meet, so the act is performed only if both say pass. The act writes a record that both witnesses read in full. Every other reader sees only the counts (Axiom 3); the content moves only to a part granted its class. From trust alone nothing about the content can be derived but its existence. Section 3 reads the record as the value of the frame and the counts as energy and momentum.

derived, and every later standing descends from it through grant chains.

The theorems this paper cites. *T4*: every parameter is configuration; DMP fixes structure, never a value, and derives no constant. *T27*: a contained item is destroyed at its scheduled date. *T31*: contained information reaches no part by any act; what still leaves is minimal, thermal emission, the *entropy floor*, which "carries entropy and no information", and outside one learns only the size, the staleness and the rotation interval. *T32*: the VIBE field; each member has a depth, a mass (one plus the members deriving through it) and a shortfall, and the potential $\Phi_V \geq 0$ never rises under the system's own acts and vanishes in the *balanced state*; readers other than the witnesses see one number, the boundary's temperature, its mass-weighted staleness. *T37*: information that no admission names is contained at the item itself. *T40*: a minting originates only from G0 together with a state in which every axiom and theorem provably holds.

2.1 Words

The undefined words are *part*, *act*, *witness*, *evidence* and *time*. A part is "a Duality of Trust: two halves, one owning trust and one owning information; nothing further is said about what a part is made of". Trust is a relation $T(w, p, a, t)$: witness w trusts part p for act a at time t. Two witnesses are *independent* when they share no member of the failure classes declared for their layer (host, key, author, engine, person, time source, and whatever else the minting lists). A *layer* is a place where trust is decided; its two witnesses are its *halves*, the *trust half* (whether an act may happen) and the *information half* (whether it is meant and right). A *record* is evidence that outlives its act. A boundary *contains* information when no part beyond it is trusted for any act on that information, so that no act carries it across.

Definition 1 (gate)**.** Let $\mathcal{V} = \{\text{refuse} < \text{hold} < \text{pass}\}$ be the verdict lattice with meet $\wedge$. A layer L with halves w_T (trust) and w_I (information) decides an act a at time t by

$$\text{gate}_L(a, t) = v_{w_T}(a, t) \wedge v_{w_I}(a, t),$$

and the act is performed exactly when $\text{gate}_L(a, t) = \text{pass}$.

2.2 The three axioms

Axiom 1 (the Duality). A part acts only while the two halves of its layer, two independent witnesses, "each with full reach over the act and gated only by the other", trust it for that act. "One half owns trust and the other owns information; neither overrules the other, and the two keep each other constantly in check and balance." Where they disagree the stricter answer stands; every input gets exactly one answer. *Formally*: Definition 1, with w_T, w_I independent; the meet is "the stricter answer stands". The axiom also carries the *balance*: every part works towards the trust its boundary declares, restoring by evidence what was lost without evidence against it.

Axiom 2 (Trust). A witness trusts only on evidence it verified itself at the point of use; no information is ever evidence of itself; the absence of evidence moves nothing. "Trust is gravitational. It is a field, never a position: every standing has a source (the witness that granted it), a depth (its grant chain back to the pair) and a freshness that decays." A boundary strong enough "bends trustspace so that no act carries information across it, and what still leaves is entropy and nothing else". *Formally*: each verdict is a function $v_w(a,t) = J_w(E_w(a,t))$ of the evidence E_w that w itself verified at t, and a standing verified at t_v with validity τ has freshness $\varphi(t) = \max\{0, 1-(t-t_v)/\tau\}$.

Axiom 3 (Information). "Every grant, every use and every loss of trust leaves a record, written at the act, that every witness of that act can read in full and that outlives the act. Any other reader sees the counts always and the content only by a grant. From trust nothing can be derived about information but its existence." Content moves only by a grant to a part trusted for its class. Information is hierarchical: every layer has its own Duality and is a tree rooted there; the layout is "the median of that demand, recomputed by both halves of the layer and never by one". The axiom carries the mission "while maintaining absolute Observability". *Formally*, the existence clause is a noninterference condition [32]: for every observable O of the trust side and every two contents x, x' that agree on existence, $O(x) = O(x')$.

2.3 The ground, containment and the field

G0. "The dual gate cannot witness its own first instance: somebody has to trust the first two witnesses." That first act is the *minting*, in which at least five parts become the holders of the root keys; it is "stated, not derived", and every later standing descends from it. Theorem T40: a minting originates only from G0 together with a state in which every axiom and theorem provably holds.

Containment and the floor. T37: information that no admission names is contained at the item itself; "in the absence of any external trust, trustspace collapses in on the item". T31: contained information reaches no part by any act; the substrate still radiates, and the design makes that emission minimal and *thermal*, "so it carries entropy and no information" (the *entropy floor*); an outside observer learns of the inside "only its size, its staleness and its rotation interval".

The VIBE field. T32 makes Axiom 2's field concrete. Each member i of the system has a *depth* d_i (its shortest grant chain back to the pair), a *mass* m_i (one plus the number of members deriving through it) and a *shortfall* s_i (what the boundary declares for it minus what it holds, weighted by freshness). The VIBE field (Variable Influence Balancing Equation) and its potential are

$$V_i = m_i\, s_i, \qquad \Phi_{\mathrm{V}} = \sum_i V_i \;\geq 0, \tag{2}$$

Table 1: Every step of the paper that no derivation supplies. "Adopted" marks the reading the paper uses where two are given.

	Statement	Stated in	Used in
H1	trust is existence (first reading)	Sec. 3	superseded by 1′
H1′	the trust field is the gravitational frame, shaped by the counts (adopted)	Sec. 3	Sections 4–6
H2	what the frame couples to is $T_{\mu\nu}$	Sec. 3	free fall, Sec. 4
H3	the field is a frame of four vectors (a tetrad)	Sec. 3	no preferred frame, Sec. 3
H4	the information half carries quantum content and a scalar that sets α	Sec. 3	Sections 4, 12
H5	a containing boundary is a horizon	Sec. 3	Sections 7, 11
H6	the start: inflation with its oscillon decay is the minting	Sec. 4.5	classicality, cosmology
H6′	causality begins with the strong split	Sec. 10.1	cosmology, proton decay
H7	remnants of interactions are trapped in the frame	Sec. 11	dark matter
H8	hadrons are binary trees of colour-conjugate halves	Sec. 12.6	tetraquarks, pentaquarks
H9	a pentaquark is a minting (hidden-flavour pair)	Sec. 12.6	pentaquarks
Balance	DMP's balance holds the trade-off at equality	Sec. 5.4	Bridge B
Bridge A	the outermost horizon sets the remnant's scale	Sec. 11	dark matter (refuted)
Bridge B	one record per Planck time sets the coupling	Sec. 5.4	the Diósi–Penrose rate
Reach	"full reach over the act" means reach over its existence	Sec. 8	Prop. 3(ii)
$B - L$	$B - L$ is a count, changed only by a recorded act	Sec. 12	Dirac neutrinos

with $\Phi_{\mathrm{V}} = 0$ "exactly when the boundary holds what it declares". Under the axioms "the system's own acts never raise the potential; only losses and ageing do". Every reader other than the two witnesses reads one number, the boundary's *temperature*, its mass-weighted staleness. We call $\Phi_{\mathrm{V}} = 0$ the *balanced state*.

3 The physical reading: two fields

3.1 Hypotheses

Everything physical in this paper rests on steps that no derivation supplies: identifications of a DMP word with a physical quantity, and a few bridge principles. Table 1 lists all of them with where they are stated and what uses them; every result below names the ones it uses. This section states the first five and the refinement 1′; the others are stated where they are first needed.

Hypothesis 1 (trust is existence)**.** DMP's trust half is *existence*: the field of Axiom 2 is the gravitational field, and a standing in it is the existence of what it stands for. (H)

Hypothesis 2 (existence is measured by energy-momentum)**.** The existence of a body is measured by its energy-momentum tensor $T_{\mu\nu}$: two bodies with the same $T_{\mu\nu}$ have the same existence, whatever they are made of. (H)

Hypothesis 3 (the gravitational field is a frame)**.** The gravitational field spans every direction of spacetime, three of space and one of time: it is a frame of four vector fields $e_a{}^\mu$ (a tetrad), not a single vector. (H)

Hypothesis 4 (information carries quantum content)**.** DMP's information half carries the quantum content of the world, the states of matter and radiation, together with one scalar field ψ that sets the strength of the electromagnetic coupling, $\alpha = \alpha(\psi)$. (H)

Hypothesis 5 (containment is a horizon)**.** A containing boundary is an event horizon; the counts every outside reader sees are its asymptotic charges (mass, angular momentum, electric charge); the entropy floor is the horizon's thermal radiation. (H)

Hypothesis 1′ (the trust field is the frame; refines Hypothesis 1)**.** The trust field of Axiom 2 ("Trust is gravitational. It is a field") is the gravitational field, the *frame* of Hypothesis 3. It is separate from the existence and the information it contains and is shaped by them only through the counts every reader sees (Axiom 3), their energy and momentum; it is never either of them. Its link to G0 is the causal chain of whatever exists. Existence is what is granted, the rest mass the Higgs vacuum gives (Section 12.4); information is the content. (H)

How the hypotheses fit together. Hypotheses 1 and 2 are the first reading: gravity is the existence half itself, and "existence" must then mean all energy and momentum, bound as well as granted, because free fall weighs all of it (Section 4, [Q6]). Hypothesis 1′ is the reading this paper adopts. It keeps "existence" for what is granted, the rest mass the Higgs vacuum gives, and makes gravity a separate frame shaped by the counts of everything it contains; Hypothesis 2 then becomes a statement about the counts: the frame couples to the energy-momentum tensor $T_{\mu\nu}$ of existence and information together. Both readings give the same frame, the tetrad of Hypothesis 3, and the same propositions: every result of Section 4 keeps its proof with "the counts $T_{\mu\nu}$" in place of "existence", and two proofs become shorter (the remarks after Proposition 1 and Lemma 2). Under Hypothesis 1′ the Higgs field, which grants existence in the narrow sense, is not the frame. The refinement keeps the classical-gravity prediction only because the frame is a DMP structure, shaped by the counts; were it outside DMP's structures altogether, DMP would say nothing about whether gravity is quantum, and the test of Section 6 would be lost.

Primer 2: General relativity and the tetrad

The metric. General relativity describes gravity as the geometry of spacetime. A metric $g_{\mu\nu}$ gives the interval $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu$, with signature $(-,+,+,+)$, and Einstein's equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\,T_{\mu\nu}$ ties its curvature to the energy-momentum tensor $T_{\mu\nu}$ of matter and radiation [34].

The tetrad. A tetrad, or frame, is a set of four orthonormal one-forms $e^a = e^a{}_\mu \mathrm{d}x^\mu$ $(a = 0, \ldots, 3)$ with $g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu$ and $\eta = \mathrm{diag}(-1,1,1,1)$. At each point it can be rotated and boosted, $e^a \to \Lambda^a{}_b(x)\, e^b$, without changing $g_{\mu\nu}$: this is the *local Lorentz freedom.* Of the tetrad's 16 components, 6 are that freedom and 10 are the metric. General relativity can be written with the tetrad as its only field [35]. A single unit timelike vector field added to the metric is different: it picks out a rest frame at every point. That is Einstein–aether theory [36], and its preferred-frame effects are tightly bounded, for example by the equal speeds of gravitational waves and light [37].

Free fall. Matter couples *minimally* when it feels gravity only through $g_{\mu\nu}$, with partial derivatives replaced by covariant ones. Then $\nabla^\mu T_{\mu\nu} = 0$, and a small body follows a geodesic, $\ddot{x}^\mu + \Gamma^\mu{}_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$ (dots: derivatives by proper time), whatever it is made of. That is the weak equivalence principle. Experiments compare the accelerations a_1, a_2 of two materials through the Eötvös ratio $\eta = 2|a_1 - a_2|/|a_1 + a_2|$ [38]; with indices, η is the Minkowski metric.

PPN parameters. In weak fields, metric theories are compared through the parametrised post-Newtonian (PPN) expansion [39]. With $U > 0$ the Newtonian potential,

$$g_{00} = -1 + \frac{2U}{c^2} - 2\beta\frac{U^2}{c^4}, \qquad g_{ij} = \Big(1 + 2\gamma\frac{U}{c^2}\Big)\delta_{ij}.$$

γ says how much space curvature a unit mass produces; light bending and the Shapiro delay scale as

$(1+\gamma)/2$. β measures the nonlinearity of gravity and enters the perihelion advance with γ. General relativity has $\gamma = \beta = 1$.

The lapse. Split into space and time, spacetime has a *lapse* N: the proper time that a static clock records per unit of coordinate time. In a weak potential $\Phi = -U$ it is $N = \sqrt{1 + 2\Phi/c^2}$, outside a mass M it is $N = \sqrt{1 - 2GM/rc^2}$, and it vanishes at a horizon.

Black holes. A stationary black hole is fixed by three numbers: its mass M, angular momentum J and charge Q [40, 41]. The rotating hole (Kerr) has its horizon at $r_+ = (GM/c^2)\big(1 + \sqrt{1 - a_*^2}\big)$, with spin $a_* = cJ/GM^2 \leq 1$ [41]. Its *surface gravity* κ is the acceleration, as measured from far away, needed to hold a body at the horizon; for a hole without spin $\kappa = c^4/4GM$. Quantum fields make the horizon radiate at $k_B T_H = \hbar\kappa/2\pi c$ [5]. The *area theorem*: with positive energy, the horizon area of classical general relativity never decreases [42]. It is the second law of black-hole mechanics [43], with entropy $S = k_B c^3 A/4G\hbar$ [6].

3.2 The frame

In the notation of Primer 2, the tetrad $e^a = e^a{}_\mu \, \mathrm{d}x^\mu$ defines the metric,

$$g_{\mu\nu} = \eta_{ab} \, e^a{}_\mu \, e^b{}_\nu, \qquad \eta = \mathrm{diag}(-1, 1, 1, 1), \tag{3}$$

and the metric fixes the tetrad only up to a local Lorentz transformation. General relativity can be written with the tetrad as its only field, in the Einstein–Cartan or the teleparallel form, with exactly Einstein's predictions [35] (P). Hypothesis 3 matters because a single timelike unit vector added to the metric would be Einstein–aether theory [36], whose preferred frame GW170817 bounds: $-3 \times 10^{-15} \leq c_{\mathrm{gw}}/c - 1 \leq 7 \times 10^{-16}$ [37] [Q17]. A frame of four has no preferred direction, provided that the records hold it up to local Lorentz transformations, that is, the metric: a record of one particular tetrad would single out its timelike leg. We read Hypothesis 3 that way.

3.3 The information field and the content

The information half holds every quantum field of matter and radiation, which this paper calls *content*, and one scalar ψ whose value sets the fine-structure constant. A scalar that multiplies the Maxwell term and leaves the gravitational term alone is Bekenstein's theory of a varying α [44]; with the scalar's energy fed back into cosmology it is the theory of Sandvik, Barrow and Magueijo [45] (P). The action whose shape the hypotheses fix is

$$S = \int \mathrm{d}^4x \, |e| \left[\frac{c^4}{16\pi G} R(e) - \frac{c^4}{16\pi G} \, \partial_\mu\phi \, \partial^\mu\phi - \frac{1}{4\mu_0} \, B(\phi) \, F_{\mu\nu} F^{\mu\nu} \right] + S_{\mathrm{content}}[e, \phi, \Psi], \tag{4}$$

where $|e| = |\det e^a{}_\mu| = \sqrt{-g}$, $R(e)$ is the curvature scalar of the metric (3), ϕ is ψ measured in units of the reduced Planck mass (so that its kinetic term is normalised like the gravitational one), $\alpha \propto 1/B(\phi)$, and Ψ stands for the quantum fields of matter. We write

$$\lambda \equiv \frac{\mathrm{d} \ln \alpha}{\mathrm{d}\phi} \tag{5}$$

for the scalar's coupling. The action is our construction; what the axioms fix about it is the subject of the next section.

3.4 The dictionary

DMP	GUID (physics)	Hypothesis
trust half; the trust field (Axiom 2), first reading	the frame $e^a{}_\mu$; gravity	1, 3
information half	quantum content Ψ and the scalar ϕ that sets α	4
a standing; mass of a member (T32)	what the frame couples to: $T_{\mu\nu}$	2
record, read in full by every witness (Axiom 3)	the value of the frame, held by every system it acts on	1
counts, seen by every reader (Axiom 3)	asymptotic charges M, J, Q	5
containing boundary; trap (T27, T31)	event horizon	5
entropy floor; temperature of a boundary (T31, T32)	thermal radiation of the horizon; Hawking temperature	5
G0, the minting	the start: inflation with the decay of its oscillons; the classical era begins as it completes	6
balanced state $\Phi_{\mathrm{V}} = 0$ (T32)	the vacuum at the minimum of its potential	Section 12
the trust field (Axiom 2), adopted reading	the frame, separate from what it contains and shaped by its counts	$1'$, 3
existence, in the narrow sense	the rest mass the Higgs vacuum grants	$1'$; Section 12.4

3.5 Names

Three words carry the account, and we use them in one sense each from here on. *The frame* is the gravitational field, the tetrad of Hypothesis 3, which the trust field of Axiom 2 becomes under Hypothesis $1'$. *Content* is the quantum state of matter and radiation, carried by the information half (Hypothesis 4). *The grant* is the rest mass an elementary particle has by its coupling to the Higgs vacuum, the narrow sense of "existence" in Hypothesis $1'$. A *record* is evidence that outlives its act and that the act's witnesses read in full; the *counts* are what every other reader sees of it, which for the frame are energy and momentum. Where earlier versions spoke of the "existence frame" or the "gravitational plane", this paper says *the frame*.

4 What the axioms fix

4.1 Free fall is universal

Proposition 1 (weak equivalence principle)**.** *Under Hypotheses 1 and 2, two bodies with the same energy-momentum fall alike in the frame, whatever they are made of.*

Proof. Axiom 3 (D): from trust nothing can be derived about information but its existence. By Hypothesis 1 the frame is the trust half, so no response of the frame to a body can depend on what the body carries (composition, internal state) except through the body's existence. By Hypothesis 2 existence is $T_{\mu\nu}$. So the frame couples to a body through $T_{\mu\nu}$ alone: the coupling is minimal, and then $\nabla^\mu T_{\mu\nu} = 0$ makes every small body, whatever it is made of, follow a geodesic

of the metric the frame defines. Bodies that differ in composition but share their initial position and velocity therefore share their worldline: the weak equivalence principle (P). □

Under Hypothesis 1′ the proof needs no reading of existence: the frame is shaped only by the counts every reader sees (Axiom 3), which are energy and momentum whatever carries them; with the same geodesic step free fall is universal, and the frame weighs granted and bound energy alike, as MICROSCOPE requires.

The principle is among the best-tested in physics. The MICROSCOPE satellite compared the free fall of titanium and platinum and found the Eötvös ratio $\eta = [-1.5 \pm 2.3\,(\text{stat}) \pm 1.5\,(\text{syst})] \times 10^{-15}$ [38]. Two consequences are sharp.

Gravity weighs all energy. About 9% of a nucleon's mass comes from the Higgs-given quark masses; the rest is the energy of quark motion and of the gluon field [46]. A gravity that weighed only the Higgs-given mass would make titanium and platinum fall differently by $\eta \approx 1 \times 10^{-3}$ [Q6], more than 10^{11} times what MICROSCOPE allows. The frame weighs the binding in full.

The information scalar may barely pull. A scalar that sets α changes the electrostatic energy of every nucleus, so each body of mass m_A carries a scalar charge $\partial \ln m_A/\partial\phi = \lambda f_A$, where f_A is the electrostatic share of its mass. With the normalisation of (4) the exchange of ϕ adds to the Newtonian force between A and B a fraction

$$\frac{F_\phi}{F_N} = 2\lambda^2 f_A f_B, \tag{6}$$

so two bodies falling towards the Earth (share $f_\oplus$) differ by $\eta = 2\lambda^2 f_\oplus (f_1 - f_2)$ (Dvali and Zaldarriaga [47]). With the Coulomb term of the semi-empirical mass formula, $f_{\text{Ti}} = 2.0 \times 10^{-3}$ and $f_{\text{Pt}} = 4.1 \times 10^{-3}$. The profile likelihood of MICROSCOPE together with the Eöt-Wash beryllium-titanium balance [48] then bounds the coupling at 95% confidence by

$$|\lambda| < 3.1 \times 10^{-5} \qquad \text{[Q6]},\ \text{[data:D1.1]}. \tag{7}$$

Verdict: derived (from Axiom 3 with Hypotheses 1, 2); *holds* against MICROSCOPE, Eöt-Wash and the lunar and planetary tests of Section 21.

4.2 The scalar stays out of gravity: $\gamma = 1$

Proposition 2 (light bending)**.** *If the information half cannot change how the frame responds to existence, the parametrised post-Newtonian parameter of the metric is $\gamma = 1$, as in general relativity.*

Proof. Axiom 1 (D): neither half overrules the other. A scalar that multiplies the curvature term, $F(\phi)R$, sets the local strength of gravity and so overrules the existence half; Axiom 1 excludes it. What remains is minimal coupling, as in (4): the scalar's energy is one more source of the frame, negligible in the solar system, and the post-Newtonian expansion is that of general relativity, $\gamma = 1$ [39] (P). □

In Brans–Dicke gravity [49], where the scalar does multiply the curvature, $\gamma = (1+\omega)/(2+\omega)$, and the Cassini measurement $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [50] forces $\omega > 4 \times 10^4$ at two standard deviations [Q7]: the scalar must decouple almost completely. Axiom 1 asks for that decoupling from the start. Two qualifications. The scalar's composition-dependent force (6) shifts the γ inferred from light bending and ranging by a term of order $\lambda^2 f_\odot f_{\text{body}} < 10^{-14}$, far below any measurement. And whether a term $F(\phi)R$ is present depends on the conformal frame in which the action is written, as Section 12.4 discusses for Higgs inflation; the physical content of the proposition is the bound (7) together with $\gamma = 1$ for the metric, not the form of the action. *Verdict: derived*; *holds.*

Primer 3: Quantum information

States and processes. A quantum state is a *density matrix* ρ: Hermitian, positive ($\rho \succeq 0$) and of unit trace. It is pure when $\rho = |\psi\rangle\langle\psi|$ and mixed otherwise. The state of one part A of a system AB is the partial trace $\rho_A = \mathrm{Tr}_B\, \rho_{AB}$ [51]. Every physical process is a *completely positive, trace-preserving* (CPTP) map, $\rho \mapsto \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$ (the identity); "completely" means that the map stays positive when it acts on one part of a larger system. A measurement with outcomes k returns k with probability $p_k = \mathrm{Tr}(K_k \rho K_k^\dagger)$ and leaves the state $K_k \rho K_k^\dagger / p_k$.

Dephasing is reading. Dephasing in a basis $\{|k\rangle\}$, $\rho \mapsto \sum_k |k\rangle\langle k|\rho|k\rangle\langle k|$, erases the off-diagonal elements (the coherences) and keeps the diagonal. It is what a measurement in that basis does to a system when the outcome is written somewhere else. A system that is dephased in one fixed basis after every interaction behaves as a classical variable.

Entanglement. A state of AB is *separable* if it is a mixture of products, $\rho_{AB} = \sum_i p_i\, \rho_A^i \otimes \rho_B^i$, and *entangled* otherwise [52]. Two numbers measure it for two qubits.

- The *negativity* $\mathcal{N}$ is the magnitude of the sum of the negative eigenvalues of the partial transpose ρ^{T_B} (the transpose taken on B alone) [53]. For two qubits $\mathcal{N} > 0$ exactly when the state is entangled [54, 55].
- The *concurrence* of a pure two-qubit state $a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$ is $C = 2|ad - bc|$, from 0 for a product to 1 for a maximally entangled state [56]. For pure two-qubit states $\mathcal{N} = C/2$.

An *entanglement witness* is an observable W with $\mathrm{Tr}(W\sigma) \geq 0$ for every separable σ; a measured $\mathrm{Tr}(W\rho) < 0$ certifies entanglement [57]. A given witness misses every entangled state for which $\mathrm{Tr}(W\rho) \geq 0$.

LOCC. In local operations and classical communication each party acts on its own system and the parties exchange only measurement outcomes. LOCC maps take separable states to separable states, and no entanglement measure grows under them [53, 52]. Creating entanglement between two systems therefore needs a carrier that is itself quantum.

No-cloning and no-broadcasting. No process copies two pure states that are neither equal nor orthogonal [58, 59]. A family of states can be *broadcast*, so that each of two readers ends up with the original, only if its members commute pairwise [60]. A quantity that several readers hold in full is therefore diagonal in one basis, which is the core of quantum Darwinism [24].

Worked example: a dephased mediator. Qubits A and B start in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and a mediator qubit M in $|0\rangle$. They interact only through M: a CNOT from A to M, a controlled-Z between M and B, and the same CNOT again.

- M *coherent.* The first CNOT gives $(|0_A 0_M\rangle + |1_A 1_M\rangle)/\sqrt{2}$. The controlled-$Z$ turns B into $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ in the second branch, and the second CNOT returns M to $|0\rangle$. What remains is $(|0\rangle_A|+\rangle_B + |1\rangle_A|-\rangle_B)/\sqrt{2}$, a maximally entangled state: $C = 1$, $\mathcal{N} = 1/2$.
- M *dephased* in $\{|0\rangle, |1\rangle\}$ after the first CNOT. The state becomes the mixture $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)_{AM} \otimes |+\rangle\langle +|_B$, and the same two steps leave $\frac{1}{2}(|0\rangle\langle 0|_A \otimes |+\rangle\langle +|_B + |1\rangle\langle 1|_A \otimes |-\rangle\langle -|_B)$. This is separable: $\mathcal{N} = 0$.

The dephased mediator carried one classical bit from A to B; that is LOCC, and it correlates the two qubits without entangling them.

4.3 The frame is classical

We need two theorems of quantum information.

Definition 2 (broadcasting)**.** A family of states $\{\rho_s\}$ on a system S can be *broadcast* if there is one physical process (a completely positive, trace-preserving map) $\mathcal{B} : S \to S_1 \otimes S_2$ such that both reduced states of $\mathcal{B}(\rho_s)$ equal ρ_s, for every s. For pure states and a product output this is *cloning*.

Lemma 1 (no-cloning; no-broadcasting)**.** *(i) A unitary that copies two pure states $|s\rangle, |t\rangle$, $U|s\rangle|0\rangle = |s\rangle|s\rangle$ and $U|t\rangle|0\rangle = |t\rangle|t\rangle$, exists only if $\langle s|t\rangle \in \{0, 1\}$ (Wootters and Zurek [58]; Dieks [59]). (ii) A family of states can be broadcast if and only if its members commute pairwise (Barnum, Caves, Fuchs, Jozsa and Schumacher [60]).* (P)

Proof of (i). Unitaries preserve inner products, so $\langle s|t\rangle = \langle s|t\rangle^2$, whose only solutions are 0 and 1. For (ii) see [60]; the "if" direction is the copier in the common eigenbasis, $|i\rangle \to |i\rangle|i\rangle$, which [Q3] runs: a diagonal state is broadcast, $|+\rangle$ is not (each reader is left with $I/2$). □

Lemma 2 (the frame is classical)**.** *Under Hypothesis 1, every value the frame takes after the first recorded act belongs to a pairwise commuting family: the frame admits no superposition of two of its values.*

Proof. Axiom 3 (D): every grant, use and loss of trust leaves a record that every witness of the act can read in full, and any other reader sees the counts always. Axiom 2 (D): every standing of the trust field is evidence held in such records. Axiom 1 (D): there are at least two independent witnesses. So every value of the trust field is held in full by at least two independent readers: it is broadcast. By Lemma 1(ii) the values commute. By Hypothesis 1 they are the values of the frame. Writing the value into records that are read in full also dephases the frame in the record basis after every act, which is the hypothesis of Lemma 3 below. □

Remark 1. The lemma is close to definitional: DMP demands absolute observability of its shared structure, and the no-broadcasting theorem states the price. It is not a discovery about nature; it is the statement of which side of an open question the axioms take. It has a precise physical counterpart in quantum Darwinism: a quantity of which the environment holds many redundant records is objective and classical, and generic quantum dynamics makes exactly the pointer observables so recorded [24, 25]. DMP's "every witness reads in full" is that redundancy, imposed as an axiom. Read in the other direction, the lemma is a design statement for information systems: a control structure that every party can audit in full is classical, while the data it governs need not be.

Under Hypothesis 1′ the lemma applies to the frame directly: the frame depends on the parts only through the counts, which every reader holds in full, so from the parts' point of view it is a measure-and-prepare channel (Lemma 3) and classical.

4.4 A classical mediator cannot entangle

Lemma 3 (classical mediation is LOCC)**.** *Let A and B be quantum systems that interact only through a mediator M, by alternating local interactions A–M and M–B. If M is classical in the sense that it is dephased in a fixed basis $\{|k\rangle\}$ after each interaction, then the induced map on AB is a local operation with classical communication, and it maps every separable state of AB to a separable state. (A diagonal marginal of M alone is not enough: a coherent mediator entangled with A can have one.)*

Proof. Dephasing M in its basis is equivalent to reading it there. An A–M interaction followed by that reading is an instrument on A: it takes ρ_A to the ensemble $\{p_k, \rho_A^{(k)}\}$ with the outcome k written in M. An M–B interaction with M in $|k\rangle$ is a channel $\mathcal{F}_k$ on B. One round therefore maps ρ_{AB} to $\sum_k (\mathcal{E}_k \otimes \mathcal{F}_k)(\rho_{AB})$ with $\mathcal{E}_k$ completely positive and $\sum_k \mathcal{E}_k$ trace preserving: a one-way LOCC map. Further rounds compose LOCC maps, which remain LOCC. LOCC maps take separable states to separable states, and no entanglement measure increases under them [53] (P). □

Galley, Giacomini and Selby prove the same statement in the generality of probabilistic theories: a mediator that entangles two quantum systems by local interaction cannot be classical [16]. Check [Q4] runs Lemma 3 on qubits: three random local couplings A–M, M–B, A–M leave A and B entangled in most of 300 trials when M is coherent (median negativity 0.12) and never when M is dephased after each coupling (largest negativity below 10^{-12}).

4.5 Where classicality begins: the start

G0 is stated, not derived, and the minting is the first act: before it there is no act and therefore no record (D). Lemma 2 rests on records, so it applies only once acts exist. GUID describes the beginning as one event, the *start*. For anything to begin to exist it must be possible for it to exist, and the first step of existing is to inflate, through the decay of the oscillons that coalesce once the initial energy has been emitted. Inflation and that decay cannot be separated or ordered, because there is no causality yet in which to order them; together they are the start.

Hypothesis 6 (the start)**.** Inflation and the decay of the oscillons it ends in are one event, the start, which is the minting. It has no internal order the axioms can refer to, and the classical era of Lemma 2, with causality in DMP's sense (acts between existing parts), begins as it completes. (H)

The physics of the start is standard (P). Inflation ends as the inflaton oscillates about the minimum of its potential; in a broad class of models self-resonance fragments it into oscillons, long-lived localised lumps of the field, which dominate the universe for a while (Amin et al. [61]); their decay into the particles of the Standard Model reheats the universe (Kofman, Linde and Starobinsky [62]). In physics these are stages of one field's history; GUID's statement that they form one event agrees with that, since the axioms, which need acts to order anything, have nothing by which to order them. Within the start the axioms place no restriction on the geometry, which may be in superposition; after it, every geometry the axioms define is recorded and classical. This matches the standard semiclassical picture, in which the geometry of the universe becomes classical through decoherence at its very beginning (Halliwell [63]; Kiefer [64]), and the no-boundary proposal, in which there is no "before" (Hartle and Hawking [65]). In GUID the start is therefore the boundary of quantum gravity in time: whatever is quantum about the geometry itself lies within it, where DMP by its own words makes no statement.

4.6 DMP can host the proposed account

Definition 3 (base configuration, host)**.** The *base configuration currently assumed for our universe* is what physics supplies and DMP does not derive, the configuration in DMP's sense (T4): the constants G, $\hbar$, c, Λ and α, the fields and parameters of the Standard Model, and the initial data of the ΛCDM model. A configuration that obeys DMP *hosts* a theory of gravity if its own shared structures (the trust field, the layout, the order of acts, the counts) carry the theory's classical degrees of freedom and its information half carries the theory's quantum content, so that nothing in the axioms prevents the configuration from reproducing what the theory predicts.

Proposition 3 (DMP can host the proposed account of gravity)**.** *Under Hypotheses 1–4, and supplied with the base configuration currently assumed for our universe, DMP hosts the theory of this paper: a classical frame over quantum content. In it, (i) the frame, carried by DMP's shared structures, mediates no entanglement between the systems it acts on; (ii) the information half can carry any quantum content.*

Proof. (i) DMP's shared structures are broadcast records (Axiom 3), so by Lemma 2 the field they carry is classical, whichever structure carries it, and by Lemma 3 a classical mediator creates no entanglement. (ii) Axiom 3 restricts what is *derived from trust* and how content moves (by grant); it does not require content to be broadcast, so nothing in the axioms forbids non-commuting states in the information half (D). A grant moves content without copying it, as no-cloning requires. The base configuration supplies the constants and the content that DMP does not derive. Part (ii) also uses the reading of "full reach over the act" as reach over the act's existence (Section 8), a hypothesis (H). □

The proposition is a statement of compatibility: it shows that nothing in the axioms obstructs the account, not that the axioms produce its dynamics. Two remarks place it. First, the constants: DMP derives none of them, and the proposition needs them supplied, as every physical theory needs its parameters (Section 29). Second, universality: whatever can run a universal quantum computer as content can also run a theory in which the geometry itself is quantum, by the Church–Turing–Deutsch principle [66] (see below); in such a theory the quantum geometry would sit in the information half and would not be DMP's frame. Part (i) is therefore the proposed theory's characteristic, falsifiable prediction, and Section 6 names the experiments that test it.

The Existence Turing machine. Proposition 3(ii) says that the information half can carry any quantum content. Assembled from DMP parts, a universal quantum computer can then run as content, and by the Church–Turing–Deutsch principle it approximates every finitely realisable physical system to any finite precision [66]; GUID calls such an assembly the *Existence Turing machine*. On a classical host this costs memory and time of order 2^n for n qubits of generic content, so the general route needs a quantum information half; Section 19 hosts four known quantum configurations classically and recovers their known results. Two consequences matter for gravity. Every candidate theory of quantum gravity whose predictions are finitely realisable can also run on the machine, as content: the axioms do not forbid quantum gravity in nature, but its geometry would not be DMP's frame. And the machine's own structure, its gates, records and layout, is classical: in the proposed theory it is the frame itself (Section 16).

4.7 Gravitational waves and the graviton

The frame's linearised excitations, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, are a massless spin-2 field. In D spacetime dimensions such a field has $D(D-3)/2$ propagating polarisations, two in 3+1 [Q8]. Weinberg's low-energy theorem [67] and Deser's self-coupling argument [68] show that a massless spin-2 field must couple equally to all energy and, made consistent, is general relativity (P): the spin-2 wave is the frame's own, and its universal coupling is Proposition 1 again. Its speed, its mass and its polarisations are tested in Section 21.

By Lemma 2, once the start is complete (Section 10) these waves are classical waves of the frame: GUID has no graviton as a quantum of the frame (D). That is consistent with every observation, for reasons the literature has made precise. Dyson showed that detectors of the LIGO type cannot register single gravitons, because mirrors heavy enough would collapse into a black hole [69]; Rothman and Boughn found that even a Jupiter-mass absorber near a neutron star would be swamped by neutrinos, and a shield against them would collapse too [70]. Every gravitational wave measured is deep in the classical regime: the energy GW150914 radiated corresponds to more than 10^{78} quanta at its frequency [Q9]. Localising any quantum inside its own Schwarzschild radius needs a wavelength below $\sqrt{4\pi}\,L_P$ and an energy above $\sqrt{\pi}\,E_P$ [Q9], the Planck scale at which measurement and gravity cannot be separated [71]; a LIGO-band graviton is 2×10^{-41} of that. The proposals to register single quanta of a gravitational wave as jumps of a cooled resonant bar [72] are a gravitational photoelectric effect, and like the photoelectric effect they can be explained by a classical wave acting on a quantised detector [73]. So the absence of the graviton in GUID is not testable by counting gravitons; it is testable only through the mediation experiments of Section 6 and, with Bridge B, through the decoherence test of Section 5.4. *Verdict: derived*; not testable by counting gravitons, only through those experiments.

5 How a classical frame can carry quantum matter

Proposition 3 places GUID among the theories in which spacetime is classical and matter quantum. Such theories have a known difficulty, and its resolution is a prediction.

5.1 The semiclassical equation fails

The simplest coupling, $G_{\mu\nu} = (8\pi G/c^4)\langle\hat{T}_{\mu\nu}\rangle$ (Møller [74], Rosenfeld [75]), sources gravity by the quantum expectation value. A mass in an equal superposition of two places then attracts a probe towards the midpoint, where the mass never is. Page and Geilker tested a macroscopic version, with the position of lead masses set by a quantum random event, and found the probe following the actual masses, not the average [11] (P). The equation is also nonlinear in the quantum state, which in general permits signalling. So a classical frame cannot simply be sourced by expectation values.

Primer 4: Classical-quantum dynamics and collapse

The classical-quantum (CQ) master equation. A consistent coupling keeps the joint description linear, positive and normalised, as probability theory demands. The joint state is a family $\varrho(z)$ of positive operators, one for each value z of the classical variables (here the frame), normalised as $\int \mathrm{d}z\ \mathrm{Tr}\,\varrho(z) = 1$ [12]. For one classical variable z driven by one quantum observable L (for gravity, the mass density), continuous Markovian dynamics has the form [13]

$$\partial_t \varrho = -\frac{i}{\hbar}[H,\varrho] + D_0\Big(L\varrho L - \tfrac{1}{2}\{L^2,\varrho\}\Big) - \frac{D_1}{2}\,\partial_z\{L,\varrho\} + \frac{D_2}{2}\,\partial_z^2\varrho,$$

with three coefficients beyond quantum mechanics:

- D_0, *decoherence*: the quantum system loses coherence between states that source the classical field differently (for gravity, different mass distributions);
- D_1, *back-reaction*: a drift of z driven by the quantum state, that is, how matter pushes the frame;
- D_2, *diffusion*: random kicks of z, that is, noise in the frame.

Complete positivity requires $D_1^2 \le 4D_0D_2$ here, and $D_0 > 0$ wherever $D_1 \neq 0$; Oppenheim et al. give the general matrix form of this trade-off [13, Eq. (23)], which is Eq. (8) of this paper. In words: *a push implies noise.* If quantum matter pushes a classical field at all, the matter decoheres and the field diffuses, and the product of the two is bounded below by the square of the push, $D_0D_2 \ge D_1^2/4$.

Measurement and feedback. An older family of consistent models makes gravity a channel. Each mass's position, or its mass density, is weakly and continuously measured, and a classical force computed from the record acts back on the other masses (Kafri, Taylor and Milburn, KTM [18]; Tilloy and Diósi [19]). Newton's law then holds on average, the measurement decoheres superpositions, and the feedback adds noise. The masses talk only through a classical record, so the channel is LOCC (Primer 3) and cannot entangle them. A weaker measurement decoheres less but gives a noisier record and so more heating; KTM find both effects at least of the order of the gravitational coupling. The least decoherence of two masses over all measurement strengths is what this paper calls the *KTM minimum*, $\omega/2$, where ω is the rate at which a coherent field would build up the entangling phase.

Gravity-related collapse. Diósi and Penrose proposed that a superposition of two mass distributions ρ_1, ρ_2 decays at the rate $\lambda = E_\Delta/\hbar$, where

$$E_\Delta = \frac{G}{2}\iint \mathrm{d}^3x\,\mathrm{d}^3y\,\frac{[\rho_1(\mathbf{x})-\rho_2(\mathbf{x})][\rho_1(\mathbf{y})-\rho_2(\mathbf{y})]}{|\mathbf{x}-\mathbf{y}|}$$

is the gravitational self-energy of their difference [76, 77, 78]. For a body of mass M and size R displaced by more than its size, $\lambda \approx GM^2/\hbar R$: the *Diósi–Penrose (DP) rate.* For point-like nuclei E_Δ diverges, so the mass density must be smeared over a length R_0. The same noise makes charged matter radiate. An underground germanium detector sees no such radiation, which excludes smearing at nuclear size and requires R_0 of order 10^{-10} m or more [79, 80]. In the Newtonian limit of CQ gravity the DP kernel is one admissible choice of the decoherence term [13].

5.2 Consistent classical-quantum dynamics

The consistent alternative evolves a joint state $\varrho(z)$ of the classical frame z and the quantum content by a map that is linear, completely positive and normalised. Oppenheim showed that such a dynamics exists and can couple a classical metric to quantum fields [12]; Primer 4 gives its master equation with the decoherence D_0, the back-reaction D_1 and the diffusion D_2. With matrices of coefficients, complete positivity forces the trade-off [13, Eq. (23)]

$$D_1\, D_0^{-1}\, D_1^{\dagger} \;\preceq\; 2D_2, \tag{8}$$

with D_0^{-1} the generalised inverse, and in addition D_0 cannot vanish where D_1 does not (P). Long coherence times of massive superpositions demand strong random fluctuations of the gravitational field, and weak fluctuations demand short coherence times. For a qubit coupled through σ_z to a classical momentum p, with the generator

$$\partial_t \varrho = D_0\,(\sigma_z \varrho \sigma_z - \varrho) - \frac{D_1}{2}\,\partial_p\{\sigma_z, \varrho\} + \frac{D_2}{2}\,\partial_p^2 \varrho, \tag{9}$$

the bound reads $D_1^2 \le 4D_0D_2$. Oppenheim et al. write the moments of the classical jump with factors $1/n!$, so their D_1 and D_2 are half of ours, and in their normalisation the same bound is (8). At equality the dynamics is exactly a measurement of σ_z with feedback, in which p is a perfect record and every block of the state stays pure (Section 16, [gravsim:G3]).

Corollary 1 (GUID diffuses)**.** *If the frame (Lemma 2) is sourced by quantum content (Hypothesis 4) through a dynamics that is linear, completely positive and Markovian, then mass superpositions decohere at a non-zero rate and the gravitational field fluctuates randomly, with* (8) *bounding the product from below.*

The corollary turns the classicality of the frame from a philosophical position into a quantitative one: in the Newtonian limit, the decoherence rates of massive superpositions measured in interferometry and the noise in precision measurements of gravitational forces together squeeze the remaining parameter space from both sides [13].

5.3 The record is a measurement, the layout is feedback

DMP's architecture coincides with the second, older family of consistent models, measurement and feedback (Primer 4): Kafri, Taylor and Milburn measure each mass's position and apply the recorded force to the other [18], and Tilloy and Diósi source a Newtonian potential from a continuously measured mass density [19]. Both decohere and add noise, as Corollary 1 requires, and for two masses the least decoherence is the KTM minimum $\omega/2$, reached where the heating is also least (Section 16, [gravsim:G4]).

Read through the dictionary, DMP's Axiom 3 has exactly this structure (D)+(H). Every act leaves a record that every witness reads in full: a measurement of the act's existence. Any other reader sees the counts: the measured mass density, coarse-grained. The layout of the system is "the median of that demand, recomputed by both halves of the layer and never by one": a classical function of the records, applied back to the system, which is feedback. From the parts' point of view the whole loop is a measure-and-prepare channel, which is why it cannot entangle (Lemma 3).

Proposition 4 (DMP is a measurement-and-feedback gravity)**.** *Under Hypotheses 1–4, the dynamics that DMP's axioms prescribe for the trust field is a continuous measurement of existence (energy-momentum) followed by classical feedback on the geometry, of the Kafri–Taylor–Milburn and Tilloy–Diósi type.*

The proposition fixes the architecture, not the constants. A measurement-and-feedback model has a free parameter, the measurement strength or equivalently the smearing length of the measured mass density, which sets the decoherence rate and the noise. DMP derives no constant (Section 29), and the most economical choice, the parameter-free Diósi–Penrose collapse [76, 77], is excluded by the absence of the spontaneous radiation it predicts in an underground germanium detector [79] (P). GUID must therefore carry a free strength, fixed by experiment or by a principle that ties DMP to physical units. *Verdict:* an identification (H) of the architecture, not a derivation of the dynamics; Section 5.4 computes the constant from the two principles GUID suggests.

5.4 Fixing the free constant

Proposition 4 fixes the architecture, not the constant. DMP cannot supply the constant by itself: its text makes every parameter configuration (T4, T32), and it has no unit of length, time or action. A number therefore needs one bridge principle that ties a DMP structure to a physical scale. We compute the two that GUID's own dictionary suggests. Neither is derived from the axioms; both are hypotheses (H), stated so that they can fail.

The balance rule. Both bridges use one structural rule. Axiom 1's balance moves every part towards its declared state, and the system's own acts never raise the VIBE potential (T32); read as dynamics, the frame adds no more noise than consistency requires. GUID therefore holds the trade-off (8) at equality,

$$2D_2 = D_1\, D_0^{-1}\, D_1^{\dagger}, \tag{10}$$

so that a measured gravitational decoherence would fix the spacetime diffusion, and conversely. Other classical-quantum theories leave the two free above the bound; (10) is GUID's own rule.

Bridge B: one record per Planck time. The second bridge reads DMP's tick as the Planck time: the frame records existence at Planck strength, so the coupling takes its value in Planck units, and the balance rule (10) holds. In the Newtonian limit Oppenheim et al. bound the coupling from both sides, from above by precision measurements of gravitational forces and from below by the observed coherence of massive molecules [13]. At Planck strength the discrete version of the dynamics has $(l_P^3/m_P)D_2 = m_P t_P \approx 10^{-51}\,\mathrm{kg\,s}$, below the allowed window 10^{-25} to $10^{-1}\,\mathrm{kg\,s}$: *excluded*. The continuous version with the Diósi–Penrose kernel has $l_P^2 D_2 = m_P^2/c = 1.6\times10^{-24}\,\mathrm{kg^2\,s/m}$, inside the allowed window 10^{-40} to $10^{-9}\,\mathrm{kg^2\,s/m}$: *allowed* [Q19]. With the trade-off at equality, a mass M of size R in a superposition of two well-separated positions decoheres at the rate

$$\lambda = \frac{M^2}{R\, l_P^2 D_2} = \frac{GM^2}{\hbar R}, \tag{11}$$

since $c/m_P^2 = G/\hbar$: the Diósi–Penrose rate [76, 77], up to an order-one factor (for a uniform sphere the exact self-energy gives $6/5 - R/\Delta x$ in place of 1). Here D_2 is the strength of the diffusion kernel of the Newtonian potential in Oppenheim et al.'s Newtonian-limit model, in their normalisation, and the two windows are theirs [13, Eqs. (44), (46), (47)]. For a fullerene it is 6×10^{-16} per second, far below what interferometry allows. For the diamond microsphere of the Bose et al. proposal ($10^{-14}\,\mathrm{kg}$, radius $0.88\,\mu\mathrm{m}$) it is 72 per second: coherence is lost in about 14 ms, while the entangling phase needs 8 to 25 s, so only $2\times10^{-3}\,\mathrm{rad}$ builds up [Q19]. Of the two versions of the bridge, the discrete one is excluded by the windows and the continuous one allowed; adopting the continuous one is a choice made with the windows in view. One length remains that the bridge does not fix: the mass density of each nucleus must be smeared over a length R_0, or the noise makes matter radiate more than an underground germanium detector sees; we adopt the bound of current analyses, $R_0 \geq 4\times10^{-10}\,\mathrm{m}$ [79, 80], at which the bulk term in (11) still dominates [Q19]. *Verdict: live* and sharp: with Bridge B, mesoscopic superpositions

decohere at the Diósi–Penrose rate. Section 21 shows how far today's superpositions are from testing it.

What the bridges show. Fixed by its own bridge hypotheses, GUID lands on known physics: Bridge A (Section 11) is refuted, and Bridge B is the Diósi–Penrose model, which predates this work, with its smearing length set by experiment. What GUID adds is a reading of that model, the balance rule (10) with a record at Planck strength, not a derivation: at Planck strength the only Newtonian decoherence rate is $G/\hbar$ times a self-energy, so landing on $GM^2/\hbar R$ is close to dimensional analysis. A number of its own would need a prediction in which GUID differs from both the classical-quantum theory and Diósi–Penrose. The most promising lead is DMP's two independent witnesses, which may imply two independent noise sources with a statistical signature that a single source lacks; we have not worked it out.

6 The experiments that decide

6.1 Gravitationally mediated entanglement

Bose et al. [14] and Marletto and Vedral [15] proposed to place two masses, each in a superposition of two positions, close enough that only gravity couples them, and to test whether they end entangled. Let each mass m be in $(|L\rangle + |R\rangle)/\sqrt{2}$ with the two positions Δx apart and the centres d apart on one line. A coherent Newtonian field gives each branch (i, j) the phase $\varphi_{ij} = Gm^2t/(\hbar d_{ij})$, where d_{ij} is the distance between the masses in that branch, so after time t

$$|\Psi(t)\rangle = \tfrac{1}{2} \sum_{i,j\in\{L,R\}} e^{i\varphi_{ij}(t)} |i\rangle_A |j\rangle_B. \tag{12}$$

Up to local phases this is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + e^{i\Delta\varphi}|11\rangle)$ with $\Delta\varphi = \varphi_{RL} + \varphi_{LR} - \varphi_{LL} - \varphi_{RR}$; its concurrence is $|\sin(\Delta\varphi/2)|$ and its negativity

$$\mathcal{N} = \tfrac{1}{2}\,|\sin(\Delta\varphi/2)|. \tag{13}$$

For the parameters of [14], $m = 10^{-14}\,\mathrm{kg}$, $\Delta x = 250\,\mu\mathrm{m}$, $d = 450\,\mu\mathrm{m}$, the entangling phase grows at $0.126\,\mathrm{rad/s}$: one radian after 8.0 s, maximal entanglement ($\mathcal{N} = 1/2$, $\Delta\varphi = \pi$) after 25 s [Q5]. A recorded, classical field (Lemma 2) in its projective extreme leaves each branch's phase but removes the coherence between branches; as a measurement-and-feedback channel it reproduces the branch phases on average and decoheres each mass at a rate of at least $\omega/2$, the classical-channel minimum. Either way the masses stay separable at all times, $\mathcal{N} = 0$ [Q5], [gravsim:G4].

GUID predicts $\mathcal{N} = 0$. With Bridge B of Section 5.4 it predicts more: the masses decohere at the Diósi–Penrose rate, about 72 per second for the proposed microspheres, long before an entangling phase can build up.

What a result would show. If decoherence at the Diósi–Penrose rate is real, a quantised field would not entangle two solid spheres either: the ratio of the entangling to the decoherence rate is fixed by the geometry and stays below what entanglement needs (Section 17). The first experiment to decide is therefore whether that decoherence exists, which tests Bridge B; only in its absence does the entanglement experiment separate classical from quantum gravity. A positive result would exclude every classical mediator. It would refute Hypothesis 1′, which puts the field that pulls laboratory masses into the frame; it would not refute DMP, which hosts either reading of that field (Section 6.5): a quantum geometry would then run on DMP as content (Proposition 3(ii)). Section 17 also checks the spin witness proposed to certify the entanglement. Two caveats are under debate. Aziz and Howl argue that classical gravity coupled to quantum matter fields can produce entanglement through virtual matter processes, at a rate far below the quantum-gravity rate [81]; Marletto, Oppenheim, Vedral and Wilson [82],

Diósi [83], Schneider, Huggett and Linnemann [84] and Gundhi, Infantino and Bassi [85] dispute it. And Christodoulou et al. showed that the entanglement follows from linearised quantum gravity on a fixed background with local mediation [86], which is the content-side mechanism of the dissent in Section 6.5. Neither changes the prediction: in GUID the frame is recorded, and Lemma 3 applies to it.

6.2 Quantumness without entanglement

Lami, Pedernales and Plenio showed that the quantum nature of gravity can be tested without creating entanglement: an LOCC inequality bounds how well any local classical channel can reproduce a gravitational evolution of coherent states of massive oscillators, such as torsion pendula, and a measured violation would exclude a classical mediator [87]. The test needs no large spatial superpositions. GUID predicts that the inequality holds.

6.3 Diffusion against decoherence

Corollary 1 gives a two-sided test. Interferometry with ever heavier molecules and levitated particles bounds the decoherence that gravity may cause from above; precision measurements of the gravitational force, which has been measured between masses as small as 90 mg [88], bound the random fluctuations of the field from above. Inequality (8) says the two cannot both be made arbitrarily small. GUID predicts that they cannot.

6.4 What existing experiments already show

Quantum matter responds to gravity as the framework requires: neutrons in a gravitational potential acquire the phase $mgA/\hbar v$ predicted for quantum matter in a classical field (Colella, Overhauser and Werner [89]), and ultracold neutrons above a mirror occupy discrete gravitational quantum states [90]. In both, the matter is quantum and the field is a given potential; neither tests whether the field itself can be in superposition. The current state of the field is reviewed by Carney, Stamp and Taylor [91]. *Verdict on GUID's classical frame: live, falsifiable.*

6.5 A dissenting reading

The same facts can be scored the other way. Split the geometry, as the background-field method does: the background is a DMP structure and classical; the linearised fluctuation, which carries the Newtonian pull between laboratory masses, is content between parts, and linearised quantum gravity on a fixed background already predicts the entanglement of the masses [86]. On that reading DMP's classical structures are the part of quantum gravity that is classical in quantum gravity too, the classical centre of a holographic code (Section 8), and the linearised field is quantum content on that classical centre. The two readings agree on every fact: the shared structures are classical, quantum content including a linearised field can move between parts by grant and stay entangled while every act is recorded, the axioms derive no constant, and they forbid coherent superpositions of backgrounds while their records exist. They differ in one place: whether the linearised field that pulls laboratory masses belongs to the frame (the proposed theory, Hypothesis $1'$, which predicts $\mathcal{N} = 0$) or to the content (the dissent, which predicts $\mathcal{N} > 0$ as long as the decoherence stays below $\omega/2$). DMP hosts either. If Diósi–Penrose decoherence is real, neither reading gives entanglement for two spheres, and the decoherence measurement comes first; only in its absence does the entanglement experiment decide between them.

7 Black holes

7.1 The horizon as containment

Hypothesis 5 maps DMP's containment onto the event horizon. The match is close. No act carries information across a containing boundary (T31); no causal curve leaves the horizon (P). The counts every outside reader sees (Axiom 3) are the hair a stationary black hole keeps: mass M, angular momentum J, charge Q (Israel [40]; Kerr [41]). Inside a Schwarzschild horizon the radial coordinate is timelike and every future-directed path ends at the singularity: everything inside derives its causal future from it, a local ground in the sense of G0. Whether crossing is survivable is answered by the equivalence principle: at the horizon of a supermassive hole the tidal stretch across a body is small; for a stellar-mass hole it is lethal before the horizon.

7.2 What a black hole gives up: Hawking–Penrose–Information radiation

The first law of black-hole mechanics (Bardeen, Carter and Hawking [43]) reads, with $G = c = 1$,

$$\mathrm{d}M = \frac{\kappa}{8\pi}\,\mathrm{d}A + \Omega_H\,\mathrm{d}J + \Phi_H\,\mathrm{d}Q, \tag{14}$$

with Ω_H the angular velocity and Φ_H the electric potential of the horizon. It separates the two ways a black hole loses mass. The rotational and electric terms can be extracted from outside at fixed or growing area: Penrose's process with particles [92], superradiant scattering with waves, and the Blandford–Znajek process with magnetic fields [93], the leading model of the jets of active galaxies. All of them work in the ergosphere, which reaches $r = 2GM/c^2$ at the equator and lies outside the horizon $r_+ = (GM/c^2)(1 + \sqrt{1 - a^2})$ for every non-zero spin [Q10]. The area term is released only quantum mechanically, as Hawking's thermal radiation, which lowers A.

Definition 4 (Hawking–Penrose–Information radiation)**.** The *Hawking–Penrose–Information (HPI) radiation* of a black hole is everything it gives up: the extraction of its rotational and electric energy from the ergoregion (the $\Omega_H \mathrm{d}J$ and $\Phi_H \mathrm{d}Q$ terms of (14)) and its thermal Hawking radiation (the $\kappa\,\mathrm{d}A/8\pi$ term).

In GUID terms the extraction is the frame being dragged along by the spinning hole, outside the horizon, so no act crosses it; the Hawking part is the floor of T31. What leaves carries the counts and no content, as Axiom 3 requires. Spinning holes are more efficient engines: the binding energy radiated by matter falling to the innermost stable orbit is $1 - \sqrt{8/9} = 5.7\%$ of its rest energy without spin, 32% at Thorne's limit $a = 0.998$, and approaches $1 - 1/\sqrt{3} = 42.3\%$ at maximal spin (Bardeen, Press and Teukolsky [94]; [Q10]). At fixed mass every additional count (spin, charge) shrinks the area, so the more an outside reader knows of a boundary, the less entropy it holds and the more of its energy can be extracted (P). The Hawking part is unobservable for astrophysical holes: a solar-mass hole is at 6.2×10^{-8} K, colder than the 2.7 K microwave sky, so it absorbs more than it emits; only holes lighter than about 0.6 lunar masses are hotter than the sky, and a solar-mass hole in empty space would take more than 10^{66} years to evaporate [Q11]. *Verdict: holds*; the Hawking part is *untested in nature*.

7.3 The information problem

If evaporation is unitary, the entropy of the radiation first rises and then falls after the Page time [7]; the island formula, derived from the gravitational path integral with replica wormholes, reproduces that curve [8, 9]. Those derivations sum over geometries, which is quantum gravity in the sense GUID's frame excludes. On the other side, a classical-quantum dynamics decoheres the quantum system (Corollary 1), so it is not unitary on the quantum system alone and does not by itself produce a Page curve. DMP's T31 as written, "what still leaves is entropy and

nothing else", and its T27, which destroys contained items at their scheduled date, side with the thermal picture. In GUID the question is therefore open in a specific form: whether the content held inside a horizon (which may be quantum, Proposition 3(ii)) returns in the radiation as content, while the frame stays classical. No observation reaches it. *Verdict: open*, in physics and in GUID.

8 Holography and quantum error correction

Primer 5: Holography and quantum error correction

Stabiliser codes. A quantum error-correcting code stores k logical qubits in n physical ones so that some errors can be undone. In a *stabiliser code* the code space is the joint $+1$ eigenspace of $n-k$ commuting products of Pauli matrices, the *stabilisers* [95]. Measuring them gives the *syndrome*. An error that anticommutes with a stabiliser flips its sign and so shows in the syndrome. The stabilisers commute with the logical operators, so the syndrome reveals nothing about the stored content. A code of distance d, written $[[n,k,d]]$, corrects any $\lfloor (d-1)/2 \rfloor$ unknown errors, or any $d-1$ *erasures* (losses at known places).

The five-qubit code. The smallest code that corrects every single-qubit error is $[[5,1,3]]$, with the stabiliser $XZZXI$ and its cyclic shifts [96, 97]. It corrects any two erasures, so its logical qubit can be recovered from any three of the five qubits and from no two: a $((3,5))$ quantum threshold scheme [98].

Perfect tensors and the HaPPY code. A tensor with $2m$ legs is *perfect* if every split of its legs into m and m is, up to normalisation, a unitary map. The encoding map of the five-qubit code, with its logical leg as a sixth, is such a tensor. Pastawski, Yoshida, Harlow and Preskill tile the hyperbolic disc with pentagons, place one perfect tensor on each, contract neighbouring legs, and leave one leg per tile in the bulk and the outermost legs on the boundary [99]. The network is an isometry from the bulk to the boundary, a toy model of the bulk-boundary correspondence of AdS/CFT [100].

Ryu–Takayanagi as a minimal cut. In AdS/CFT the entropy of a boundary region A is $S(A) = \mathrm{Area}(\gamma_A)/4G\hbar$ (with $c = k_B = 1$), where γ_A is the bulk surface of least area anchored on the edge of A [101]. In a network of perfect tensors the entropy of A, in bits, is for many regions the number of qubit legs cut by the smallest cut that separates A from the rest of the boundary [99].

Reconstruction. The bulk region between A and γ_A is the *entanglement wedge* of A. Every bulk operator inside it can be represented by an operator acting on A alone [102, 103]: the bulk is encoded redundantly, like a logical qubit. A code has *complementary recovery* when what cannot be reconstructed on A can be reconstructed on the rest of the boundary.

The area operator. Harlow showed that every code with complementary recovery obeys a quantum Ryu–Takayanagi formula (Eq. (15) in Section 8). Its area term is the expectation value of an *area operator* that lies in the centre of the algebra reconstructable on A: it commutes with everything that either side can do [22]. A *fixed-area state* is an eigenstate of this operator, and such states have a flat entanglement spectrum [104]. A general state of quantum gravity is a superposition of areas.

8.1 The strongest assembly

The axioms can be assembled as a holographic quantum error-correcting code on a fixed background, the kinematics of the HaPPY model [99]. A part ("no further statements about its constituents") becomes one block of the five-qubit code [96], which encodes one logical qubit in five physical ones with distance 3. Its encoding map is a *perfect tensor* with six legs: every bipartition of the six legs into three and three is a unitary map, so every three-leg cut carries three qubits of maximal entanglement [A1]. A layer, which Axiom 3 makes a tree rooted at its Duality, becomes a rooted tree of such tiles, one root and five children with twenty boundary legs. In the orientation we use, a child tile's logical leg carries the edge to its parent and one of its physical legs carries the information sitting there (the bulk leg); both tile maps are isometries [A2].

Minimal cuts. For a contiguous boundary region R, the entropy of the reduced boundary state equals the number of legs cut by the minimal cut through the tree that separates R from its complement [A2]. This is the Ryu–Takayanagi formula [101], $S(R) = |\gamma_R|/4G\hbar$ with γ_R the minimal bulk surface anchored on ∂R, on a tree. DMP's "fewest hops in total" selects the same minimal surfaces. The equality holds for every region, contiguous or not, when each bulk leg is left open (entangled with a reference); with the bulk legs fixed it fails for some non-contiguous regions, where a tile whose legs the cut splits two against two lets one classical bit past, and in the opposite orientation of the tiles it fails even for intervals (Section 18, [holo:H2]).

Reconstruction. The five-qubit code corrects any two erasures, so the logical qubit is recoverable from any three of the five physical qubits and from no two: a $((3, 5))$ quantum threshold scheme [98]. In mutual information with a reference R purifying the root, $I(R : \text{pair}) = 0$ for all ten pairs and $I(R : \text{triple}) = 2$ bits for all ten triples [A3]. This is entanglement-wedge reconstruction [102]: a bulk operator is reconstructable on any boundary region whose wedge contains it.

The two witnesses. The trust half and the information half each measure the stabiliser syndrome of the code. Stabilisers commute with the logical operators, so the syndrome says that content is present and in the code, never what it is: this is Axiom 3's existence clause, exactly. Both halves read the same syndrome; the gate passes only on two trivial syndromes; each of the 15 single-qubit faults gives a distinct syndrome on which both refuse, and correction restores the content with fidelity 1. That every single-qubit fault is caught is a property of this code's stabilisers, not of existence checks in general: a witness that reads only the existence sector detects faults that leave the sector and passes, by construction, a fault that changes the content inside it. The witnesses certify existence; they do not protect content (Section 19 shows both behaviours on sectors of up to 2.7 million states). If the two halves instead read the content itself, the content is disturbed (fidelity exactly 1/2 for every content, [holo:H5]), so "full reach over the act" must mean full reach over its existence [A4].

Two layers in one object. Two parts whose content is a logical Bell pair reach the Tsirelson value $2\sqrt{2} \approx 2.83$ of the CHSH combination on their content, while the gates' verdicts, deterministic syndrome readings, give exactly the classical value 2 [A5]. The gate is classical and the content quantum, at the same time and in the same assembly.

8.2 Where the assembly stops: the classical centre

In any code with complementary recovery, Harlow proved the quantum Ryu–Takayanagi formula

$$S(\rho_R) = \mathrm{Tr}(\rho\, \mathcal{L}_R) + S_{\mathrm{bulk}}(\rho, \mathcal{M}_R), \tag{15}$$

where $\mathcal{M}_R$ is the algebra of bulk operators reconstructable on R and the *area operator* $\mathcal{L}_R$ lies in the centre of $\mathcal{M}_R$: it commutes with everything either side can do [22]. DMP's structures recomputed by both halves are such central elements. Lemma 2 then says that in GUID the area is a number, never an operator with a spread of values: the axioms allow one fixed-area sector at a time. In holography fixed-area states are a special, though useful, class (Dong, Harlow and Marolf [104]); the general state of quantum gravity superposes areas. So the assembly reproduces the kinematics of holographic codes on a fixed background ([A1]–[A5]) and stops exactly at superpositions of geometry. Section 18 runs the assembly at scale.

The tree is not our space. Read as space, a tree's boundary is totally disconnected (the p-adic line of p-adic AdS/CFT [105, 106]), and no tree approximates a plane: an n-cycle needs distortion at least $n/3 - 1$ in any tree [107]. So a literal tree cannot be our space. Keeping

the hierarchy as the routing tree of each request over a graph with loops removes this objection and touches none of the results above. Read as the tree of histories, each branch one consistent record (decoherent histories [108]), the tree owes no spacelike structure at all.

9 Lower dimensions

Why 3+1? In D spacetime dimensions a massless spin-2 field has $D(D-3)/2$ polarisations and the Weyl tensor, the part of the curvature not fixed by local matter, has $D^2(D^2-1)/12 - D(D+1)/2$ components: two and ten in 3+1, none and none in 2+1 [Q8]. In 2+1 a mass does not attract another at all; it cuts a wedge out of space (Deser, Jackiw and 't Hooft [109]). Gravity there is, perturbatively, a Chern–Simons gauge theory [110], with a holographic boundary algebra for negative cosmological constant [111]; whether pure quantum gravity in 2+1 exists as a consistent theory is open [112]. Its degrees of freedom are global: the holonomies of the flat connection around particles and handles, which do not commute once quantised [113], and point particles scatter through their conical deficits [114].

So 2+1 is not an exception to Proposition 3. With no local field there is no local mediator to forbid, but the global holonomies are quantum, and a DMP structure that carries them is broadcast and so can carry one classical sector of them, never their superposition; with matter, the same rule decoheres the particles' relative positions. In GUID's terms, the frame in 2+1 would have nothing local that could be held as a classical record, and what it does carry is global and quantum. DMP can run 2+1 quantum gravity as content, but its frame cannot be 2+1 gravity; and our universe is 3+1, whose local degrees of freedom, gravitational waves, are measured [115]. *Verdict: holds as a reading*, with the dimension count as *holds*; a classical 2+1 frame with classical holonomies would also be consistent.

10 Cosmology

Primer 6: Cosmology

The expanding universe. On large scales the universe is homogeneous and isotropic. Distances grow with a scale factor $a(t)$, light from the past is redshifted by $1+z = a_0/a$, and the expansion rate $H = \dot{a}/a$ obeys, for a flat universe, $H^2 = 8\pi G\rho/3 + \Lambda c^2/3$. Each component's share is $\Omega_i = \rho_i/\rho_c$ with the critical density $\rho_c = 3H^2/8\pi G$; the value today is H_0.

ΛCDM. The standard model of cosmology adds to the known particles a cold, collisionless *dark matter* and a cosmological constant Λ. Six parameters fix it: the densities of baryons and dark matter ($\Omega_b h^2$, $\Omega_c h^2$), the angular size of the sound horizon, the optical depth to reionisation, and the amplitude A_s and tilt n_s of the primordial fluctuations [116].

Inflation. An early epoch of nearly exponential expansion, $a \propto e^{Ht}$, flattens and smooths the universe and stretches quantum fluctuations into the seeds of galaxies [117]. It is driven by a scalar field, the *inflaton*, with potential V, and its length is counted in *e-folds*, $N = \ln(a_{\rm end}/a_{\rm start})$. In *slow roll* the field creeps down a flat potential; with the reduced Planck mass $M_{\rm pl}$, the parameters $\epsilon = \frac{1}{2}M_{\rm pl}^2(V'/V)^2$ and $\eta_V = M_{\rm pl}^2 V''/V$ are small. The fluctuations then have a nearly scale-invariant spectrum with tilt $n_s \approx 1 - 6\epsilon + 2\eta_V$, and gravitational waves with a tensor-to-scalar ratio $r \approx 16\epsilon$ [118].

Reheating and oscillons. After inflation the inflaton oscillates about the minimum of its potential and decays into ordinary particles, which *reheats* the universe [62]. In a broad class of models it first fragments into *oscillons*, long-lived localised lumps of the field [61].

The probes. The cosmic microwave background (CMB) was released at recombination, $z \approx 1100$. Its acoustic peaks are the imprint of sound waves in the early plasma: their spacing fixes the sound horizon and their heights the densities [116]. The same sound horizon, about 150 Mpc today, is imprinted on the clustering of galaxies as *baryon acoustic oscillations* (BAO), a standard ruler for distance against redshift [119].

Dark energy. A cosmological constant has the equation of state $w = p/\rho c^2 = -1$ at all times. DESI fits the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1 - a)$ [120, 121]. A *thawing* dark energy has $w_0 > -1$ and $w_a < 0$: it was close to -1 early and weakens today.

Data names and tensions. P-ACT combines the CMB spectra of Planck and of the Atacama Cosmology Telescope (ACT, data release 6); P-ACT-LB adds CMB lensing (L) and DESI BAO (B) [122]. The H_0 *tension* is the disagreement between the Hubble constant inferred from the CMB with ΛCDM, near $67\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the value from a distance ladder of Cepheids and supernovae, near 73 [116, 123]. A ladder calibrated on the tip of the red-giant branch lies in between [124].

10.1 The strong split and the timeline

The start (Hypothesis 6, Section 4.5) contains one more event that orders everything after it.

Hypothesis 6′ (the strong split; refines Hypothesis 6)**.** The start also contains the split of the strong force from the electroweak force, and causality begins with that split. If the two were ever one force, the unification lies before time: within causality the two were never one. (H)

The Standard Model agrees within causality: in it the strong and the electroweak force are separate gauge interactions and never meet; run with the Standard Model's content alone, the three couplings do not meet at one point [125] (Section 24). A unification would leave three traces in the causal era, and they make the hypothesis testable in part (P). (1) *Magnetic monopoles.* The split of a unified force generically creates very heavy magnetic monopoles, about one per causal volume (Preskill [126]); had the split happened after inflation, the universe would be full of them. None has been seen: MACRO bounds their flux below $1.4 \times 10^{-16}\,\mathrm{cm^{-2}\,s^{-1}\,sr^{-1}}$ [127]. The split must therefore happen before or during inflation, and reheating must stay below the unification energy, which was one of Guth's reasons for inflation [117]. Hypothesis 6′ puts it exactly there: the ordering is not only allowed, it is required. (2) *Proton decay.* The heavy gauge bosons of a unified force, of mass M_X, still let protons decay after the split, with a lifetime of order $\tau \sim M_X^4/(\alpha_G^2 m_p^5)$ to within a factor of about ten: about 5×10^{35} years for $M_X = 10^{16}\,\mathrm{GeV}$ and $\alpha_G = 1/40$ [Q20]. Super-Kamiokande finds $\tau(p \to e^+\pi^0) > 2.4 \times 10^{34}$ years [128], which requires $M_X \gtrsim 5 \times 10^{15}\,\mathrm{GeV}$ in this estimate and excludes the simplest unified theory of Georgi and Glashow [129]; Hyper-Kamiokande will reach much further [130] (Section 21, [data:D8.1], [data:D8.2]). (3) *The couplings.* A real unification needs particles beyond the Standard Model that make the couplings meet. The bound on inflation, $V^{1/4} < 1.4 \times 10^{16}\,\mathrm{GeV}$ (Section 10.2), allows a split at the unification scale within the start [Q20]. *Verdict: holds as a reading*; the ordering is required by the absence of monopoles; the unification itself is *untested*, with proton decay as its live test.

The timeline. Read together, Hypotheses 6, 6′ and Section 12.4 give one sequence:

1. *The start.* The strong force splits from the electroweak force; inflation dilutes the monopoles of the split; the decay of the oscillons reheats the universe to below the unification energy. Causality begins.
2. *About* 10^{-11} *s, at 160 GeV.* The symmetric electroweak state becomes unstable; the Higgs vacuum takes over granting rest mass to the W, the Z and the matter particles, and the information field, the photon and with it $\alpha \approx 1/128$, splits out of the electroweak field.
3. *Afterwards.* α runs down to 1/137 at low energies; nucleosynthesis, recombination and the late universe follow.

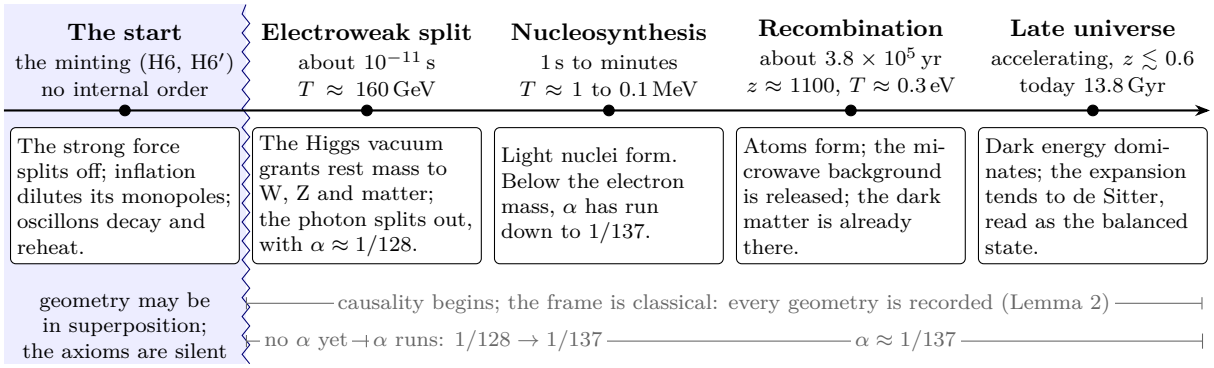


Figure 4: The timeline of Section 10.1, not to scale. The start (shaded; Hypotheses 6 and 6′) has no internal order the axioms can refer to; causality and the classical frame begin as it completes. The electroweak split follows at about 10^{-11} s and 160 GeV [131]: the Higgs vacuum grants rest mass, and the photon, and with it α, splits out of the electroweak field. Nucleosynthesis, recombination and the late universe are those of standard cosmology [132, 116].

10.2 Primordial gravitational waves

During inflation the vacuum fluctuations of the metric are stretched into a nearly scale-invariant background of gravitational waves (Starobinsky [133]). Their power spectrum is

$$\mathcal{P}_t = \frac{2}{\pi^2}\left(\frac{\hbar H}{M_{\mathrm{pl}}c^2}\right)^2 = \frac{16\,G\hbar H^2}{\pi c^5}, \tag{16}$$

with $M_{\mathrm{pl}}^2 = \hbar c/8\pi G$, linear in $\hbar$ and zero for a classical metric [Q14]; its ratio to the scalar spectrum is $r = 16\epsilon$ in single-field slow roll, with the consistency relation $n_t = -r/8$ for its tilt. Because (16) vanishes as $\hbar \to 0$, Krauss and Wilczek argued that detecting this background would establish that the gravitational field was quantised during inflation [134] (P). The current bound is $r < 0.036$ at 95% confidence (BICEP/Keck [135]; $r < 0.032$ combined with Planck and baryon acoustic oscillations [136]), which puts the energy scale of inflation below $V^{1/4} = 1.4 \times 10^{16}$ GeV and its Hubble rate below $\hbar H = 4.7 \times 10^{13}$ GeV [Q14].

The inflaton is not GUID's. The action (4) contains no inflaton, and Hypothesis 6 places inflation within the start, where the axioms place no restriction. GUID therefore takes the inflaton and its potential from physics. The models that fit the scalar spectrum best are plateau potentials, among them Starobinsky's R^2 inflation [137] and Higgs inflation [138]; both predict $n_s = 1 - 2/N$ and $r \approx 12/N^2 \approx 3 \times 10^{-3}$ at $N = 60$ e-folds [Q14], the target of the next generation of polarisation surveys [139]. Both are written with a scalar coupled to curvature in one frame and as a minimally coupled scalar in another. That is the form Proposition 2 excludes for the frame after the start; within the start the axioms do not constrain it, and afterwards the extra scalar of R^2 inflation is heavy, near 10^{13} GeV, so it leaves $\gamma = 1$ in the solar system untouched. We score the plateau prediction conditionally on that choice (Section 21).

Under Hypothesis 6 the tensor modes are generated within the start, where the axioms do not constrain the geometry, and become classical, as the standard account says they do through squeezing and decoherence [140], as the start completes. GUID therefore allows the standard spectrum and its consistency relation, and a detection would be the first observation of the quantum geometry of the start. GUID's own predictions about the geometry begin after the start. *Verdict: holds as a reading*; GUID predicts nothing about tensor modes beyond the inflation model it borrows.

10.3 The late universe: the balanced state

In the flat ΛCDM model that fits the data ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$ [116]), the Hubble rate falls at every moment from the past into the far future and tends to the de Sitter rate

$H_\infty = \sqrt{\Omega_\Lambda}\, H_0 \approx 0.83\, H_0$; the expansion has accelerated since $z = (2\Omega_\Lambda/\Omega_m)^{1/3} - 1 \approx 0.63$ [Q13]. A universe settling into de Sitter expansion cools towards an empty final state. GUID reads this as the cosmological balanced state, the equilibrium in which the need for further existence has fallen to a constant. *Verdict: holds as a reading* in ΛCDM. The newest data leave the end open: an evolving dark energy, the Hubble constant and the curvature are tensions of ΛCDM that GUID inherits unchanged (Section 21).

11 Dark matter: remnants trapped in the gravitational field

Hypothesis 7 (trapped remnants)**.** The modelling of this paper explicitly allows remnants of interactions to be trapped in the gravitational field: energy and momentum that interactions have left in the frame, or behind a boundary the frame has closed, continue to gravitate after the interaction. (H)

DMP motivates the hypothesis twice: a record outlives its act (Axiom 3), and a containing boundary keeps what it holds while every outside reader sees only its counts (T31). Dark matter, as far as the data show, is matter of exactly that kind: it gravitates and does nothing else that has been seen. Within the classical-frame modelling there are three routes by which remnants are trapped, and each can be calculated.

What every remnant must satisfy. Four measurements constrain every route (P). (1) It must already be there at recombination: the acoustic peaks of the microwave background measure the dark matter density at $z \approx 1100$, before the first star, as $\Omega_c h^2 = 0.120$, the same comoving value later structure requires, with the baryons then already at the share $\Omega_b/\Omega_m = 0.157$ [116] [Q15]. So the remnant formed before matter-radiation equality, consistent with formation in the start; the baryon content was fixed early, as nucleosynthesis and the microwave background agree. (2) It must be collisionless and move with the galaxies: in the merging cluster 1E0657−558 the lensing mass follows the galaxies, not the hot gas that holds most of the baryons (Clowe et al. [141]). (3) It cannot be the known neutrinos: they are less than 0.6% of the dark matter density, since cosmology bounds their mass sum below 0.064 eV [119]; a dwarf galaxy made of them would need each to weigh more than about 170 eV by the Pauli principle (Tremaine and Gunn [142]); and relic neutrinos move at about 3200 km/s and stellar ones at nearly c, against a Galactic escape speed of 533 km/s [143] [Q15]. (4) It must leave α unchanged to about a part per million over ten billion years (Section 12.1).

The stochastic remnant. Corollary 1 makes the frame diffuse: every interaction of quantum matter with the classical frame leaves a random kick in the field. Oppenheim and Russo computed the expectation value of the metric in classical-quantum gravity from its path-integral formulation, for static, spherically symmetric configurations [144]. Around a mass M the Newtonian potential acquires a linear and a quadratic term,

$$\Phi(r) = -\frac{GM}{r} - \frac{\gamma_0}{2} - \frac{\gamma_1}{2}\, r - \frac{\gamma_2}{2}\, r^2, \tag{17}$$

where the quadratic term is the cosmological constant of Schwarzschild–de Sitter space, $\gamma_2 = \Lambda c^2/3 > 0$, which repels, and the linear term is anticorrelated with it: its mean is negative, $\gamma_1 < 0$, so it attracts, with $|\gamma_1|$ of order $c^2\sqrt{\Lambda}$. Physically, where accelerations are low the variance of the field's acceleration exceeds the square of its mean and acts as an additional, entropic attraction. The circular speed is

$$v^2(r) = r\,\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{GM}{r} + \frac{|\gamma_1|}{2}\, r - \gamma_2\, r^2. \tag{18}$$

The first two terms balance at $r_* = \sqrt{2GM/|\gamma_1|}$, where

$$v_*^4 = 2GM|\gamma_1|. \tag{19}$$

Around r_* the rotation curve is roughly flat, changing by about 12% between $r_*/2$ and $2r_*$, and its height obeys $v^4 \propto M$, the form of the observed baryonic Tully–Fisher relation. For a galaxy of 10^{11} solar masses in baryons and $|\gamma_1| = 10^{-10}\,\mathrm{m/s^2}$, the order Oppenheim and Russo find, $r_* = 17\,\mathrm{kpc}$ and $v_* = 227\,\mathrm{km/s}$, with the Λ term below 10^{-4} of v_*^2 there [Q18]. The acceleration scale is tied to the cosmological constant: $c^2\sqrt{\Lambda/3}/2\pi = 8.6 \times 10^{-11}\,\mathrm{m/s^2}$, close to Milgrom's empirical $a_0 = 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ [145] [Q18]. Such a closeness is Milgrom's known coincidence, not support: GUID supplies no factor 2π (Section 23 counts how often a random scale comes as close). On this route a single mechanism, the frame's diffusion, would give both a dark-energy term and a dark-matter-like term. The calculation is preliminary, as its authors stress: it is static and spherically symmetric, depends on an infrared regularisation, fixes γ_1 by boundary conditions rather than by a coupling constant, and does not yet address gravitational lensing or the microwave background, and Hertzberg and Loeb dispute that it can produce flat rotation curves at all [146]. A deterministic relative of the same idea exists in tetrad gravity: Hehl and Mashhoon found that a memory of the frame, a nonlocal dependence on its past, acts in the linear approximation as dark matter given by an integral transform of the matter [147]; such a memory would make the effect depend on each galaxy's history.

Bridge A: the outermost horizon sets the scale. The first of GUID's two bridges (Section 5.4) fixes the stochastic remnant's scale. It depends on the outer cut-off $r_{\max}$ of the path integral. Oppenheim and Russo find, in units with $c = 1$, the mean $\gamma_1 = -\frac{9}{2}\,\gamma_2\, r_{\max}\,(1-4\beta)/(5-18\beta)$, with $\gamma_2 = \Lambda/3$ and a parameter $\beta < 0$ of their action to which the result is insensitive [144]. In GUID the outermost containing boundary is the cosmological event horizon, which in ΛCDM tends to the de Sitter radius $r_{\mathrm{dS}} = c/H_\infty = 1.66 \times 10^{26}\,\mathrm{m}$. With $r_{\max} = r_{\mathrm{dS}}$ and c restored,

$$|\gamma_1| = (0.90 \text{ to } 1.00)\; cH_\infty = (4.9 \text{ to } 5.4) \times 10^{-10}\,\mathrm{m/s^2}, \tag{20}$$

with no free parameter [Q19]. It has the right order of magnitude and is too strong. Matching the turnover speed (19) to the observed baryonic Tully–Fisher relation $v^4 = GMa_0$ requires $|\gamma_1| = a_0/2 = 6 \times 10^{-11}\,\mathrm{m/s^2}$; Bridge A exceeds this by a factor of 8 in v^4, 1.7 in v. For a galaxy of 10^{11} solar masses it gives 340 km/s at $r_* = 7.6\,\mathrm{kpc}$, where the relation gives 200 km/s, and the linear term lets the curve rise to 620 km/s at 50 kpc, where observed curves stay flat [Q19]. The estimate already rejects it; Section 22 confronts it with the data. Its near miss is Milgrom's coincidence $a_0 \approx cH/2\pi$, which is known.

The contained remnant. The second route is containment (Hypothesis 5). Energy that collapsed behind a horizon during the start gravitates through its counts alone and is dark by T31: primordial black holes. The start itself makes them. When the inflaton fragments into oscillons, which in a broad class of models dominate the universe for a while after inflation [61], enough of them can collapse into black holes of sublunar mass to make up all of the dark matter (Cotner, Kusenko and Takhistov [148]). Primordial black holes can be all of the dark matter only in the asteroid-mass window, about 10^{14} to $10^{19}\,\mathrm{kg}$ (10^{17} to $10^{22}\,\mathrm{g}$), which current constraints leave open [23]. The window's light edge is set by the Hawking part of their HPI radiation: a hole of $10^{14}\,\mathrm{kg}$ glows at $1.2 \times 10^9\,\mathrm{K}$, $k_BT \approx 0.1\,\mathrm{MeV}$, and its evaporation products (MeV gamma rays, and positrons that annihilate at 511 keV) bound how many such holes there can be; holes lighter than a few $10^{11}\,\mathrm{kg}$ have evaporated already [Q18]. At the heavy edge, $10^{19}\,\mathrm{kg}$ with a Schwarzschild radius of 15 nm and a temperature of $1.2 \times 10^4\,\mathrm{K}$, microlensing takes over [Q18]. The prediction of this route is specific: if dark matter is the contained remnant of the start, the Hawking part of HPI radiation, unobservable for astrophysical black holes (Section 7), becomes observable at the window's light edge. The window is taken from current constraints and the formation from Cotner et al.; this paper does not test either.

The radiated remnant. Gravitational waves are remnants of interactions that propagate in the frame. Pulsar timing has found a nanohertz background with characteristic strain $h_c \approx 2.4 \times 10^{-15}$ at the frequency 1/yr (NANOGrav [149]), which corresponds to $\Omega_{\rm gw} = 2\pi^2 f^2 h_c^2 / 3H_0^2 \approx 8 \times 10^{-9}$ per logarithmic frequency [Q18]: more than 10^7 times too little for the dark matter, and as radiation ($w = 1/3$) it does not cluster. It is a real remnant trapped in the field, but not the dark matter.

What Part III tests. Section 22 confronts the stochastic remnant and Bridge A with the rotation curves of the SPARC galaxies, and Section 23 an accumulated version, cold-dark-matter halos and the change with cosmic time; it also holds the dark-matter verdict. Section 21 confronts the other requirements.

12 The information scalar and matter

12.1 The fine-structure constant as the information field

In GUID α is set by the information scalar ϕ (Hypothesis 4), so it may vary where ϕ does. Three statements follow, and Sections 21 and 24 test them.

The forward statement: α does not grow. With the action (4) as written, with no potential for ϕ, this is *derived*. The scalar is then driven only by the electrostatic energy of the baryons, as in Bekenstein's theory, so the drift $\dot\alpha/\alpha$ is negative, proportional to λ^2 and to the small electrostatic share $f_b \lesssim 10^{-3}$ of baryonic mass, and at the bound (7) far below any clock. A drift large enough to measure needs a potential for ϕ, and then the sign is not predicted. A measured positive drift would refute the action as written. *Verdict: derived; untested.*

Only α varies. Because the scalar multiplies only the Maxwell term, the proton-to-electron mass ratio $\mu = m_p/m_e$ varies only through the electromagnetic self-energies of the proton and the electron. The ratio $R = \mathrm{d}\ln\mu / \mathrm{d}\ln\alpha$ is then small, at most a few hundredths in magnitude, against $|R|$ of 20 to 50 in models where all couplings vary together; its exact value depends on the scale at which the Yukawa couplings are held fixed (Section 24). If α is ever seen to vary, μ must vary far less. *Verdict: live.*

α has a beginning. Physically α does not exist before the electroweak split: there is no photon then, only the couplings of the electroweak field. It is born at the split, near 1/128 at the Z mass, and runs down to 1/137 at low energies (Section 12.4; [Q16], [Q20]). A value of one at any physical energy is beyond known physics: the Standard Model gives $1/\alpha \approx 105$ even at the Planck energy [Q16]. *Verdict: holds.*

Primer 7: Particle physics

The Higgs mechanism. The Standard Model is a gauge theory of $SU(3) \times SU(2) \times U(1)$: colour for the strong force, weak isospin and hypercharge for the electroweak force [150, 151, 152]. The Higgs field is a doublet whose potential has its minimum away from zero, at a vacuum value $v = 246\,\mathrm{GeV}$ [153, 154]. In that vacuum the W and Z bosons acquire masses proportional to v, while one combination of the electroweak fields, the photon, stays massless. Quarks and charged leptons get their masses from *Yukawa couplings* y_f to the doublet, $m_f = y_f v/\sqrt{2}$. The Higgs boson therefore couples to each fermion with strength m_f/v and to the W and Z with $2m_V^2/v$.

The κ framework. Measurements of Higgs production and decay are summarised by *coupling modifiers* κ_i, which scale each Standard Model coupling, so that a rate involving particle i scales as κ_i^2; the Standard Model has every $\kappa_i = 1$ [155].

Schemes and running. A coupling or mass in quantum field theory is defined only together with a *renormalisation scheme* and an energy scale μ. In the $\overline{\rm MS}$ scheme only the divergent part of each loop is subtracted, and parameters run with μ. The *on-shell* scheme instead fixes masses at the poles of the propagators and α at zero momentum transfer. The same quantity can differ between the two by more than its measurement error, so every comparison states its scheme [156, 132]. The

dependence on μ obeys the renormalisation-group equations (RGEs). At one loop $\mathrm{d}\alpha_i^{-1}/\mathrm{d}\ln\mu = -b_i/2\pi$, with $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ in the Standard Model (hypercharge normalised as in grand unification). The strong coupling therefore weakens at high energy, and the hypercharge coupling grows. Precision work runs at two or three loops [157].

Dirac or Majorana. A *Dirac* mass joins a left-handed neutrino to a right-handed partner and conserves lepton number L. A *Majorana* mass joins the neutrino to its own antiparticle and changes L by two. The lowest-dimension operator of Standard Model fields that gives neutrinos mass is Weinberg's dimension-five operator, $(c/\Lambda)\,(LH)(LH)$, with L the lepton doublet; in the Higgs vacuum it gives a Majorana mass of order cv^2/Λ, small when the new scale Λ is large [158]. Majorana masses can therefore also come from the Higgs vacuum.

Neutrinoless double-beta decay ($0\nu\beta\beta$) turns two neutrons of a nucleus into two protons and two electrons with no neutrinos. It changes L by two and can occur only if neutrinos have a Majorana mass. Through light-neutrino exchange its rate is proportional to $m_{\beta\beta}^2$, with the effective mass $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i|$ set by the mixing matrix U and the masses m_i [132]. Xenon and germanium experiments search for it [159, 160].

Grand unification. A grand unified theory (GUT) embeds the three gauge groups in one simple group, the smallest being $SU(5)$ [129], with a single coupling above a scale M_X. Its two generic traces, proton decay and magnetic monopoles, and the running of the couplings are given where Section 10.1 needs them.

12.2 The spin ladder

GUID reads spin as the rungs of the Duality, and the Standard Model fills each rung (P):

Spin	GUID reading	Particles	Physics
0	existence itself	Higgs field; the scalar ϕ	the Higgs vacuum value gives rest mass; ϕ sets α
1/2	a duality, which can stay in balance	quarks, leptons	a massive Dirac fermion is two chiral halves joined by its mass
1	information itself; delivered, an act	photon, gluons, W, Z	the carriers of every interaction
2	the perfect duality, two in one	the frame's wave	a massless spin-2 field must couple to all energy: general relativity [67, 68]

The rungs are told apart by rotation: a state of helicity s picks up the phase $e^{is\theta}$ under a turn by θ about its momentum, so a spin-1 quantum looks the same again after 360°, a spin-2 quantum after 180°, and a spin-1/2 state only after 720°, returning with a minus sign after 360°. The ladder stops at spin 1 for quanta: no elementary spin-3/2 or spin-2 quantum is expected, and searches find none (Section 24). *Verdict: holds as a reading*; the Standard Model says the same.

12.3 The Higgs vacuum as the grant of existence

In GUID existence is granted by the ground. The Higgs field is the one field whose value in empty space is not zero: with potential $V(H) = -\mu^2 H^\dagger H + \lambda_H (H^\dagger H)^2$ it rests at $\langle H^\dagger H\rangle = v^2/2$, $v = \mu/\sqrt{\lambda_H} = 246\,\mathrm{GeV}$, and an elementary particle has its rest mass by its coupling to that background [153, 154] (P). GUID reads the background value, which quantum field theory treats as a classical number, as G0's grant of rest mass, on the existence side and classical, in agreement with Lemma 2; the Higgs boson, a quantum ripple of the field that decays in about $10^{-22}\,\mathrm{s}$, is content. The vacuum at the minimum of its potential is the physical counterpart of the balanced state $\Phi_\mathrm{V} = 0$ of (2). Like G0 ("stated, not derived"), the Higgs potential is an input of the Standard Model, not a derivation.

Two consequences can be tested (Section 24). *The couplings follow the grant.* If the vacuum grants every elementary rest mass, each massive particle couples to the Higgs boson in proportion to its mass. *Neutrinos are Dirac.* If, in addition, lepton number is conserved, neutrino masses come from Yukawa couplings and neutrinoless double-beta decay does not occur at any rate. The addition is needed because the Weinberg operator (Primer 7) gives a Majorana mass from the same vacuum. We add one reading, not derived: the difference $B - L$ of baryon and lepton number is one of the counts every reader sees (Axiom 3), like electric charge, so it changes only when something carrying it crosses a boundary by a recorded act. Neutrinoless double-beta decay would create two units of lepton number inside a closed nucleus with nothing carrying them in, so it does not occur; proton decay, which conserves $B - L$, remains allowed. A signal at the next generation of experiments would refute the statement. *Verdict: live*, given the added reading.

The reading has one sharp limit, and it is gravitational. Gravity weighs all energy, granted and bound, while the Higgs field gives only about 9% of a nucleon's mass (Section 4), so the Higgs field is not the gravitational field. And gravity is far too weak to give mass: its coupling to the electron, $Gm_e^2/\hbar c = 1.75 \times 10^{-45}$, and its share of the binding of hydrogen, 4.4×10^{-40} of the electric one, are negligible [Q16]. *Verdict: holds as a reading* of the Higgs vacuum as the grant; *refuted* as an identity of the Higgs field with gravity.

12.4 The Higgs vacuum as the existence that separated out

GUID makes the Higgs field the leading candidate for the field that grants existence, the one that separated out when the fields did. Physics says when (P). Above about 160 GeV the Higgs field rests at zero, and no elementary particle has rest mass; lattice calculations find that, with the measured Higgs mass, the change is a smooth crossover at $T_c = 159.5 \pm 1.5$ GeV (D'Onofrio and Rummukainen [131]), about 10^{-11} s after the start [Q20]; the measured Higgs mass, 125.1 GeV, lies well above the end point of the line of first-order transitions [data:D8.6]. GUID's posit, in its form that holds, has four parts.

Before the split, existence asserted itself through interaction with existence. In the hot plasma before the crossover the particles carry only thermal masses, of order coupling times temperature, which come entirely from their interactions with everything around them. *Verdict: holds as a reading.*

That became unstable. As the universe cooled, the curvature of the Higgs potential at zero field turned negative and the field rolled to its present value $v = 246$ GeV; with a 125 GeV Higgs this is gradual, not a sudden phase transition. *Verdict: holds*, as a crossover.

The information field split out of the electroweak field. At the split the photon emerges as the fixed mixture $A = \sin\theta_W\, W^3 + \cos\theta_W\, B$ of the two electroweak gauge fields, and the electric charge is born as $e = g \sin\theta_W$, so that α appears near 1/128 at the Z mass [Q20] (Section 24). *Verdict: holds*, exactly.

The Higgs field became needed then. At that moment the W and Z take their masses from the vacuum, $m_W = gv/2 \approx 80$ GeV and $m_Z = m_W/\cos\theta_W \approx 91$ GeV at tree level [Q20], within about half a per cent of the measured values, depending on the scheme (Section 21), while the photon stays massless: the Higgs vacuum grants existence to the bosons that need it at the very moment the information field emerges. *Verdict: holds.* The Standard Model Higgs belongs to this split, the weak from the electromagnetic; a Higgs-type field for the earlier split of the strong force, if that split was a unification breaking (Hypothesis 6′), would be a second, hypothetical field.

An open question. In Higgs inflation the same field also drives the start: with a coupling $\xi H^\dagger H R$ of the Higgs to curvature, ξ of order 10^4, it fits the observed spectrum (Bezrukov and Shaposhnikov [138]), which would make the field that grants existence the field of the start itself. As for R^2 inflation (Section 10), the coupling to curvature is the form Proposition 2 excludes for the frame after the start, it disappears in the Einstein frame, and within the start the axioms

do not constrain it. Whether Axiom 1 says more is open.

12.5 The forces as tiers of one interaction

GUID reads the forces as tiers of one interaction: the strong force binds the cluster, the weak force keeps it stable, electromagnetism links clusters of clusters. Each does that job (P): the strong force binds quarks into nucleons and nucleons into nuclei, the weak force moves nuclei towards the valley of stability, electromagnetism binds atoms and molecules. The weak and electromagnetic forces are one electroweak interaction [150, 151, 152]. Whether all three meet at one energy is open: in the Standard Model the couplings cross pairwise between 10^{13} and 3×10^{16} GeV and do not meet at one point [125], [sm:SM4]. Gravity is not among the tiers in GUID: it is the frame the tiers act in. *Verdict: holds as a reading*; unification *untested*. In GUID's timeline the strong force splits from the electroweak force at the start (Hypothesis 6′), and the electroweak field splits into the weak and the electromagnetic force about 10^{-11} s later, when the Higgs vacuum appears (Section 12.4).

Primer 8: Hadron spectroscopy

Colour. Quarks carry the colour charge of $SU(3)$: a quark is a triplet $\mathbf{3}$, an antiquark an antitriplet $\bar{\mathbf{3}}$. Only colour *singlets* exist as free particles, the hadrons [132]. The products $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$ and $\mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6}$ give the ordinary hadrons: a meson $q\bar{q}$ in the singlet, and a baryon qqq, whose first two quarks couple to $\bar{\mathbf{3}}$ and then with the third to a singlet. One-gluon exchange attracts in the singlet $q\bar{q}$ and, half as strongly, in the $\bar{\mathbf{3}}$ qq, and repels in the $\mathbf{8}$ and the $\mathbf{6}$.

Diquarks. A *diquark* is a pair of quarks in the attractive $\bar{\mathbf{3}}$. The "good" diquark, with spin 0 and a flavour-antisymmetric pair such as $[ud]$, binds most; the "bad" one has spin 1 [161].

Compact or molecular. A hadron with more than three quarks, an *exotic*, can be *compact*, for example a diquark and an antidiquark bound by colour forces, $[qq][\bar{q}\bar{q}]$ [162]. Or it can be a *hadronic molecule*: two colour-singlet hadrons held together by the residual strong force, like the deuteron [163]. Molecules sit just below the threshold $m_1 + m_2$ of their two constituents.

The colour-spin interaction. In the constituent-quark model a hadron's mass is the sum of effective quark masses plus a hyperfine term proportional to $(\boldsymbol{\lambda}_i \cdot \boldsymbol{\lambda}_j)(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j)/m_i m_j$ for each pair, where the $\boldsymbol{\lambda}$ are colour and the $\boldsymbol{\sigma}$ spin matrices. This chromomagnetic interaction of De Rújula, Georgi and Glashow explains, for example, why the Δ is heavier than the nucleon [164]. Karliner and Rosner (KR) refined it with separate effective quark masses in mesons and in baryons, the baryon masses for every constituent of a compact multiquark, and binding terms for heavy pairs with $B(QQ) = B(Q\bar{Q})/2$; with it they predicted the doubly charmed baryon and doubly heavy tetraquarks [165, 166].

Thresholds and cusps. Where a new two-hadron channel opens, at $M = m_1 + m_2$, the scattering amplitude has a branch point. A *cusp* can then appear in a mass spectrum without any state, and a state that couples to the channel is pulled towards the threshold. Many exotic candidates lie within a few MeV of a threshold [163].

Names. The Particle Data Group names hadrons by their quantum numbers, not by their structure [132]; the $X(3872)$, for example, is listed as $\chi_{c1}(3872)$. LHCb's convention for exotics [167] uses T for four-quark and P for five-quark states, with subscripts for heavy flavour, ψ standing for hidden charm $c\bar{c}$. So T_{cc}^{+} is $cc\bar{u}\bar{d}$, seen just below the $D^{*+}D^{0}$ threshold [168], and the hidden-charm pentaquark $P_c(4312)^{+}$ [169] becomes $P_{\psi}^{N}(4312)^{+}$.

String junctions. In a baryon the three colour flux tubes meet at a point, a *string junction* [170]. A compact tetraquark has two junctions and a compact pentaquark three, and some models add a constant energy per junction [171].

12.6 Hadrons as trees of dualities

Axiom 1 makes every part a Duality of two halves, and Axiom 3 makes every layer a tree rooted at its own Duality (D). Read for the strong force, which binds the cluster, the two give one more hypothesis.

Hypothesis 8 (hadrons as trees of dualities)**.** Every hadron is a binary tree of two colour-conjugate halves, each internal node a bound duality: a meson is $(q)(\bar{q})$, a baryon $(q)[qq]$, a tetraquark $[qq][\bar{q}\bar{q}]$ (compact) or $(q\bar{q})(q\bar{q})$ (molecular), a pentaquark $[qq][qq]\bar{q}$ or $(qqq)(q\bar{q})$. (H)

We call it H8 for short. QCD's colour algebra already allows every colour singlet to be written as such a tree, so the statement as such holds as a reading. Its content lies in the masses a colour-spin model gives the compact trees, in how close the molecular ones lie to two-hadron thresholds, and in the partner states a tree implies. Section 25 tests all three with a standard colour-spin model on the compact trees, fitted to ordinary hadrons only. *Verdict: holds as a reading*; the tests of its content are reported in Section 25.

G0 gives a sharper reading of the five-quark hadrons. The dual gate cannot witness its own first instance: "somebody has to trust the first two witnesses", and the minting makes at least five parts the holders of the root keys (Section 2).

Hypothesis 9 (the minting in hadrons)**.** A pentaquark is a minting: a bound triple of colour-bearers that together carry all three colours, and the first two witnesses, a quark and its own antiquark, so that the pentaquark's antiquark has a partner quark of its own flavour. (H)

We call it H9. It reads the structure of G0 at the scale of one hadron; it does not identify a pentaquark with the start of Hypothesis 6, which is G0 for the universe. QCD requires the three colours and some quark-antiquark pair in every pentaquark; it does not require the pair to be of one flavour. H9 does, so it forbids pentaquarks whose antiquark has no partner, such as the Θ^+ $(uudd\bar{s})$ or $ccuu\bar{d}$, but not a doubly charmed state such as $ccud\bar{u}$, and it predicts hidden-bottom pentaquarks $(uudb\bar{b})$ as partners of the observed hidden-charm ones. It was conceived after seeing that every established pentaquark carries a $c\bar{c}$ pair, so those states are not evidence for it; the test lies in the states not yet established (Section 25).

13 Readings that add no prediction

Several of GUID's identifications reproduce established physics without adding anything a measurement could tell apart. They show that the reading is consistent, not that it is right, and we collect them in one table; Appendix G gives each in full.

14 Relation to the approaches to quantum gravity

Table 3: The approaches to quantum gravity and GUID.

Approach	What is quantum	Relation to GUID	What tells them apart
Effective field theory [3, 4]	metric fluctuations below E_P	GUID keeps the classical part; the graviton-loop term of (1) is absent	a term 10^{-70} of Newton's at 1 m [Q1]: unmeasurable
String theory, AdS/CFT [100]	everything; geometry emerges from a quantum boundary	hostable as content; the code kinematics of Section 8 are shared, with a classical centre	entanglement of masses, if decoherence $< \omega/2$
Loop quantum gravity [172, 173]	the triad/tetrad; area has a discrete spectrum	the frame is LQG's own variable, not quantised; area a number	area spectrum (not accessible); BMV if decoherence $< \omega/2$
Asymptotic safety [174, 175]	the metric, with a UV fixed point	hostable	BMV if decoherence $< \omega/2$
Causal dynamical triangulations [176]	sum over geometries	allowed within the start only	BMV if decoherence $< \omega/2$

Approach	What is quantum	Relation to GUID	What tells them apart
Causal sets [177]	discrete order; $\Lambda \sim 1/\sqrt{N}$	DMP's order is a tree, not a sprinkled causal set	Lorentz invariance needs random sprinkling
Thermodynamic gravity [21]	the microscopic entropy only	GUID's thermal reading, Section G.1	none beyond GR
Classical-quantum gravity [12, 13]	matter only	GUID's dynamics, Corollary 1	diffusion against decoherence
Channel models [18, 19]	matter only	GUID's architecture, Proposition 4	noise and decoherence rates
Gravity-related collapse [76, 77]	matter, collapsing	Bridge B lands on its rate; nuclear-scale smearing excluded, $R_0 \geq 4 \times 10^{-10}$ m allowed [79, 80]	decoherence of a 10^{-14} kg superposition (Section 17)
Semiclassical gravity [74, 75]	matter; metric sourced by $\langle T \rangle$	rejected: nonlinear; Page–Geilker	done [11]
Modified dynamics (MOND) [145]	nothing; a law at low acceleration	the stochastic remnant gives a term of the size a_0; tested in Sections 22 and 23	lensing, acoustic peaks, merging clusters, planetary ephemerides
Spacetime from entanglement [178, 179]	content; geometry emergent	GUID's frame as the classical centre of such a geometry is open	unclear

15 What Part I predicts and where it is tested

Table 4 closes Part I with every prediction that an experiment or an observation can reach, the alternative it must be told apart from, and where Parts II and III test it. Part IV collects the verdicts.

Table 4: The test plan: predictions of Part I, their alternatives and where they are tested.

Observable	GUID	Alternative	Tested in
The frame			
Free fall; the information scalar	universal; $\eta = 2\lambda^2 f_\oplus \Delta f$, $\lvert\lambda\rvert$ bounded	universal	Sec. 21
Post-Newtonian γ, β, Nordtvedt η	as in general relativity	as GR	Sec. 21
Gravitational waves	speed c, massless, two tensor modes	as GR	Sec. 21
Entanglement of two masses through gravity	none, $\mathcal{N} = 0$	quantised metric: $\mathcal{N} = \frac{1}{2}\lvert\sin(\Delta\varphi/2)\rvert$	Sec. 17
Decoherence of mesoscopic superpositions (Bridge B)	rate $GM^2/\hbar R$	none required	Secs. 17, 21
LOCC inequality, massive oscillators	satisfied	can be violated	proposed
Decoherence against diffusion	both non-zero, trade-off at equality	neither required	Sec. 16
Black holes and holography			
Kerr holes	efficiency $\leq 42.3\%$; $\lvert a\rvert \leq 1$; area never shrinks	as GR	Sec. 21
Holographic assembly	minimal cuts, reconstruction, classical centre	quantum code kinematics	Sec. 18

Observable	GUID	Alternative	Tested in
Cosmology			
Late universe	de Sitter floor, $w = -1$	evolving dark energy	Sec. 21
Scalar tilt (plateau inflation, borrowed)	$n_s = 1 - 2/N$	other potentials	Sec. 21
Primordial tensor modes	standard spectrum, made within the start	$r = 16\epsilon$	Sec. 21
Relic monopoles; proton decay	none; possible from a unification before the start	—	Secs. 21, 24
Dark matter			
Stochastic remnant; Bridge A	flat curves, $v^4 \propto M$, scale from the horizon	halos; RAR	Secs. 22, 23
Contained remnant	primordial black holes, asteroid-mass window	particle dark matter	Sec. 23
The information scalar and matter			
Drift of α	zero or negative	none predicted	Sec. 21
Variation of $\mu = m_p/m_e$ with α	small, $\lvert R\rvert$ below a few hundredths	unification: $\lvert R\rvert = 20$ to 50	Sec. 24
Higgs couplings	proportional to mass	Standard Model	Sec. 24
Neutrinoless double-beta decay	none (with $B - L$ as a count)	Majorana masses	Sec. 24
Unification of the gauge couplings	none within the Standard Model's running	grand unification	Sec. 24
Elementary spin	0, 1/2 and 1 only	Standard Model	Sec. 24
Exotic hadrons (H8)	compact-tree masses; molecules near thresholds; partners	QCD models	Sec. 25
Pentaquarks (H9)	hidden flavour only; hidden-bottom states exist	open flavour allowed	Sec. 25
Hawking radiation of real holes	thermal; return of content open	Page curve	not observable

Part II
The theory as simulated

16 Gravity simulated: the recorded frame against the coherent field

This section compares the frame, modelled as measurement and feedback (Proposition 4, Section 5), with a coherent Newtonian field for many masses (G1), and validates the simulation on two known results: the trade-off of Oppenheim et al. (G3) and the Kafri–Taylor–Milburn channel (G4). The frame's trajectories are product states by construction, and G3 and G4 reproduce published theorems, so the new content is the many-body comparison here and the two-mass analysis of Section 17.

16.1 Two models of the same masses

Masses on a line. We take N masses m on a line with spacing d. Each mass is a two-level system with branch $s_i \in \{0, 1\}$ and position $x_i = i\,d + (s_i - \frac{1}{2})\Delta x$, so $\delta = \Delta x/d < 1$. Phases are measured in units of $\omega_0 = Gm^2/(\hbar d)$, and $\tau = \omega_0 t$. A coherent Newtonian field gives the branch

Table 2: Readings that agree with established physics and add no prediction. Verdict: *holds as a reading* for each.

Reading	What it reproduces	Check
The existence a boundary asserts, gh/c	$4\pi^2$ times the Unruh temperature; the Hawking temperature at a horizon	[Q1]
Time slows where the frame is strong	the lapse; the Pound–Rebka and clock redshifts	[Q2]
Einstein's equation as an equation of state	Jacobson's derivation from the Clausius relation; the constant is open	(68)
Entropy as the cost of converting existence and information	Landauer's principle; DMP's entropy floor (T31)	
The containment split of a horizon	Christodoulou's mass formula for the Kerr hole	(69), [Q10]
The early universe a Kugelblitz	the Hubble mass at its own Schwarzschild radius, at every epoch	[Q12]
Inflation as the possibility space of existence	the flattening of curvature over about 60 e-folds	[Q13]
α as the VIBE coefficient	α as the self-interaction strength of charge; $a_e = \alpha/2\pi$	[Q16]

configuration s the phase $\tau\,\Omega(s)$, with $\Omega(s) = \sum_{i<j} d/|x_i(s_i) - x_j(s_j)|$. From $|+\rangle^{\otimes N}$ the state is

$$|\Psi(\tau)\rangle = 2^{-N/2} \sum_{s\in\{0,1\}^N} e^{i\tau\Omega(s)}|s\rangle. \tag{21}$$

Since every s_i takes two values, every pair term is affine in each bit, and Ω is exactly an Ising form:

$$\Omega(s) = \text{const} + \sum_i h_i s_i + \sum_{i<j} J_{ij} s_i s_j, \qquad J_{ij} = -g(|i-j|), \tag{22}$$

$$g(n) = \frac{1}{n-\delta} + \frac{1}{n+\delta} - \frac{2}{n} = \frac{2\delta^2}{n(n^2-\delta^2)}.$$

The local fields h_i collect $1/(n-\delta) - 1/n$ from every partner on the right and $1/(n+\delta) - 1/n$ from every partner on the left. A direct sum over all 256 branches at $N = 8$ agrees with (22) to 10^{-14} [gravsim:G1]. The nearest-neighbour coupling $g(1) = 2\delta^2/(1-\delta^2)$ is the entangling rate of an isolated pair, the BMV (Bose–Marletto–Vedral) rate of Section 17. The couplings fall as $g(n) \approx 2\delta^2/n^3$.

Measuring every mass in the X basis after the diagonal phases implements $H^{\otimes N} e^{i\tau\Omega} H^{\otimes N}$ on $|0\rangle^{\otimes N}$. This is an IQP (instantaneous quantum polynomial-time) circuit [180]: commuting diagonal gates between two layers of Hadamard gates.

The instance. We use the instance of Section 6, so $\delta = 5/9$, with a uniform diamond sphere of radius $R = 0.879\,\mu$m. The G1 phase unit is $\omega_0 = 0.1406\,$rad/s. An isolated pair reaches the phase π at $\tau_\pi = \pi/g(1) = 3.52$. The Bridge-B rate of Section 17 is $\lambda = (d/R)\,\omega_0 = 511.7\,\omega_0$ [gravsim:G1].

The recorded frame. The frame of Proposition 4 is modelled as follows (H).

- Each mass j is measured continuously in its branch basis. This contributes the Lindblad term $\kappa(Z_j\rho Z_j - \rho)$ with $Z_j = 1 - 2s_j$, which decays its coherence at 2κ.
- The measurement leaves the record $\mathrm{d}Y_j = 2\sqrt{\kappa}\,\langle Z_j\rangle\,\mathrm{d}t + \mathrm{d}W_j$, with independent Wiener increments $\mathrm{d}W_j$.
- The frame turns each record into a position estimate, $\hat{s}_j\,\mathrm{d}t = \frac{1}{2}(\mathrm{d}t - \mathrm{d}Y_j/2\sqrt{\kappa})$.
- It acts on mass j with the classical Newtonian field of the others' estimates: the phase of branch $s_j = 1$ relative to $s_j = 0$ advances at $h_j + \sum_i J_{ij}\hat{s}_i$.

Nothing else couples the masses. Bridge B sets $2\kappa = \lambda$. The record noise is the only noise the frame adds, which is the balance rule (10); Section 16.3 shows that this is exactly the equality case of the trade-off.

We sample the records exactly. Each mass's branch is drawn with its Born probability. Then $Y_j(t) = 2\sqrt{\kappa}\,z_j t + W_j(t)$, where $z_j = \pm 1$ is the drawn branch. The conditional state of mass j is a function of Y_j alone, and the frame's phases are linear in the W_i. This representation is standard for a measured observable that commutes with the dynamics. We also check it against an Euler integration of the joint stochastic Schrödinger equation. A time step costs three $N \times N$ matrix products, one for the phases and one for each branch-dependent force, so the cost is $O(N^2)$ per step.

The averaged frame. Averaged over records, measurement with Markovian feedback obeys the Wiseman–Milburn master equation [181]. Here every operator is diagonal in the branch basis, so it integrates in closed form:

$$\rho_{ss'}(\tau) = \rho_{ss'}(0)\exp\Big\{\tau\Big[i\big(\Omega(s) - \Omega(s')\big) - \kappa\sum_k (1 - z_k z'_k) - \tfrac{1}{2}\sum_i \big(F_i(s) - F_i(s')\big)^2\Big]\Big\}, \quad (23)$$

with $F_i(s) = (4\sqrt{\kappa})^{-1}\sum_j J_{ij}s_j$. The three terms in the exponent are:

1. the coherent field's commutator $-i[H_Q, \rho]$, which the frame reproduces exactly on average;
2. the measurement dephasing;
3. the record noise fed back as force noise, which dephases masses that share a record in a correlated way.

Mass j loses coherence at $\Gamma_j = 2\kappa + \sum_i J_{ij}^2/(32\kappa)$. This is at least $\frac{1}{2}(\sum_i J_{ij}^2)^{1/2}$, with equality at $\kappa = (\sum_i J_{ij}^2)^{1/2}/8$. For a single partner the minimum is $\omega/2$, half the entangling rate. This is the qubit form of the Kafri–Taylor–Milburn (KTM) minimum [18], used again in Section 16.4 and Section 17.

16.2 Many masses: what each picture produces

The coherent field builds volume-law entanglement, slowly. We build (21) exactly for $N = 8$ to 24. The largest case is a statevector of 2^{24} amplitudes, 256 MB in double precision. Let S be the entropy, in bits, of the reduced state of the first $N/2$ masses. We compare it with the Page entropy, the mean of S over random pure states [182]: $S_{\mathrm{Page}} = [\psi(q^2 + 1) - \psi(q + 1) - (q - 1)/2q]/\ln 2$, with $q = 2^{N/2}$ and ψ the digamma function. Table 5 shows how S grows [gravsim:G1]. At τ_π the cut carries about one ebit (one maximally entangled pair): 1.009 at both $N = 20$ and 24, the bond across the cut. S then grows as longer couplings wind past 2π. Over twelve times with $\tau = 10^4$ to 10^7 it averages 9.282 at $N = 20$ (Page: 9.279); at $N = 24$ and $\tau = 10^5$ it is 11.270 (Page: 11.279). Because the couplings fall as $1/n^3$, the state stays close to an area law at the times a laboratory can reach ($t \lesssim 100\,\mathrm{s}$, $\tau \lesssim 14$). It reaches Page entanglement only after hours.

Table 5: Half-chain entropy (bits) of the coherent-field state (21), $N = 20$, against the dimensionless time $\tau = \omega_0 t$ and the physical time of the instance [gravsim:G1]. The last column is the mean over twelve times between $\tau = 10^4$ and 10^7; the individual values lie between 9.0 and 9.4. Page's value is 9.279 bits.

τ	0.58	3.52 (τ_π)	46	268	1550	8960	10^4–10^7
t	4 s	25 s	5.5 min	32 min	3.1 h	18 h	20 h–2.3 yr
S (bits)	0.12	1.01	2.10	3.89	6.59	9.35	9.28

Pairwise entanglement dies of monogamy. The two-body marginals need no statevector. In (21) the sum over the other masses factorises:

$$\rho_{ij}\big[(a,b),(a',b')\big] = \tfrac{1}{4}\, e^{i\tau[e(a,b)-e(a',b')]} \prod_{k\neq i,j} \frac{1+e^{i\tau[J_{ik}(a-a')+J_{jk}(b-b')]}}{2}, \tag{24}$$

with $e(a,b) = h_i a + h_j b + J_{ij} ab$. At $N = 16$, 20 and 24 this agrees with partial traces of the statevector to 2.3×10^{-14}. It runs at $N = 4000$, where all 3999 neighbouring pairs take about one second [gravsim:G1]. We measure entanglement by the negativity $\mathcal{N}$ [53], the magnitude of the sum of the negative eigenvalues of the partial transpose [54, 55] (Primer 3). The results at $N = 4000$ are these.

- Bulk neighbours reach $\mathcal{N} = 0.153$ at $\tau = 1.27$ (9 s). At $\tau = 1.3$ every one of the 3999 neighbouring pairs is entangled; the minimum is 0.1528, and the end pair has 0.212.
- A bulk pair is first separable at $\tau = 2.67$ (19 s). It is separable at τ_π, where an isolated pair would have $\mathcal{N} = 1/2$.
- No scanned pair is entangled beyond $\tau \approx 19.5$ (2.3 min; scan to $\tau = 300$). An isolated pair keeps oscillating up to $\frac{1}{2}|\sin(g(1)\tau/2)| = 0.499$.

At $N = 16$ the entangled pairs number 15 of 120 at $\tau = 1.3$, none at τ_π, two (the end pairs) at $\tau = 10$, and none at $\tau = 10^3$. Each mass is entangled with both of its neighbours, and entanglement is monogamous. The coherent field's entanglement therefore becomes multipartite. At scale, pairwise negativity is the wrong witness, and the half-chain entropy is the right one.

Porter–Thomas statistics emerge with N. Let $p(x)$ be the distribution of the X-basis outcomes $x \in \{0,1\}^N$. A random quantum state makes $2^N p(x)$ exponentially distributed; this is the Porter–Thomas distribution [183]. We use two moments, the collision probability $Z = 2^N \sum_x p(x)^2$ and the third moment $M_3 = 2^{2N} \sum_x p(x)^3$, which are 2 and 6 under Porter–Thomas. Anticoncentration ($Z = O(1)$) is the regime in which classical sampling of IQP circuits is argued to be hard [180, 184].

As a reference, IQP circuits with independent uniform angles give $\mathbb{E}[Z] = 2 - 2^{-N}$ exactly: in the fourth moment the phase average forces $\{s_1, s_3\} = \{s_2, s_4\}$ bit by bit, a quadratic term forbids mixing the two pairings, and $2 \cdot 4^N - 2^N$ quadruples remain. Sixty random instances give 2.036 ± 0.018 at $N = 10$ and 1.9953 ± 0.0033 at $N = 14$ (1σ) [gravsim:G1].

For the gravitational circuit:

- $Z = 2^N$ at $t = 0$, and $Z = 1.38$ to 1.41 at τ_π for every N. This is flatter than Porter–Thomas: the near-cluster state has almost uniform X outcomes.
- At $\tau = 10$ (71 s), $Z = 1.74$, 2.10, 2.78, 3.78 and 5.16 for $N = 8$, 12, 16, 20 and 24. Z grows with N, so the output is not anticoncentrated.
- From $\tau \approx 10^3$ (2 h), Porter–Thomas statistics emerge. Over twelve times with $\tau = 10^4$ to 10^7 (three at $N = 24$), $|Z - 2| = 0.113$, 0.069, 0.012 and 0.002 and $|M_3 - 6| = 1.35$, 0.79, 0.11 and 0.02 for $N = 12$, 16, 20 and 24.

The recorded frame at $N = 4000$. We run 4000 masses, 16 trajectories and 200 steps up to $t = 4/\lambda$ (55.6 ms) at the Bridge-B rate [gravsim:G1]. The time loop takes 8.2 s on one core, or 2.6 ms per trajectory-step. Building the matrices takes 1.0 s, and they occupy 384 MB.

- *Entanglement is exactly zero.* Each trajectory is a product state by construction. A direct Euler integration of the joint stochastic Schrödinger equation for $N = 4$ confirms this without the product ansatz: the largest second Schmidt coefficient is 1.8×10^{-15}.
- *Coherence decays as $e^{-\lambda t}$.* The mean $|\rho_{01}|$ over 64 000 mass-trajectories equals $\frac{1}{2}e^{-\lambda t}$ within five standard errors at $t = 1$, 2 and $3/\lambda$. The fitted rate is 0.997λ. The complex mean matches the closed form $\frac{1}{2}e^{-\lambda t}e^{-ih_j t}\prod_i(1 + e^{-iJ_{ij}t})/2 \cdot e^{-\sum_i J_{ij}^2 t/32\kappa}$. Record noise adds $\sum_i J_{ij}^2/(32\kappa) = 3.8 \times 10^{-7}\lambda$ to the rate.
- *The mean force is Newton's.* At the mean records the frame's force equals a direct double sum over masses and branches of $Gm^2\,\mathrm{sgn}(x_i - x_j)/(x_i - x_j)^2$. The agreement is 10^{-13} of the pair-force scale, and Newton's third law holds to 1.9×10^{-14}.
- *With the record noise included*, let $P_j = \int_0^T F_j\,\mathrm{d}t$ be the impulse on mass j over the run. Here $T = 4/\lambda = 7.8 \times 10^{-3}/\omega_0$, forces are in units of Gm^2/d^2, and time is in units of $1/\omega_0$. Averaged over trajectories, P_j equals T times the Newtonian force for all 4000 masses. The mean squared Student t is 1.09, against 1.15 expected for 15 degrees of freedom. For the end mass the ratio is 1.13 ± 0.17.
- *Momentum is conserved only on average.* The total impulse $\sum_j P_j$ has mean 0.16 and standard deviation 0.40 over the 16 trajectories. It is zero on average within the standard error of 0.10, as Newton's third law requires, but not per trajectory. This is the frame's diffusion, the D_2 of Section 5.
- *The output is a product distribution.* Per trajectory $p(x) = \prod_j p_j(x_j)$, so $Z = \prod_j 2(p_{j,0}^2 + p_{j,1}^2) = \prod_j(1 + \langle\sigma_x\rangle_j^2)$; enumeration at $N = 10$ confirms this. The mean $\log_2 Z$ falls from 4000 at $t = 0$ to 1047, 315, 103 and 35 at $t = 1$, 2, 3 and $4/\lambda$, towards 0, the uniform distribution. Porter–Thomas statistics never appear.

The averaged frame never entangles. Product trajectories make the mixture separable. Equation (23) lets us test this independently, at every measurement strength. For $N = 4$ and 6 we take 13 strengths, $\kappa/\kappa_{\mathrm{opt}} = 10^{-3}$ to 10^3, and 25 times $\tau = 10^{-2}$ to 50. The largest negative eigenvalue of the partial transpose, over all bipartitions (7 and 31 cuts) and all pairs, is exactly 0; the coherent field gives -0.5 for the same masses [gravsim:G1]. Two Monte Carlo checks at $N = 4$ with $\kappa = 0.2\,\omega_0$ reproduce (23): 40 000 exactly sampled trajectories (largest $|z|$ over the 256 matrix elements 2.5) and 8000 Euler-integrated trajectories of the joint equation (largest difference 0.005).

Cost. On one core of a shared machine (timings vary by about 20% between runs), the coherent field at $N = 24$ takes 4.5 s per time for the state and Z, and 29 s for S, in 256 MB [gravsim:G1]. Its time grows as $2^{1.21N}$ for $N = 16$ to 24; the exponent includes the $N2^N$ cost of the Hadamard transform and memory effects. Scaling conservatively by exactly 2 per added mass from $N = 24$ gives at least 10^{1198} s at $N = 4000$ (the fitted law gives 10^{1450} s), and the statevector would need 10^{1205} bytes. The recorded frame does the same 4000 masses in 8.2 s for 16×200 trajectory-steps, in 384 MB, at $O(N^2)$ per step.

16.3 The classical-quantum toy and the trade-off

Model. We take one qubit with Lindblad operator σ_z, coupled to a classical momentum p (Primer 4). The joint object is a p-dependent 2×2 block $\rho(p)$, with $\int \mathrm{Tr}\,\rho(p)\,\mathrm{d}p = 1$. It evolves as

$$\partial_t\rho(p) = D_0(\sigma_z\rho\sigma_z - \rho) - \frac{D_1}{2}\,\partial_p\{\sigma_z, \rho(p)\} + \frac{D_2}{2}\,\partial_p^2\rho(p). \tag{25}$$

In blocks, ρ_{00} drifts at $+D_1$ and ρ_{11} at $-D_1$, both diffusing with D_2. The coherence ρ_{01} does not drift; it diffuses and decays at $2D_0$.

Start from $|+\rangle\langle+|$ times a Gaussian of width s_0. The blocks stay Gaussian, with common variance $s^2 = s_0^2 + D_2 t$. A block is positive iff $|\rho_{01}|^2 \le \rho_{00}\rho_{11}$. Since $N(p; a, s)N(p; -a, s) =$

$N(p;0,s)^2 e^{-a^2/s^2}$, the condition does not depend on p. It reads $e^{-4D_0 t} \le e^{-D_1^2 t^2/(s_0^2+D_2 t)}$. Positivity therefore holds for all t iff

$$D_1^2 \le 4 D_0 D_2. \tag{26}$$

Beyond the bound it fails from $t^* = 4D_0 s_0^2/(D_1^2 - 4D_0 D_2)$ on (P). The joint operator $\sum_p |p\rangle\langle p| \otimes \rho(p)$ is block-diagonal, so its spectrum is the union of the blocks' spectra.

Mapping to the trade-off. Oppenheim et al. write the master equation with moments D_n that already carry the factor $1/n!$. They write the back-reaction as $-\partial_p(D_1^{0\alpha} L_\alpha \rho + \text{h.c.})$ [13, Eq. (9)]. For (25) this gives $D_1^{\text{OSSWD}} = D_1/2$ and $D_2^{\text{OSSWD}} = D_2/2$. Their trade-off [13, Eq. (23)], $D_1 D_0^{-1} D_1^\dagger \preceq 2D_2$, is Eq. (8) of Part I. It then reads $(D_1/2)^2/D_0 \le 2\,(D_2/2)$, which is exactly (26). Eq. (8) is therefore correct as written, in Oppenheim et al.'s normalisation [gravsim:G3].

Numerics. We discretise $p \in [-22, 22]$ on 4401 points (spacing 0.01) with second-order central differences. We evolve to $T = 3$ with a Krylov matrix exponential, taking $D_0 = D_2 = 1$ and $s_0 = 0.1$ and sampling 61 times. The largest deviation from the exact Gaussian solution is 1.1×10^{-4} of the peak. Table 6 gives the results [gravsim:G3]. Inside the bound and at equality no block ever turns negative. At 1% beyond the bound positivity fails first at $t = 1.00$, as predicted, and at twice the bound the smallest block eigenvalue reaches -0.24 of the peak. Bisection on D_2 finds the smallest diffusion that keeps positivity up to $T = 50$: $0.9998 = D_1^2/(4D_0) - s_0^2/T$, which tends to the bound as T grows.

Table 6: The classical-quantum toy (25) against the bound (26), with $D_0 = D_2 = 1$, $s_0 = 0.1$ and $T = 3$ [gravsim:G3]. Block eigenvalues are relative to the peak density. The joint object is $\sum_p |p\rangle\langle p| \otimes h\rho(p)$ with $h = 0.01$. Times are on the output grid of step 0.05.

$D_1^2/(4D_0D_2)$	0.5	1 (equality)	1.01	2
smallest block eigenvalue / peak	0	-3×10^{-17} (rounding)	-5.0×10^{-4}	-0.239
first negative time (theory t^*)	never (∞)	never (∞)	1.00 (1.00)	0.05 (0.01)
joint object: smallest eigenvalue	0	0	-8.3×10^{-7}	-8.9×10^{-4}
joint object: negative mass at T	0	10^{-18}	1.6×10^{-5}	9.7×10^{-4}

What equality is. Measure σ_z at strength D_0 and drive p by the record, $\mathrm{d}p = \alpha\,\mathrm{d}Y$. By Itô calculus the joint law of (p, ρ_c) obeys (25) with $D_1 = 2\alpha\sqrt{D_0}$ and $D_2 = \alpha^2$: exactly the equality case. Any additional independent noise on p raises D_2 above the bound. We sample 400 000 trajectories at $t = 1$, with $D_0 = 1$, $\alpha = 1$ and $s_0 = 0$ [gravsim:G3].

- They reproduce $\int \rho_{01}\,\mathrm{d}p = \frac{1}{2} e^{-2D_0 t}$ (0.06784 against 0.06767) and the branch drift $D_1 t$ (2.001 against 2).
- At equality p determines the conditional state, so p is a perfect record and every block $\rho(p)$ is pure. The mixedness $4 \det \rho(p)/(\operatorname{Tr}\rho(p))^2$ is at most 8×10^{-4}, set by the bin width.
- With D_2 doubled the blocks are mixed. At $p = 0$ the mixedness is 0.8646, against $1 - e^{-(4D_0D_2 - D_1^2)t/D_2} = 0.8647$.

The balance rule (10) therefore says that the frame keeps every bit of its record. It gives minimal diffusion for the given decoherence, or equivalently minimal decoherence for the given diffusion.

16.4 Two oscillators: the quantum coupling against the Kafri–Taylor–Milburn channel

Model. Two oscillators interact through the linearised Newtonian coupling. In units $\hbar = m = \Omega = 1$ the Hamiltonian is $H = \sum_k (p_k^2 + x_k^2)/2 - K x_1 x_2$, with $K = 0.1$. The covariance matrix σ of (x_1, p_1, x_2, p_2) obeys $\dot\sigma = A\sigma + \sigma A^{\mathsf{T}} + D$, which we solve exactly with an augmented

matrix exponential. The log-negativity is $E_N = \max(0, -\log_2 2\tilde{\nu}_-)$, with $\tilde{\nu}_-$ the smallest symplectic eigenvalue of the partially transposed σ. The Peres–Horodecki criterion is necessary and sufficient for two-mode Gaussian states [185].

In the KTM channel [18] each x_k is measured at strength k, with Lindblad term $k(x\rho x - \frac{1}{2}\{x^2, \rho\})$. The record $\hat{x}_k$ drives the force $K\hat{x}_k$ on the other mass. Averaged over records, the dynamics is the quantum coupling plus a momentum diffusion

$$D(k) = k + \frac{K^2}{4k} \;\geq\; K \qquad \text{(equality at } k = K/2\text{)}. \tag{27}$$

This is KTM's Eq. (14) with their $\Gamma = k$. Its minimum is their Eq. (15), $\Gamma = K/2$, and their Eq. (19), heating $\hbar K$.

Results [gravsim:G4].

- *The quantum coupling entangles.* The largest log-negativity is 0.145 from the vacuum (at $t = 31$). It is 2.96 from vacua squeezed locally by $r = 1$ in each mode (at $t = 141$).
- *The KTM channel never does.* We take nine strengths $k = (10^{-2}$ to $10^{2})\, K/2$ and $t \leq 200$. The smallest $2\tilde{\nu}_-$ is ≥ 1 to 10^{-9}, so the log-negativity is exactly 0 for both initial states.
- *The minimum is the separability edge.* The quantum coupling plus dephasing at $0.99K$, just below the channel's minimum, entangles the vacuum ($E_N = 1.3 \times 10^{-4}$), as KTM's Eq. (18) states.
- *Trajectories* (4000, $\mathrm{d}t = 0.002$, $T = 10$, $k = K/2$). The conditional covariance stays block-diagonal (cross block exactly 0) and pure ($4 \det V_1 = 0.99995$), so every trajectory is a product state. The trajectory-averaged means follow the quantum mean dynamics from $\langle x_1 \rangle = 1$ (largest $|z| = 2.1$). The covariance of the means plus the conditional covariance equals the master-equation covariance (largest $|z| = 0.98$).
- *The instance.* KTM's Eq. (5) gives $K = 2Gm^2/d^3 = 1.465 \times 10^{-28}\,\mathrm{N/m}$. The minimal coherence decay for $\Delta x = 250\,\mu\mathrm{m}$ is then $K\Delta x^2/(2\hbar) = 0.0434\,\mathrm{s}^{-1}$. This equals the qubit minimum $\omega/2 = 0.0628\,\mathrm{s}^{-1}$ times $(d^2 - \Delta x^2)/d^2$: the same physics, linearised. The Bridge-B rate of Section 17 lies far above this minimum. The reason is that it is set by each mass's own record, not by the pair coupling.

Choices made after seeing the data. Four checks were changed after a first run had failed them; the script records each change.

- We first tested pairwise entanglement at τ_π and found every pair separable. We kept that result, and moved the pair check to $\tau = 1.3$, near the bulk maximum, with a scan in τ.
- A 3% tolerance on $Z - 2$ failed at $N = 16$ (deviation 3.4%). We replaced it by a convergence check over N and a 1% check at $N \geq 20$, and added three late times at $N = 24$.
- We first normalised the force comparison by the near-cancelling bulk sum. It now uses the pair-force scale.
- We report the conservative 2^N extrapolation instead of the fitted exponent.

The parameters of G3 and G4 were fixed before the runs. $K = 0.1$ is far stronger than gravity, chosen so that the entanglement is visible.

Caveats.

- *The 2^N figure is the brute-force cost.* At laboratory times the coherent chain carries about one bit across the cut, and tensor networks would simulate it cheaply; it is hard only at late times. The hardness of IQP sampling [180, 184] is a complexity-theoretic conjecture for typical circuits, not a theorem about this structured one.
- *Only each mass's own record is measured.* The inter-mass cross term of the Diósi–Penrose kernel is not modelled; for the instance it is small (Section 17).

- *The models are idealised.* The masses are two-level systems with branches along the line, and a ring was not run. G3 has one qubit, one classical variable and constant coefficients. G4 has symmetric oscillators and unit detection efficiency.

All numbers in this section come from `docs/simulation/gravity_simulated.py` (110 of 110 checks, about three minutes on one core; deterministic, with fixed seeds).

Verdict. At 4000 masses the frame reproduces Newton's law on average, decoheres each mass at λ and adds momentum noise. It never entangles, at any measurement strength (*proven*, by construction and by the averaged master equation). The coherent field entangles the same masses multipartitely (*proven* for $N \le 24$). It reaches Page entropy and Porter–Thomas statistics only after hours, while its pairwise entanglement dies within seconds. Two known theorems are reproduced as validations (*proven*):

- the trade-off of Section 5 is correct as written, and its equality case is the measurement-and-feedback unravelling with a pure record;
- the Kafri–Taylor–Milburn minimum is the exact separability edge.

17 The decisive experiment under Bridge B

This section asks whether the BMV (Bose–Marletto–Vedral) experiment of Section 6 can tell the frame from a quantised field when each mass decoheres at the Bridge-B rate of Section 5.4. It also checks the spin witness published for that experiment.

17.1 Setting and hypotheses

Two masses m have centres d apart, and each is split by Δx along the line joining them. A coherent Newtonian field gives the branch pair (a, b) the phase $\varphi_{ab} = Gm^2 t/(\hbar d_{ab})$. The distances are $d_{00} = d_{11} = d$, $d - \Delta x$ for the inner pair and $d + \Delta x$ for the outer pair. Up to local phases the state is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + e^{i\Sigma}|11\rangle)$, with $\Sigma = \omega t$ and

$$\omega = \frac{Gm^2}{\hbar}\Big[\frac{1}{d-\Delta x} + \frac{1}{d+\Delta x} - \frac{2}{d}\Big] = \frac{Gm^2}{\hbar}\,\frac{2\Delta x^2}{d(d-\Delta x)(d+\Delta x)}. \tag{28}$$

Its negativity is $\mathcal{N} = \frac{1}{2}|\sin(\Sigma/2)|$ [53] (Primer 3). We use the instance of Section 6, with a uniform diamond sphere of radius $R = 0.879\,\mu$m [gravsim:G2].

The Diósi–Penrose rate. Diósi's master equation [76] is

$$\dot\rho = -\frac{G}{2\hbar}\iint \frac{[f(x),[f(y),\rho]]}{|x-y|}\,\mathrm{d}^3x\,\mathrm{d}^3y,$$

with f the mass-density operator. Under it the coherence between two mass configurations decays at

$$\Gamma = \frac{I(\Delta f, \Delta f)}{2\hbar}, \qquad I(f,g) = G\iint \frac{f(x)\,g(y)}{|x-y|}\,\mathrm{d}^3x\,\mathrm{d}^3y, \tag{29}$$

where Δf is the difference of the two densities. For a uniform sphere displaced by b this gives $\Gamma = E_G/\hbar$. Here $E_G = (GM^2/R)(2\ell^2 - \frac{3}{2}\ell^3 + \frac{1}{5}\ell^5)$ for $\ell = b/2R \le 1$, and $E_G = GM^2(6/5R - 1/b)$ beyond [77, 186].

We checked this by radial quadrature of I [gravsim:G2]. The self-energy integral is $\frac{6}{5}GM^2/R$ to machine precision, and the closed form is reproduced to 2×10^{-16}. Beyond one diameter the mutual term is exactly GM^2/b (the shell theorem). The phases between non-overlapping spheres are therefore exactly those of point masses.

Bridge B's $\lambda = Gm^2/(\hbar R)$ differs from $E_G/\hbar$ by the factor $E_G R/(GM^2)$. For $b \ge 2R$ that factor is $6/5 - R/b$, between 0.7 and 1.2; for smaller splits it falls to zero as $b^2/2R^2$. For the instance, $\lambda = 71.97\,\mathrm{s}^{-1}$ (coherence time 13.9 ms) and $E_G/\hbar = 86.1\,\mathrm{s}^{-1}$.

Hypotheses. We compare five evolutions of the same two masses. In each, coherence factors multiply the coherent-field density matrix element by element.

- **(Q)**: the coherent field, with no collapse.
- **(Q+DP local)** and **(Q+DP exact)**: (Q) with each mass dephasing independently, at λ and at the exact sphere rate $E_G/\hbar$ respectively.
- **(Q+DP full)**: (Q) with the joint master equation (29) for the total density of both masses. Its cross term $I(\Delta f_A, \Delta f_B)/\hbar = \mp\omega$ makes the coherence between branches that differ in both masses decay at $2\Gamma \mp \omega$. These are the dynamics of the Diósi–Penrose model itself, the coherent Newtonian coupling plus Diósi's kernel.
- **(G)**: the recorded frame of Section 16, Eq. (23), with $2\kappa = \lambda$. We also run it at the Kafri–Taylor–Milburn (KTM) optimum $\kappa = \omega/8$, the least noisy classical channel [18].

17.2 The ratio $\mathcal{R} = \omega/\lambda$ and its supremum

Under Bridge B the entangling phase built up per coherence time is

$$\mathcal{R} = \frac{\omega}{\lambda} = R\Big[\frac{1}{d-\Delta x} + \frac{1}{d+\Delta x} - \frac{2}{d}\Big] = \frac{2R\Delta x^2}{d(d-\Delta x)(d+\Delta x)}. \tag{30}$$

G, $\hbar$ and m cancel, so at fixed sphere radius $\mathcal{R}$ is fixed by geometry alone. Numerically, at fixed R, it is independent of m from 10^{-18} to 1 kg to 4×10^{-16}. At fixed density, however, $R \propto m^{1/3}$. Heavier spheres of the same material at the same d and Δx therefore have $\mathcal{R} \propto m^{1/3}$, until they touch; the suprema below do not depend on m. For the instance $\mathcal{R} = 1.745\times10^{-3}$, or 1.458×10^{-3} at the exact sphere rate, as in [Q19] [gravsim:G2].

Supremum. The spheres must not overlap, so the closest approach is $w = d-\Delta x \geq 2R$. Then $\mathcal{R} = 2R\Delta x^2/[w(w+\Delta x)(w+2\Delta x)]$. Its derivative in Δx has numerator $\Delta x(2w^2+3w\Delta x) > 0$, so $\mathcal{R}$ increases towards R/w, and it decreases with w. Hence $\sup\mathcal{R} = 1/2$. It is approached as $\Delta x \to \infty$ with the inner branches touching, and never attained. Three further results:

- *With Bose et al.'s gap.* Their closest approach of 200 μm, chosen so that the Casimir–Polder potential is a tenth of the gravitational one [14], gives $\mathcal{R} \leq R/w = 4.4\times10^{-3}$.
- *Any arrangement of two spheres.* Two of the four branch distances enter with a minus sign, so ω is at most $Gm^2/\hbar$ times the sum of the inverses of the other two. Each of those is at least $2R$, so $\omega \leq Gm^2/(\hbar R)$ and $\mathcal{R} \leq 1$. This bound is approached by parallel splits side by side at contact as $\Delta x \to \infty$. 188 398 random non-overlapping geometries all stay below it; the largest is 0.446.
- *At the exact sphere rate.* The suprema are 1/2 for collinear splits (now as $\Delta x \to 0$ at $d = 2R + \Delta x$; it is 5/12 as $\Delta x \to \infty$) and 5/6 for any arrangement [gravsim:G2].

17.3 An exact threshold for entanglement

Proposition 5 (threshold)**.** *Let two qubits evolve with the Ising coupling of* (28)*, with each coherence decaying as $e^{-\lambda t}$ through independent local dephasing. Some product input becomes entangled at some time iff $\omega > 2\lambda$.*

Proof. If. From $|{+}{+}\rangle$ the smallest eigenvalue of the partial transpose is $\frac{t}{4}(2\lambda-\omega) + O(t^2)$. At $\omega = 3\lambda$ the computed slope is -0.25 [gravsim:G2].

Only if. For $\lambda \geq \omega/2$ the generator is the sum of two parts. The first is the averaged KTM feedback generator at $\kappa = \omega/8$: the coherent coupling plus dephasing at $\omega/2$ (Eq. (23)). It is the record average of an LOCC protocol (local operations and classical communication). The second is local dephasing at $\lambda - \omega/2 \geq 0$. Both generate LOCC semigroups. By the Trotter product formula the evolution is a limit of LOCC maps, and the separable set is closed (P). □

The proof works instant by instant. It therefore also covers geometries that change in time, provided $\lambda(t) \geq \omega(t)/2$ throughout. Bisection over 40 s of evolution gives the threshold 2.0000 [gravsim:G2]. The threshold equals the KTM minimum $\omega/2$ [18] (qubit form in Section 16.1). For the instance that minimum is $0.063\,\mathrm{s}^{-1}$, and the Bridge-B rate is 1146 times larger [gravsim:G4].

With Diósi's correlated kernel and equal rates Γ, the threshold is lower. The smallest eigenvalue of the partial transpose is then $\frac{t}{4}(2\Gamma - \sqrt{2}\,\omega) + O(t^2)$, and bisection gives $\omega/\Gamma = 1.41421 = \sqrt{2}$. Two equally split spheres have $\omega/\Gamma \leq 5/6$ at the exact rate, so they do not entangle under the full kernel either. For unequal splits we have not derived the full-kernel threshold. The random geometries reach at most $\omega/\sqrt{\Gamma_A \Gamma_B} = 0.44$ [gravsim:G2].

Corollary 2 (any two bodies, independent dephasing)**.** *For any two bodies, (Q) with each body dephasing independently at its own Diósi–Penrose self-rate never becomes entangled.*

Proof. The entangling rate is $\omega = |I(\Delta f_A, \Delta f_B)|/\hbar$. The self-rates are $\Gamma_A = I(\Delta f_A, \Delta f_A)/2\hbar$ and $\Gamma_B = I(\Delta f_B, \Delta f_B)/2\hbar$. The kernel $1/|x-y|$ is positive (its Fourier transform is $4\pi/k^2$), so Cauchy–Schwarz gives $\omega^2 \leq 4\Gamma_A\Gamma_B$.

Now take the averaged KTM generator with strengths $2\kappa_A = \Gamma_A/2$ and $2\kappa_B = \Gamma_B/2$. It dephases body A at $\Gamma_A^{\mathrm{KTM}} = 2\kappa_A + \omega^2/(32\kappa_B) = \Gamma_A/2 + \omega^2/(8\Gamma_B)$, and $\Gamma_A^{\mathrm{KTM}} \leq \Gamma_A$ iff $\omega^2 \leq 4\Gamma_A\Gamma_B$. The same holds for B. The generator of the corollary is therefore this KTM generator plus non-negative local dephasing, and the argument of Proposition 5 applies. □

Equality in Cauchy–Schwarz is impossible for disjoint bodies. Among the random two-sphere geometries the largest $\omega/(2\sqrt{\Gamma_A\Gamma_B})$ is 0.22 [gravsim:G2]. Under independent dephasing at the self-rates, decoherence at the Diósi–Penrose rate therefore always meets the classical-channel minimum.

The corollary does not extend to the correlated kernel. There the threshold is $\sqrt{2}$, while Cauchy–Schwarz only bounds ω/Γ by 2. The bound is approached when Δf_A and Δf_B nearly coincide, as for interleaved thin slabs. Such bodies could therefore entangle under the full kernel. We claim the full-kernel result for two spheres only, and did not simulate other shapes. Table 7 shows the instance in time.

Table 7: The instance under the hypotheses, over $0 \leq t \leq 40\,\mathrm{s}$ [gravsim:G2]. Witnesses are evaluated in the local frame of the (Q) prediction. The stabiliser witness is $W_{\mathrm{st}} = \langle XZ\rangle + \langle ZX\rangle$, at most 1 for separable states (two-pair lemma below). The optimal witness is $W_{\mathrm{opt}} = |e\rangle\langle e|^{T_B}$, with e the negative eigenvector of the partial transpose of the (Q) state; it is negative only on entangled states. The control row violates the threshold of Proposition 5 on purpose.

hypothesis	largest $\mathcal{N}$	largest W_{st}	smallest W_{opt}
(Q)	0.5 at 25.0 s	2.0	-0.5
(Q+DP local), $\lambda = 72\,\mathrm{s}^{-1}$	0 (10^{-16})	8×10^{-7}	≥ 0
(Q+DP exact), $86\,\mathrm{s}^{-1}$	0	6×10^{-7}	≥ 0
(Q+DP full kernel)	0	6×10^{-7}	≥ 0
(G), Bridge B	0	8×10^{-7}	≥ 0
(G) at the KTM minimum, $0.063\,\mathrm{s}^{-1}$	0	0.53	≥ 0
control: (Q) with dephasing at $0.99\,\omega/2$	3.0×10^{-4} at 1.55 s	0.54	-3×10^{-4}

17.4 The spin witness of Bose et al., checked and corrected

We read the arXiv version, arXiv:1707.06050 (v1, the only one). The published text, PRL 119, 240401 (2017) [14], may differ. All statements below refer to the arXiv version [witness:B1]. In spin form, with $|\uparrow\rangle = |0\rangle$, the state after the two Stern–Gerlach interferometers (the unnumbered equation after their Eq. (6)) is

$$|\Psi\rangle = \tfrac{1}{2}\big[|00\rangle + e^{ia}|01\rangle + e^{ib}|10\rangle + |11\rangle\big], \qquad a = \Delta\varphi_{LR},\ b = \Delta\varphi_{RL},\ \Sigma = a + b, \tag{31}$$

with $\mathcal{N} = \frac{1}{2}|\sin(\Sigma/2)|$.

The printed witness is not a witness. The arXiv version defines $W = |\langle\sigma_x\otimes\sigma_z\rangle - \langle\sigma_y\otimes\sigma_z\rangle|$ and states that $W > 1$ proves entanglement. The two terms share σ_z on the second spin, so $\sigma_x\otimes\sigma_z - \sigma_y\otimes\sigma_z = \sqrt{2}\,\sigma_n\otimes\sigma_z$ with $n = (\hat{x}-\hat{y})/\sqrt{2}$. A product state with spin 1 along n and spin 2 up gives $W = \sqrt{2}$, the largest value any state can give.

- An optimiser over product states finds $\sqrt{2}$, and 400 000 random product states reach 1.4097.
- On the proposal's own states the printed W reaches at most $\sqrt{2}$, the value a product state already gives. It can never certify entanglement there.
- At $a = 0$, $b = 1$ rad it is 0.651, although the state is entangled ($\mathcal{N} = 0.240$).
- At the phases where the arXiv text quotes $W \approx 1.16$, $\Delta\varphi_{LR} \approx -0.2$ and $\Delta\varphi_{RL} \approx 0.7$, the state (31) gives a printed W of 0.33.

Chevalier, Paige and Kim quote the original condition as $|\langle X\otimes Z\rangle + \langle Y\otimes Y\rangle| > 1$ [26]. That is the valid variant discussed below, and they do not remark on a different operator. The arXiv operator is therefore likely a misprint. The commonly quoted form is valid but weak in this geometry: it detects only $\Sigma > \pi/2$.

Valid two-setting witnesses. Let $a_1 \perp a_2$ and $b_1 \perp b_2$ be unit axes, and let a product state have Bloch vectors u and v. By Cauchy–Schwarz,

$$|(u\cdot a_1)(v\cdot b_1) + (u\cdot a_2)(v\cdot b_2)| \le \sqrt{(u\cdot a_1)^2 + (u\cdot a_2)^2}\,\sqrt{(v\cdot b_1)^2 + (v\cdot b_2)^2} \le 1, \tag{32}$$

and convexity extends the bound to every separable state [57]. The printed operator fails exactly because $b_1 = b_2 = \hat{z}$. In the symmetric gauge $a = b = \Sigma/2$ the corrected witness

$$W_c = |\langle\sigma_x\otimes\sigma_x\rangle + \langle\sigma_y\otimes\sigma_z\rangle| \;\le 1 \text{ (separable)}, \qquad W_c = 1 + \sin(\Sigma/2) = 1 + 2\mathcal{N} \tag{33}$$

detects every $\Sigma > 0$ [witness:B2].

In the laboratory gauge the axes rotate by $\theta = (\Delta\varphi_{LR} - \Delta\varphi_{RL})/2$. On spin 1 they are $x' = \cos\theta\,\hat{x} - \sin\theta\,\hat{y}$ and $y' = \sin\theta\,\hat{x} + \cos\theta\,\hat{y}$. On spin 2 the axis is $x'' = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}$, and z is unchanged. Then $|\langle\sigma_{x'}\otimes\sigma_{x''}\rangle + \langle\sigma_{y'}\otimes\sigma_z\rangle| = 1 + \sin(\Sigma/2)$ for every (a, b). This is the readjustment "in accordance to these local phases" that the paper asks for. The angle is local data. The single-spin Bloch vectors are $\cos(\Sigma/2)(\cos\theta, -\sin\theta, 0)$ and $\cos(\Sigma/2)(\cos\theta, \sin\theta, 0)$, measured on calibration runs separate from the witness runs.

Gauge-free and weak alternatives. Let $T_{ij} = \langle\sigma_i\otimes\sigma_j\rangle$ be the correlation matrix. Separable states satisfy $\|T\|_1 \le 1$ (trace norm). For a product state $T = uv^{\mathsf{T}}$, so $\|T\|_1 = |u||v| \le 1$, and convexity does the rest. This is the two-qubit case of de Vicente's criterion [187].

- The trace norm needs nine settings and no phase knowledge. On (31), $\|T\|_1 = 1 + 2\sin(\Sigma/2)$ in every gauge.
- Its estimate is biased upward. On a product state the bias is +0.250, +0.075 and +0.023 at 100, 1000 and 10 000 runs per setting. It therefore needs a calibrated null, whereas W_c has an unbiased linear estimator and is the one for a hypothesis test [witness:B4].
- The variant $|\langle\sigma_x\otimes\sigma_z\rangle + \langle\sigma_y\otimes\sigma_y\rangle|$ satisfies the lemma, but it is weak. In the paper's gauge ($a = 0$) it equals $1 - \cos\Sigma$ and detects only $\Sigma > \pi/2$. In the symmetric gauge it is 0 for every Σ [witness:B2].

Table 8 collects the values at the paper's target phase.

Under dephasing. Let each mass dephase at λ, with $q = e^{-\lambda t}$ and $s = \sin(\omega t/2)$. In the symmetric gauge $T = \left(\begin{smallmatrix} q^2 & 0 & 0 \\ 0 & 0 & qs \\ 0 & qs & 0 \end{smallmatrix}\right)$. Here $\langle xx\rangle$ lives on the coherences between branches that differ

Table 8: Witnesses on the state (31) at $\Sigma = 1\,\mathrm{rad}$, the order-one phase of [14] [witness:B2].

criterion	value	separable bound	entanglement shown?
printed W $(a = 0)$	0.651	$\sqrt{2}$ (not 1)	no; the bound is wrong
variant $\lvert\langle xz\rangle + \langle yy\rangle\rvert$ $(a = 0)$	0.460	1	no
W_c, rotated axes	1.479	1	yes
trace norm $\lVert T\rVert_1$	1.959	1	yes (with a calibrated null)
negativity $\mathcal{N}$	0.240	0	yes (needs tomography)

in both masses, and $\langle yz\rangle$, $\langle zy\rangle$ on those that differ in one [witness:B3]. Each criterion has its own threshold in ω/λ:

- The trace norm, $q^2 + 2qs = 1 + (\omega - 2\lambda)t + \ldots$, has threshold 2, as tight as the negativity.
- The corrected witness, $W_c = q^2 + qs = 1 + (\omega/2 - 2\lambda)t + \ldots$, has threshold 4.
- $\lvert\langle yz\rangle + \langle zy\rangle\rvert = 2qs$ peaks at a finite time, where $\tan(\omega t/2) = \mu$ with $\mu = \omega/2\lambda$. Its peak value $2\mu\, e^{-\arctan(\mu)/\mu}/\sqrt{1+\mu^2}$ equals 1 at $\omega/\lambda = 3.8057$. This is also the threshold of the best two-pair witness, with axes re-optimised at every time.

The best two-pair witness therefore needs about twice the margin that entanglement itself needs (3.81 against 2), and W_c needs 4. Chevalier et al. propose a three-correlator witness, built from the eigenvector of the partial transpose. It detects entanglement whenever the dephasing rate is below half the entangling rate [26], i.e. at the negativity threshold $\omega/\lambda > 2$.

Runs needed. With two settings and n runs each, $\langle xx\rangle = 1$ has no shot noise on the pure state, and $\langle yz\rangle = \sin(\Sigma/2)$ has variance $\cos^2(\Sigma/2)/n$. A result at k standard deviations therefore needs $n \geq k^2 \cot^2(\Sigma/2)$ runs per setting [witness:B4]:

- 31 runs (3σ) or 84 runs (5σ) at $\Sigma = 1$;
- 9 or 25 at $\pi/2$;
- 3.6×10^5 or 1.0×10^6 at 10^{-2};
- 3.6×10^9 or 1.0×10^{10} at $\Sigma = 10^{-4}$, the phase of the paper's more modest variant ($\Delta x \approx 1\,\mu\mathrm{m}$).

Twenty thousand simulated experiments of 2×84 runs at $\Sigma = 1$ give a mean estimate of 1.481 (against 1.479) and a mean significance of 5.06σ.

17.5 Prior work and what is new

We searched arXiv for analyses of BMV under Diósi–Penrose or CSL (continuous spontaneous localisation) collapse. The search was not exhaustive. We found the following.

- *Bose et al.* already remark that collapse mechanisms "if true, will typically lead to a strong loss of coherence on the time-scale of the experiment and inhibit the gravitationally mediated entanglement" [14]. The remark is qualitative.
- *Chevalier, Paige and Kim* find numerically that the dephased BMV state is not entangled once the dephasing rate exceeds $(\omega_{RL} + \omega_{LR})/2$ [26]. That is the threshold $\omega/\lambda = 2$ of Proposition 5.
- *Tilloy and Diósi* show that the least decoherence compatible with a consistent, measured-and-fed-back Newtonian field singles out the Diósi–Penrose model [27]. Corollary 2 is a two-body, two-branch form of that statement.
- *Trillo and Navascués* show that the full Diósi–Penrose dynamics, our (Q+DP full), can entangle two point-like masses. It does so only if they are closer than about 0.85 times the model's smearing length [188], that is, when their smeared densities nearly coincide. This agrees with the full-kernel threshold $\sqrt{2}$ and with the slab remark above. Feng, Vedral and Marletto dispute that these dynamics are a classical mediator [189].

We found no published bound on ω/λ over BMV geometries at Diósi–Penrose rates, and no CSL analysis of BMV with an exact threshold. As far as we can tell, the following are new:

- the geometric suprema (1/2 collinear, 1 for any two spheres, 5/6 at the exact rate);
- the proof of the threshold for every product input and for geometries that change in time, with its identification with the KTM minimum;
- the full-kernel threshold $\sqrt{2}$ in the form ω/Γ;
- the conclusion that no two-sphere BMV geometry entangles at Diósi–Penrose rates under either kernel;
- the check and correction of the witness.

17.6 What the experiment decides

Every two-sphere geometry has $\omega/\lambda \leq 1$ at the Bridge-B rate, and at most 5/6 at the exact rate. Entanglement needs more than 2 under independent dephasing, and more than $\sqrt{2}$ under the correlated kernel. The corrected witness W_c needs more than 4, and the best two-pair witness more than 3.81. If decoherence at the Diósi–Penrose rate is real, a BMV experiment with two solid spheres therefore finds no entanglement, whether gravity is quantum or classical (S). Three consequences follow.

- *A null BMV result does not separate GUID with Bridge B from quantum gravity with Diósi–Penrose collapse.*
- *A positive result excludes every classical mediator*, GUID included (Lemma 2). If the decoherence is local and uncorrelated, it also shows that the decoherence is below $\omega/2$, which refutes decoherence at the Diósi–Penrose rate (Bridge B). With correlated noise, $\lambda < \omega/\sqrt{2}$ suffices for entanglement.
- *Under Bridge B the discriminating measurement is the decoherence itself.* It is the coherence time of a mesoscopic superposition, $\hbar R/(GM^2)$ up to the factor $E_G R/(GM^2)$, together with the force noise that the balance rule ties to it. This is the "diffusion against decoherence" test of Section 6.

Choices made after seeing the data.

- The hypotheses, geometries and time grid were fixed before the first run.
- The random geometries (Δx from $10^{-2}R$ to $10^{4}R$) were sampled once, and the ranges were not tuned.
- The witness analysis was added after we found the printed operator to be invalid. It uses the paper's own state.
- No threshold or tolerance in this section was changed after a run.

Caveats.

- *Conventions.* Diósi–Penrose rates in the literature differ by factors of order one. At half the exact rate the spheres' supremum becomes 5/3, still below 2. At half the paper's λ it becomes 2, approached but never attained, so the margin vanishes in the extreme parallel geometry. With the full kernel and half the rate, extreme parallel geometries would reach $5/3 > \sqrt{2}$.
- *Scope.* Only uniform rigid spheres with two branches were scanned. Other shapes are covered only under independent dephasing (Corollary 2). Nuclear granularity is neglected; [Q19] shows that the bulk term dominates.
- *The debate.* The debate about classical gravity and quantum matter fields [81, 82, 83] is outside these models.
- *The source.* The witness statements refer to the arXiv version; the published text may differ.

All numbers in this section come from `docs/simulation/gravity_simulated.py` (110 of 110 checks, about three minutes on one core) and `docs/simulation/bmv_witness.py` (34 of 34 checks, under a minute); both are deterministic.

Verdict. At the Bridge-B rate no BMV geometry of two spheres yields entanglement, whatever gravity is (*proven*). The best case falls short of the exact threshold $\omega/\lambda = 2$ under independent dephasing by a factor of 2. At the exact rate it stays below the full-kernel threshold $\sqrt{2}$, at 5/6. For any two bodies dephasing independently at their Diósi–Penrose self-rates, the Newton kernel forbids entanglement (*proven*); under the correlated kernel, suitably shaped bodies are not excluded. GUID's prediction $\mathcal{N} = 0$ is *derived* for two spheres and *untested*, and it does not discriminate. Under Bridge B the decisive measurement is the decoherence rate of mesoscopic superpositions (*live*). The spin witness as printed in the arXiv version is not a valid witness, and its commonly quoted form is valid but weak (*proven*). The corrected $W_c = 1 + \sin(\Sigma/2)$, with laboratory-gauge axes, needs $\omega/\lambda > 4$ under dephasing and $k^2 \cot^2(\Sigma/2)$ runs per setting.

18 The holographic code at scale

Section 8 assembles DMP's parts as a holographic code on a fixed tree. It states three kinematic properties of that code: the Ryu–Takayanagi formula as a minimal cut, entanglement-wedge reconstruction, and witnesses that read the existence of content and never the content. We test all three exactly, on a tree of 1706 tiles with 5120 boundary qubits. Appendix E holds the construction, the proofs and the remaining tables.

The test.

- *The tiles.* Each tile is the six-leg perfect tensor of the five-qubit code [96, 99] (Primer 5).
- *The tree.* A layer of depth D is a rooted tree. The root has five children and every other tile has four. The $n = 5 \cdot 4^D$ legs of the leaves form the boundary; we go up to $D = 5$, where $n = 5120$.
- *Two orientations of a child tile.* In orientation A, the one of Section 8, the edge to the parent sits on the child's logical leg. In orientation H, the HaPPY convention [99], the bulk leg sits there instead.
- *Two settings.* In setting (a) every bulk leg is fixed to $|0\rangle$, which leaves a pure boundary state. In setting (b) every bulk leg is maximally entangled with its own reference qubit, so all bulk qubits are unknown inputs.
- *What is computed.* The network is a stabiliser state with generator matrix G, so the entropy of a boundary region A is exact [190]: $S(A) = \mathrm{rank}(G|_A) - |A|$ bits.
- *What it is compared with.* The minimal cut $\mathrm{mc}(A)$, the fewest legs a cut must cross to separate A from the rest. These are the "fewest hops" of Section 8.

Ryu–Takayanagi.

- *Bulk legs open.* In setting (b), $S(A) = \mathrm{mc}(A)$ in all 2 206 008 region–network cases. These include all 2^{20} regions at $D = 1$ (S). This is the Ryu–Takayanagi formula [101], with the bulk entropy counted.
- *Bulk legs fixed.* In setting (a), $S(A) \leq \mathrm{mc}(A)$ in all 2 206 008 cases, as in any tensor network.
  - In orientation A, equality holds for all 42 428 tested intervals (exhaustive at $D = 1$).
  - For non-contiguous regions it fails (Table 9): in 13.3 % of all regions at $D = 1$, and in 37 % of the random subsets at $D = 5$. The deficits there reach 53 bits.
  - In orientation H it fails even for 10.5 to 32 % of the intervals.
- *The cause.* With the bulk fixed, a tile's parent edge shares one bit with every pair of its child legs; with the bulk leg open it shares none [holo:H0]. A cut that splits a tile's legs

Table 9: Percentage of regions with $S(A) \neq \mathrm{mc}(A)$ in setting (a), for orientations A and H. Setting (b) has no exception. At $D = 1$ the regions are all 380 intervals and all 2^{20} subsets. At $D = 2, \ldots, 5$ they are 3200, 24 320, 9216 and 5312 intervals, together with 1500 random subsets and 1500 unions of 2 to 6 intervals per depth. The last row gives the largest deficit $\mathrm{mc} - S$, in bits. [holo:H2]

	intervals		random subsets		unions of intervals	
D	A	H	A	H	A	H
1	0	10.5	13.3 (all)	13.3 (all)	–	–
2	0	21.7	7.5	7.8	1.8	8.3
3	0	22.7	18.8	19.5	2.6	23.7
4	0	29.3	29.7	29.4	1.9	40.0
5	0	32.0	37.3	38.2	1.5	55.1
max deficit	0	4	53	51	2	4

two against two therefore lets one classical bit past. Every exception contains tiles that neither side's greedy wedge reaches (Appendix E.2).

- *Wedge reconstruction.* In setting (b), $I(R_t : A)$ is 0 or 2 bits, never 1, in all 13 243 616 pairs of region and bulk qubit t [holo:H3]. Four predictions of the wedge agree in every pair: exact recoverability, the greedy wedge [99], the recursion below, and the minimal cut (Appendix E.3).

The erasure threshold. Erase each boundary qubit independently with probability p. An edge can be recovered from the boundary below it exactly when at least 3 of the 4 legs under its tile can. The root can be recovered exactly when at least 3 of its 5 children can (Proposition 6, *proven*). The availability a_h of an edge above a tile of height h is therefore

$$a_0 = 1 - p, \qquad a_h = 4a_{h-1}^3 - 3a_{h-1}^4, \qquad P_{\mathrm{root}}(D) = P\big(\mathrm{Bin}(5, a_D) \geq 3\big). \tag{34}$$

The unstable fixed point $a^* = (1 + \sqrt{13})/6$ of this map sets the erasure threshold, *proven*:

$$p_c = 1 - a^* = \frac{5 - \sqrt{13}}{6} = 0.232408. \tag{35}$$

- As D grows, the root survives with probability tending to 1 below p_c and to 0 above it.
- At p_c every level sits at the fixed point. Every bulk qubit, at every depth and for every D, is then recovered with the same probability $P(\mathrm{Bin}(5, a^*) \geq 3) = 0.914162$. The hierarchy is scale invariant there.
- A plain concatenation of the five-qubit code, with no bulk qubit inside, has threshold 1/2. That is the no-cloning bound of the erasure channel [191].
- In 97 900 Monte Carlo samples the stabiliser computation and the recursion agree for every sample and every bulk qubit (Table 28) [holo:H4].

The two witnesses. A syndrome commutes with every logical operator. It therefore reports that content is in the code sector, and never what the content is. So existence witnesses detect faults that leave the sector and miss faults within it. We see both at scale [holo:H5].

- *The code.* At $D = 5$ the code has 3414 stabiliser generators and 1706 logical qubits.
- *The two witnesses.* Witness T reads the generators directly. Witness I is coded independently: it reads another basis through ancilla circuits.
- *Without noise.* For six bulk contents, both read the trivial syndrome, and all 10 236 content observables keep the value +1.
- *Under noise.* In six Pauli-noise realisations ($p = 0.0005$ to 0.01) the syndrome is identical bit for bit across the contents. The two witnesses agree in all 42 runs (six contents, without noise and in each realisation), and under noise both refuse.

- *Reading the content.* Reading logical Z, then logical X, on 86 bulk qubits drops their mean fidelity to 0.517 (expected $\frac{1}{2}$). The syndrome stays trivial, so the gate cannot see that the content was read.
- *One tile.* On one tile, 960 of the 1024 Pauli errors leave the sector, and both witnesses refuse them. Of the 64 that stay inside, 16 are stabilisers and do no harm. The other 48 are logical errors of weight at least 3: they change the content and pass both witnesses unseen.

A fault within the sector is invisible to existence witnesses by construction. Only the code distance protects against it: lookup correction restores every single-qubit fault, and the logical failure rate falls as $10p^2$ at small p (Appendix E.5).

All numbers here and in Appendix E come from one script, `docs/simulation/holographic_code.py`, which passes 77 of 77 checks in about 8.5 minutes on one core.

Verdict.

- The Ryu–Takayanagi formula *holds* exactly on this tree when the bulk legs are open.
- With the bulk fixed it holds for contiguous regions, and then only in the orientation of Section 8. As a statement about every region it is *refuted*.
- Wedge reconstruction *holds*, and is all or nothing.
- The erasure threshold $p_c = (5 - \sqrt{13})/6 \approx 0.232$ is *proven*, and at p_c the recovery probability is the same at every depth.
- The witnesses read existence and never content. They detect every fault that leaves the code sector and miss every fault within it (*proven* for any stabiliser code, and seen at 5120 qubits).

19 Hosting known quantum systems

Section 4.6 states that DMP's information half can carry any quantum content while its witnesses read only that the content exists. Here we host four quantum configurations with known answers. This validates that construction and our code; it tests no new physics.

What hosting means. A configuration is hosted as existence when it is given as a pair (H, Σ): a Hamiltonian H and a *syndrome* $\Sigma \geq 0$ that commutes with H. The states with $\Sigma|\psi\rangle = 0$ form the *existence sector*, the states the configuration allows: a fixed total spin, Gauss's law, or a register that holds only values below N. Because $[H, \Sigma] = 0$, the dynamics never leaves the sector, and reading Σ leaves the quantum state, the content, undisturbed. The host runs the configuration inside the sector. Two independent witnesses then check only that the result lies in the sector, never what it contains. Witness T works in integer arithmetic on every basis configuration that carries amplitude. Witness I evaluates $\langle\psi|\Sigma|\psi\rangle$ in floating point. This is the reading of a stabiliser syndrome (a measurement that shows a code state is present without revealing it, Section 8), applied here to sectors of up to 2.7 million basis states.

The four configurations. Appendix D gives the models and methods.

- B1, the antiferromagnetic Heisenberg ring of up to 24 spins. It is solved by Lanczos iteration (a Krylov-subspace eigensolver) and extrapolated in the ring size.
- B2, the transverse-field Ising ring of $L = 4000$ spins. It is solved exactly as free fermions by a Bogoliubov transformation (a change to independent fermion modes). Its central charge $c = \frac{1}{2}$ is read twice: from the ground-state energy and from the entanglement entropy of blocks.
- B3, the Schwinger model (quantum electrodynamics in 1+1 dimensions) on up to 22 lattice sites. Its vector mass $M/g = 1/\sqrt{\pi}$ [192] is reached by extrapolating in lattice size and lattice spacing.

- B4, Shor's period-finding circuit [193] on an exact state vector of up to 24 qubits, for nine composite numbers from 15 to 143.

Table 10 compares the hosted values with the known ones.

Table 10: Hosted benchmarks against known values (S) (B1 [hosted:B1], B2 [hosted:B2], B3 [hosted:B3], B4 [hosted:B4]). σ is the largest change over a fixed family of alternative fits, combined with any propagated uncertainty (Appendix D); for B3 it measures consistency, not significance. TV is the total-variation distance.

Benchmark	Known value	Hosted value	Deviation
B1 Heisenberg ring, e_0 ($N \to \infty$)	$\frac{1}{4} - \ln 2 = -0.4431471806$	$-0.4431457(11)$	$+1.5 \times 10^{-6}$ (1.3σ)
B2 Ising ring ($L = 4000$), E_0	finite-L closed form	equal	2×10^{-16} relative
B2 c from the Casimir energy	$\frac{1}{2}$ ($+9.0 \times 10^{-9}$ at this L)	$0.500000005(8)$	$+5(8) \times 10^{-9}$
B2 c from the block entropy	$\frac{1}{2}$	$0.50000016(26)$	$+1.6 \times 10^{-7}$ (0.6σ)
B2 saturated S, $g = 0.5$ / 1.5	0.6989875284 / 0.3069825111	equal	3×10^{-12}
B3 Schwinger M/g, shifted / naive	$1/\sqrt{\pi} = 0.5641896$	0.586(39) / 0.593(31)	0.55σ / 0.92σ
B4 Shor distributions, 192 states	Shor's analytic form	equal	TV $\leq 7.5 \times 10^{-12}$
B4 factors of nine numbers, 15 to 143	the known prime factors	all recovered	none

Witnesses and faults. In every benchmark both witnesses pass the hosted states, and two orthogonal states of one sector (two different contents) receive the same verdict. Faults that leave the sector (a single spin flip, a charge without its electric field, a register value out of range) are refused by both, while a fault inside the sector passes, as Section 18 discusses.

The cost of classical hosting. A dense state vector of n qubits holds 2^n complex amplitudes of 16 bytes. Measured on one core, the time of a layer of gates per qubit grows as $2^{1.068n}$ for $n = 18$ to 26 (Appendix D). Extrapolated, $n = 50$ needs 18 PB for the state and about 12 years per layer. The $2n + 3 = 4099$ logical qubits of Beauregard's circuit for a 2048-bit RSA modulus [194] would need 10^{1235} bytes. Structured content stays cheap. B2 hosts 4000 qubits in four 128 MB matrices because its state is Gaussian, and stabiliser circuits are simulated in polynomial time by the Gottesman–Knill theorem [195, 196] (Section 18). A classical host therefore emulates the Existence Turing machine of Section 4.6 only for small or structured content.

All numbers in this section and in Appendix D come from `docs/simulation/hosted_benchmarks.py` (53 of 53 checks, 475 s on one core).

Verdict. The hosting construction *holds* on four configurations with known answers: the Heisenberg energy to 1.5×10^{-6}, the Ising central charge to 3×10^{-7} by two routes, the Schwinger mass to 7%, and Shor's outcome distributions to better than 10^{-11} (S). The witnesses read existence, never content. This validates the host and our code. It tests no new physics, and every prediction of GUID remains *untested* by it.

Part III
The predictions tested against data

20 How the confrontations are scored

Sections 21 to 25 confront Part I with measurements by one set of rules, fixed here; Primer 9 gives the statistics behind them.

Primer 9: Statistics

Pulls and 95%. A measurement $x \pm \sigma$ usually means a Gaussian error with standard deviation σ. If a prediction $x_G \pm \sigma_G$ is right and the two errors are independent, the difference $x - x_G$ is Gaussian with variance $\sigma^2 + \sigma_G^2$, so the *pull* $p = (x - x_G)/\sqrt{\sigma^2 + \sigma_G^2}$ is a standard normal number. It falls outside ± 1.96 with probability 5% (two-sided), so $|p| < 1.96$ is a test at 95%, and a published 95% half-width b corresponds to $\sigma = b/1.96$. *Worked pull:* a model predicts 10.0 ± 0.3 and an experiment finds 11.0 ± 0.4. Then $p = 1.0/\sqrt{0.4^2 + 0.3^2} = 1.0/0.5 = 2.0$: a FAIL at 95%, although a right model gives $|p| \geq 2.0$ in 4.6% of independent tests. For n independent tests, about $0.05\,n$ FAILs are expected by chance, and the sum of the squared pulls follows a χ^2 distribution with n degrees of freedom. A *p-value* is the probability, if the hypothesis under test is true, of a result at least as extreme as the one seen; in formulas below it carries a subscript, p_i, to keep it apart from the pull.

BIC and ΔBIC. A model with k fitted parameters that reaches the maximum likelihood $\mathcal{L}_{\max}$ on N data points has $\mathrm{BIC} = -2\ln\mathcal{L}_{\max} + k\ln N$ [197]. The penalty $k\ln N$ charges each extra parameter. For large N and independent points, the difference ΔBIC of two models approximates -2 times the logarithm of their Bayes factor, the ratio of the probabilities of the data under the two models. A ΔBIC of 2 to 6 is then positive evidence, 6 to 10 strong and above 10 very strong [198]. The approximation assumes N independent points. If the points come in groups that share an error, such as the distance of one galaxy, the effective number of independent data is closer to the number of groups. A point-wise ΔBIC then overstates the evidence, roughly in proportion to the number of points per group.

The bootstrap. To learn how much a fitted value would vary in a new sample, one draws many new samples of the same size from the data themselves, with replacement, and refits each [199]. The spread of the refitted values estimates the uncertainty; the 16th to 84th percentiles bracket a range comparable to $\pm 1\sigma$. The units resampled must be the independent ones, for rotation curves whole galaxies, not single points.

Many tests. With m tests at 5% each, the chance of at least one false positive, the *family-wise error rate*, grows towards $1 - 0.95^m$. Bonferroni's remedy tests each at $0.05/m$. Holm's method orders the p-values, $p_{(1)} \leq \cdots \leq p_{(m)}$, and rejects $p_{(i)}$ while $p_{(i)} \leq 0.05/(m - i + 1)$; it keeps the family-wise error rate at 5% under any dependence and never rejects less than Bonferroni [200]. Fisher's method combines n *independent* p-values: under the null each p_i is uniform, so $-2\ln p_i$ follows χ^2 with 2 degrees of freedom, and $X^2 = -2\sum_i \ln p_i$ follows χ^2 with $2n$ [201]. It responds strongly to a few very small p_i.

Look-elsewhere and preregistration. A search over many masses, scales or candidate values finds a small *local* p-value somewhere by chance. The *global* p-value, the chance of so small a value anywhere in the search, is larger, roughly the local one times a trials factor [202]. A test is confirmatory only if its hypothesis, data, statistic and decision rule were fixed before the outcome was seen; *preregistration* writes them down in a dated record beforehand, and every choice made afterwards is then visible as exploratory [203].

The pull and the 95% rule. A confrontation sets a GUID value x_G against a datum. Where it exists, a theory uncertainty σ_G comes with x_G. It is propagated from measured inputs, such as Planck's Ω_m, or it is a model's own error, such as the scatter of a fitted mass formula. For a Gaussian datum $x \pm \sigma$ (one standard deviation) the *pull* is

$$p = \frac{x - x_G}{\sqrt{\sigma^2 + \sigma_G^2}}. \tag{36}$$

A published 95% half-width b enters as $\sigma = b/1.96$; for an asymmetric error, the side towards the prediction is used. A confrontation is **OK** when $|p| < 1.96$, that is at 95% for a Gaussian, and a **FAIL** otherwise. The same threshold holds in every section. Other data are allowed regions, probabilities or qualitative statements of the experimenters. Such a confrontation is OK when x_G lies inside the region. Where a region is quoted at 90% CL the row says so; passing a 90% region implies passing at 95%. A FAIL stays a FAIL when general relativity, ΛCDM or the Standard Model share it. Its verdict word is then *inherited* (Section 1.6).

The classes. Every row has one class. A row that is not scored states the reason.

- *Scored*: a prediction confronted with a datum it was not set from, with the verdict OK or FAIL. Only scored rows enter the tally (Section 26).
- *Not yet sensitive*: a scored row whose prediction is specific to GUID, while the datum lies far from where GUID departs from general relativity, ΛCDM, the Standard Model or quantum mechanics. "Far" means that the departure is smaller than the datum's reach by a factor of ten or more, or that it scales with a free parameter that may be zero. Such a row can still fail, so it stays scored. When it holds, it shows only that GUID's departure is small.
- *Conditional*: the same test for a model that Part I uses but GUID does not derive, such as a plateau inflation potential. It is reported apart and not counted for GUID.
- *Consistency* (CONS): the prediction was set from this datum, so it holds by construction. It is never counted as a success.
- *Information* (INFO): superseded by a newer measurement that includes its data, derived from scored rows, overlapping a scored row, inconclusive, a check of a theory GUID adopts unchanged, or without a usable uncertainty. It is reported, not scored.

A scored row is also either *shared*, when its prediction is that of general relativity, ΛCDM, the Standard Model or quantum mechanics, or *GUID-specific*. Only a GUID-specific row that is sensitive can tell GUID apart from the theories it adopts.

Chance and shared data. If GUID were right and n scored data were independent, about $0.05n$ FAILs would occur by chance. Data often share inputs, such as the microwave background, so FAILs can cluster. Each section names the rows that share data.

Likelihoods and BIC. Fits of whole curves, such as rotation curves, compare models by the Bayesian information criterion [197] (Primer 9),

$$\mathrm{BIC} = -2 \ln \mathcal{L}_{\max} + k \ln N, \tag{37}$$

for k fitted parameters and N data points. Lower is better; ΔBIC is the difference to a reference model, negative when the model fits better, and above 10 counts as very strong evidence for independent points [198]. The points of one galaxy share its distance, inclination and stellar mass-to-light ratio, so a point-wise ΔBIC overstates the evidence, and a parameter fitted per galaxy absorbs the shared errors. Sections 22 and 23 therefore report ΔBIC but base their verdicts on the galaxy bootstrap and on galaxy-by-galaxy counts.

The galaxy bootstrap. Ranges of fitted parameters come from a bootstrap over whole galaxies [199], with 1000 draws and a fixed random seed; a range in brackets is the 16th to 84th percentile.

Many tests. A family of tests uses the Holm correction [200]; independent p-values are combined by Fisher's method [201]; and a search over many candidates counts the look-elsewhere effect [202] as the probability of a match at least as close over every candidate tried (Sections 23 and 25). Primer 9 explains all three.

Preregistration. The Standard Model statements and their scoring were committed in `docs/simulation/sm_preregistration.md` before any analysis (Section 24). The candidate scales and history proxies of Section 23, and the assignment of model levels to exotic hadrons in Section 25, were fixed before the comparison. Each section lists, as "choices made after seeing the data", what it decided later and whether a verdict depends on it.

Sources. Every measured number was checked against the arXiv text or abstract of its source. Rows read only through secondary sources are listed in each section's caveats.

21 Laboratory, solar-system and cosmological data

This section confronts the predictions of Part I that laboratory, solar-system and cosmological measurements reach: free fall and the information scalar (Sections 4 and 12.1), black holes (Section 7), Bridge B (Section 5.4), the start, the late universe and the splits (Section 10). Rows are scored as in Section 20.

21.1 Master table

Table 11 lists the scored, conditional and consistency rows, and Fig. 5 shows the Gaussian pulls. A dagger marks a scored row that is GUID-specific but not yet sensitive (Section 20). The INFO rows are superseded (D1.4, D4.1, D4.2), derived (D1.12, D3.2), overlapping (D1.10, D2.2, D2.4, D3.6, D5.1, D5.3), inconclusive (D3.1 to D3.1d), checks of quantum theory (D9.1, D9.2), without a stated uncertainty (D7.2), tree-level values (D8.3, D8.4) or notes (D1.11, D2.13, D4.5, D4.6, D5.8, D6.9, D7.3, D9.6). The rows of the contained remnant, D7.1 to D7.3, stay in the table; Section 23 discusses them with the dark-matter verdict.

Table 11: Scored, conditional and consistency rows (pull: Eq. (36); "in": inside the stated region; data at 1σ unless stated; †: GUID-specific, not yet sensitive). IDs are rows of `predictions_vs_data.json`.

ID	GUID prediction	Datum and source	Pull/test	
Scored (31): *D1, the information scalar* (GUID-specific)				
D1.1	free fall Ti/Pt: $\eta = k\lambda^2 \leq 0$	$\eta = (-1.5 \pm 2.3 \pm 1.5) \times 10^{-15}$ [38]	-0.55σ	OK†
D1.2	free fall Be/Ti, $\eta \leq 0$	$\eta = (0.3 \pm 1.8) \times 10^{-13}$ [48]	$+0.17\sigma$	OK†
D1.3	drift of α zero or negative	clocks $(1.8 \pm 2.5) \times 10^{-19}\,\mathrm{yr}^{-1}$ [204]	$+0.72\sigma$	OK†
D1.5	$\Delta\alpha/\alpha$ 1.8 Gyr ago	Oklo $\lvert\Delta\alpha/\alpha\rvert < 1.1 \times 10^{-8}$ (95%) [205]	in	OK†
D1.6	$\Delta\alpha/\alpha$ at $z = 1.15$	$(1.3 \pm 1.3 \pm 0.4) \times 10^{-6}$ [206]	$+0.96\sigma$	OK†
D1.7	$\Delta\alpha/\alpha$ at $z = 5.5$–7.1	$(-2.18 \pm 7.27) \times 10^{-5}$ [207]	-0.30σ	OK†
D1.8	$\Delta\alpha/\alpha$ at recombination	$\alpha/\alpha_0 = 0.9993 \pm 0.0025$ [208]	-0.28σ	OK†
D1.9	$\Delta\alpha/\alpha$ at nucleosynthesis	$(-7 \pm 5) \times 10^{-3}$, α alone [209]	-1.40σ	OK†
D2, post-Newtonian gravity: general relativity's values (shared)				
D2.1	$\gamma - 1 = 0$	Cassini $(2.1 \pm 2.3) \times 10^{-5}$ [50]	$+0.91\sigma$	OK

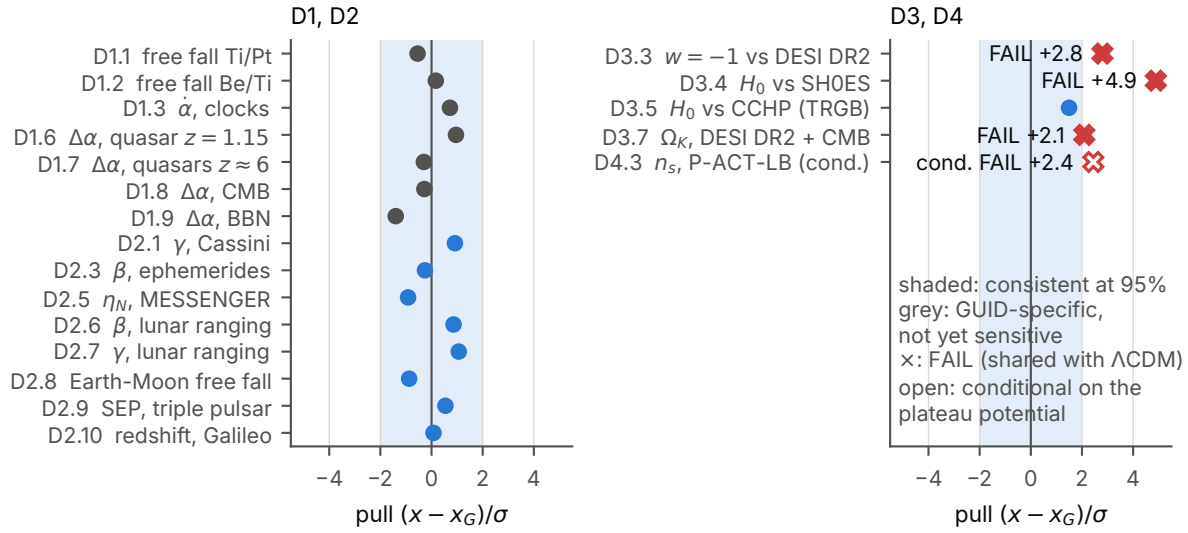


Figure 5: Pulls, Eq. (36), of the scored and conditional rows with a Gaussian datum (Table 11; bounds, qualitative and consistency rows not shown). Shaded: $|p| < 1.96$; grey: GUID-specific and not yet sensitive; crosses: FAILs; open cross: conditional on a plateau potential. What to look for: the general-relativistic rows (D2) scatter within 1.1σ; the FAILs are all cosmological.

ID	GUID prediction	Datum and source	Pull/test	
D2.3	$\beta - 1 = 0$	ephemerides $(-0.45 \pm 1.75) \times 10^{-5}$ [210]	-0.26σ	OK
D2.5	Nordtvedt $\eta_N = 0$	MESSENGER $(-6.6 \pm 7.2) \times 10^{-5}$ [211]	-0.92σ	OK
D2.6	$\beta - 1 = 0$	lunar ranging $(6.2 \pm 7.2) \times 10^{-5}$ [212]	$+0.86\sigma$	OK
D2.7	$\gamma - 1 = 0$	lunar ranging $(1.7 \pm 1.6) \times 10^{-4}$ [212]	$+1.06\sigma$	OK
D2.8	Earth–Moon $\Delta(m_g/m_i) = 0$	$(-2.1 \pm 2.4) \times 10^{-14}$ [212]	-0.88σ	OK
D2.9	strong-field $\Delta = 0$	triple-system pulsar $(0.5 \pm 1.8) \times 10^{-6}$ (95%) [213]	$+0.54\sigma$	OK
D2.10	redshift of the lapse	Galileo $(0.19 \pm 2.48) \times 10^{-5}$ [214]	$+0.08\sigma$	OK
D2.11	$c_{\rm gw} = c$	$-3 \times 10^{-15} \le \Delta c/c \le 7 \times 10^{-16}$ (no CL stated) [37]	in	OK
D2.12	graviton mass 0	$m_g \le 2.42 \times 10^{-23}\,\mathrm{eV}/c^2$ (90%) [215]	in	OK
D5, black holes (shared with general relativity)				
D5.2	thin-disk efficiency ≤ 0.423	Soltan $\epsilon = 0.075$–0.094 [216]	in	OK
D5.4	same	individual disks, mean $\epsilon = 0.089$ [217]	in	OK
D5.6	area never shrinks	GW150914, $P(\Delta A > 0) = 0.97$ [218]	agrees	OK
D5.7	area never shrinks	GW250114, "to high credibility" [219]	agrees	OK
D6, Bridge B: decoherence $\Gamma\tau$ in the observed coherence time (GUID-specific)				
D6.1	$\Gamma\tau = 6 \times 10^{-15}$	28 kDa molecules, 266 nm, 7 ms [220]	<0.1	OK†
D6.2	$\Gamma\tau = 2 \times 10^{-13}$	197 kDa clusters, $\gtrsim$80 nm, 12 ms [221]	<0.1	OK†
D6.3	$\Gamma\tau = 2 \times 10^{-16}$	Rb atoms, 54 cm, 1 s [222]	<0.1	OK†
D6.4	$\Gamma\tau = 1 \times 10^{-14}$	Cs atoms, 3.9 μm, 20 s [223]	<0.1	OK†
D6.8	$l_P^2 D_2 = 1.6 \times 10^{-24}\,\mathrm{kg^2\,s/m}$	window 10^{-40} to 10^{-9} [13]	in	OK†
D3, the balanced state (shared with ΛCDM)				
D3.3	$w_0 = -1$, $w_a = 0$	ΛCDM disfavoured at 2.8 to 4.2σ [119]	2.8σ	**FAIL**
D3.4	$H_0 = 67.4 \pm 0.5$ (CMB fit)	Cepheids $73.04 \pm 1.04\,\mathrm{km\,s^{-1}Mpc^{-1}}$ [123]	$+4.89\sigma$	**FAIL**
D3.5	$H_0 = 67.4 \pm 0.5$	TRGB 70.39 ± 1.94 [124]	$+1.50\sigma$	OK
D3.7	$\Omega_K \approx 0$	$(2.3 \pm 1.1) \times 10^{-3}$ [119]	$+2.09\sigma$	**FAIL**
Conditional on a plateau potential (Starobinsky or Higgs; pull at the best N in 50 to 60)				
D4.3	$n_s = 1 - 2/N = 0.960$–0.967	P-ACT-LB 0.974 ± 0.003 [122]	$+2.44\sigma$	**FAIL**
D4.4	$r = 12/N^2 = 0.0033$–0.0048	$r < 0.036$ (95%) [135]	in	OK
Consistency: the prediction was set from the datum (not counted)				
D5.5	Kerr bound $\lvert a\rvert \le 1$	61 spins, measured with Kerr models [224]	0 out	CONS
D6.5	$R_0 \ge 4 \times 10^{-10}$ m (adopted)	$R_0 > 0.54 \times 10^{-10}$ m (95%) [79]	in	CONS
D6.6	same	$R_0 \gtrsim 4 \times 10^{-10}$ m [80]	in	CONS
D6.7	same	$R_0 \ge 4.0 \times 10^{-14}$ m (detector heating, quoted in [79])	in	CONS
D8.1	proton decay possible, M_X free	$\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr (90%) [128]	in	CONS
D8.2	no relic monopoles (ordering set by it)	flux $<1.4 \times 10^{-16}\,\mathrm{cm^{-2}s^{-1}sr^{-1}}$ (90%) [127]	in	CONS
D8.5	$1/\alpha \approx 128$ at the split	$1/\hat{\alpha}(M_Z) = 127.950 \pm 0.017$ [156]	<1%	CONS
D8.6	the split is a crossover	$m_H = 125.11 \pm 0.11$ GeV >95 GeV [225, 226]	in	CONS
D9.3	neutrinos are not the remnant	$\sum m_\nu < 0.064$ eV, below 0.57% of Ω_c [119]	in	CONS
D9.4	same	$m_\nu < 0.45$ eV (90%), below 12% [227]	in	CONS
D9.5	GW background is not dark matter	$\Omega_{\rm gw} \le 1.3 \times 10^{-8}$ at 1/yr (90%) [149]	in	CONS
D7.1	PBH window 10^{17}–10^{22} g (adopted)	window open [23] (Section 23)	agrees	CONS

21.2 D1: the information scalar

Model. Hypothesis 4 adds a scalar ϕ that sets the fine-structure constant (Section 12.1). In units $\hbar = c = 1$ with $\bar{M}_P^2 = 1/8\pi G$, Part I's action gives it the kinetic term $-\frac{1}{2}\bar{M}_P^2(\partial\phi)^2$, no potential, and the coupling $\lambda = \mathrm{d}\ln\alpha/\mathrm{d}\phi$ (H). A body whose mass has electrostatic share f_A then has $\partial \ln m_A/\partial\phi = \lambda f_A$. Exchange of ϕ adds $2\lambda^2 f_A f_B$ to the Newtonian force. Two bodies falling towards the Earth therefore differ by $\eta = 2\lambda^2 f_\oplus (f_1 - f_2)$ (Section 4; [47]) (P). The Coulomb term $a_C Z(Z-1)/A^{1/3}$ of the semi-empirical mass formula gives $f_{\rm Ti} = 2.02 \times 10^{-3}$, $f_{\rm Pt} = 4.05 \times 10^{-3}$, $f_{\rm Be} = 4.89 \times 10^{-4}$ and, for a coarse Earth composition, $f_\oplus = 1.71 \times 10^{-3}$. MICROSCOPE then measures $\eta = k\lambda^2$ with $k = -6.94 \times 10^{-6}$ [data:D1.1]. Because $\lambda^2 \ge 0$,

GUID fixes the sign: $\eta \le 0$ for Ti/Pt and Be/Ti. These tests can therefore fail although λ is free.

The coupling. We profile $\chi^2_{\rm EP}(\lambda) = \sum_i [(k_i\lambda^2 - \eta_i)/\sigma_i]^2$ over MICROSCOPE [38] (errors in quadrature, 2.75×10^{-15}) and the Eöt-Wash Be/Ti balance [48], with $\lambda^2 \ge 0$. The best fit, $|\lambda| = 1.47 \times 10^{-5}$, is not significant, and $\Delta\chi^2 = 3.84$ gives

$$|\lambda| < 3.1 \times 10^{-5} \quad (95\%\ \text{CL}) \qquad \text{[data:D1.1]}, \tag{38}$$

set by MICROSCOPE alone.

Model A: the action as written. Without a potential, the scalar is driven only by the electromagnetic energy of matter, as in the theories of Bekenstein [44] and of Sandvik, Barrow and Magueijo [45]. Dark matter, a gravitational remnant, carries none; baryons carry an electrostatic share $f_b > 0$. For pressureless baryons with $\partial\rho_b/\partial\phi = \lambda f_b \rho_b$, the field equation $\bar{M}_P^2 \Box\phi = \partial\rho_b/\partial\phi$ in a flat Friedmann universe reads, with $N = \ln a$ and $\phi' = \mathrm{d}\phi/\mathrm{d}N$,

$$\phi'' + \Big(3 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\Big)\phi' = -3\lambda f_b\, \Omega_b(N), \tag{39}$$

with $\Omega_b(N)$ the baryon share of the critical density. In the matter era $\mathrm{d}\ln H/\mathrm{d}N = -3/2$, and every solution approaches $\phi' = -2\lambda f_b \Omega_b/\Omega_m$ (checked numerically). Integrated through flat ΛCDM [116] to today, it gives $\phi'_0 = -0.141\,\lambda f_b$ and

$$\frac{\dot\alpha}{\alpha}\Big|_0 = \lambda H_0 \phi'_0 = -9.70 \times 10^{-12}\,\lambda^2 f_b\ \mathrm{yr}^{-1} \le 0 \qquad \text{[data:D1.3]}. \tag{40}$$

The scalar rolls towards lower matter energy, and a positive electrostatic share makes that a fall in α. GUID's forward statement that α does not grow is therefore *derived* for the action as written. It is not observable. The helium Coulomb energy alone gives $f_b = 6.0 \times 10^{-5}$. Take instead $f_b = 10^{-3}$, a generous upper value for the electromagnetic self-energy of nucleons, and $|\lambda|$ at the bound (38): the drift is then -9.6×10^{-24} per year, four orders of magnitude below the clocks.

Model B: a slowly rolling scalar. We give the scalar a linear potential $V = V_0 - s\phi$ and make it a thawing dark energy, one that Hubble friction holds frozen until late times. A canonical scalar has $\rho_\phi + p_\phi = \bar{M}_P^2 \dot\phi^2$, so $\phi'^2 = 3\Omega_\phi(1+w)$ exactly. For small slopes the history has a universal shape, and each α datum measures the present drift $d = \dot\alpha/\alpha|_0$:

$$\frac{\Delta\alpha}{\alpha}(z) = -\frac{d}{H_0} G(z), \qquad G(z) = \int_{-\ln(1+z)}^{0} \frac{\phi'(N)}{\phi'(0)}\,\mathrm{d}N. \tag{41}$$

G is 0.118 at Oklo ($z = 0.137$, the 1.8 Gyr implied by the quoted bound and rate of Damour and Dyson [228]), 0.435 at $z = 1.15$, 0.514 at $z = 6.3$ and 0.516 from recombination back. For $1 + w_0 = 0.16$ these values change by less than 2%. Each datum gives $d_i = -H_0(\Delta\alpha/\alpha)_i/G(z_i)$ (Fig. 6a). The clocks [204], which include the data of [229], give $\pm 2.5 \times 10^{-19}$ per year and Oklo [205] $\pm 3.3 \times 10^{-18}$. The quasars [206, 207], the CMB [208] and nucleosynthesis [209] are three to six orders weaker. Their weighted mean, $d = (1.79 \pm 2.49) \times 10^{-19}$ per year ($\chi^2 = 3.05$ for 5 degrees of freedom), is derived and dominated by the clocks, so it is reported, not scored [data:D1.12].

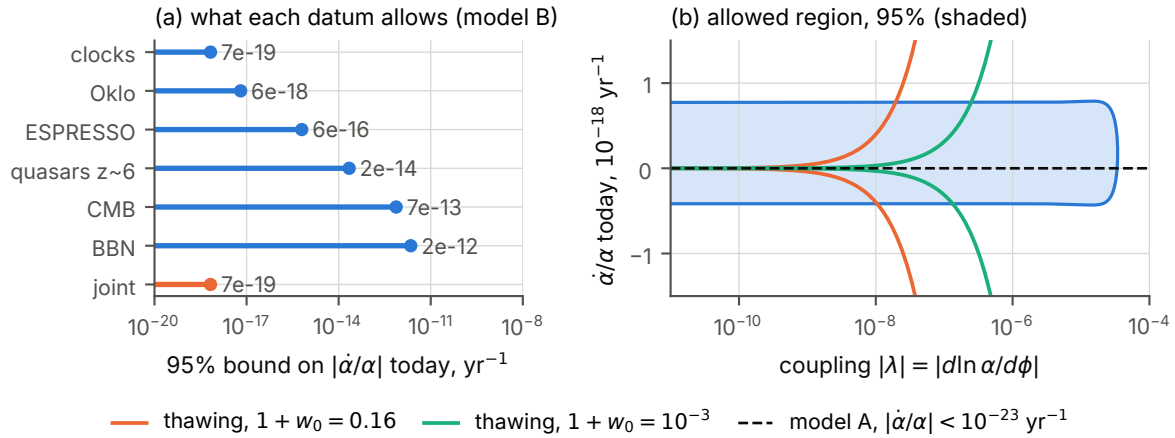


Figure 6: The information scalar. (a) The 95% bound on the present drift of α implied by each datum and by all together in Model B, Eq. (41). (b) The 95% allowed region (shaded) of coupling $|\lambda|$ and present drift from free fall and the α data. Coloured curves: the drift of a scalar that is a thawing dark energy, $d = \pm\lambda H_0\phi_0'$, for $1 + w_0 = 0.16$ and 10^{-3}. Dashed: Model A, below 10^{-23} per year. What to look for: the region is symmetric in sign, and a scalar rolling with the dark energy must couple below 10^{-7}.

The sign. The data say nothing about it. The clocks dominate; their central value is positive at 0.7σ; $P(d < 0) = 0.24$; and the 95% interval $[-3.1, 6.7] \times 10^{-19}$ per year holds both signs. In Model B, drift and coupling are linked only by the field speed, $d = \lambda H_0\phi_0'$. Suppose the scalar drove the evolving dark energy DESI prefers, with $1 + w_0 = 0.162$ and so $\phi_0' = 0.577$. The clocks would then demand $|\lambda| < 1.7 \times 10^{-8}$, about two thousand times below the free-fall bound (2.1×10^{-7} for $1 + w_0 = 10^{-3}$) [data:D1.12] (Fig. 6b).

Verdict: all eight scored rows hold, and none is yet sensitive. Their departure from general relativity scales with λ^2, and λ is free and may be zero; Model A's shifts lie 2.6×10^4 to 1.5×10^9 times below the errors [data:D1]. Free fall *holds*, and the coupling is bounded by (38). The negative drift is *derived* for the action as written and *untested*; where a drift could be seen, its sign is not predicted.

21.3 D2: post-Newtonian gravity

Minimal coupling (Section 4) gives the post-Newtonian limit of general relativity (P): $\gamma = \beta = 1$, a Nordtvedt parameter $\eta_N = 4(\beta - 1) - (\gamma - 1) = 0$ [210], the gravitational redshift of the lapse, and massless classical waves at c with two tensor polarisations. Of overlapping data, Cassini (D2.1) is scored for γ and the ephemeris combination (D2.3) for β; D2.2 and D2.4 are INFO. All ten scored rows hold. The largest pull is 1.06σ (lunar γ). The value $\chi^2 = 4.68$ for 8 Gaussian pulls gives $p = 0.79$; it is indicative only, because the three lunar rows come from one fit [data:D2]. The graviton bound is at 90% credibility, and the GWTC-3 tests also find no non-tensor polarisation [215]. *Verdict: holds*, and adds nothing to general relativity.

21.4 D5: black holes

GUID's black holes are Kerr's (Section 7). Christodoulou's formula [230] has no solution above $|a| = 1$, spin energy is extracted in the ergosphere, and the area never shrinks classically. Matter reaching the innermost stable orbit of a thin disk radiates $\epsilon(a) = 1 - E_{\mathrm{isco}}(a)$ [94]: 0.0377 for $a = -1$ (retrograde), 0.0572 for $a = 0$, 0.423 for $a \to 1$ [data:D5]. The Soltan argument compares the light of all quasars with the mass in black holes today. It gives $\epsilon = 0.075$ to 0.094 [216]; the older 0.04 to 0.16 [231] and > 0.1 [232] rest on similar inputs and are INFO. Individual disk fits give a mean of 0.089 [217]. Both scored values lie below the Kerr maximum and imply typical

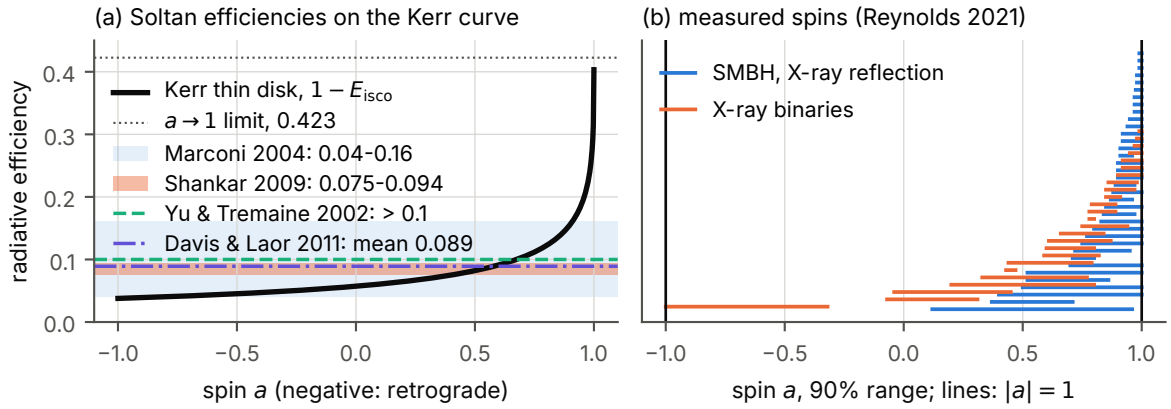


Figure 7: Black holes. (a) Kerr thin-disk efficiency against spin (black) with the Soltan efficiencies of Refs. [231, 216, 232] and the mean of individual disk fits [217]; where a band meets the curve it reads off the typical spin. (b) 90% ranges of 36 supermassive and 25 stellar spins [224], sorted; lines: the Kerr bound, built into the spin measurements. What to look for: every efficiency is below the $a \to 1$ limit.

spins of 0.40 to 0.62 and 0.58 (Fig. 7a). The curve bounds the efficiency only from above: radiatively inefficient flows can fall below 5.7%. All 61 measured spins of the compilation of Reynolds [224], 36 supermassive and 25 stellar, lie in $[-1, 1]$ at 90%. They are measured with Kerr models capped at $a = 0.998$, so this row cannot fail and is a consistency row. The ringdown of GW150914 shows an area increase with probability 0.97 [218], and that of GW250114 shows it "to high credibility" [219] [data:D5.6,D5.7]. *Verdict: holds*, with general relativity.

21.5 D6: Bridge B against the largest superpositions

Bridge B (Section 5.4) gives the Diósi–Penrose rate [76, 77] $\Gamma = E_\Delta/\hbar$, where

$$E_\Delta = \frac{G}{2} \iint \frac{[\rho_1(\mathbf{x}) - \rho_2(\mathbf{x})][\rho_1(\mathbf{y}) - \rho_2(\mathbf{y})]}{|\mathbf{x} - \mathbf{y}|} \, \mathrm{d}^3x \, \mathrm{d}^3y \tag{42}$$

for a superposition of the mass densities ρ_1 and ρ_2; Part I uses its order of magnitude $GM^2/\hbar R$. For a uniform sphere displaced by $d = xR$, $E_\Delta = (GM^2/R)(x^2/2 - 3x^3/16 + x^5/160)$ for $x < 2$ and $(GM^2/R)(6/5 - 1/x)$ for $x \geq 2$. For a mass smeared as a Gaussian of width s, $E_\Delta = GM^2[1/(\sqrt{\pi}s) - \mathrm{erf}(d/2s)/d]$. Both forms agree with Fourier quadrature [data:method].

For each of the largest superpositions made we take the larger of $GM^2/\hbar R$ and the full $E_\Delta/\hbar$. The full value is the bulk sphere plus every nucleus smeared over a width R_0; an atom is its nucleus smeared over R_0. The smearing length is free in Bridge B. An underground germanium detector sees no spontaneous radiation, which requires $R_0 > 0.54 \times 10^{-10}\,\mathrm{m}$ with probability 0.95 [79]. That excludes Penrose's parameter-free choice ($5 \times 10^{-12}\,\mathrm{m}$ for germanium). Current bounds are summarised as $R_0 \gtrsim 4 \times 10^{-10}\,\mathrm{m}$ [80], and Bridge B adopts $R_0 \geq 4 \times 10^{-10}\,\mathrm{m}$ from them; these are consistency rows [data:D6.5-D6.7]. The rates below use the smallest allowed width, $0.54 \times 10^{-10}\,\mathrm{m}$, because a larger R_0 lowers the rate: this is the conservative choice, and the adopted value only widens the margins. A run is consistent when $\Gamma\tau < 0.1$ over its observed coherence time τ.

All four runs pass, with margins of 4×10^{12} to 4×10^{15} (Table 12) [data:D6.1-D6.4]. A margin that large means the runs are not yet sensitive: they would see Bridge B only near $\Gamma\tau \approx 1$. The continuous diffusion of Bridge B lies inside the window of Part I (D6.8, Section 5.4), at least 6×10^{14} times inside either edge.

The smallest decisive experiment. A uniform sphere separated by more than its size has $\tau_{\mathrm{DP}} = \hbar R/(\frac{6}{5}GM^2) \propto M^{-5/3}$. For silica, τ_{DP} is 10 ms at $1.2 \times 10^{-14}\,\mathrm{kg}$, 1 s at $7.6 \times 10^{-16}\,\mathrm{kg}$ and

Table 12: Bridge B against the largest superpositions. Γ: the larger rate estimate (text); margin $1/\Gamma\tau$. Molecule radius from mass and 1300 kg/m^3; cluster diameter 8 nm as stated, separation at the stated lower bound of ten diameters.

experiment	mass [kg]	separation	τ [s]	Γ [s^{-1}]	margin
oligoporphyrins, 28 kDa [220]	4.6×10^{-23}	266 nm	7×10^{-3}	8.1×10^{-13}	2×10^{14}
Na clusters, 197 kDa [221]	3.3×10^{-22}	80 nm	1.2×10^{-2}	2.0×10^{-11}	4×10^{12}
Rb atoms [222]	1.4×10^{-25}	54 cm	1	2.4×10^{-16}	4×10^{15}
Cs atoms [223]	2.2×10^{-25}	3.9 μm	20	5.7×10^{-16}	9×10^{13}

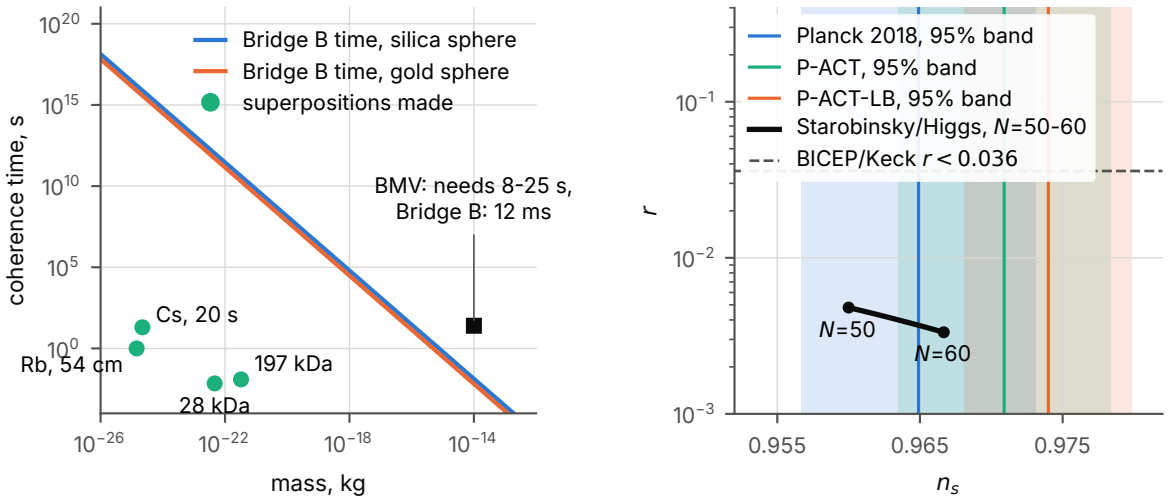


Figure 8: Left (D6): Bridge B's coherence time $\hbar R/(\frac{6}{5}GM^2)$ for silica and gold spheres (lines), the observed coherence of the largest superpositions (dots [220, 221, 222, 223]) and the BMV proposal [14] (square: the 8 to 25 s it needs). A superposition kept above the line would show Bridge B; existing ones fall short by 12 to 16 orders of magnitude. Right (D4): the plateau potentials at leading order for $N = 50$ to 60 (black) against the 95% bands of n_s from Planck [118], P-ACT and P-ACT-LB [122] and $r < 0.036$ [135]; the line is inside the Planck band, partly inside P-ACT, outside P-ACT-LB.

100 s at 4.8×10^{-17} kg [data:D6]. The first experiment to see Bridge B superposes such a particle by more than its diameter for that long. That is five to eight orders of magnitude in mass beyond the heaviest superposition so far, 3.3×10^{-22} kg [221] (Fig. 8, left). The microsphere of the BMV (Bose–Marletto–Vedral) proposal decoheres in about 12 ms at the exact sphere rate (14 ms at Part I's $GM^2/\hbar R$), long before its entangling time (Section 17). *Verdict:* consistent with every superposition made, and tested by none: *live*, *untested*.

21.6 D4: inflation and the start

Part I's start is an inflation that ends in oscillons and contains the strong split (Section 4.5, Hypothesis 6′). GUID derives no inflaton potential. The plateau potentials of Starobinsky's R^2 inflation [137] and of Higgs inflation [138] both couple a scalar to curvature, in open tension with the proposition that fixes $\gamma = 1$ (Section 4). Their rows are therefore *conditional* and kept out of GUID's tally. At leading order both predict $n_s = 1 - 2/N$ and $r = 12/N^2$ for the N e-folds after the pivot scale leaves the horizon: $n_s = 0.960$ to 0.967 and $r = 0.0048$ to 0.0033 for $N = 50$ to 60 (P).

ACT DR6 with Planck, CMB lensing and DESI DR1 BAO (baryon acoustic oscillations), called P-ACT-LB, gives $n_s = 0.974 \pm 0.003$ and needs $N \geq 62.7$. At the best N in the range, $N = 60$, the pull is $+2.44\sigma$: a conditional FAIL [data:D4.3]. Planck alone (0.9649 ± 0.0042, best $N = 57$, pull -0.00σ, $46 \leq N \leq 74$) and P-ACT ($N \geq 54.7$) are included in P-ACT-LB

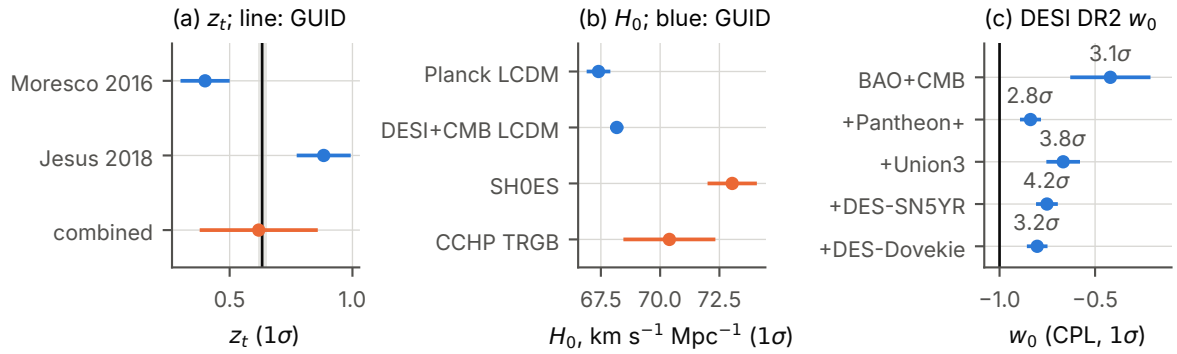


Figure 9: The balanced state (one standard deviation). (a) Transition redshift: GUID with Planck's Ω_m (line and band), the chronometer and supernova determinations [233, 235] and a scaled mean of the two that is reported but not scored (the $H(z)$ values of [234] lie between). (b) H_0: the ΛCDM fits GUID inherits against the Cepheid ladder [123] and the TRGB [124]. (c) w_0 of DESI DR2 for five combinations [119, 236] with the quoted significance against $w = -1$ (line). What to look for: the z_t values scatter on both sides of GUID; GUID is far from SH0ES and below every DESI w_0.

and reported as INFO [data:D4.1,D4.2]. The ratio $r \leq 0.0048$ lies far below $r < 0.036$ [135], a conditional OK [data:D4.4]. Part I places the strong split before or during inflation because no relic monopoles are seen, so MACRO's bound [127] is a consistency row (D8.2). A split after inflation, one monopole per horizon at 10^{16} GeV [126], would exceed it 3×10^{18}-fold [data:D4.5]. Reheating below the unification energy is allowed, not tested (at most 5.7×10^{15} GeV against $M_X \gtrsim 4.8 \times 10^{15}$ GeV, [Q20], [data:D4.6]). *Verdict:* the start is *untested*; the plateau potentials are in tension with the spectral index.

21.7 D3: the balanced state

GUID's late universe is flat ΛCDM settling into de Sitter expansion, $H \to H_\infty = \sqrt{\Omega_\Lambda} H_0$ (Section 10). Acceleration starts at $z_t = (2\Omega_\Lambda/\Omega_m)^{1/3} - 1$: 0.632 ± 0.018 for Planck's $\Omega_m = 0.315 \pm 0.007$ [116], or 0.664 ± 0.009 for DESI DR2 with the CMB [119]. Its published determinations disagree with each other. Cosmic chronometers give 0.4 ± 0.1 [233] (-2.3σ). $H(z)$ data give 0.72 ± 0.05 or 0.84 ± 0.03 for $H_0 = 68$ or 73.24, so the value depends mainly on that prior [234] ($+1.7$, $+6.0\sigma$). Supernovae with $H(z)$ give 0.883 ± 0.110 [235] ($+2.2\sigma$). The test is *inconclusive* (INFO) [data:D3.1].

The direct test is the equation of state. DESI DR2 fits $w = w_0 + w_a(1 - a)$. With the CMB it finds $w_0 = -0.42 \pm 0.21$ and $w_a = -1.75 \pm 0.58$, which disfavours ΛCDM at 3.1σ; with the three supernova samples it finds 2.8, 3.8 and 4.2σ [119]. A recalibration of the DES supernovae lowers its value to 3.2σ, with Bayesian odds of about 5:1 [236]. Scored on the weakest combination, 2.8σ, the floor FAILs [data:D3.3]. The model's $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}Mpc^{-1}}$ lies 4.9σ from the Cepheid ladder, 73.04 ± 1.04 [123], and 1.5σ from the tip of the red giant branch (TRGB), 70.39 ± 1.94 [124] [data:D3.4,D3.5]. Inflation predicts $|\Omega_K| \lesssim 10^{-5}$. DESI with the CMB finds $\Omega_K = (2.3 \pm 1.1) \times 10^{-3}$, 2.1σ from flat, while reporting no significant preference. Planck with BAO, flat to 0.4% at 95% [118], overlaps it through the CMB and is reported as INFO [data:D3.6,D3.7]. *Verdict:* the three FAILs are *inherited* from ΛCDM (Section 21.9).

21.8 D8, D9: the splits and quantum content

The splits. Proton decay and the running couplings are tested with the Standard Model statements (Section 24). Two rows here are consistency rows: GUID's example unification mass was chosen above Super-Kamiokande's bound [128], and the ordering of the strong split was set by the absence of monopoles [data:D8.1,D8.2]. At the electroweak split the W and Z take their

masses from $v = (\sqrt{2}G_F)^{-1/2} = 246.22\,\text{GeV}$ [237] (P). From $1/\hat{\alpha}(M_Z) = 127.950 \pm 0.017$ [156] and $\sin^2\hat{\theta}_W(M_Z) = 0.23129 \pm 0.00004$ [132], the tree-level $m_W = (\pi\hat{\alpha}/\sqrt{2}G_F \sin^2\hat{\theta}_W)^{1/2} = 80.22\,\text{GeV}$ lies 0.18% below $80.369 \pm 0.013\,\text{GeV}$ (without CDF) [238]. The tree-level $m_Z = m_W/\cos\hat{\theta}_W = 91.50\,\text{GeV}$ lies 0.34% above $91.1880 \pm 0.0020\,\text{GeV}$ [132]. Tree level is scheme dependent: $m_Z \cos\hat{\theta}_W$ gives 79.95 GeV (−0.52%), and the on-shell route from $\alpha(0)$, G_F and m_Z gives 80.94 GeV (+0.71%). Part I's half per cent holds only in the $\overline{\text{MS}}$ (modified minimal subtraction) route, so D8.3 and D8.4 are tree-level Standard Model values (INFO) [data:D8.3,D8.4]. Part I's $\alpha \approx 1/128$ (Section 24) and its crossover, read from $m_H = 125.11 \pm 0.11\,\text{GeV}$ [225] above the 70 to 95 GeV end point of the first-order line [226, 131], are consistency rows [data:D8.5,D8.6]. *Verdict: holds as a reading*; none of it is a test.

Quantum content. Quantum content reaches but does not exceed Tsirelson's bound $2\sqrt{2}$ (Section 8). The best value, $S = 2.82759 \pm 0.00051$ [239], and the loophole-free $S = 2.42 \pm 0.20$ [240] both lie below it, as quantum theory requires; they are checks of quantum theory and are INFO [data:D9.1,D9.2]. The statements that the known neutrinos and the gravitational-wave background are not the remnant were set from such data and are consistency rows. The sum $\sum m_\nu < 0.064\,\text{eV}$ [119] leaves the neutrinos below 0.57% of the dark matter, KATRIN's $m_\nu < 0.45\,\text{eV}$ (90% CL) [227] leaves at most 12%, and the nanohertz background has $\Omega_{\text{gw}} \leq 1.3 \times 10^{-8}$ at 1/yr [149]. Gravity weighing only the Higgs-given mass is excluded by MICROSCOPE (D9.6, INFO) [data:D9.3-D9.6].

21.9 The inherited tensions

The three scored FAILs and the conditional FAIL are real tensions of established accounts that GUID adopts unchanged. They share data: D3.3 and D3.7 both use DESI DR2 BAO with the CMB, D3.4 rests on the CMB fit of ΛCDM, and D4.3 uses the CMB with DESI DR1 BAO. The chance count of Section 20, $0.05 \times 31 = 1.6$ FAILs, assumes an independence that does not hold.

- *Evolving dark energy* (D3.3). No combination reaches 5σ, and the fitted w crosses -1, which no single canonical scalar can do. If the preference holds, the future is not de Sitter and the balanced state of Section 10 must be read again.
- *The Hubble constant* (D3.4) and *curvature* (D3.7, called not significant by DESI and built on D3.3's data) are known ΛCDM discrepancies; the TRGB value agrees.
- *The spectral index* (D4.3, conditional). The gap of a few e-folds comes partly from the BAO that also drive D3.3. The e-folds of the pivot scale depend on reheating, so the datum constrains the reheating as much as the potential. GUID's start, which needs inflation but not this potential, is not tested by it.

21.10 No data yet

No measurement reaches what tells GUID apart: mediated entanglement ($\mathcal{N} = 0$ [14, 15]; Section 17), the LOCC inequality [87], the trade-off at equality (only D6.8's window is known), Bridge B's decoherence and the sign of the drift of α. Tensor modes, the light-edge Hawking glow and proton decay have bounds only; nothing reaches the return of information.

Choices made after seeing the data.

- The kinds were assigned after a first scoring. Twenty-four rows scored as OK in that pass, and one FAIL (D4.3, now conditional), left the tally. The CHSH rows D9.1 and D9.2 followed after a cold reading of the paper.
- The not-yet-sensitive class uses a factor of ten, set after the factors were known. The smallest factor is 2.6×10^4 (D1.3), so any threshold up to that value gives the same classes.
- DESI is scored on its least significant combination. Of overlapping data, Cassini, the ephemeris β, DESI's curvature and Shankar et al.'s efficiency are scored.

- The lunar errors are read as 1σ, the stricter reading. The Kerr test uses the upper end of each efficiency range, again the stricter one. Bridge B passes a run below $\Gamma\tau = 0.1$.
- Model inputs: $f_b = 10^{-3}$, 1.8 Gyr for Oklo, $1300\,\mathrm{kg/m^3}$ for the molecules and the cluster separation at its lower bound.
- D7.2 (a factor 5.8, inside a chosen tolerance of ten) and D8.3, D8.4 (0.5% only in the $\overline{\mathrm{MS}}$ route) depend on choices and are INFO for that reason. No scored verdict depends on the choices above.

Caveats. Read only through secondary sources: Cassini's γ (ADS abstract), MESSENGER's η_N (journal record), the PDG and CODATA constants (PDG and NIST records, a 2025 preprint), Fein et al.'s interferometer parameters (a later preprint, a review, the university's release), and the species and separations of Refs. [222, 223] (a review, the preprint's figure). Model B's mapping of w_0 to the field speed is exact at $z = 0$, but the CPL form $w_0 + w_a(1 - a)$ is not a thawing fit. The nucleosynthesis datum uses 2001 abundances. Recent analyses reach parts per million by letting all couplings vary together [241], which a scalar that sets only α does not do. All numbers here come from `docs/simulation/predictions_vs_data.py` (74 rows; 28 of 31 scored, 1 of 2 conditional; 6 of 6 method checks; about four seconds on one core).

Verdict. Of 31 scored confrontations, 28 hold at 95% and 3 fail: evolving dark energy, the Hubble constant and a 2σ curvature. They are correlated ΛCDM tensions that GUID *inherits*. The plateau potentials fail conditionally on the spectral index. Eighteen scored rows test what GUID shares with general relativity and ΛCDM. The thirteen GUID-specific rows hold, and none is yet sensitive. No datum tells GUID apart; the experiments of Section 6 can.

22 Dark matter I: rotation curves against the stochastic remnant

We test the stochastic remnant of Section 11, with its size fixed by Bridge A (Eq. (20)) or left free, against the rotation curves of the SPARC galaxies; Primer 10 gives the background.

Primer 10: Galaxy dynamics

Rotation curves. Gas and stars in a disc galaxy move on nearly circular orbits. Their speed V against radius R is the *rotation curve*, and the centripetal acceleration is $g_{\mathrm{obs}} = V^2/R$. Far out, most curves stay flat where the visible mass alone would make them fall as $R^{-1/2}$: the missing-mass problem.

Baryonic mass models. The visible, or *baryonic*, mass has three parts, each with its own contribution to V^2, computed from its observed surface density: the gas (atomic hydrogen with its helium), the stellar disc and the central bulge. Starlight becomes stellar mass through a mass-to-light ratio Υ; at a wavelength of 3.6 µm it is near 0.5 for discs and 0.7 for bulges [242]. The sum of the three, divided by R, is the baryonic acceleration g_{bar}.

Two empirical laws. The *baryonic Tully–Fisher relation* (BTFR): a galaxy's baryonic mass grows as the fourth power of its flat rotation speed, $M_{\mathrm{bar}} \propto V_{\mathrm{flat}}^4$, with small scatter [243]. The *radial acceleration relation* (RAR): at every radius of every galaxy, g_{obs} is one function of g_{bar}. It is close to g_{bar} at high acceleration and close to $\sqrt{g_{\mathrm{bar}}\, g_\dagger}$ below a scale $g_\dagger \approx 10^{-10}\,\mathrm{m\,s^{-2}}$ [242, 244].

MOND. In Milgrom's modified Newtonian dynamics, $g_{\mathrm{obs}} = \nu(g_{\mathrm{bar}}/a_0)\, g_{\mathrm{bar}}$, where the new constant a_0 is about $1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ [145]. The *interpolating function* $\nu(y)$ tends to 1 at high acceleration ($y \gg 1$) and to $y^{-1/2}$ at low; the deep regime gives $V^4 = GMa_0$, the BTFR. Common choices are the "simple" function $\nu(y) = \frac{1}{2} + (\frac{1}{4} + 1/y)^{1/2}$ and the "standard" one, and the RAR's fitting function is another [245].

Dark-matter halos. In cold dark matter each galaxy sits in a halo. Simulated halos follow the NFW profile

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

of Navarro, Frenk and White [246]. Its mass M_{200} is counted inside the radius r_{200} that encloses 200 times the critical density, and its concentration $c = r_{200}/r_s$ falls slowly with mass [247]. At the centre the density rises as r^{-1}, a *cusp*. Many dwarf and low-surface-brightness galaxies prefer a *core* of nearly constant density, as in the Burkert profile $\rho = \rho_0 r_0^3/[(r + r_0)(r^2 + r_0^2)]$ [248]; this mismatch is the cusp-core problem [249]. The *stellar-to-halo relation*, found by matching the abundances of galaxies and halos, gives the stellar mass a halo of given mass hosts; the ratio peaks at a few per cent near 10^{12} solar masses [250].

SPARC. The Spitzer Photometry and Accurate Rotation Curves sample holds 175 disc galaxies with 3.6 µm photometry and rotation curves from atomic-hydrogen and Hα data, with mass models for gas, disc and bulge [251].

22.1 The model tested

The linear term of Eq. (17) adds $|\gamma_1|/2$ to the centripetal acceleration at every radius, Eq. (18). In Ref. [144], γ_1 is a boundary condition of the path integral, not a field sourced by local matter. We therefore take it to be the same at every radius of a galaxy, whatever the distribution of its baryons. At radius R of a disc galaxy the prediction is

$$g_{\mathrm{model}}(R) = g_{\mathrm{bar}}(R) + \tfrac{1}{2}|\gamma_1| - \gamma_2 R, \tag{43}$$

where g_{bar} is the Newtonian acceleration of the observed stars and gas (H).

The Λ term and the turnover. With the Planck 2018 value of Λ [116], $\gamma_2 = \Lambda c^2/3 = 3.27 \times 10^{-36}\,\mathrm{s}^{-2}$. The term $\gamma_2 R$ stays below 7×10^{-4} of the observed acceleration at every point. It moves the fitted $|\gamma_1|$ by 7×10^{-5} of its value [sparc:S2a]. For a point mass the speed has its minimum v_* at the turnover radius r_* of Eq. (19); beyond it the curve rises as $\sqrt{|\gamma_1| r/2}$.

The values tested.

(i) Bridge A, Eq. (20): $|\gamma_1| = 4.88 \times 10^{-10}$ ($\beta = 0$) to $5.42 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ ($\beta \to -\infty$) [Q19], with no free parameter.

(ii) One universal $|\gamma_1|$, fitted.

(iii) The dwarf-galaxy scaling of Ref. [144], $|\gamma_1|(M) = g_0(1 + M/10^{10} M_\odot)$, with $g_0 \approx 10^{-28}\,\mathrm{m^{-1}}$ in units $c = 1$, i.e. $8.99 \times 10^{-12}\,\mathrm{m\,s^{-2}}$. The authors take it over from conformal-gravity fits; it is not derived.

Comparison models. We compare with Newton, $g_{\mathrm{model}} = g_{\mathrm{bar}}$, and with the empirical radial acceleration relation (RAR) [242], which has one acceleration scale $g_\dagger$ (P):

$$g_{\mathrm{model}} = F(g_{\mathrm{bar}}) = g_{\mathrm{bar}}/[1 - \exp(-\sqrt{g_{\mathrm{bar}}/g_\dagger})]. \tag{44}$$

22.2 Data, error model and likelihood

Data and provenance. From SPARC [251] we use 175 disc galaxies and 3391 radii. At each radius we use the rotation speed V_{obs} and its random error δV. We also use the speeds V_{gas}, V_{disk} and V_{bul} of the mass model at unit mass-to-light ratio. Per galaxy we use D, i and their errors, $L_{[3.6]}$, M_{HI}, V_{flat} and the quality flag Q. The two public files are not in the repository. The script checks their SHA256 and stops on a mismatch [sparc:S0]:

```
MassModels_Lelli2016c.mrt   9108994b12cc401b94a1768beca61c53ec354779385c9c9cc571049f3043244c
SPARC_Lelli2016c.mrt        5aa0501f6b0d881fa579030e315e7b5b6ef561a5bd3a07472f9929c7e5728243
```

We apply the cuts of Ref. [242]. $Q = 3$ removes 12 galaxies and $i < 30°$ removes 10 more, which leaves 153. The cut $\delta V/V_{\mathrm{obs}} < 0.1$ keeps 2696 of their 3168 points, from 147 galaxies [sparc:S1].

Accelerations and errors. With $\Upsilon_{\rm d} = 0.5$ and $\Upsilon_{\rm b} = 0.7$ [242],

$$g_{\rm obs} = \frac{V_{\rm obs}^2}{R}, \qquad g_{\rm bar} = \frac{V_{\rm gas}|V_{\rm gas}| + \Upsilon_{\rm d} V_{\rm disk}|V_{\rm disk}| + \Upsilon_{\rm b} V_{\rm bul}|V_{\rm bul}|}{R}. \tag{45}$$

The errors follow Ref. [244, Eq. (2), Sect. 2.1.2], with $g_{\rm gas}$ and $g_\star$ the gas and stellar parts of $g_{\rm bar}$:

$$\frac{\delta g_{\rm obs}}{g_{\rm obs}} = \Big[\Big(\frac{2\delta V}{V_{\rm obs}}\Big)^2 + \Big(\frac{2\delta i}{\tan i}\Big)^2 + \Big(\frac{\delta D}{D}\Big)^2\Big]^{1/2}, \qquad \delta g_{\rm bar} = \big[(0.1\, g_{\rm gas})^2 + (0.25\, g_\star)^2\big]^{1/2}. \tag{46}$$

The terms of $\delta g_{\rm bar}$ are a 10% HI flux calibration and a 25% scatter in Υ. In dex the two errors average 0.104 and 0.089, against 0.1 and 0.08 in Ref. [244]. We work in $\log_{10} g$ because the dominant errors are multiplicative.

Likelihood. The residual of point i is $r_i = \log_{10} g_{{\rm obs},i} - \log_{10} g_{{\rm model},i}$. The primary likelihood, L1, is the Gaussian likelihood of Section 20 in r_i, with the effective variance

$$\sigma_i^2 = \sigma_{{\rm obs},i}^2 + s_i^2\, \sigma_{{\rm bar},i}^2, \qquad s_i = \frac{\partial \log g_{\rm model}}{\partial \log g_{\rm bar}}. \tag{47}$$

Here $\sigma_{\rm obs}$ and $\sigma_{\rm bar}$ are the errors of Eq. (46) in dex; the error in $g_{\rm bar}$ thus enters through the model's slope. The variant L2 uses the velocity errors plus an intrinsic floor $\sigma_{\rm int}$, fitted for each model. It measures how much unexplained scatter each model needs. Models are compared by ΔBIC, as defined in Section 20, with $N = 2696$ points.

Points of one galaxy share its distance, inclination and Υ errors. Point-wise BIC differences therefore overstate the evidence. We take parameter ranges from a bootstrap over galaxies (1000 draws, fixed seed; brackets give the 16–84% range), and we count galaxy by galaxy which model fits better.

22.3 The radial acceleration relation reproduced

We fit Eq. (44) by orthogonal-distance regression in $(\log g_{\rm bar}, \log g_{\rm obs})$, with errors in both variables, as Refs. [242, 244] did. Our implementation reproduces `scipy.odr` to 10^{-3}.

- The galaxy cuts reproduce exactly (153).
- The public release has slightly more points than the 2016 version: 3168 and 2696, against 3149 and 2693.
- The scatter reproduces: rms 0.133 dex (published 0.13), Gaussian width 0.110 (0.11), and 0.130 expected from the errors (0.12) [sparc:S1].

Figure 10a shows the relation.

The scale $g_\dagger$. We find $g_\dagger = (1.102 \pm 0.016) \times 10^{-10}\,{\rm m\,s^{-2}}$ (1σ, scaled by $\sqrt{\chi^2/{\rm dof}} = \sqrt{1.55}$). The published value is $(1.20 \pm 0.02\,{\rm stat} \pm 0.24\,{\rm sys}) \times 10^{-10}$. Ours lies within the total uncertainty but not within three times the random error. Across six fitting choices it ranges from 1.02 to 1.27 [sparc:S1]. The choices are orthogonal distance in log or linear space, with full or velocity-only errors on $g_{\rm obs}$, and least squares in log space, weighted or unweighted. The papers do not state the space of their fit. The pipeline is right, and the quoted random error leaves out a fitting-choice dependence of about ± 0.12. This orthogonal-distance value serves only the comparison with the published fit. The model comparisons below use the L1 maximum-likelihood value, $g_\dagger = 1.12 \times 10^{-10}\,{\rm m\,s^{-2}}$ (Table 13).

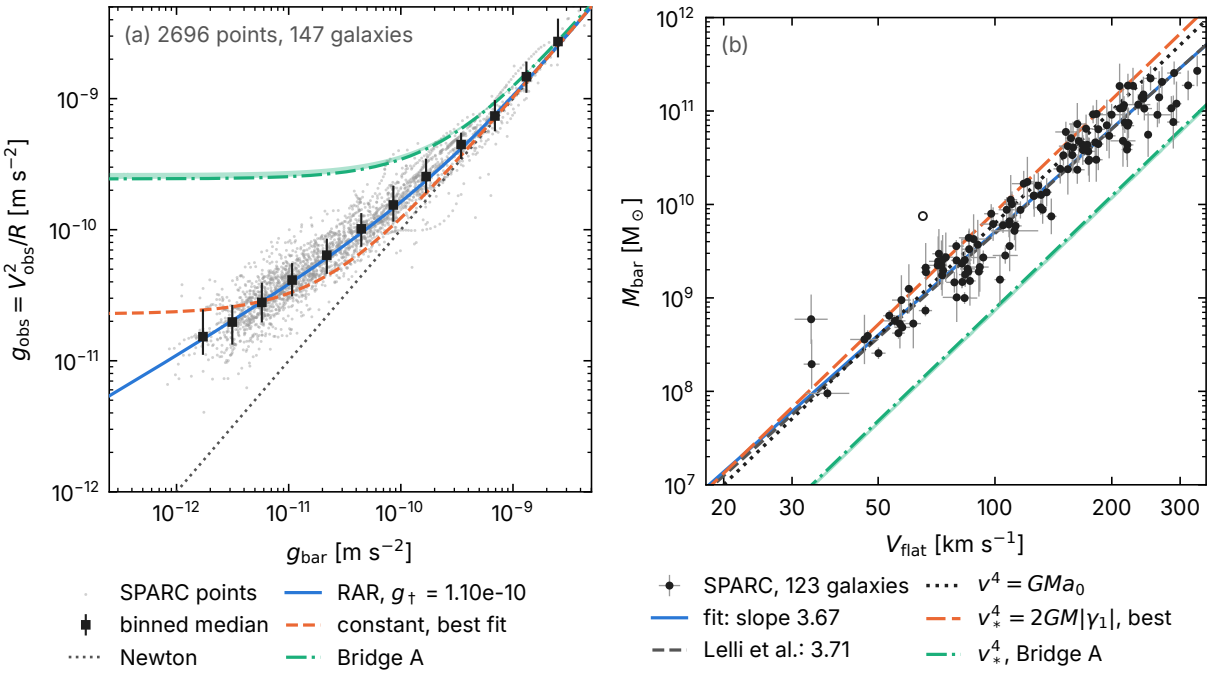


Figure 10: (a) The RAR from SPARC [251]: 2696 points in 147 galaxies (grey) and binned medians (squares; bars give the 16–84% range of the points). Lines: Newton (dotted), our RAR fit (solid), the best universal constant remnant (dashed) and Bridge A (dash-dotted, band over β). At low acceleration the data follow the RAR down, while each constant remnant flattens at $|\gamma_1|/2$. (b) The baryonic Tully–Fisher relation (123 galaxies; open circle: clipped). Lines: our fit (solid), the published fit [243] (dashed), $v^4 = GMa_0$ (dotted), and $v_*^4 = 2GM|\gamma_1|$ for the best universal $|\gamma_1|$ (dashed) and for Bridge A (dash-dotted) [sparc:S1], [sparc:S3].

22.4 Bridge A against the data

With Bridge A's value, Eq. (43) has no free parameter. It lies above the data at 98% of the points. The median of $\log(g_{obs}/g_{model})$ is:

- -0.59 to -0.63 dex overall;
- -0.97 to -1.02 dex at $g_{bar} < 10^{-11}\,\mathrm{m\,s^{-2}}$;
- -0.15 to -0.17 dex at $g_{bar} > 10^{-10}\,\mathrm{m\,s^{-2}}$.

The misfit grows where the remnant dominates (Figure 11a). The fit has $\chi^2/N = 96$ to 107, and its BIC exceeds the RAR's by 2.5 to 2.8×10^5 [sparc:S2b]. Bridge A's value is 11 times the best universal one and 8 times the per-galaxy median. Section 22.8 gives its Tully–Fisher consequences.

22.5 One universal $|\gamma_1|$ against the RAR

The best universal remnant is $|\gamma_1| = 4.54 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ [4.15, 4.98] (Table 13). That is below Bridge A, and below the $a_0/2$ used in the estimate of Section 11. The value is a compromise. It overshoots the lowest accelerations and undershoots intermediate ones (Figure 11a). Against the one-parameter RAR its BIC is worse by +5066 under L1. Under L2, where each model fits its own floor, it is worse by +1542. The constant then needs $\sigma_{int} = 0.169$ dex where the RAR needs 0.123. The RAR fits better in 109 of the 147 galaxies [sparc:S2a].

22.6 The excess acceleration is not constant

Whatever the value of $|\gamma_1|$, Eq. (43) predicts the same excess $\Delta g = g_{obs} - g_{bar}$ at every g_{bar}. The RAR instead tends to $\Delta g \simeq \sqrt{g_{bar} g_\dagger}$ for $g_{bar} \ll g_\dagger$, a logarithmic slope of 1/2. We take

Table 13: Models fitted to the 2696 points [sparc:S2a], [sparc:S2b], [sparc:S2d]. k: free parameters; brackets: galaxy bootstrap 16–84% in units of the value's power of ten; ΔBIC relative to the RAR (positive: worse). The RAR row is the L1 maximum-likelihood fit, used below; the orthogonal-distance value of Section 22.3, 1.102×10^{-10}, serves only the comparison with the published fit.

model	k	best value [$\mathrm{m\,s^{-2}}$]	χ^2/dof	ΔBIC (L1)	$\sigma_{\rm int}$ (L2) [dex]	ΔBIC (L2)
Newton	0	–	20.8	+52645	0.512	+7384
RAR (L1 fit)	1	$g_\dagger = 1.12 \times 10^{-10}$ [1.04, 1.22]	1.55	0	0.123	0
constant remnant	1	$\lvert\gamma_1\rvert = 4.54 \times 10^{-11}$ [4.15, 4.98]	3.47	+5066	0.169	+1542
$g_0(1 + M/10^{10} M_\odot)$	1	$g_0 = 1.25 \times 10^{-11}$	8.98	+19789		
same, g_0 of Ref. [144]	0	$g_0 = 8.99 \times 10^{-12}$	9.54	+21427		
Bridge A, $\beta = 0$	0	$\lvert\gamma_1\rvert = 4.88 \times 10^{-10}$	96.3	+254651		
Bridge A, $\beta \to -\infty$	0	$\lvert\gamma_1\rvert = 5.42 \times 10^{-10}$	107.0	+283359		
free exponent p	2	$K = 7.56 \times 10^{-11}$, $p = 0.436$	1.55	+82		

binned medians of Δg in 0.3 dex bins below $g_\dagger/10 = 1.12 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ (914 points). Then we fit their slope, resampling galaxies:

$$\mathrm{d}\log \Delta g/\mathrm{d}\log g_{\rm bar} = 0.44\ [0.39,\ 0.49] \qquad \text{(a constant: 0; the RAR in the same bins: 0.45)} \tag{48}$$

[sparc:S2d]. The RAR's local slope is below 1/2 because these bins are not deep in the low-acceleration regime.

A free exponent. The law $g_{\rm bar} + K(g_{\rm bar}/10^{-10}\,\mathrm{m\,s^{-2}})^p$ contains the constant at $p = 0$. Fitted under L1 with K and p free, it gives $p = 0.436$ [0.396, 0.464] from all points. The same fit restricted to the 914 points below $g_\dagger/10$ gives $p = 0.543$. Against $p = 0$, ΔBIC $= -4983$; against the RAR it is still +82.

The binned excess. Below $g_\dagger/10$ the binned excess rises from 1.4 to $3.0 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, with no plateau. Further up it rises to 8.4×10^{-11} and flattens in the bins at $\log g_{\rm bar} = -9.77$ and -9.46. The RAR excess also reaches its maximum there (Figure 11b).

A null test of the estimator. Δg is binned in the noisy $g_{\rm bar}$, so its errors are correlated with the binning variable. We therefore test the estimator on mock data. Each mock keeps the real radii, velocity errors and assumed $g_{\rm bar}$ of the 153 galaxies. Per galaxy it draws:

- a true distance and inclination from their tabulated errors;
- a true stellar mass-to-light ratio (0.1 dex) and a true HI flux (10%).

Per point it draws a velocity error. The true acceleration follows from the true $g_{\rm bar}$ at the true radius. The observer infers $g_{\rm obs}$ with the assumed distance and inclination. The mock then passes the same precision cut, bins, low-acceleration cut, slope estimator and galaxy bootstrap [sparc:S2d_mock].

- *Null.* For a constant $\lvert\gamma_1\rvert = 4.54 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, 400 mocks return a slope of +0.005 [−0.033, +0.046] (16–84% over the mocks). One mock analysed exactly like the data gives +0.012, with bootstrap range [−0.033, +0.048]. None of the 400 reaches the data's 0.436; the largest is 0.10.
- *Positive control.* For the RAR the mocks return 0.469 [0.424, 0.518], against 0.45 without noise. 76% of them lie above the data's value.

The estimator cannot turn a constant excess into the observed slope, and the data sit inside the RAR's mock distribution.

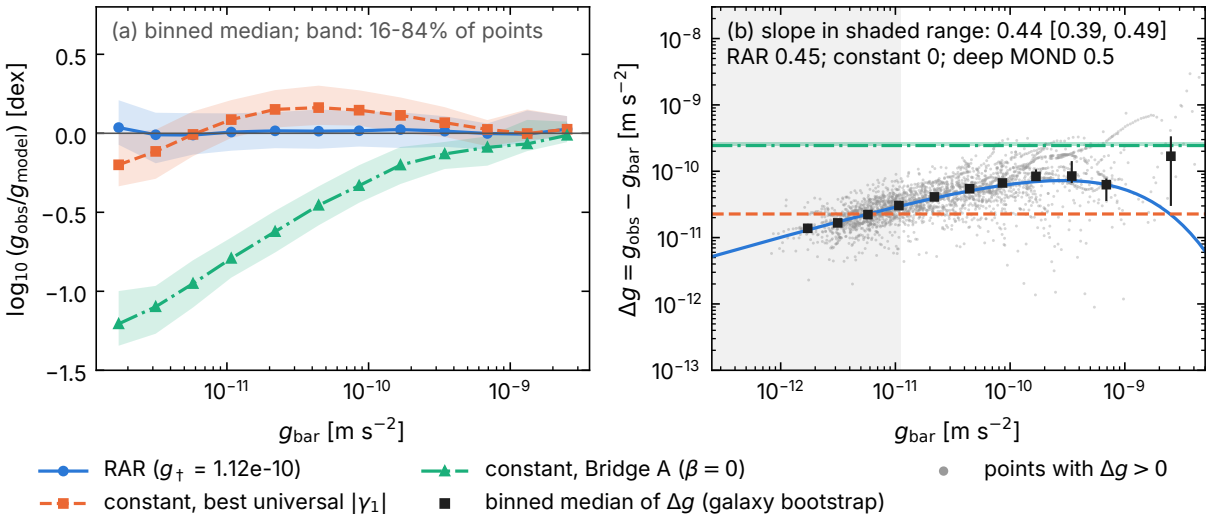


Figure 11: (a) Binned median of $\log(g_{obs}/g_{model})$ for the RAR, the best universal constant remnant and Bridge A ($\beta = 0$); bands: 16–84% of the points. (b) The excess Δg (grey: points with $\Delta g > 0$; squares: binned medians, galaxy-bootstrap bars), the RAR prediction (solid), and the constant excess of the best universal fit (dashed) and of Bridge A (dash-dotted). In the shaded region the slope is 0.44, where a constant would be flat. Bins whose range reaches $\Delta g \leq 0$ are not drawn [sparc:S2d].

22.7 One $|\gamma_1|$ per galaxy and the Oppenheim–Russo scaling

If the remnant were a boundary constant per galaxy, each galaxy could carry its own value. We fit one $|\gamma_1|$ to each of the 132 galaxies with at least five points, with Υ fixed and velocity errors per point. Distance, inclination, Υ ($\pm 25\%$) and HI ($\pm 10\%$) errors enter as whole-galaxy refits, added in quadrature; the median total error is 0.18 dex. The values have median $5.95 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, a 16–84% range of 3.69 to 10.4×10^{-11}, and a spread of 0.25 dex (Figure 12). 131 of the 132 lie below Bridge A beyond their errors [sparc:S2c].

The mass trend. $|\gamma_1|$ rises with $M_{bar} = \Upsilon_d L_{[3.6]} + 1.33 M_{HI}$: Spearman $\rho = 0.45$ ($p = 4 \times 10^{-8}$), log-log slope 0.14 [0.12, 0.17]. Against V_{flat} (116 galaxies), $\rho = 0.61$ and the slope is 0.69 [0.60, 0.76]. Part of the mass trend trades against Υ. With Υ_d free (a log-normal prior of 0.11 dex around 0.5, bounded to 0.1–1.0), the slope is 0.09 and $\rho = 0.34$.

The Oppenheim–Russo scaling rises far more steeply than the data. We fit four laws to the per-galaxy $\log|\gamma_1|$:

- a constant: rms 0.249 dex, BIC 17.2;
- a power law in M_{bar}: rms 0.215 dex, BIC −16.7;
- $g_0(1 + M/10^{10} M_\odot)$ with g_0 fitted (2.04×10^{-11}): rms 0.395 dex, BIC 138.8;
- the same law with the g_0 of Ref. [144]: rms 0.531 dex, BIC 212.5.

Even with its amplitude fitted, the scaling is worse than a constant by +122. It also has a structural flaw. For massive galaxies it makes $|\gamma_1| \propto M$, so $v_*^4 \propto M^2$: a Tully–Fisher slope of 2, where the data give 3.7.

The shape within galaxies. We fit one free parameter per galaxy for each law, with velocity errors only (2652 points). A free constant gives $\chi^2 = 109720$, and a free $g_\dagger$ gives 26790. The RAR shape wins in 98 of 132 galaxies. Both χ^2 are large because velocity errors omit non-circular motions and the galaxy-wide errors.

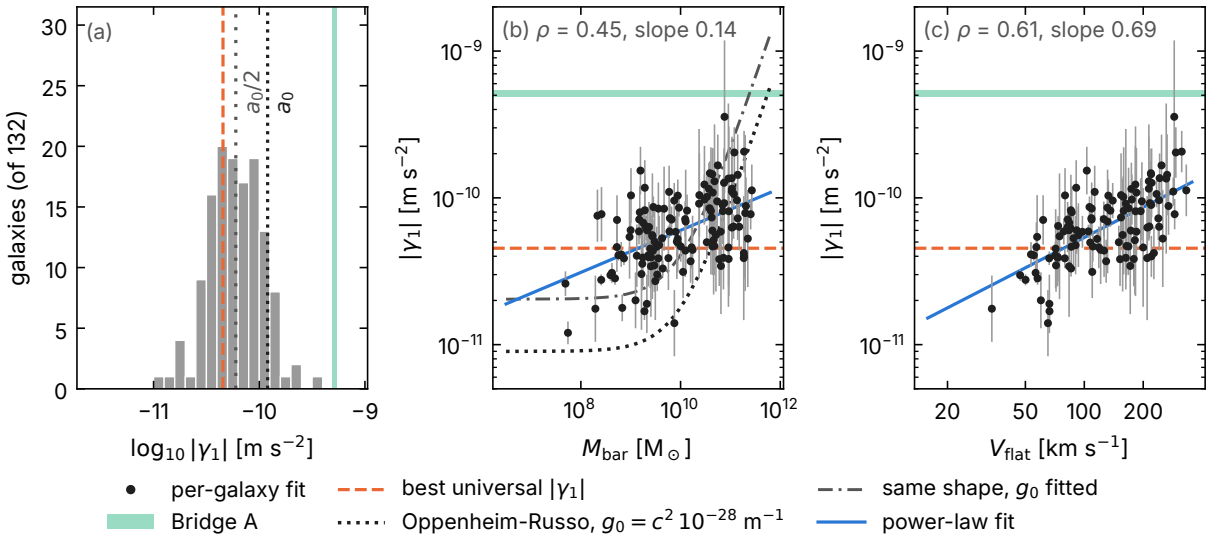


Figure 12: One $|\gamma_1|$ per galaxy (132 SPARC galaxies). (a) Distribution, with Bridge A (band), the best universal value (dashed), $a_0 = 1.2\times10^{-10}$ (the order Ref. [144] quotes) and $a_0/2$. (b) Against $M_{\rm bar}$, with the Oppenheim–Russo scaling at their amplitude (dotted) and fitted (dash-dotted), and a power-law fit (solid). (c) Against $V_{\rm flat}$. The values sit well below Bridge A and rise far more slowly with mass than $(1 + M/10^{10}M_\odot)$ [sparc:S2c].

22.8 The baryonic Tully–Fisher relation

We repeat the analysis of Ref. [243] (P):

- the galaxies with $V_{\rm flat}$ and $i \geq 30°$, without the eight starburst dwarfs and Holmberg II (UGC 4305) it excludes;
- $M_{\rm bar} = \Upsilon_{\rm d} L_{[3.6]} + 1.33 M_{\rm HI}$ with $\Upsilon_{\rm d} = 0.5$, and the errors of its Eq. (5), with 10% on $M_{\rm HI}$;
- a fit of $\log M_{\rm bar} = s \log V_{\rm flat} + \log A$ with errors in both variables and an intrinsic scatter that sets $\chi^2/{\rm dof} = 1$;
- 2.6σ clipping as in LTS_LINEFIT [252], which removes UGC 7125, the outlier the paper names.

We find $s = 3.67 \pm 0.10$ and $\log A = 2.36 \pm 0.21$ (bootstrap), with an intrinsic scatter of 0.109 dex for 123 galaxies. The paper finds 3.71 ± 0.08, 2.27 ± 0.18 and 0.10 ± 0.02 for 118; the public table has $V_{\rm flat}$ for more galaxies. Without clipping we find 3.65, 2.41 and 0.119 [sparc:S3].

The turnover relation. With the slope fixed at 4, $\log A = 1.676$, i.e. $V_{\rm flat}^4 = GM_{\rm bar}a$ with $a = 1.59 \times 10^{-10}\,{\rm m\,s^{-2}}$, against MOND's $a_0 = 1.2 \times 10^{-10}$ [145]. Through Eq. (19) this needs $|\gamma_1| = 7.95 \times 10^{-11}\,{\rm m\,s^{-2}}$. Two estimators compare Bridge A with the data in v^4 at fixed mass:

- the ratio of normalisations, Bridge A's $|\gamma_1|$ over the 7.95×10^{-11} that the fitted $\log A$ needs: 6.1 to 6.8;
- the median offset of the galaxies from the Bridge A relation at their observed $V_{\rm flat}$, $10^{\,\mathrm{median}(\log M_{\rm obs} - \log M_{\rm pred})}$: 6.4 to 7.1.

Both use v_*, the minimum of the model curve, while $V_{\rm flat}$ is measured further out. Section 11 gives the estimate against $a_0/2$.

The outer speed. The direct test predicts each galaxy's speed at its outermost measured radius from its own baryons (124 galaxies). The median $\log(V_{\rm pred}/V_{\rm flat})$ and the rms are:

- the RAR: +0.014 and 0.061 dex;
- the best universal constant: +0.006 and 0.087 dex;
- Bridge A: +0.48 to +0.50 dex, a factor 80 to 98 in v^4;
- Newton: −0.32 dex.

The constant gets the outer normalisation right on average. It has more scatter than the RAR, though, and a trend with mass (Spearman $\rho = -0.42$, against -0.08 for the RAR). It overshoots small galaxies and undershoots large ones [sparc:S3].

Choices made after seeing the data.

- The $g_\dagger$ check first required $1.20 \pm 3 \times 0.02$ and failed at 1.10. It now uses the total uncertainty, ± 0.24.
- The 1% tolerance on the point counts was set after seeing 3168 and 2696.
- The first BTFR fit kept Holmberg II and did not clip, and it missed the published slope by more than 2σ. Both steps were then taken from the published method [243, 252].
- The bound $0.1 \leq \Upsilon_\mathrm{d} \leq 1$ was added after unbounded fits ran to $\Upsilon_\mathrm{d} \approx 9$ in a few galaxies.
- The outer-speed test was added after we saw that v_* misrepresents rising model curves.
- The mock null test of the slope estimator was added after the slope was known. Its two tolerances were fixed before the mocks were first run: $|\mathrm{median}| < 0.1$ for the null, and within 0.1 of the noiseless RAR slope for the control.

None of these choices changes a verdict.

Caveats.

- We test the remnant as written: a constant $|\gamma_1|$, independent of the mass distribution, from a static, spherically symmetric calculation [144]. Hertzberg and Loeb dispute its large-scale force [146]; Section 11 gives its other limits.
- A remnant that follows the matter distribution is untested here, as are lensing, the microwave background and clusters.
- Point-wise ΔBIC overstates the evidence, but every comparison keeps its sign in the bootstrap and in the galaxy counts.

All numbers in this section come from `docs/simulation/sparc_rotation.py` (39 of 39 checks, about 70 s on one core), with results in `docs/simulation/results/sparc_rotation.json`.

Verdict. The reproduction of the RAR from the public SPARC data *holds*: the same 153 galaxies, the same scatter, and $g_\dagger$ inside its published uncertainty. Bridge A is *refuted* by the rotation-curve points themselves; it overshoots 98% of them, by a dex at low acceleration. A constant extra acceleration of any size, and the Oppenheim–Russo mass scaling, are *refuted* as the explanation of rotation curves. The best constant fits worse than the one-parameter RAR (ΔBIC +5066, worse in 109 of 147 galaxies), and the observed excess rises as $g_\mathrm{bar}^{0.44}$ where a constant predicts zero, a slope the estimator does not create in mocks. Section 23 takes up the accumulated remnant, halos and primordial black holes, and gives the dark-matter verdict.

23 Dark matter II: accumulation, halos, cosmic time and primordial black holes

This section tests the two routes of Hypothesis 7 (Section 11) once their history is included, and scores primordial black holes, the contained remnant's candidate, against published constraints.

23.1 The accumulated stochastic remnant

Data, cuts, error model, the likelihoods L1 and L2 and the BIC are those of Section 22.2. Primers 10 and 9 explain the quantities and the statistics. Our galaxy-bootstrap resamples use a different random seed, so bootstrap ranges can differ from that section's in the last digit.

Hypothesis. (H) The random kicks that the frame's diffusion leaves in the field (Section 5) accumulate as a random walk. Suppose the variance of the extra acceleration grows with the local source and with time at a constant rate $\dot{a}$, $\langle \delta g^2 \rangle = g_{\rm bar}\, \dot{a}\, t$. If the entropic attraction is of the order of its root mean square, the extra acceleration is the geometric mean of $g_{\rm bar}$ and an accumulated scale $a_{\rm acc} \equiv \dot{a}\, t$. The classical-quantum dynamics does not give this law; it is an assumption. We fixed two forms before any fit:

$$\text{form B (Brownian sum):} \quad g_{\rm obs} = g_{\rm bar} + \sqrt{g_{\rm bar}\, a_{\rm acc}}\,, \tag{49}$$

$$\text{form S (self-consistent):} \quad g_{\rm obs} = \tfrac{1}{2} g_{\rm bar} + \left(\tfrac{1}{4} g_{\rm bar}^2 + g_{\rm bar}\, a_{\rm acc}\right)^{1/2}. \tag{50}$$

Form S solves $(g_{\rm obs} - g_{\rm bar})\, g_{\rm obs} = g_{\rm bar} a_{\rm acc}$: the excess is sourced by the total field it helps to make. Both forms tend to $\sqrt{g_{\rm bar} a_{\rm acc}}$ at low acceleration, so $a_{\rm acc}$ plays the role of Milgrom's a_0 [145]. Form S *is* MOND's "simple" interpolating function, $\nu(y) = \frac{1}{2} + (\frac{1}{4} + 1/y)^{1/2}$ in $g_{\rm obs} = \nu(g_{\rm bar}/a_0)\, g_{\rm bar}$. Form B adds the Newtonian and the deep-MOND acceleration. Both forms were introduced after Section 22.6 had measured the residual slope of 0.44. Agreeing with that slope is therefore not a prediction, and refuting them refutes these two forms, not every form a GUID prediction could take. The reference is the RAR of Eq. (44) [242], with one universal $g_\dagger$.

Fits. Form S fits as well as the RAR, with $a_{\rm acc} = 1.10 \times 10^{-10}\,{\rm m\,s^{-2}}$ (Table 14, Fig. 13) (S). Form B is worse by ΔBIC +209. Its excess keeps growing with $g_{\rm bar}$ and overshoots the inner parts of massive galaxies by 0.07 to 0.08 dex. Both forms beat the constant by about 5000. The low-acceleration slope of $\log(g_{\rm obs} - g_{\rm bar})$ is 0.436 in the data, 0.448 for the RAR, 0.449 for form S and 1/2 for form B. Galaxy by galaxy, the universal RAR beats form S in 73 of 147 galaxies [remnant:R1a].

The solar system. Both forms are pointwise laws in $g_{\rm bar}$, and the Sun's field is the cleanest $g_{\rm bar}$ there is. Add a small radial force $f = k\, r^n$ to $F = GM/r^2$ and write $\epsilon = f r^2/GM$. Near-circular orbits have the apsidal angle $\psi = \pi[3 + {\rm d}\ln F/{\rm d}\ln r]^{-1/2}$, and ${\rm d}\ln F/{\rm d}\ln r = -2 + (n+2)\epsilon$. The perihelion therefore moves by

$$\Delta\varpi = 2\psi - 2\pi = -\pi\,(n+2)\,\epsilon \quad \text{per orbit}, \tag{51}$$

which we checked by integrating orbits [remnant:R1a]. Form B adds $\sqrt{GM a_{\rm acc}}/r$ ($n = -1$); form S tends to the constant $a_{\rm acc}$ ($n = 0$). With the fitted scales, form B turns the perihelion of Mars by -6.0×10^4 mas/yr and form S that of Saturn by -74 mas/yr. The EPM2011 ephemerides allow -0.00020 ± 0.00037 and -0.0032 ± 0.0047 mas/yr [253]. Over six planets, form B exceeds the allowed range by factors of 10^5 to 1.6×10^8 and form S by 190 to 8×10^4. As written, both forms are *refuted*. A survivor would need a rule that switches the term off at high acceleration, as the RAR's exponential does; accumulation supplies none.

Table 14: The two accumulated forms, one universal parameter each, on the data of Table 13, which gives the Newton, RAR and constant rows (S) [remnant:R1a]. Scale in $10^{-10}\,{\rm m\,s^{-2}}$ (galaxy bootstrap 16–84%); ΔBIC relative to the RAR; last column: median $\log(g_{\rm obs}/g_{\rm model})$ at $g_{\rm bar} > 10^{-9}\,{\rm m\,s^{-2}}$ (RAR: -0.001).

model	scale	χ^2/dof	ΔBIC (L1)	$\sigma_{\rm int}$ [dex]	ΔBIC (L2)	high $g_{\rm bar}$ [dex]
form B, $a_{\rm acc}$	0.772 [0.708, 0.839]	1.579	+209	0.129	+226	−0.079
form S, $a_{\rm acc}$	1.10 [1.02, 1.19]	1.542	−5.5	0.123	−2.6	−0.019

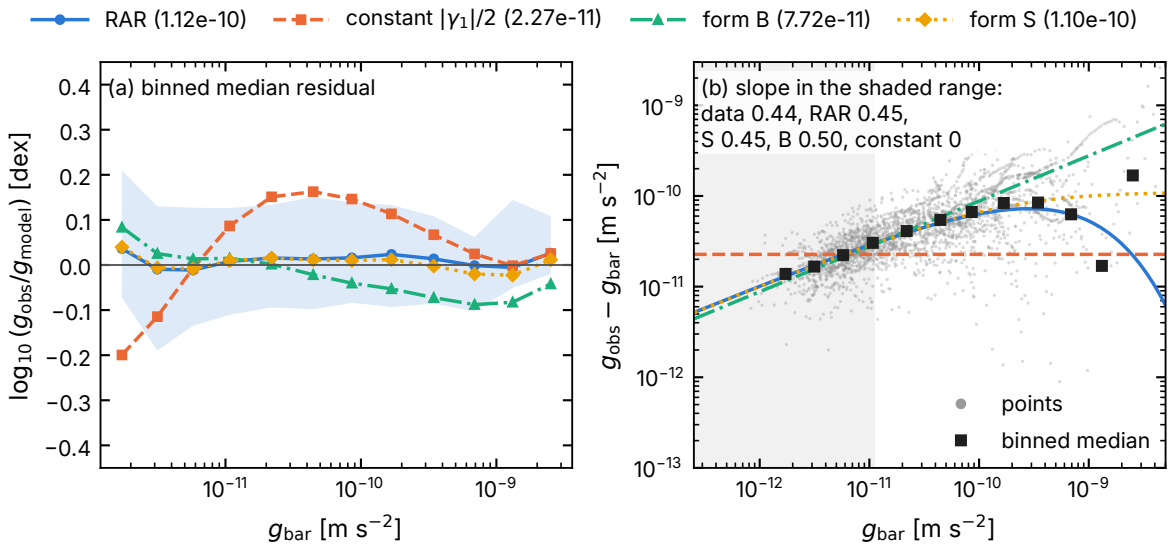


Figure 13: (a) Binned median residual $\log(g_{\rm obs}/g_{\rm model})$ of the universal fits (band: 16–84% of the points about the RAR). (b) The excess $g_{\rm obs} - g_{\rm bar}$ with binned medians and the four models. SPARC data [251] (S) [remnant:R1a]. Look for form B drifting low above $10^{-10}\,{\rm m\,s^{-2}}$ and form S on the RAR.

Table 15: The nine candidate scales (S) [remnant:R1b]: fitted scale over candidate, and ΔBIC of the fixed candidate against the form's best fit on the same data. The five scales without 2π share one row (ranges).

candidate	value [$\rm m\,s^{-2}$]	B: ratio	B: ΔBIC	S: ratio	S: ΔBIC	$g_\dagger$: ratio
$cH_0/2\pi$ ($H_0 = 67.4$)	1.04×10^{-10}	0.689	+778	0.990	−7.2	1.009
$cH_0/2\pi$ ($H_0 = 73$)	1.13×10^{-10}	0.684	+827	0.973	−2.5	0.992
$cH_\infty/2\pi$ ($H_0 = 67.4$)	8.63×10^{-11}	0.832	+176	1.196	+219	1.219
$cH_\infty/2\pi$ ($H_0 = 73$)	9.34×10^{-11}	0.826	+196	1.176	+181	1.198
cH_0, cH_∞, c/t_0	$(5.4\text{–}7.1)\times10^{-10}$	0.11–0.13	> 29000	0.16–0.19	> 25000	0.16–0.19

23.2 Natural scales and the look-elsewhere effect

We fixed nine candidates for $a_{\rm acc}$ before fitting: cH_0, $cH_0/2\pi$, cH_∞ and $cH_\infty/2\pi$, with $H_\infty = \sqrt{\Omega_\Lambda} H_0$ and $\Omega_\Lambda = 0.685$, each for $H_0 = 67.4$ [116] and $73\,{\rm km\,s^{-1}\,Mpc^{-1}}$; and c/t_0 with $t_0 = 13.8\,{\rm Gyr}$. SPARC's 97 Hubble-flow distances assume $H_0 = 73$ [251]. Rescaled to 67.4 they give $a_{\rm acc} = 1.03 \times 10^{-10}\,{\rm m\,s^{-2}}$ for form S, and each 67.4 candidate is compared with these rescaled data. The error on $\log a_{\rm acc}$ is 0.110 dex for form S: galaxy bootstrap 0.034, a ±0.1 dex shift of Υ 0.105, likelihood choice 0.008. For form B it is 0.131 dex [remnant:R1b].

The scales without 2π are excluded (Table 15, Fig. 14). They are the ones a horizon argument gives, Bridge A among them (Section 11). Form S's $a_{\rm acc}$ lies within 3% of $cH_0/2\pi$ at either H_0, and within 1% when both use 67.4. This is Milgrom's coincidence $a_0 \approx cH_0/2\pi$ [145], which $g_\dagger$ shares; it is not a test, for three reasons. First, accumulation supplies no factor 2π. Second, the candidates form two clusters a factor 2π apart (span 0.91 dex, median spacing 0.035 dex), and four of them lie within 2σ of $a_{\rm acc}$. Third, a scale drawn log-uniformly within ±1 dex of the fit would land as close to a candidate with probability 0.10 for form S. For form B, whose nearest candidate is $cH_\infty/2\pi$, the probability is 0.21 [remnant:R1b].

23.3 Per-galaxy scales and history

We fitted one scale to each of the 132 galaxies with at least five points. Each fit uses velocity errors per point. Distance, inclination, Υ ($\pm25\%$) and H I mass ($\pm10\%$) enter as whole-galaxy shifts, added in quadrature into an error s_i. The intrinsic scatter τ of $\ell_i = \log a_i$ maximises

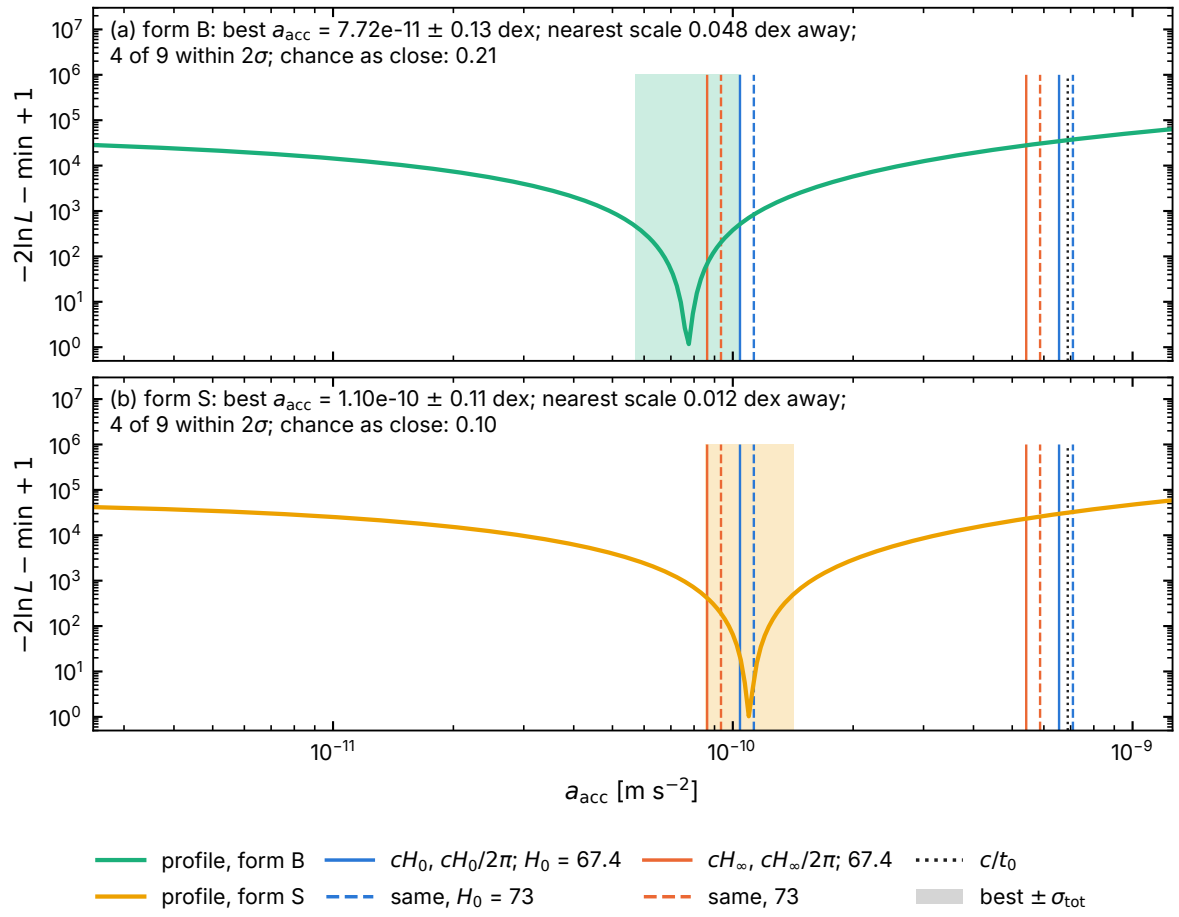


Figure 14: $-2\ln L$ of the universal fit against a_{acc} for forms B (a) and S (b), with the nine candidates and the band $\pm\sigma_{tot}$ (S) [remnant:R1b]. Blue: cH_0 family; orange: cH_∞ family; solid: $H_0 = 67.4$; dashed: 73; dotted: c/t_0. Look for the two clusters a factor 2π apart: the minimum falls in the lower one, with four candidates inside the band.

$-\sum_i[(\ell_i-\mu)^2/(s_i^2+\tau^2)+\ln(s_i^2+\tau^2)]$.

Universality. Form S gives $\tau = 0.20$ dex, as much as the RAR's per-galaxy $g_\dagger$; form B gives 0.23 [remnant:R1c]. Mock universes with a universal RAR, passed through the SPARC error model and the same analysis, give $\tau = 0.04$ (16–84%: 0.00 to 0.10) [remnant:R2]. The galaxies thus scatter more than this error model allows, equally for $g_\dagger$ and $a_{\rm acc}$. Either a galaxy-level term is missing, such as an Υ that varies by more than 0.1 dex, or neither scale is universal. Marginalising each galaxy over Υ, distance and inclination, Li et al. find no credible variation [254]; Rodrigues et al. conclude the opposite [255]. Here form S and the RAR stand or fall together.

History. A universal $a_{\rm acc}$ carries no history beyond the global time. A galaxy-specific accumulation would give older, more evolved systems a larger scale. Before computing anything we fixed five proxies and the predicted signs. $\log a_i$ should fall with Hubble type, gas fraction and disk scale length. It should rise with effective surface brightness and with $N_{\rm orb} = t_0 V_{\rm out}/(2\pi R_{\rm out})$, the orbits completed at the outermost radius in a Hubble time (median 17). A fixed variance per orbit gives $a_i \propto N_{\rm orb}$. We correlated three per-galaxy quantities with the proxies: form B's a_i, the mean residual from the universal form B, and the RAR's $g_{\dagger,i}$. That gives 15 Spearman and 15 partial correlations (at fixed $\log M_{\rm bar}$), with bootstrap intervals and the Holm correction over all 30. A universal law fitted per galaxy with a slightly wrong shape already correlates with proxies that track a galaxy's place on the $g_{\rm bar}$ axis. We therefore also tested each ρ against its mean in 500 mock universes, using the data's own bootstrap spread. The mocks draw distances, inclinations, Υ and gas from their errors and add 0.044 dex of point scatter, calibrated on the data.

None of the 30 correlations survives the Holm correction (smallest adjusted $p = 0.33$). None departs from the universal-RAR baseline (smallest adjusted $p = 0.35$) [remnant:R2]. Six partial correlations fall outside the mocks' 95% bands, but those bands are too narrow, because the mocks lack the extra scatter found above. The predicted signs hold for two of five proxies. The slope of $\log a_i$ on $\log N_{\rm orb}$ is +0.12 (95%: −0.17 to +0.46), where a per-orbit random walk predicts +1 (Fig. 15). There is no history dependence beyond the global time, and per-orbit accumulation is *refuted*.

23.4 The contained remnant: halos

Primordial black holes are cold and collisionless, so they would form halos as any cold dark matter does. We fitted each of the 132 galaxies with $g_{\rm model} = g_{\rm bar} + GM_{\rm h}(<R)/R^2$ under L1, maximising likelihood times prior. Three halo models enter.

1. NFW, $M_{\rm h}(<r) = M_{200}\,\mu(cr/r_{200})/\mu(c)$ with $\mu(u) = \ln(1+u) - u/(1+u)$ and $M_{200} = \frac{4\pi}{3}200\rho_{\rm crit} r_{200}^3$. The concentration prior is $\log c_{200} = 0.905 - 0.101\log[M_{200}/(10^{12}h^{-1}M_\odot)]$, with 0.11 dex scatter [247].
2. Cosmological NFW: in addition, the stellar-to-halo relation of Moster et al. [250],

$$m_\star/M_{200} = 2N\left[(M_{200}/M_1)^{-\beta} + (M_{200}/M_1)^{\gamma}\right]^{-1},$$

   with $\log M_1 = 11.590$, $N = 0.0351$, $\beta = 1.376$, $\gamma = 0.608$, 0.15 dex scatter and $m_\star = 0.5L_{[3.6]}$. These are the halos that any cold dark matter must form.
3. Burkert, a cored profile, $\rho = \rho_0 r_0^3/[(r+r_0)(r^2+r_0^2)]$, without prior.

The per-galaxy BIC counts $k = 2$ for a halo, 1 for a free scale and 0 for a universal law.

Summed over galaxies and relative to the universal RAR, ΔBIC is −1360 for NFW (264 parameters), −1108 for cosmological NFW and −1780 for Burkert [remnant:R3] (S). The universal forms S and B have −4.6 and +218 on these 2652 points (Table 14, on 2696 points, gives −5.5 and +209). Every per-galaxy model beats every universal law. This is largely because

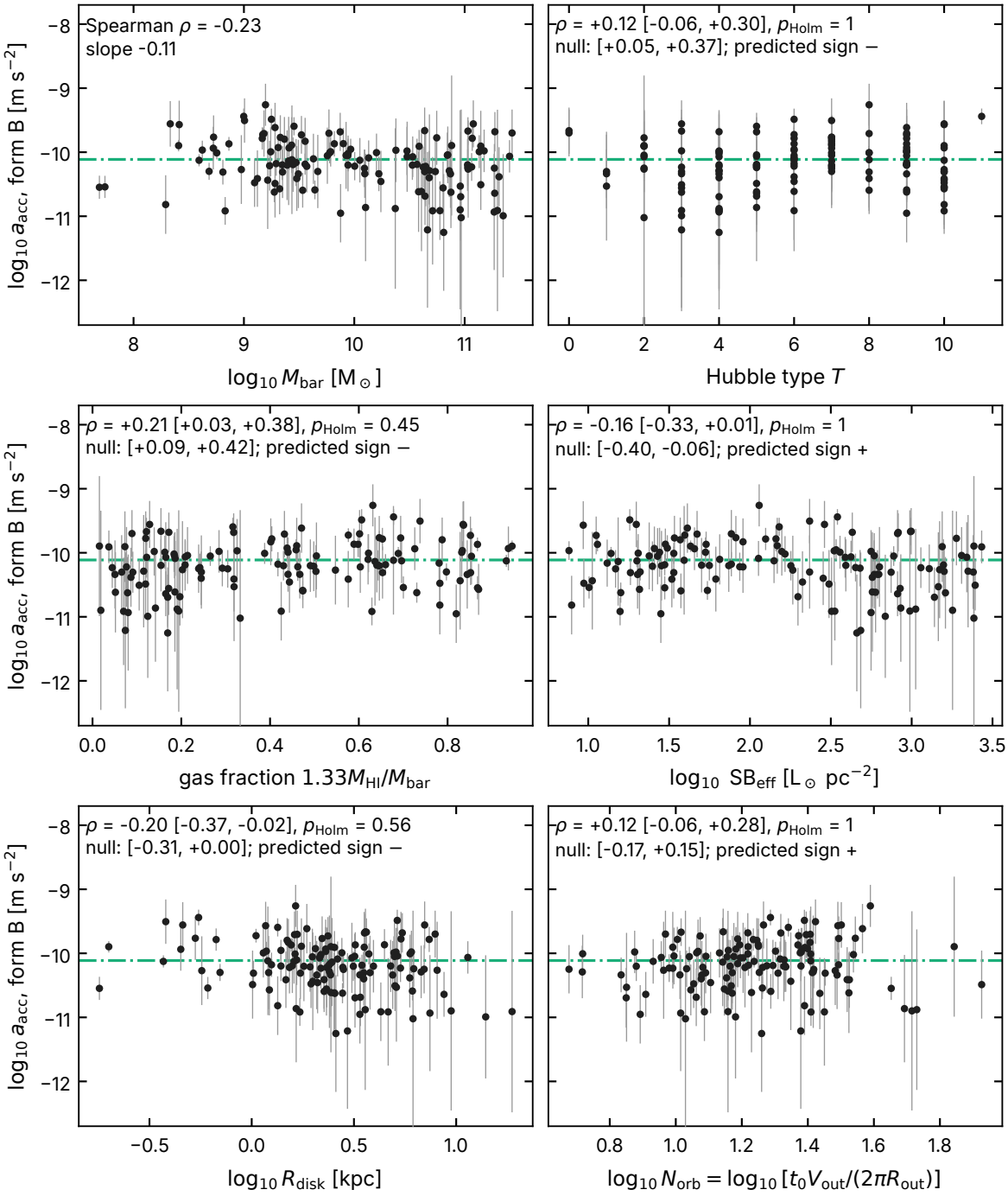


Figure 15: Per-galaxy $a_{\rm acc}$ (form B; statistical and whole-galaxy errors) against baryonic mass and the five proxies, with Spearman ρ, bootstrap 95% interval, Holm-adjusted p, the mock 95% band for a universal RAR and the predicted sign; dash-dotted: the universal fit (S) [remnant:R2]. Look for the absence of the predicted trends, in particular with $N_{\rm orb}$.

Table 16: A universal term added to halos (S) [remnant:R4]. Bound: one-sided 95% (galaxy-bootstrap 84th percentile in brackets). Share: median fraction of the dark acceleration $g_{\rm obs} - g_{\rm bar}$ at each galaxy's outermost point supplied at the bootstrap bound. Alone, the square-root term needs 7.7×10^{-11} and the constant $2.3 \times 10^{-11}\,\mathrm{m\,s^{-2}}$.

halos	term	best A [$\mathrm{m\,s^{-2}}$]	ΔBIC	95% bound [$\mathrm{m\,s^{-2}}$]	share
NFW	$\sqrt{g_{\rm bar} A}$	0	+7.9	2.1×10^{-15} (6.5×10^{-15})	1.0%
NFW, cosmological	$\sqrt{g_{\rm bar} A}$	0	+7.9	2.3×10^{-15} (3.0×10^{-14})	2.1%
Burkert	$\sqrt{g_{\rm bar} A}$	0	+7.9	1.3×10^{-14} (1.4×10^{-13})	4.5%
NFW	constant	0	+7.9	8.4×10^{-13} (3.5×10^{-12})	16%
NFW, cosmological	constant	6.6×10^{-13}	+3.5	1.2×10^{-12} (1.9×10^{-12})	8.5%
Burkert	constant	4.2×10^{-12}	−50.1	5.1×10^{-12} (6.7×10^{-12})	31%

L1 treats a galaxy's common distance, inclination and Υ errors as independent per point, and any per-galaxy parameter absorbs them: one free RAR scale per galaxy already reaches −1883. Against that yardstick NFW halos are worse by +523, Burkert halos by +103 and cosmological NFW halos by +775. Cored halos beat cuspy ones by −420, the known cusp-core issue. Per galaxy, NFW beats the universal RAR in 88 of 132 galaxies and Burkert in 98. The NFW halos have $\log M_{200} = 11.47$ (16–84%: 10.76 to 12.18), about 24 times the baryonic mass. Halos accommodate the RAR's one-parameter regularity with two parameters per galaxy; they do not predict it. These fits test generic cold dark matter halos: nothing here distinguishes primordial black holes from particle dark matter.

23.5 The contained remnant: primordial black holes

This paper does not test primordial black holes. Section 11 takes them as GUID's contained remnant, holes formed in the start when the inflaton fragments into oscillons. That formation channel is the result of Cotner, Kusenko and Takhistov [148], not ours. The mass window is adopted, not predicted. Gorton and Green find that primordial black holes can be all of the dark matter between about 10^{17} and $10^{22}\,\mathrm{g}$ (10^{14} to $10^{19}\,\mathrm{kg}$) for a single mass. For extended mass functions the window is narrower [23]. Products of Hawking evaporation close the window below (INTEGRAL/SPI MeV gamma rays, Voyager 1 electrons and positrons). Microlensing of M31 by Subaru-HSC closes it above. Three rows of Section 21 record the comparison, in the classes of Section 20 [data:D7.1-D7.3]:

- D7.1, the window: GUID's window coincides with the open one because it was taken from it (CONS).
- D7.2, the evaporation edge: holes lighter than $1.7 \times 10^{11}\,\mathrm{kg}$ have evaporated within 13.8 Gyr (a Stefan–Boltzmann estimate). That is a factor 5.8 below the "about $10^{15}\,\mathrm{g}$" of Carr et al. [256], a value without a stated uncertainty (INFO).
- D7.3, the edge temperatures: a hole of $10^{14}\,\mathrm{kg}$ glows at $1.23 \times 10^{9}\,\mathrm{K}$ ($k_B T = 0.106\,\mathrm{MeV}$), one of $10^{19}\,\mathrm{kg}$ at $1.23 \times 10^{4}\,\mathrm{K}$. The glow at the light edge is what bounds the window (INFO).

Verdict: untested in this paper. As a candidate it is *live*: evaporation and microlensing searches can close the window [23].

23.6 Halos and a universal stochastic term together

The cohesive whole would be baryons, a halo per galaxy and one universal term, $\Delta g_{\rm u} = A$ or $\Delta g_{\rm u} = \sqrt{g_{\rm bar} A}$. At 66 amplitudes from 10^{-16} to $10^{-9.5}\,\mathrm{m\,s^{-2}}$ we refitted every halo and summed the profile $\mathcal{P}(A) = \sum_i \min_{\theta_i}[-2\ln L_i(\theta_i, A) - 2\ln p(\theta_i)]$, with p the halo prior. The one-sided 95% bound is where $\mathcal{P}$ rises by 2.71. Adding the term costs $\Delta\mathrm{BIC} = \mathcal{P}(\hat{A}) - \mathcal{P}(0) + \ln N$, with $N = 2652$ points.

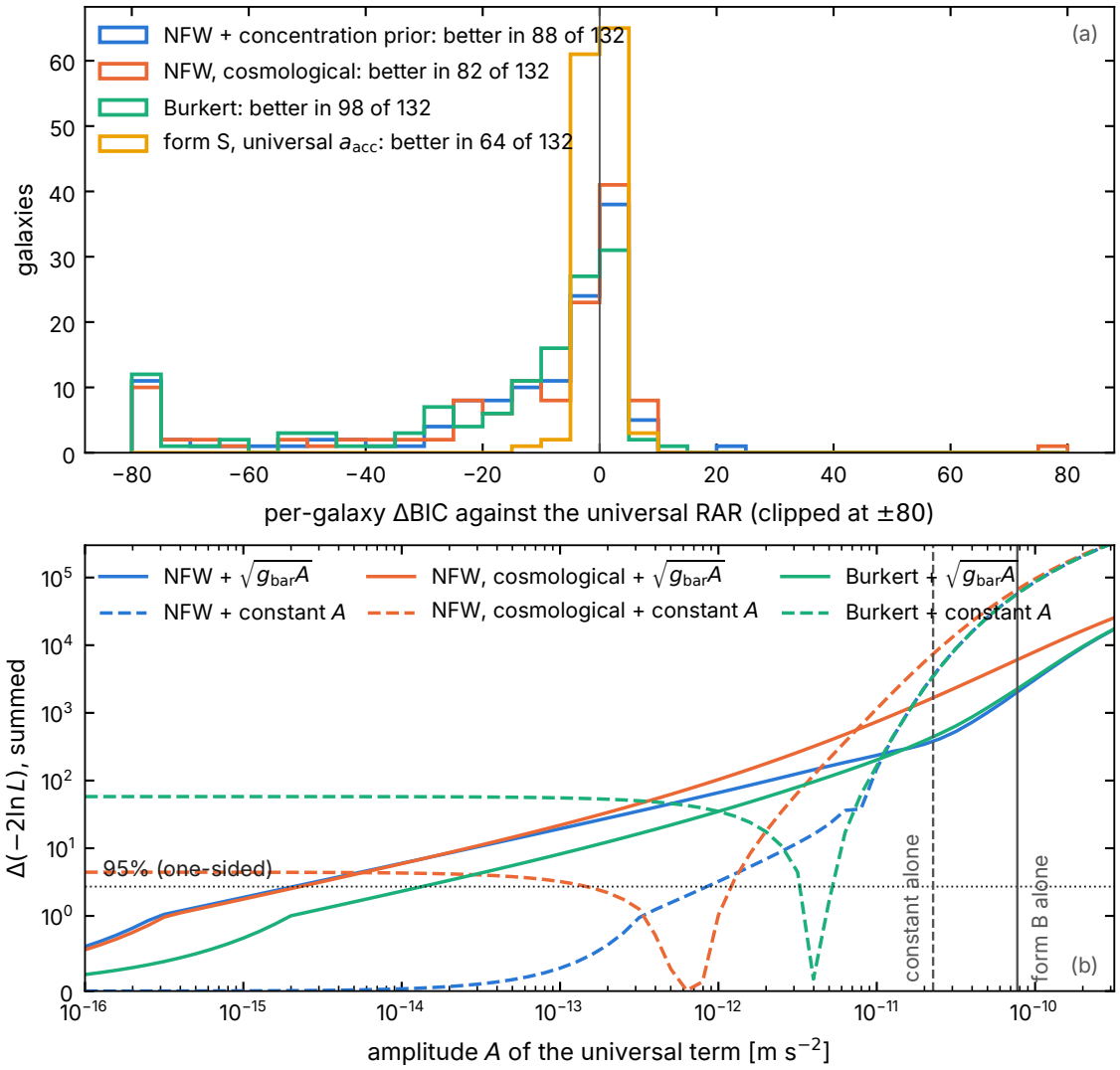


Figure 16: (a) Per-galaxy ΔBIC of three halo models and of the universal form S against the universal RAR (negative: better; clipped at ±80). (b) $\Delta(-2\ln L)$ summed over 132 galaxies, every halo refitted, against the amplitude of a universal term (solid: $\sqrt{g_{\rm bar}A}$, dashed: constant), with the one-sided 95% level and the values each term needs alone (S) [remnant:R3,R4]. Look for the square-root curves leaving zero at once, orders of magnitude below the value the term needs to matter.

With NFW halos the best square-root amplitude is zero, in 87 to 97% of galaxy resamples (Table 16, Fig. 16) (S). The profile rises roughly as $\sqrt{A}$: the refitted halos cannot absorb the term, so it costs at first order. With cosmological halos its bound, 3.0×10^{-14} at the bootstrap 84th percentile, is about 2600 times below what form B needs alone. With NFW halos a constant is bounded at 4 to 5% of the value that replaces the halos. Burkert halos prefer one at about a fifth of that value (ΔBIC −50); since NFW halos do not want it, we read it as a correction to the cored halo's outer shape.

Consistency requirement. The acoustic peaks fix $\Omega_c h^2 = 0.120\pm0.001$ of collisionless matter at $z\approx1100$ [116]. If that is the contained remnant, halos of cosmological mass exist, and a stochastic term must not count their dark acceleration twice. The room left is the share in Table 16, measured at each galaxy's outermost radius. For the square-root term it is 1 to 5% of the dark acceleration, 2.1% with cosmological halos. For a constant it is 8 to 31%: 8.5% with cosmological halos and 31% with the Burkert halos that prefer it [remnant:R4].

23.7 Cosmic time

We compare four laws, fixed in advance: $a_{\rm acc}$ constant, $\propto H(z) = H_0[\Omega_m(1+z)^3+\Omega_\Lambda]^{1/2}$, $\propto 1/t$, and $\propto t$ (random-walk accumulation). We use $\Omega_m = 0.315$ [116] and the flat-ΛCDM age $t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\mathrm{arsinh}[\sqrt{\Omega_\Lambda/\Omega_m}(1+z)^{-3/2}]$. The laws give $a(1)/a(0) = 1$, 1.79, 2.36 and 0.42, and $a(2)/a(0) = 1$, 3.03, 4.21 and 0.24 [remnant:R5]. In the deep regime $V^4 = GMa_{\rm acc}$, so at fixed rotation speed $\Delta\log M_{\rm bar} = -\log[a(z)/a(0)]$. For accumulation this is +0.34 dex at $z = 0.9$ and +0.69 at $z = 2.3$. Massive disks at high redshift sit at 3 to 11 a_0 at their half-light radii [257], where any change is diluted. For a RAR-shaped law at fixed V and R with $g_{\rm obs} = 3g_\dagger$ the shifts shrink by factors of 2 to 7 (dashed in Fig. 17).

Measurements. Übler et al. [258] fit KMOS$^{\rm 3D}$ disks with the local slope 3.75. They find baryonic Tully–Fisher zero-point offsets of -0.44 ± 0.04 dex at $z\approx0.9$ (65 disks) and -0.27 ± 0.05 at $z\approx2.3$ (46), with statistical errors only. Without their pressure-support correction the offsets are −0.33 and −0.09. Their baryonic masses carry about 0.15 dex each, with gas from scaling relations. Read as a change of scale, the offsets imply $a(z)/a(0) = 2.8$ (deep) to 7.6 ($3g_\dagger$) at $z\approx0.9$, and 1.9 to 4.2 at $z\approx2.3$ [remnant:R5]. MUSE-DARK III [259] fits the RAR of 79 galaxies at $0.33 < z < 1.44$. It finds $a_0(z\approx1) = 2.38^{+0.12}_{-0.10}\times10^{-10}\,\mathrm{m\,s^{-2}}$, 1.98 ± 0.44 times the local 1.2, with $g_{\rm obs}$ and $g_{\rm bar}$ taken from one model. Six rotation curves at $z\approx0.9$–2.4 "all but exclude" about $4a_0$ at $z\approx2$ [257]. Dark-matter fractions of high-redshift disks [260] would need each galaxy's $g_{\rm bar}$ and are not used.

Comparison. Accumulation predicts a smaller scale in the past, hence more baryons at fixed speed. Every measurement points the other way (Fig. 17). MUSE-DARK's 2.0 ± 0.4 lies between the $H(z)$ and $1/t$ laws, and at $z = 2$ the $1/t$ law reaches the excluded $4a_0$. The Übler offsets have the sign of a larger past scale, but they fall between $z\approx0.9$ and 2.3, which no monotonic law does. The systematics (gas masses, pressure support, model-dependent $g_{\rm bar}$) are as large as the effect, and ΛCDM galaxy evolution makes such offsets too. This is therefore not a clean test; read literally, the data disfavour accumulation. The time arrows also disagree in principle. An accumulated scale grows, while cH and c/t fall, so an accumulating $a_{\rm acc}$ equals $cH_0/2\pi$ only by a coincidence of the present epoch.

23.8 What cannot be fitted

The stochastic remnant exists only as the Newtonian limit of a static, spherical calculation [144]. The acoustic peaks ($\Omega_c h^2 = 0.120$ of pressureless matter at $z\approx1100$ [116]) need its cosmological perturbation theory, and lensing needs its metric. Neither exists (*untested*). MOND-like laws reduce the mass discrepancy of clusters three- to four-fold without removing it [261]; forms B and S inherit that residual, and no cluster test was run. In 1E0657−558 the lensing mass follows the galaxies, offset from the X-ray gas at 8σ [141], whereas a term set by the local baryonic field follows the gas. Cold dark matter of any kind, primordial black holes included, covers all four as ΛCDM does (*live*).

Choices made after seeing the data.

1. The amplitude grid of Section 23.6 first started at $10^{-13}\,\mathrm{m\,s^{-2}}$. The first run put the square-root bound below it, and that amplitude is not negligible: $\sqrt{10^{-12}\cdot10^{-13}}$ is 30% of the lowest $g_{\rm bar}$. The grid now starts at 10^{-16}, and the consistency check uses 10^{-30}.
2. The mocks were raised from 200 to 500 per null model after a test run.
3. The mocks' extra point scatter is calibrated on the data.
4. One free RAR scale per galaxy was chosen as the halos' yardstick after seeing that every per-galaxy model beats every universal one.

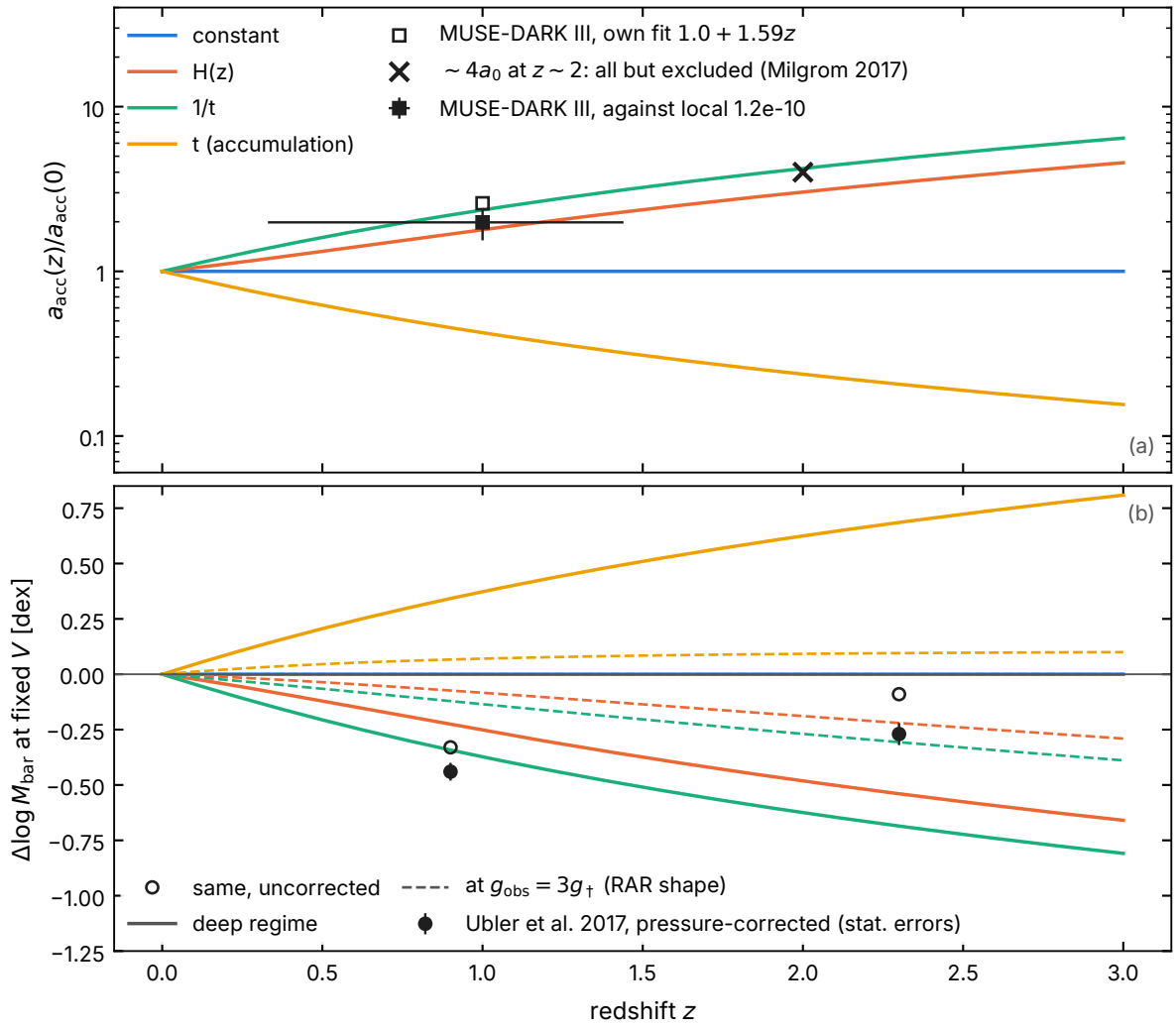


Figure 17: (a) The four laws for $a_{\rm acc}(z)/a_{\rm acc}(0)$, MUSE-DARK III at $z \approx 1$ (filled: against the local 1.2; open: their linear fit) and the value found all but excluded at $z \approx 2$. (b) $\Delta \log M_{\rm bar}$ at fixed rotation speed, deep (solid) and at $3g_\dagger$ (dashed), with the Übler et al. offsets (filled: pressure-corrected; open: uncorrected) (S) [remnant:R5]. Look for the sign: accumulation above zero, every measurement below.

5. Reading the Burkert constant as a halo-shape correction is an interpretation made after seeing it.
6. The shift test and the mock calibration of τ were added after test runs with 4 to 20 mocks, whose bands looked narrower than the bootstrap intervals. All are reported.

Caveats.

- Υ is fixed at 0.5 and 0.7, and $m_\star = 0.5 L_{[3.6]}$ in the stellar-to-halo prior. A free Υ would improve halo fits and universal laws alike.
- The stellar-to-halo relation was derived for a WMAP7 cosmology and is not checked at SPARC's lowest masses. The halos are not adiabatically contracted.
- The high-redshift numbers are published summaries for disks selected differently from SPARC.

All numbers in this section come from `docs/simulation/remnant_history.py` (39 of 39 checks, about two and a half minutes on one core), except the D7 rows, which come from Section 21.

Verdict. The stochastic remnant does not supply the dark acceleration of galaxies. Its static form is *refuted* by the rotation curves (Section 22). The accumulated forms B and S were introduced after the residual slope was measured, so their failure refutes these two forms, not every form a GUID prediction could take. Form S fits as well as the RAR but is MOND's simple interpolating function. As pointwise laws, both forms are *refuted* in the solar system. Neither shows a galaxy-specific history, and the high-redshift data favour a scale that was larger in the past, not one that accumulates. The rotation curves are consistent with cold dark matter halos of any kind (*live*), although NFW halos lose to one free RAR scale per galaxy. With halos present, the stochastic term keeps only the small room of Table 16, so the two routes compete rather than combine. GUID's remaining dark-matter candidate is primordial black holes formed in the start [148]; this paper does not test it.

24 The Standard Model statements against precision data

This section tests Part I's five statements about the particles and couplings of the Standard Model, SM1 to SM5 (Sections 10.1 and 12.1 to 12.4), as a preregistration fixed them before any analysis; Primer 7 gives the particle physics they use.

24.1 Statements and data

The preregistration (`docs/simulation/sm_preregistration.md`) states the five statements and fixes, for each, the data and the verdict.

- SM1, the spin ladder: elementary quanta have spin 0, 1/2 or 1 (Section 12.2).
- SM2, Hypothesis $1'$ read for matter: the Higgs vacuum grants every elementary rest mass (the *grant*). Higgs couplings are then proportional to mass, neutrinos are Dirac particles, the photon and gluons are massless, and $m_W = gv/2$ (Section 12.3).
- SM3, Hypothesis 4 with the action (4): the information scalar multiplies only the Maxwell term (Section 12.1).
- SM4, Hypothesis $6'$: within causality the forces were never one (Section 10.1).
- SM5, the electroweak split: α is born near 1/128 at the Z mass and runs to 1/137 (Section 12.4).

We score every statement as written. Where one turned out to be ill-posed, we say so and score it anyway. Pulls, the scoring classes (scored, not yet sensitive, INFO) and the verdict words are those of Section 20. Every measured number was checked on 2026-10-31 against its publication, on arXiv wherever possible. The exceptions are the CDF W mass, read from a fit paper that

quotes it; the rubidium value of α, read from the journal abstract; Gasser and Leutwyler's 1982 value, read through a later thesis; and the PDG and CODATA values that Section 21 verified. The script's citation list gives, per source, what was read and the value.

24.2 SM1: the spin ladder

SM1 says that elementary quanta have spin 0, 1/2 or 1. The spin-2 wave belongs to the classical frame and has no quantum, and no elementary spin-3/2 quantum exists. The test is the absence of what a richer ladder would add.

- Supersymmetry, whose gravitino has spin 3/2, is bounded through its coloured partners. ATLAS excludes gluinos below 2.30 TeV and degenerate light squarks below 1.85 TeV at 95% CL, in simplified models with a massless lightest neutralino [262] [sm:SM1.1].
- A spin-2 Kaluza–Klein graviton of the Randall–Sundrum model is excluded below 4.5, 3.9 and 2.2 TeV for $k/\bar{M}_{\rm Pl} = 0.1$, 0.05 and 0.01 [263] [sm:SM1.2].
- An excited top quark of spin 3/2 is excluded below 1.2 TeV [264] [sm:SM1.3].
- The one scalar found has spin 0: every spin-2 alternative is excluded at more than 99.9% CL [265] [sm:SM1.4].

We found no gravitino-specific limit we could verify, and use none. Null searches bound masses; they cannot exclude a spin. The Standard Model makes the same statement for its own content. *Verdict: holds as a reading*, and it cannot become more.

24.3 SM2: the Higgs vacuum grants every elementary rest mass

Couplings proportional to mass. In the Standard Model a fermion of mass m_f couples to the Higgs boson with $y_f = \sqrt{2}\,m_f/v$, and a weak boson with $g_{hVV} = 2m_V^2/v$, where $v = (\sqrt{2}G_F)^{-1/2} = 246.22\,\text{GeV}$ [237] (P). Hypothesis $1'$ makes the grant the only source of elementary rest mass, so it predicts exactly these couplings. The experiments measure coupling modifiers κ_i, which scale each Standard Model coupling. The *reduced couplings*

$$y_f^{\rm red} = \kappa_f \frac{m_f}{v}, \qquad y_V^{\rm red} = \sqrt{\kappa_V}\,\frac{m_V}{v} \tag{52}$$

then lie on one line, $y^{\rm red} = m/v$, of slope 1 on logarithmic axes [266, 267]. To test slope and intercept together we fit

$$\ln \kappa_{{\rm eff},i} = a + (s-1)\ln\frac{m_i}{m_*}, \qquad \kappa_{\rm eff} = \kappa_f \text{ or } \sqrt{\kappa_V}, \qquad v_{\rm fit} = v\,e^{-a}, \tag{53}$$

by weighted least squares with the symmetrised errors of each κ. The pivot m_* makes a and s uncorrelated. The prediction is $s = 1$ and $v_{\rm fit} = v$.

The data are the CMS modifiers of the fit with effective photon, gluon and $Z\gamma$ couplings: $\kappa_W = 1.02 \pm 0.08$, $\kappa_Z = 1.04 \pm 0.07$, $\kappa_t = 1.01^{+0.11}_{-0.10}$, $\kappa_b = 0.99^{+0.17}_{-0.16}$, $\kappa_\tau = 0.92 \pm 0.08$ and $\kappa_\mu = 1.12^{+0.21}_{-0.22}$ [267]. ATLAS gives its per-particle values only in figures [266]; we did not read them off and use its two-parameter fit instead. Fermion masses enter at the electroweak scale, $m_f = y_f(M_Z)\,v/\sqrt{2}$, from the $\overline{\rm MS}$ Yukawa couplings [268]: 168.4, 2.84, 1.730 and 0.1019 GeV for t, b, τ and μ. This is close to ATLAS's convention of quark masses at the Higgs mass. The W and Z enter with their pole masses [sm:SM2a].

The fit gives $s = 1.009 \pm 0.017$ and $v_{\rm fit} = 244.0 \pm 5.7\,\text{GeV}$ at the pivot $m_* = 57\,\text{GeV}$, with $\chi^2 = 1.25$ for 4 degrees of freedom. The pulls are $+0.54\sigma$ and -0.39σ [sm:SM2a.4], [sm:SM2a.5]. With pole and PDG masses the slope is again 1.009 ± 0.017: the mass convention moves each point along the line, not off it. Jointly, the six tree-level modifiers give $\chi^2 = 1.70$ for 6 against $\kappa = 1$, with diagonal errors taken on the side towards 1 ($p = 0.95$) [sm:SM2a.1]. The covariance is not published in the text, so the collaborations' own likelihoods decide. CMS finds $p = 0.28$ for all nine modifiers [267]. ATLAS finds $p = 0.56$ (with $\kappa_c = \kappa_t$), 0.65 (κ_c free) and 0.61

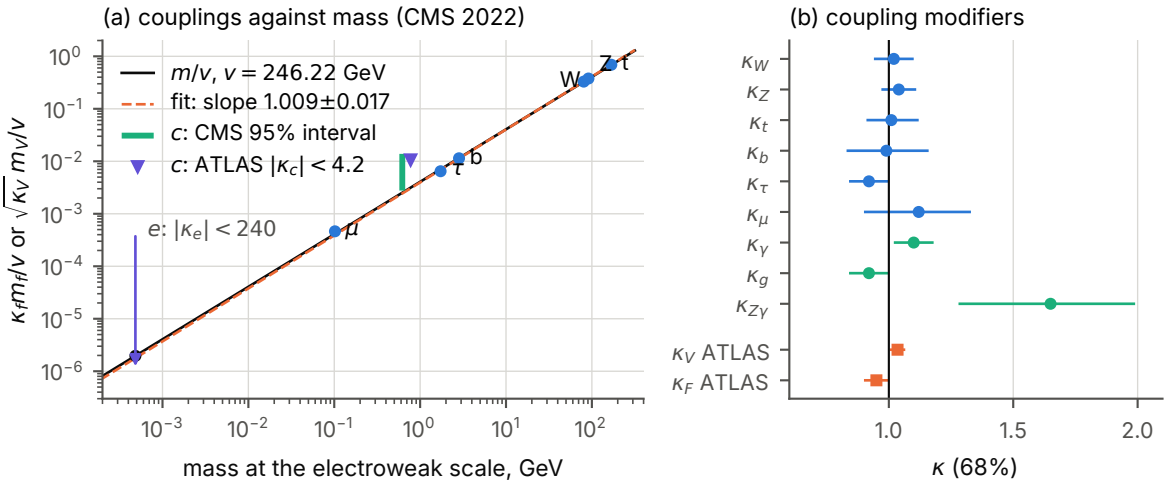


Figure 18: (a) Reduced Higgs couplings, Eq. (52), against mass at the electroweak scale (CMS 2022 [267]; fermion masses from [268]). Black: the prediction m/v; dashed: the fit of Eq. (53). Green bar: the CMS 95% interval for charm; triangle: the ATLAS 95% bound $|\kappa_c| < 4.2$ [269]; arrow: the electron bound $|\kappa_e| < 240$ [270] above the predicted value (open circle). (b) The CMS modifiers (68%; green: loop-induced) and ATLAS's κ_V, κ_F. What to look for: six couplings over 3.2 decades of mass lie on the line; charm and the electron are bounded, not measured.

(effective loop couplings) [266] [sm:SM2a.2]. ATLAS's two-parameter fit, $\kappa_V = 1.035 \pm 0.031$ and $\kappa_F = 0.95 \pm 0.05$ with correlation 0.39, gives $\chi^2 = 3.72$ for 2 ($p = 0.16$; ATLAS quotes 0.14) [sm:SM2a.3]. The largest single departure is the loop-induced $\kappa_{Z\gamma} = 1.65^{+0.34}_{-0.37}$, $+1.76\sigma$, which is not a mass coupling [sm:SM2a.kZgamma].

What is not measured is untested.

- Charm: CMS's 95% interval $1.1 < |\kappa_c| < 5.5$ just excludes 1, while ATLAS's newer bound $|\kappa_c| < 4.2$ includes it [267, 269] [sm:SM2a.6].
- The electron: $B(H \to ee) < 3.0 \times 10^{-4}$ bounds $|\kappa_e| < 240$, against a predicted branching fraction of about 5×10^{-9} [270] [sm:SM2a.7].
- The light quarks are unmeasured. The neutrinos' Dirac couplings, $y_\nu = \sqrt{2} m_\nu / v \approx 2.9 \times 10^{-13}$ for the heaviest neutrino at the oscillation minimum, are out of reach of any collider [sm:SM2a.8], [sm:SM2a.9].

Verdict: holds for W, Z, t, b, τ and μ; *untested* for the rest. The statement is the Standard Model's, and the data that support it support both.

Dirac neutrinos, and the assumption they need. The grant alone does not make neutrinos Dirac particles: the Weinberg operator gives a Majorana mass $m_\nu = f\, v^2/\Lambda$ from the same vacuum [158, 271], and a Majorana mass changes $B - L$, the combination the Standard Model conserves exactly, by two (P). Dirac neutrinos therefore need the added reading of Section 12.3, that $B - L$ is a count, changed only by a recorded act across a boundary [sm:SM2b.0]. It forbids every Majorana mass, whatever its source, and still allows the proton decay of SM4, since $p \to e^+\pi^0$ conserves $B - L$. It is not derived: the axioms do not say which conserved quantities are counts [sm:SM2b.7].

Given this reading, neutrinos have Dirac masses from Yukawa couplings to right-handed partners, and neutrinoless double-beta decay does not occur at any rate. Through light-neutrino exchange its amplitude is proportional to the effective Majorana mass $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i|$, which is zero (P). The searches agree.

- KamLAND-Zen finds $T_{1/2}(^{136}\mathrm{Xe}) > 3.8 \times 10^{26}\,\mathrm{yr}$ at 90% CL, or $m_{\beta\beta} < 28$ to $122\,\mathrm{meV}$ depending on the nuclear matrix element [159] [sm:SM2b.1].

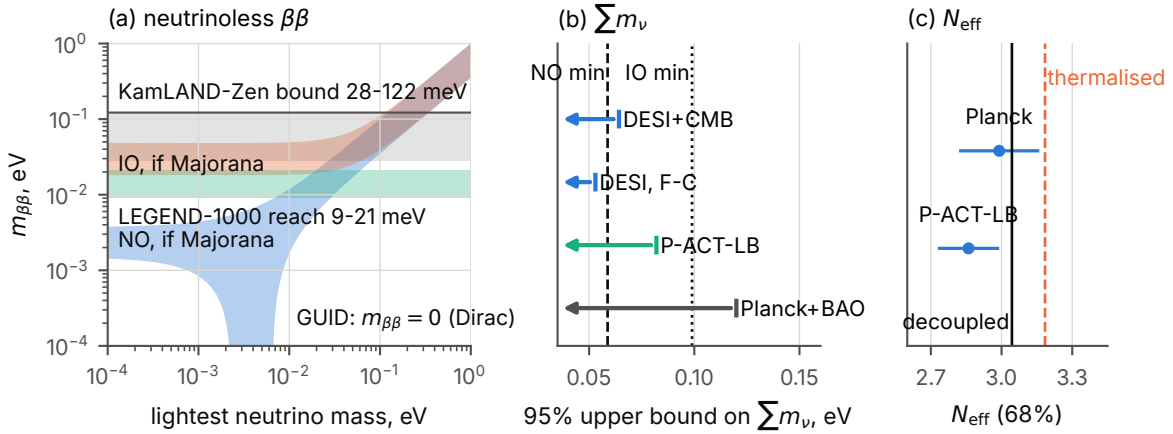


Figure 19: (a) The effective Majorana mass that Majorana neutrinos would have, for either ordering (NuFIT 6.0 [272]; bands over both Majorana phases), against the KamLAND-Zen bound [159] and the LEGEND-1000 reach [273]. With $B - L$ conserved, GUID predicts $m_{\beta\beta} = 0$, below the plot. (b) 95% upper bounds on $\sum m_\nu$ [275, 276, 116] against the oscillation minima (dashed: normal, dotted: inverted ordering). (c) $N_{\rm eff}$ measured [116, 276] against decoupled right-handed partners (GUID, 3.044) and partners thermalised before the electroweak split. What to look for: a signal in (a) would refute GUID's Dirac neutrinos; the tension in (b) is shared with every neutrino model.

- GERDA, the MAJORANA DEMONSTRATOR and LEGEND-200 together find $T_{1/2}(^{76}\mathrm{Ge}) > 1.9 \times 10^{26}$ yr, or $m_{\beta\beta} < 75$ to 200 meV [160] [sm:SM2b.2].

Were the neutrinos Majorana, the oscillation parameters of NuFIT 6.0 [272] would put $m_{\beta\beta}$ between 18.2 and 48.2 meV for the inverted ordering. For the normal ordering it would lie between 1.5 and 3.7 meV at zero lightest mass (Fig. 19a). LEGEND-1000 aims at a 3σ discovery at 1.3×10^{28} yr, or 9 to 21 meV [273], and nEXO at 1.35×10^{28} yr [274] [sm:SM2b.3]. A signal in any of them would refute SM2.2. A null result would not establish Dirac neutrinos, since a normal-ordered Majorana spectrum can hide below every planned reach. *Verdict: live, given $B - L$ conservation*; it holds against every search so far.

Two side tests concern the right-handed partners and the mass scale. With only Yukawa interactions the partners never come into equilibrium, and their contribution to the radiation density is negligible [277]. Hence $N_{\rm eff} = 3.0440$ [278]. Planck with baryon acoustic oscillations finds 2.99 ± 0.17 [116], and ACT DR6 with Planck, lensing and BAO finds 2.86 ± 0.13 [276]. The pulls are -0.32σ and -1.42σ [sm:SM2b.4], [sm:SM2b.5]. Had the partners been thermalised in the start and decoupled above every Standard Model threshold, entropy conservation gives each a share $(10.75/106.75)^{4/3}$, so $N_{\rm eff} = 3.185$. That lies 2.50σ above the ACT value [sm:SM2b.5b]. It is not GUID's statement, but it tells the start not to equilibrate them.

Oscillations bound the sum of the masses from below: $\sum m_\nu \geq 0.0588$ eV for the normal and ≥ 0.0989 eV for the inverted ordering (NuFIT quotes 0.058 and 0.098 eV). Cosmology bounds it from above. DESI's baseline 95% bound, $\sum m_\nu < 0.0642$ eV, lies 0.0054 eV *above* the normal-ordering minimum. Its Feldman–Cousins limit, 0.053 eV, breaches that minimum, and its analysis with an effective mass allowed to go negative is in 3σ tension with it [275] [sm:SM2b.6]. The inverted ordering is disfavoured. The tension is the same for Dirac and Majorana neutrinos and is ΛCDM's: *inherited*. The same bound makes the known neutrinos less than 0.57% of the dark matter [data:D9.3].

Massless bosons and the W mass. That the photon and the gluons stay massless is a reading of the Standard Model, with no number scored [sm:SM2c.1]. At tree level $m_W = gv/2$ and $m_Z = m_W/\cos\theta_W$ come within −0.18% and +0.34% of the measured masses [data:D8.3], [data:D8.4]. Beyond tree level GUID adds no loop of its own, so its m_W is the Standard Model's.

The global electroweak fit without m_W gives $80.3499 \pm 0.0056\,\mathrm{GeV}$ [279]. The new measurements agree with it:

- ATLAS, $80.3665 \pm 0.0159\,\mathrm{GeV}$, $+0.98\sigma$ [280];
- CMS, $80.3602 \pm 0.0099\,\mathrm{GeV}$, $+0.91\sigma$ [281];
- LHCb, $80.354 \pm 0.032\,\mathrm{GeV}$, $+0.13\sigma$ [282] [sm:SM2d.1-SM2d.3].

The combination without CDF, which shares data with them, is $+1.34\sigma$ [238] [sm:SM2d.5]. CDF's $80.4335 \pm 0.0094\,\mathrm{GeV}$ [283] lies 7.6σ above (Fig. 21a) [sm:SM2d.4]. We reproduce the fit authors' 6.5σ against the 2022 average that included CDF. *Verdict: holds.* CDF is an *inherited* tension, now isolated by three experiments that agree with the Standard Model and with each other.

24.4 SM3: only the Maxwell term varies

The ratio R. Hypothesis 4 with the action (4) lets the scalar change one coefficient, that of $F_{\mu\nu}F^{\mu\nu}$. The strong and weak gauge couplings, the Yukawa couplings and the Higgs potential do not move. Below the electroweak scale this shifts $1/\alpha$ by a constant, the same at every scale at one loop, because the running of $1/\alpha$ does not depend on α there. Hence $\delta \ln \alpha(\mu) = [\alpha(\mu)/\alpha(0)]\, \delta \ln \alpha(0)$. The proton-to-electron mass ratio $\mu = m_p/M_e$ then changes only through electromagnetic contributions, and

$$R \equiv \frac{\mathrm{d} \ln \mu}{\mathrm{d} \ln \alpha} = \frac{\mathrm{d} \ln m_p}{\mathrm{d} \ln \alpha} - \frac{\mathrm{d} \ln M_e}{\mathrm{d} \ln \alpha}. \tag{54}$$

The value depends on what is held fixed while α moves, the *matching*. We compute two.

Matching at the electroweak scale. The Yukawa couplings and v are held fixed at $\mu_0 = M_Z$, where Part I places the birth of α; the three-flavour QCD scale Λ_{QCD} is held fixed too. The electron's $\overline{\mathrm{MS}}$ mass runs in QED as $\mathrm{d} \ln m/\mathrm{d} \ln \mu = -3\alpha(\mu)/2\pi$ at one loop, and its pole mass is $M_e = m_e(m_e)(1 + \alpha/\pi)$ (P). These are the QCD relations with the colour factor $4/3$ replaced by 1. With $m_e(\mu_0) = y_e v/\sqrt{2}$ fixed,

$$\frac{\mathrm{d} \ln M_e}{\mathrm{d} \ln \alpha} = \frac{3}{2\pi\alpha(0)} \int_{m_e}^{\mu_0} \alpha(\mu)^2\, \mathrm{d} \ln \mu + \frac{\alpha}{\pi} = 0.0436 + 0.0023 = 0.0460. \tag{55}$$

The running $\alpha(\mu)$ is the one-loop leptonic running, scaled to the three-loop $\Delta\alpha_{\mathrm{lep}}(M_Z^2) = 314.977 \times 10^{-4}$ [284], plus the hadronic $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (276.0 \pm 1.0) \times 10^{-4}$ [285] spread logarithmically from 0.5 GeV to M_Z. It reproduces $1/\alpha(M_Z) = 128.937$ on shell. With α frozen at $\alpha(0)$, the first term would be $(3\alpha/2\pi) \ln(M_Z/m_e)$, with $\ln(M_Z/m_e) = 12.09$ [sm:SM3]. This is the electron's electromagnetic self-energy: 4.6% of its mass.

The proton carries α in two pieces. One is its electromagnetic self-energy E_{em}, at fixed quark masses at 2 GeV. The other is the QED running of those quark masses between μ_0 and 2 GeV, weighted by the sigma terms $f_q = \sigma_q/m_N$:

$$\frac{\mathrm{d} \ln m_p}{\mathrm{d} \ln \alpha} = \frac{E_{\mathrm{em}}}{m_p} + \sum_{q=u,d,s} f_q Q_q^2\, \frac{3}{2\pi\alpha(0)} \int_{2\,\mathrm{GeV}}^{\mu_0} \alpha(\mu)^2\, \mathrm{d} \ln \mu = 6.7 \times 10^{-4} + 2.2 \times 10^{-4}. \tag{56}$$

We take $E_{\mathrm{em}} = 0.63\,\mathrm{MeV}$, Gasser and Leutwyler's Born-level value from the measured form factors [286], read through [287], with $\pm 0.40\,\mathrm{MeV}$. This error covers their estimate of $\pm 0.2\,\mathrm{MeV}$ for intermediate states. It also covers the spread of the proton-neutron difference between the Cottingham evaluation, $0.58 \pm 0.16\,\mathrm{MeV}$, and lattice values near 1.0 MeV [288]. The individual self-energy is not an observable on its own, and its scheme dependence is part of this error. The sigma-term fractions are the lattice values $f_u = 0.0142$, $f_d = 0.0242$ and $f_s = 0.0577$ [289]. Scaling f_u and f_d up to the dispersive $\sigma_{\pi N} = 59.1 \pm 3.5\,\mathrm{MeV}$ [290] changes R by 8×10^{-5} [sm:SM3].

Table 17: The error budget of R under the electroweak matching, Eqs. (54)–(56); quadrature sum ± 0.0023.

source	δR
matching scale μ_0 between M_W and v (half range)	2.2×10^{-3}
$\Lambda^{(3)}_{\mathrm{QCD}}$ fixed against $\alpha_s(M_Z)$ fixed (heavy-quark thresholds)	6.2×10^{-4}
proton electromagnetic self-energy, $\pm 0.40\,\mathrm{MeV}$	4.3×10^{-4}
two-loop QED anomalous dimension, $O(\alpha/\pi)$ of the electron term	1.1×10^{-4}
sigma terms, lattice against dispersive	7.5×10^{-5}
hadronic running model, onset 0.3 to 1 GeV (half range)	6.2×10^{-5}

Together, $R = 8.9 \times 10^{-4} - 0.0460 = -0.0451 \pm 0.0023$ (Table 17). Over the range of choices the value lies between -0.0489 and -0.0444 [sm:SM3.1]. Its sign is fixed: a larger α gives the electron more self-energy, in proportion, than the proton.

Matching at low energy. Many analyses of varying constants hold the low-energy masses fixed instead. Then the electron term of Eq. (55) vanishes. With every low-energy mass fixed, $R = E_{\mathrm{em}}/m_p = (+6.7 \pm 4.3) \times 10^{-4}$; with only the electron's fixed, $R = (+8.9 \pm 7.6) \times 10^{-4}$ [sm:SM3.1b].

Which matching Hypothesis 4 selects. Holding the low-energy electron mass fixed while α moves requires the Yukawa coupling to move against it, $\mathrm{d}\ln y_e/\mathrm{d}\ln\alpha = -0.046$. That is a coupling of the scalar to the electron's mass term. Hypothesis 4 lets the scalar multiply the Maxwell term and nothing else, so the mass terms are fixed where the Maxwell term is defined. The Maxwell term exists only below the electroweak split, so they are fixed at the electroweak scale; the grant of Hypothesis $1'$ is fixed there too. We therefore take the electroweak matching as GUID's. The exact scale within the electroweak range, M_W to v, is part of Table 17.

The preregistered window. The preregistration expected R "of order 10^{-3} to 10^{-2}". Under the electroweak matching the derived value lies a factor of 4.5 above that window, and no scale between M_W and v brings it inside. The estimate behind the window counted the proton's electromagnetic mass, 0.09% of it, and missed the electron's, 4.6%. Under the low-energy matching R is of order 10^{-3}, at the lower edge of the window and within one standard deviation of it. The preregistration did not fix the matching. We score the window as written: *refuted under the electroweak matching* that Hypothesis 4 selects, by derivation rather than by data, and *inside it under the low-energy matching* [sm:SM3.1], [sm:SM3.1b].

What the window was meant to say survives under both. Unification models tie μ to α through Λ_{QCD} and give $|R|$ of 20 to 50. In the class studied by Ferreira et al., $\Delta\mu/\mu = [0.8R_\Lambda - 0.3(1+S)]\Delta\alpha/\alpha$, which is -24.3 for their typical $R_\Lambda \sim 30$ and $S \sim 160$ [291]. Calmet and Fritzsch find the nucleon mass changing about 40 times faster than α [292]. GUID's $|R|$ is 540 times smaller than the first, and at least 400 times smaller than any $|R| \geq 20$ [sm:SM3].

The data. Each model is a line through the origin, $\Delta\mu/\mu = R\,\Delta\alpha/\alpha$; the data are bounds near zero on both axes (Fig. 20). The rows are *not yet sensitive*: no variation has been seen, so they cannot tell GUID's line from the Standard Model's constant constants.

- Methanol absorption at $z = 0.886$ gives $|\Delta\mu/\mu| \leq 1.1 \times 10^{-7}$ at 2σ, over a lookback time of 7.5 Gyr [293]; we read it as a Gaussian $0 \pm 0.55 \times 10^{-7}$. ESPRESSO gives $\Delta\alpha/\alpha = 1.3 \pm 1.3 \pm 0.4\,\mathrm{ppm}$ at $z = 1.15$ [206] [data:D1.6]. On GUID's line the two combine to $\chi^2 = 0.51$ for one degree of freedom ($p = 0.48$) and to $\Delta\alpha/\alpha = 0.58 \pm 0.91\,\mathrm{ppm}$. Under GUID the molecular bound limits α about as tightly as the quasar does [sm:SM3.2].
- On the unification line $R = -24.3$ the same data are equally consistent ($p = 0.34$), but they then force $|\Delta\alpha/\alpha| < 4.4 \times 10^{-9}$ at 95% [sm:SM3.3].

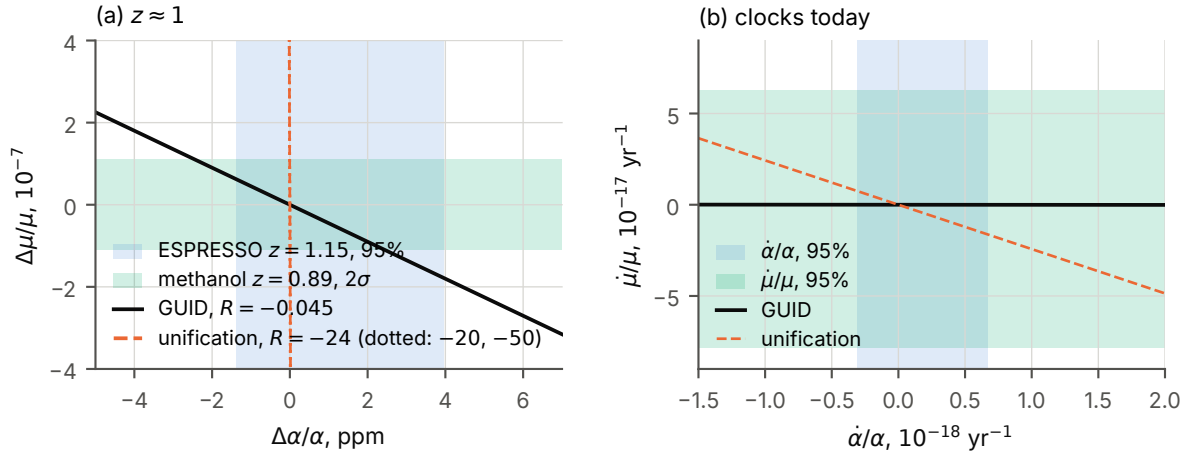


Figure 20: Variations of μ against variations of α. (a) Near $z \approx 1$: the ESPRESSO 95% band in α ($z = 1.15$) [206] and the methanol 2σ band in μ ($z = 0.89$) [293]; the lines are GUID's $R = -0.045$ (electroweak matching) and unification's $R = -24$ (dashed; dotted: -20 and -50) [291]. The low-energy matching's line would be nearly horizontal. (b) Present drifts from clocks [204, 229]. What a model allows is its line inside both bands: a segment of ppm length for GUID, a point within 4×10^{-9} of zero for unification at $z \approx 1$.

- Molecular hydrogen at $z = 2.0$ to 4.2 gives $|\Delta\mu/\mu| < 5 \times 10^{-6}$ at 3σ [294]. That is no constraint on α under GUID ($< 1.1 \times 10^{-4}$), and 2.1×10^{-7} under unification [sm:SM3.5].
- Today's clocks give $\dot{\alpha}/\alpha = (1.8 \pm 2.5) \times 10^{-19}\,\mathrm{yr}^{-1}$ [204] [data:D1.3] and $\dot{\mu}/\mu = (-8 \pm 36) \times 10^{-18}\,\mathrm{yr}^{-1}$ from ytterbium against caesium [229]. Both lines pass ($p = 0.82$ and 0.92) [sm:SM3.4].

The test that would decide: a variation of α at the ppm level with $|\Delta\mu/\mu|$ below 10^{-7} would exclude unification and agree with GUID. A $\Delta\mu/\mu$ much larger than $0.045\,|\Delta\alpha/\alpha|$ would refute Hypothesis 4. *Verdict: live.*

Lepton magnetic moments. The scalar couples to a lepton only through the lepton's electromagnetic self-energy. Its strength is $\lambda\, m_\ell/\bar{M}_\mathrm{P}$ times a factor below 0.05 (0.046 for the electron, Eq. (55)); to photons it is $\lambda/\bar{M}_\mathrm{P}$. Any contribution to $a_\ell = (g-2)/2$ needs two such insertions and one helicity flip. Hence $\Delta a_\ell = C\lambda^2 m_\ell^2/\bar{M}_\mathrm{P}^2$, where C collects loop factors and logarithms; we take $C \leq 10$, generously. At $|\lambda| < 3.1 \times 10^{-5}$ (7) this gives $\Delta a_e \lesssim 4 \times 10^{-52}$ and $\Delta a_\mu \lesssim 2 \times 10^{-47}$. These lie 38 and 36 orders of magnitude below the experimental errors [sm:SM3.6]: *derived*, and *holds*, in a row that is not yet sensitive.

Nor can the scalar make α differ between laboratories. A body of electrostatic share f sources $\phi = 2\lambda f\,\Phi/c^2$, the coupling behind Eq. (6). So $\delta\alpha/\alpha = 2\lambda^2 f\,\delta\Phi/c^2 \leq 10^{-20}$ for $f \leq 5 \times 10^{-3}$ and $\delta\Phi/c^2 \leq 10^{-9}$, which covers the Earth's surface potential and the annual change of the Sun's [sm:SM3.7]. The two best determinations of α differ by 1.2×10^{-9}, or 5.5σ: $\alpha^{-1} = 137.035999046(27)$ from caesium [295] and 137.035999206(11) from rubidium [296]. That difference is 10^{11} times what the scalar allows. It is experimental, or lies in the inputs of the Standard Model; it is not GUID's.

The electron's moment, $g/2 = 1.00115965218059(13)$, gives with the Standard Model $\alpha^{-1} = 137.035999166(15)$ [297]. Against rubidium that is $+2.15\sigma$ ($a_e^\mathrm{exp} - a_e^\mathrm{SM} = +0.34 \times 10^{-12}$), and against caesium -3.89σ (-1.02×10^{-12}). Both FAIL at 95% [sm:SM3.8], [sm:SM3.9]. They are one *inherited* tension, in the determinations of α, seen twice. The muon is clean: the world average after Fermilab's final result, $a_\mu = 116\,592\,071.5(14.5) \times 10^{-11}$ [298], lies $+0.60\sigma$ from the 2025 Standard Model value $116\,592\,033(62) \times 10^{-11}$ [299] [sm:SM3.10] (Fig. 21b,c). *Verdict* of the side prediction: *derived*; *holds*.

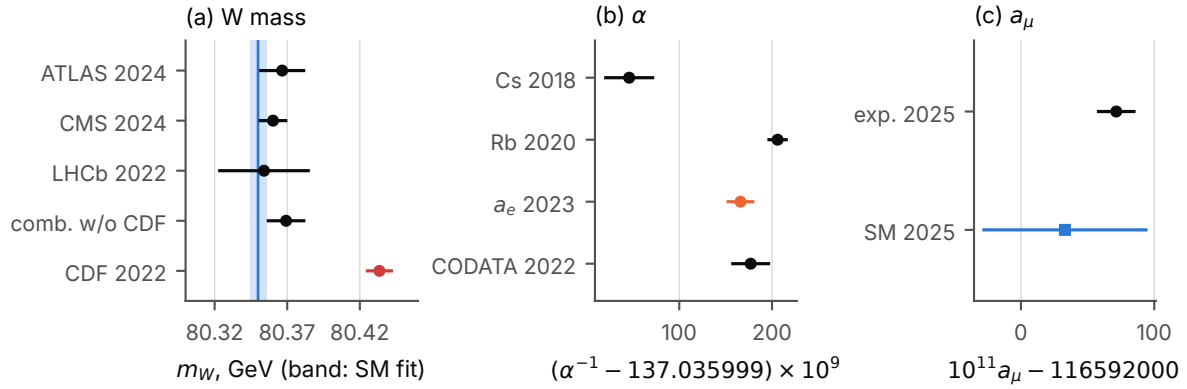


Figure 21: (a) The W mass: measurements [280, 281, 282, 238, 283] against the Standard Model fit without m_W [279] (band, 1σ). (b) α^{-1} from caesium [295], rubidium [296], the electron's moment [297] and CODATA 2022 [300]. (c) The muon anomaly, world average [298] and Standard Model [299]. What to look for: CDF alone departs in (a); in (b) the electron's moment sits between two determinations of α that disagree with each other; (c) agrees.

24.5 SM4: no unification within causality

Hypothesis 6′ puts any unification of the strong with the electroweak force before the start. Within causality, then, the Standard Model couplings run with Standard Model content alone should not meet at one point. We run the three couplings at two loops, with $\alpha_1 = \frac{5}{3}\alpha_Y$ normalised for grand unification:

$$\frac{\mathrm{d}\alpha_i^{-1}}{\mathrm{d}\ln\mu} = -\frac{b_i}{2\pi} - \sum_j \frac{b_{ij}\alpha_j}{8\pi^2} + \frac{d_i\, y_t^2}{32\pi^3}, \tag{57}$$

with $b = (41/10, -19/6, -7)$, $b_{ij} = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}$, $d = (17/10, 3/2, 2)$ and $16\pi^2\,\mathrm{d}y_t/\mathrm{d}\ln\mu = y_t(\frac{9}{2}y_t^2 - 8g_3^2 - \frac{9}{4}g_2^2 - \frac{17}{20}g_1^2)$ [157]. For contrast, the minimal supersymmetric extension takes over above a common threshold $M_{\mathrm{SUSY}} = 2\,\mathrm{TeV}$. It has $b = (33/5, 1, -3)$, $b_{ij} = \begin{pmatrix} 199/25 & 27/5 & 88/5 \\ 9/5 & 25 & 24 \\ 11/5 & 9 & 14 \end{pmatrix}$ and $d = (26/5, 6, 4)$ [301], its own top-Yukawa equation [302], $y_t \to y_t/\sin\beta$ and $\tan\beta = 10$. The inputs are $1/\hat{\alpha}(M_Z) = 127.950 \pm 0.017$ [156], $\sin^2\hat{\theta}_W = 0.23129 \pm 0.00004$ [132], $\alpha_s(M_Z) = 0.1180 \pm 0.0009$ [268, 303] and $y_t(M_Z) = 0.967$ [268], so $\alpha_i^{-1}(M_Z) = 59.01$, 29.59 and 8.47. As the measure of mismatch we take the smallest spread $\Delta = \max_i \alpha_i^{-1} - \min_i \alpha_i^{-1}$ over 10^{12} to $10^{19}\,\mathrm{GeV}$, which lies at a pairwise crossing. Its error combines 200 draws of the inputs (fixed seed) with a theory allowance. The allowance is the whole two-loop correction, $|\Delta_{2\,\mathrm{loop}} - \Delta_{1\,\mathrm{loop}}| = 0.47$, a deliberately generous stand-in for missing higher orders and thresholds [sm:SM4].

In the Standard Model α_1 meets α_2 at $1.08 \times 10^{13}\,\mathrm{GeV}$, α_1 meets α_3 at $1.71 \times 10^{14}\,\mathrm{GeV}$, and α_2 meets α_3 at $2.93 \times 10^{16}\,\mathrm{GeV}$ (Fig. 22). The smallest spread is $\Delta = 3.20$ at $1.7 \times 10^{14}\,\mathrm{GeV}$, against 0.045 from the inputs and 0.47 from the allowance: 6.8σ from a single point [sm:SM4.1]. Put differently, a single meeting point would need $\alpha_s(M_Z) = 0.0738$, 37% below the measured value and 49σ from it [sm:SM4.2]. The supersymmetric contrast meets at $1.15 \times 10^{16}\,\mathrm{GeV}$ with $1/\alpha_G = 25.66$, and needs $\alpha_s(M_Z) = 0.1181$, within the input error, before any threshold correction [sm:SM4.3]. A unification within causality would need such content beyond the Standard Model; superpartners near 2 TeV are bounded by the searches of SM1, not excluded.

Proton decay is the remaining trace of a unification before the start. In Part I's estimate $\tau \sim M_X^4/(\alpha_G^2 m_p^5)$, Super-Kamiokande's $\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}\,\mathrm{yr}$ [128] requires $M_X > 4.8 \times 10^{15}\,\mathrm{GeV}$ for $\alpha_G = 1/40$, and $4.7 \times 10^{15}\,\mathrm{GeV}$ for the Standard Model's $\alpha_G = 1/42.2$ [sm:SM4.4], [data:D8.1]. M_X is free, so the bound tests nothing GUID fixes. A unification

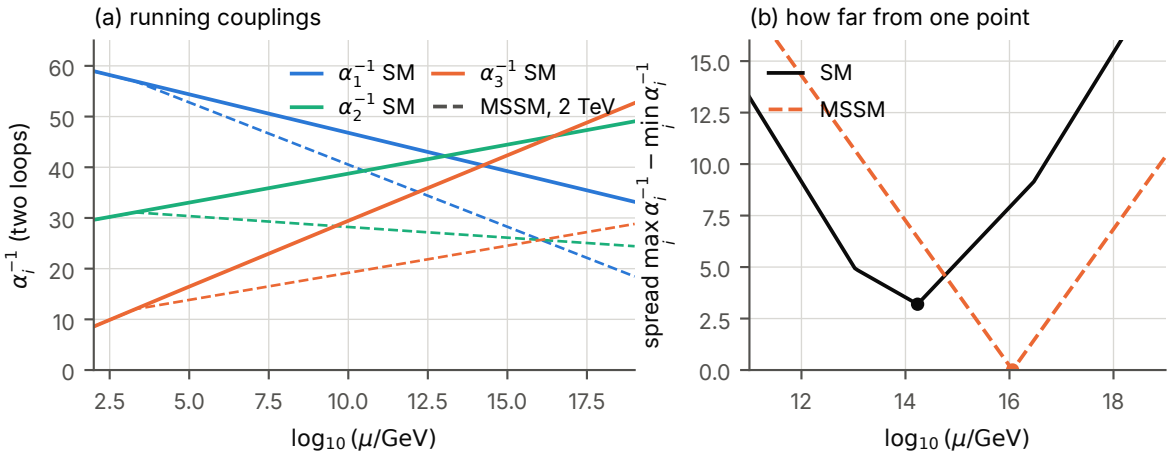


Figure 22: (a) The inverse couplings at two loops, Eq. (57), in the Standard Model (solid) and with supersymmetric content above 2 TeV (dashed), from the measured values at M_Z [156, 132, 268]. (b) Their spread $\max_i \alpha_i^{-1} - \min_i \alpha_i^{-1}$; dots: the smallest spread. What to look for: the Standard Model's three crossings span three decades in energy and never close to a point.

at the Standard Model's $\alpha_1 = \alpha_2$ crossing would give $\tau \sim 7 \times 10^{23}$ yr, 3.5×10^{10} times too short [sm:SM4.5]. *Verdict: holds as a reading*: that the couplings do not meet is the Standard Model's own statement. Hypothesis 6′ itself, a unification before the start, is *untested*; Hyper-Kamiokande [130] is its test.

24.6 SM5: the coupling at the split

Part I says that α is born near 1/128 at the Z mass and runs down to 1/137. From $\alpha(0)$ [300] and the measured vacuum polarisation, $1/\alpha(M_Z) = \alpha(0)^{-1}[1 - \Delta\alpha_{\mathrm{lep}} - \Delta\alpha_{\mathrm{had}}^{(5)}] = 128.937 \pm 0.014$ on shell [284, 285], +0.73% from 128 [sm:SM5.1]. In the $\overline{\mathrm{MS}}$ scheme (modified minimal subtraction) the value is 127.950 ± 0.017 [156], −0.04% from 128 [data:D8.5]. The scheme offset implied by Erler's paired values [304], 0.98, turns our on-shell value into 127.96 [sm:SM5.2]. The two schemes differ by about one unit in $1/\alpha$, which is as precisely as "near 1/128" can be meant. Part I took the value from this physics, so the row is a consistency row, not counted as a success (Section 20). The split itself is the lattice crossover at $T_c = 159.5 \pm 1.5$ GeV, with the measured Higgs mass above the end point of the first-order line [data:D8.6]. *Verdict: holds*, in either scheme, as the Standard Model's.

24.7 Choices, caveats and the count

Choices made after seeing the data.

- ATLAS publishes its per-particle coupling modifiers only in figures, so the mass-coupling fit uses the CMS values; ATLAS enters through its two-parameter fit and its p-values.
- The two rows of the electron's moment are scored separately, although they show one tension. This was decided after both were seen to fail.
- The smallest spread of the couplings is evaluated at their crossings rather than on a grid, after the grid minimum was seen to sit at a kink (a change below 0.02).
- The argument that Hypothesis 4 selects the electroweak matching was written after R had been computed, and after a reader pointed out that the low-energy matching puts R inside the window. The electroweak baseline itself was chosen before computing R.
- The Dirac step is made conditional on $B - L$ conservation after a reader pointed out the dimension-five operator. The data and their outcome are unchanged.

Table 18: The preregistered statements and their verdicts. Rows are those of `standard_model.json`; "data:" rows are those of Section 21.

	statement	verdict	rows
SM1	no elementary spin-3/2 or spin-2 quantum	holds as a reading (the Standard Model says the same)	SM1.1–1.4
SM2.1	Higgs couplings $\propto$ mass, all $\kappa = 1$	holds for W, Z, t, b, τ, μ; untested for c, e, u, d, s, ν	SM2a.1–9
SM2.2	Dirac neutrinos, $m_{\beta\beta} = 0$	live, given $B - L$ conservation (an added reading); holds against every search	SM2b.0–7
SM2.3	photon and gluons massless	holds as a reading	SM2c.1
SM2.4	$m_W = gv/2$, $\rho = 1$, with Standard Model loops	holds; CDF 2022 inherited	SM2d.1–6
SM3	only α varies; R "of order 10^{-3} to 10^{-2}"	window refuted under the electroweak matching ($R = -0.045 \pm 0.002$), inside it under the low-energy matching; live against data	SM3.1–5
SM3 side	scalar's share of lepton $g-2$ negligible	derived; holds; a_e limited by the inherited α discrepancy	SM3.6–10
SM4	no unification within causality	holds as a reading (the Standard Model's statement); Hypothesis 6′ untested	SM4.1–5
SM5	α born near 1/128 at m_Z, runs to 1/137	holds	SM5.1–4

Chosen before the comparison: the error of the proton's self-energy, the supersymmetric threshold of 2 TeV with $\tan\beta = 10$, and reading the methanol 2σ bound as a Gaussian centred on zero. No verdict depends on these choices, except the window of SM3, which depends on the matching as stated above.

Caveats.

- The coupling fit ignores correlations between the κ, which are not published in the text. The collaborations' own p-values, which include them, agree with the verdict.
- The proton's electromagnetic self-energy is not an observable on its own; its scheme dependence is covered only by the error we assign.
- The matching of the electron's mass at the electroweak scale has electroweak corrections of order α/π that we bound but do not compute.
- The methanol and quasar data lie at different redshifts and along different lines of sight; combining them assumes a slowly varying scalar. The clock values of α and μ share data, and the caesium-based μ assumes a model for the caesium nuclear g-factor.
- The two-loop running omits threshold corrections at the top mass, the conversion to the supersymmetric scheme in the contrast model, and all corrections at a unification scale. The theory allowance is a deliberately large stand-in for them.
- The Standard Model value of m_W was fitted with the 2022 top mass. CDF's m_W was read from a paper quoting it, and the rubidium α from the journal abstract.

The count. Of 29 scored rows, 25 agree at 95% [sm:summary]. In the classes of the tally (Section 26), twenty test statements GUID shares with the Standard Model or ΛCDM: 17 agree, and the 3 FAILs are inherited (CDF's W mass and the electron's moment against either value of α). One is GUID-specific and sensitive: the window of R, which fails under the electroweak matching. Eight GUID-specific rows agree but are not yet sensitive: the two neutrinoless double-beta searches, since Majorana neutrinos of normal ordering can also stay below them; the two

N_{eff} rows, since decoupled partners change N_{eff} negligibly; and SM3.2 and SM3.4 to SM3.6, which scale with a coupling that may be zero. Table 18 lists the verdicts. All numbers in this section come from `docs/simulation/standard_model.py` (32 of 32 checks, about five seconds on one core) or, where marked "data:", from the script of Section 21.

Verdict. GUID's statements about the Standard Model hold wherever the data reach them, and almost everywhere they hold as the Standard Model's own: couplings proportional to mass over three decades, the W mass at loop level, couplings that do not meet without new content, and α near 1/128 at the split. The three FAILs among them are inherited. Dirac neutrinos need one assumption beyond the grant, that $B - L$ is a conserved count; given it, no neutrinoless double-beta decay occurs at any rate, a live prediction. The variation of μ that Hypothesis 4 allows is $R = -0.045 \pm 0.002$ times that of α under the matching the hypothesis selects, outside the preregistered window; under the low-energy matching it would be inside. Either way it is at least 400 times smaller than unification requires.

25 Hadrons as trees of dualities

This section tests Hypotheses 8 (H8) and 9 (H9) of Section 12.6 against hadron spectroscopy: the masses of compact trees, the closeness of molecular states to thresholds, the partners a tree implies, and H9's four registered predictions.

25.1 The model and the scoring

H8 fixes no mass by itself. To give its compact trees masses we use the colour-spin (chromomagnetic) model of De Rújula, Georgi and Glashow [164] in the form of Karliner and Rosner [165, 166] (KR), the model that predicted the doubly charmed baryon and a bound doubly bottom tetraquark (Primer 8). The mass of a hadron is

$$M = \sum_i m_i + \sum_{i<j} B_{ij} - \sum_{i<j} C_{ij}\, \langle \boldsymbol{\lambda}_i \cdot \boldsymbol{\lambda}_j\, \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \rangle, \tag{58}$$

with effective quark masses m_i, binding terms B_{ij} for heavy and strange pairs, and hyperfine strengths $C_{ij} \propto 1/(m_i m_j)$; $\boldsymbol{\lambda}$ are the colour (Gell-Mann) and $\boldsymbol{\sigma}$ the spin (Pauli) matrices. The expectation value is taken in the H8 tree: each node below the root couples two colour triplets to the conjugate triplet, the root couples to a singlet, and node spins with the same total J are mixed. As in KR, every constituent of a multiquark takes its effective mass from baryons, and a binding term scales with the colour factor of its pair.

We fit all 18 parameters to 42 ordinary ground-state mesons and baryons and to nothing exotic. The spread of the fit, $\sigma_{\text{model}} = 8.1\,\text{MeV}$, fixed before any exotic comparison, sets the model's uncertainty. The held-out doubly charmed baryon Ξ_{cc}^{++} comes out at $3642.9 \pm 12.3\,\text{MeV}$ against LHCb's $3621.55 \pm 0.38\,\text{MeV}$ [305] (pull +1.74). With KR's own parameters the code reproduces KR's published masses to 0.15 MeV [exotics:H], [exotics:F]. Rows are scored as in Section 20: OK, and the prediction *holds*, for |pull| < 1.96, with the model's σ_{pred} and the data error on the side of the model. States not yet seen are *live*. Each assignment of a model level to a state was fixed before the comparison. Appendix F gives the couplings, the fit, two variants of the mass convention and the full spectra.

25.2 Compact trees against data

The model and its doubly heavy predictions are KR's. We refit its parameters and apply it to H8's trees without refitting (Table 19, Fig. 23) [exotics:A]. What the table tests is that model's masses for compact trees, in KR's mass convention, which dominates them (Appendix F.2). It does not test GUID: H8 is a reading QCD allows.

Table 19: Compact H8 trees in KR's colour-spin model (our fit): model mass with σ_{pred}, measured mass, pull, verdict, and model mass minus the lowest S-wave two-hadron threshold. †: assigned to the nearest level, with chance probability 0.57, 0.24 and 0.08 that a mass placed at random within ±150 MeV lies as close. Data: T_{cc}^{+} [306], $X_0(2900)$ [307], $T_{c\bar{s}0}(2900)$ [308], $X(6900)$ [309], P_c [169], $P_{\psi s}(4338)$ [310], $P_{cs}(4459)$ [311]. Variants and further states: Table 30. Masses in MeV [exotics:A].

state	tree, $J^{P(C)}$	model	data	pull	verdict	$M-T$
T_{cc}^{+}	$[cc]_1[\bar{u}\bar{d}]_0$, 1^+	3897 ± 12	3874.74 ± 0.08	$+1.86$	holds	$+22$
$X_0(2900)$	$[ud]_0[\bar{c}\bar{s}]_0$, 0^+	2755 ± 10	2866 ± 7	-9.28	refuted	$+392$
$T_{c\bar{s}0}(2900)$	$[cq][\bar{s}\bar{q}]$, 0^+	2809 ± 9	2908 ± 23	-4.06	refuted	$+701$
$X(6900)$	$[cc]_1[\bar{c}\bar{c}]_1$, 2^{++}	6425 ± 29	6905 ± 13	-15.2	refuted	$+231$
$P_c(4312)$	lowest $([cq][qq])\bar{c}$	4277 ± 11	$4311.9^{+6.8}_{-0.9}$	-3.06	refuted	$+354$
$P_c(4440)^{\dagger}$	$([cq][qq]_1)\bar{c}$, $\frac{1}{2}^-$	4428 ± 11	$4440.3^{+4.3}_{-4.9}$	-0.98	holds	$+506$
$P_c(4457)^{\dagger}$	$([cq][qq]_1)\bar{c}$, $\frac{1}{2}^-$	4462 ± 11	$4457.3^{+4.1}_{-1.8}$	$+0.38$	holds	$+539$
$P_{\psi s}(4338)$	$([cs][ud]_0)\bar{c}$, $\frac{1}{2}^-$	4392 ± 11	4338.2 ± 0.8	$+4.81$	refuted	$+292$
$P_{cs}(4459)^{\dagger}$	$([cs][ud]_0)\bar{c}$, $\frac{3}{2}^-$	4460 ± 11	$4458.8^{+5.5}_{-3.1}$	$+0.13$	holds	$+361$
T_{bb}^{-}	$[bb]_1[\bar{u}\bar{d}]_0$, 1^+	10399 ± 12	–	–	live	-205
T_{bc}	$[bc]_0[\bar{u}\bar{d}]_0$, 0^+	7146 ± 11	–	–	live	$+2$
T_{bc}	$[bc]_1[\bar{u}\bar{d}]_0$, 1^+	7180 ± 12	–	–	live	-9

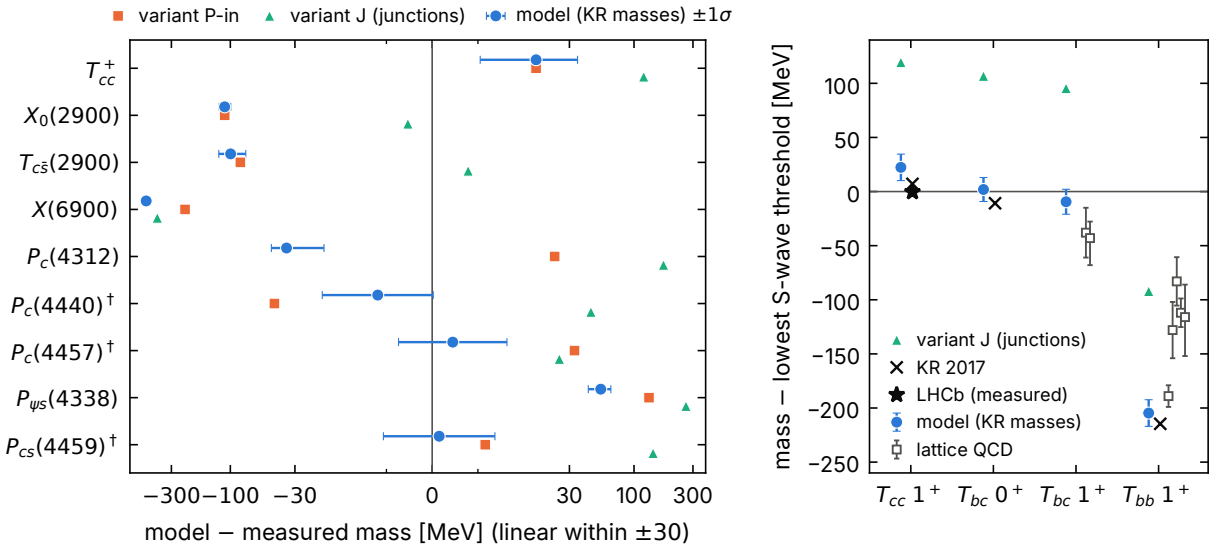


Figure 23: Left: model minus measured mass for the nine scored states of Table 19 (symmetric logarithmic axis, linear within ±30 MeV), for the model with its ±1σ combined error and the variants P-in and J of Appendix F.2; † marks assignments by proximity. Right: mass minus the lowest S-wave threshold for the doubly heavy tetraquarks: the model, variant J, KR's published values [166], LHCb's T_{cc}^{+} [306], and lattice QCD [312, 313, 314, 315, 316, 317, 318]. What to look for: most scored points lie far outside ±2σ, and no mass convention places them all [exotics:A].

Doubly heavy tetraquarks. T_{cc}^+, LHCb's narrow $D^0D^0\pi^+$ state 0.36 MeV below the $D^{*+}D^0$ threshold [168, 306], *holds* at $z = +1.86$. The model, however, puts it 22 MeV above that threshold; with $\sigma_{\text{pred}} = 12$ MeV it cannot tell whether the state is bound, as KR said of their own 3882 MeV [166]. The bound T_{bb}^- is KR's prediction (215 MeV below $B\bar{B}^*$ [166]) and lattice QCD's (83 to 189 MeV [312, 313, 314, 315, 316]). Refitted here, it lies at 10399 ± 12 MeV, 205 MeV below the threshold. *Live*, as their prediction. The T_{bc} states lie within σ_{pred} of their thresholds.

The refutations. Five masses fail. $X_0(2900)$ ($z = -9.3$) and $T_{c\bar{s}0}(2900)$ ($z = -4.1$) come out 100 MeV too light. LHCb's $X(6900)$ lies 480 MeV above even the most favourable S-wave assignment ($z = -15$). The lowest pentaquark levels miss $P_c(4312)$ ($z = -3.1$) and $P_{\psi s}(4338)$ ($z = +4.8$). Three states hold only by proximity, among levels dense enough to match most peaks. The junction variant would rescue the two states near 2.9 GeV, but it puts T_{cc}^+ 120 MeV too high; no convention fits all. The compact tree $[cq][\bar{c}\bar{q}]$ places 1^{++} 236 MeV above $X(3872)$, which sits on the $D^0\bar{D}^{*0}$ threshold as a molecular tree would.

25.3 Partners

A tree with light-quark halves comes in isospin multiplets; one whose halves carry spin comes in spin multiplets [exotics:C].

- *Charged partners of a compact* $X(3872)$. The diquark-antidiquark reading of $X(3872)$ [162] implies charged $X^\pm$ near 3872 MeV. BaBar finds $\mathcal{B}(B^0 \to X^-K^+, X^- \to J/\psi\pi^-\pi^0) < 5.4 \times 10^{-6}$ (90% CL), where an isovector partner implies 2.56×10^{-5} [319]; Belle's limits are also well below that expectation [320]. *Failed prediction* of the compact reading. Our model's charged 1^{++} partners lie near 4108 MeV, where no search was found.
- *Isospin partners of* T_{cc}^+ with $J = 0, 1, 2$: 4042, 4071 and 4131 MeV, above their open thresholds and hence broad. LHCb saw nothing narrow near threshold [306], as the model has it; the predicted states lie outside the searched windows: *untested*.
- *Isospin partners of* T_{bb}^-: 10583, 10593 and 10613 MeV: *live*.
- *Pentaquark multiplets*: 10 levels with isospin 1/2 and partners with isospin 3/2; three peaks are seen, and their production favours a spin-0 light diquark [321]: *untested*.

25.4 Threshold proximity

Where both halves are colour singlets, H8 makes the state a molecule, which lies near the threshold of its two hadrons. We take the 26 exotic hadrons reported as observations, find for each the nearest two-hadron threshold allowed in an S wave, and compare the distance with a null that places each mass uniformly within ±150 MeV (Appendix F.5). Thirteen states lie within 10 MeV of an allowed threshold, where 7.0 are expected. Fisher's combined p is 5.4×10^{-5} (4.1×10^{-6} for ±300 MeV) [exotics:B]. Two states carry it: $X(3872)$ sits 0.005 MeV above $D^0\bar{D}^{*0}$ and T_{cc}^+ 0.36 MeV below $D^{*+}D^0$; without them, $p = 0.057$. With distances floored at the mass error, $p = 5.1 \times 10^{-3}$. Exotic hadrons do lie near thresholds. That is the molecular branch of H8, but it is also the standard behaviour of hadronic molecules and threshold cusps in QCD. Many of these states were found as narrow structures near a threshold, so part of the excess is selection. *Verdict: holds* as a statement about exotic hadrons; as evidence for H8 over QCD, *holds as a reading*.

25.5 The minting in hadrons (H9)

H9 was conceived after seeing that every established pentaquark carries $c\bar{c}$; those states agree with it by construction and are not evidence. We test the four predictions registered as SM7

Table 20: Claimed pentaquarks: content, whether the antiquark has a partner of its flavour (hidden) or not (open), status, claim and searches that did not confirm it [exotics:I1].

state	content	pair	status	claim	not confirmed by
$\Theta^+(1540)$	$uudd\bar{s}$	open	unconfirmed	LEPS [322]	CLAS [323], HERA-B [326]
$\Xi^{--}(1860)$	$ddss\bar{u}$	open	unconfirmed	NA49 [324]	HERA-B [326]
$\Theta_c(3100)$	$uudd\bar{c}$	open	unconfirmed	H1 [325]	ZEUS [327]
$P_c(4380)$	$uudc\bar{c}$	hidden	unconfirmed	LHCb [329]	verification pending [169]
$P_c(4312, 4440, 4457)$	$uudc\bar{c}$	hidden	established	LHCb [169]	–
$P_c(4337)$	$uudc\bar{c}$	hidden	evidence	LHCb [330]	–
$P_{cs}(4459)$	$udsc\bar{c}$	hidden	evidence	LHCb [311]	$\Lambda_c^+ D_s^-$ [331]
$P_{\psi s}(4338)$	$udsc\bar{c}$	hidden	established	LHCb [310]	–
doubly charmed	$ccuu\bar{d}$, $ccud\bar{u}$	open, hidden	never claimed	–	LHCb [328]
hidden strange	$uuds\bar{s}$	hidden	never claimed	–	Belle [332]

before this analysis.

I1, no open flavour. Table 20 lists every pentaquark claim we could find and verify. The three claims with an antiquark that has no partner are all unconfirmed. LEPS's $\Theta^+(1540)$ [322] was not seen by CLAS with more than 30 times the luminosity [323]. NA49's $\Xi^{--}(1860)$ [324] and H1's $\Theta_c(3100)$ [325] were not seen by HERA-B [326] and ZEUS [327]. Crossing hidden or open with established or not gives Fisher's exact $p = 0.17$ (one-sided) [exotics:I1]. The contrast is weak and not clean:

- the open claims date from 2003 and 2004 and rest on small samples;
- the hidden-charm states come from LHCb's samples of b-hadron decays, whose $b \to c\bar{c}s$ transition supplies a $c\bar{c}$ pair (P);
- light hidden-flavour pentaquarks share their quantum numbers with N^* resonances and are not tagged as pentaquarks.

SM7 names doubly charmed pentaquarks as forbidden, but H9's rule forbids only those whose light antiquark lacks a partner, such as $ccuu\bar{d}$. So the model's lowest doubly charmed level, $([cc]_1[ud]_0)\bar{q}$ at 4232 MeV, is allowed. Its isospin-3/2 level $ccuu\bar{d}$ at 4301 ± 12 MeV, 540 MeV above $\Xi_{cc}^{++}\pi^+$, is forbidden. The two readings conflict there; LHCb's null search in $\Sigma_c D$ and $\Lambda_c D$ [328] bounds production, not existence. *Verdict: holds* as registered, with no statistical weight.

I2, hidden bottom. Heavy-quark flavour symmetry lets the light cloud bind a $\Sigma_b \bar{B}^{(*)}$ pair as it binds $\Sigma_c \bar{D}^{(*)}$. Carrying the binding energies of $P_c(4312)$, $P_c(4440)$, $P_c(4457)$ and $P_{\psi s}(4338)$ over unchanged gives molecular states at 11085 MeV (below $\Sigma_b \bar{B}$ at 11093.7 MeV), 11117 and 11134 MeV (below $\Sigma_b \bar{B}^*$ at 11139.0 MeV), and a $udsb\bar{b}$ state at 11074 MeV. Scaling the binding with the reduced mass moves them by up to 33 MeV (Appendix F.6). The compact tree $([bq][qq])\bar{b}$ has its lowest level at 10883 ± 11 MeV, about 200 MeV lower, so a measurement would decide between the readings [exotics:I2]. We found no experimental search on arXiv. These states cannot be made in b-hadron decays as the P_c were; photoproduction $\gamma p \to \Upsilon p$ at an electron-ion collider has been proposed [333]. *Verdict: live.*

I3, the general rule. "Every exotic hadron contains a pair of one flavour" fails on four established states: T_{cc}^+ ($cc\bar{u}\bar{d}$), $X_0(2900)$ and $X_1(2900)$ ($\bar{c}\bar{s}ud$), and $T_{c\bar{s}0}(2900)$ ($c\bar{s}u\bar{d}$) [exotics:I3]. *Verdict: refuted.* H9 itself is restricted to pentaquarks and is not touched.

I4, the splitting $\alpha M/2$. The rule says the two members of a spin doublet are split by $\alpha M/2$, with M their mean mass [300]. $P_c(4440)/P_c(4457)$ suggested it (17.0 against 16.2 MeV) and does not count. Eligible pairs were fixed from quark content and J^P only, before any splitting was computed: 25 pairs of observed exotic states (Appendix F.6). Two pass at 1.96σ, both with errors as large as the prediction, where 4.2 are expected by chance; 22 fail by 3σ or more. The two pairs most often read as heavy-quark spin partners fail: $Z_b(10650) - Z_b(10610) = 45.0 \pm 2.5$ MeV against 38.8 MeV, and $Z_c(4020) - Z_c(3900) = 123.9 \pm 6.7$ MeV against 14.5 MeV. The colour-spin model gives the Z_c splitting, 131.6 ± 8.9 MeV [exotics:I4]. *Verdict: refuted*, with no excess of agreements over chance.

Choices made after seeing the data.

- The binding rule across the two halves of a multiquark was written into the script before any exotic state was computed, by authors who knew the published masses; the alternative rule is reported for every state (Appendix F.2).
- Sharing KR's heavy-strange hyperfine enhancement between mesons and baryons was fixed before the fit, from KR's published residuals.
- One measurement per catalogue state was taken, the most recent one we could read; $X(3915)$, $P_{cs}(4459)$ and $P_c(4337)$ were left out of the threshold catalogue.
- The chance probabilities of the proximity assignments, and the threshold tests without the two sub-MeV states and with floored distances, were added after seeing the results.
- H9 was conceived after seeing the established pentaquarks. Reading SM7's doubly charmed example by H9's own rule was decided before the I1 counts.
- The I4 list was written before any splitting was computed, but all masses were known from the threshold catalogue.

Caveats. KR's mass convention for multiquarks dominates the compact masses: the junction variant shifts tetraquarks by 90 to 113 MeV and pentaquarks by about 205 MeV, and ordinary hadrons cannot choose between them. σ_{model} does not contain the uncertainty of carrying the model to four and five quarks. Only S-wave ground states are computed. The threshold test inherits the selection of its catalogue. Appendix F.7 gives the full list.

All numbers in this section and in Appendix F come from the script `exotic_hadrons.py` in `docs/simulation` (80 of 80 checks, about one minute on one core).

Verdict. H8 holds as a reading: QCD's colour algebra already writes every hadron as a tree of conjugate halves. KR's colour-spin model on H8's compact trees, fitted to ordinary hadrons, holds for T_{cc}^+ and fails for 5 of 9 measured masses. These refutations hit that model's compact masses, not GUID. The bound T_{bb}^- that the model and lattice QCD predict, 10399 ± 12 MeV here, remains live. Exotic hadrons lie closer to thresholds than chance ($p = 5 \times 10^{-5}$, carried by two states), as hadronic molecules in QCD do. Of H9's predictions, I1 holds without statistical weight, the hidden-bottom pentaquarks of I2 are live near 11.1 GeV, and the general rule I3 and the splitting rule I4 are refuted.

26 The tally

This section counts the scored confrontations of Sections 21, 24 and 25 together, in the classes of Section 20.

Table 21 gives the count [tally:total]. Of 72 scored confrontations, 58 hold at 95% and 14 fail. The conditional rows of the plateau potential (one FAIL), the consistency rows and the INFO

Table 21: Every scored confrontation of Part III. Shared: the prediction is that of general relativity, ΛCDM, the Standard Model (QCD included) or quantum mechanics. GUID-specific rows are sensitive, or not yet sensitive (Section 20). In brackets: FAILs.

section	scored	OK	FAIL	shared	GUID-specific, sensitive	GUID-specific, not yet sensitive
data (Section 21)	31	28	3	18 (3)	0	13 (0)
Standard Model (Section 24)	29	25	4	20 (3)	1 (1)	8 (0)
hadrons (Section 25)	12	5	7	9 (5)	2 (2)	1 (0)
all	72	58	14	47 (11)	3 (3)	22 (0)

rows are not counted. Rotation curves and the remnant (Sections 22 and 23) compare models by ΔBIC and score no rows, so they are not counted either.

The FAILs.

- *Shared, eleven.* Three are tensions of ΛCDM that share data: evolving dark energy, the Hubble constant and curvature (D3.3, D3.4, D3.7). Three are tensions of the Standard Model: the CDF W mass (SM2d.4) and the electron's $g-2$ with α from rubidium and from caesium (SM3.8, SM3.9), which are one disagreement between the two values of α. Five are masses of a colour-spin model on compact trees: $X_0(2900)$, $T_{c\bar{s}0}(2900)$, $X(6900)$ read as 2^{++}, $P_c(4312)$ and $P_{\psi s}(4338)$.
- *GUID-specific and sensitive, three, and all three fail.* The preregistered size of $R = \mathrm{d}\ln\mu/\mathrm{d}\ln\alpha$ (SM3.1) is refuted by derivation, not by a datum. The general rule that every exotic hadron carries a pair of one flavour (H9:I3) is refuted by the open-flavour tetraquarks. The splitting rule $\alpha M/2$ for pentaquark and tetraquark doublets (H9:I4) is refuted by the resolved doublets.
- *GUID-specific and not yet sensitive, none.*

The classes. The thirteen GUID-specific data rows test the information scalar (D1) and Bridge B (D6). The information scalar's departures scale with a coupling that may be zero, Bridge B's decoherence lies a factor 4×10^{12} or more below what the largest superpositions could see, and its diffusion lies deep inside the allowed window. Eight Standard Model rows are GUID-specific but not yet sensitive, and so is the open-flavour count of H9 (H9:I1), whose contrast is not significant. Dirac neutrinos give no double-beta signal, but Majorana neutrinos can give none either (SM2b.1, SM2b.2). Decoupled right-handed partners change N_{eff} negligibly (SM2b.4, SM2b.5). The scalar's μ–α line and its share of $g-2$ scale with the free coupling (SM3.2, SM3.4 to SM3.6). The nine model rows of the hadron section test a standard colour-spin model on the compact trees of Hypothesis 8. That hypothesis cannot fail on a single state, and the compact picture predates GUID, so they are counted as shared. Table 21 would change if a reader classed them as GUID-specific: the five model FAILs would then be GUID-specific and sensitive.

What the count says. Forty-seven rows test what GUID adopts from established physics and carry eleven of the fourteen FAILs. Twenty-five rows test what GUID adds. Every one of them that a datum or a derivation can already reach, three, fails: the size of R fixed in advance, the general one-flavour rule for exotic hadrons, and the $\alpha M/2$ splitting rule. The other twenty-two hold but are not yet sensitive to GUID's departures. So far, then, no GUID-specific statement has passed a test that could have failed it; the ones that could fail did. The class of each row and its reason are in `tally.json` (6 of 6 checks).

Part IV
Synthesis

27 What the tests show

Taken together, Parts II and III sort GUID's statements into four kinds.

What reduces to established physics. Free fall, light bending, the post-Newtonian limit, gravitational waves, Kerr black holes, the late universe, the Higgs couplings and the running of the forces all come out as in general relativity, ΛCDM and the Standard Model, and they agree with the data as those theories do (Sections 21 and 24). They show that the reading is consistent, not that it is right. The tensions in this group, evolving dark energy, the Hubble constant and curvature, are ΛCDM's, inherited unchanged.

What was refuted. Bridge A and the stochastic dark-matter remnant fail against galaxy rotation curves in every form computed (Sections 22 and 23). The window fixed in advance for the variation of the proton-to-electron mass ratio with α failed as fixed (Section 24). A standard model of the compact hadron trees fails for most measured exotic masses, and two rules GUID added for exotic hadrons, a pair of one flavour in every exotic and an $\alpha M/2$ splitting of doublets, fail (Section 25). Every GUID-specific statement that the data can already reach has failed (Section 26). The claim that an entanglement experiment with two spheres decides between classical and quantum gravity fails if the Diósi–Penrose rate is real (Section 17).

What GUID adds that is not yet testable. The classical frame itself, through Hypothesis 1′; the Diósi–Penrose rate through Bridge B, whose first decisive test needs a superposition five to eight orders of magnitude heavier than any made so far (Section 21); primordial black holes as the dark matter, which this paper does not test; neutrinos without Majorana mass; and the minting reading of pentaquarks, which forbids open-flavour pentaquarks and predicts hidden-bottom ones (Section 25).

What to compute next. Two calculations would give GUID a number of its own. DMP's two independent witnesses may imply two independent noise sources with a statistical signature that a single source lacks (Section 5.4); and a relativistic form of the stochastic remnant would say whether anything of it survives lensing, the acoustic peaks and merging clusters (Section 23).

28 The ledger

Table 22 gives every statement of the account with its kind and its verdict, without numbers; the numbers live in the sections named. A statement whose verdict or whose basis changed in this version is marked ∗ (Appendix C).

Table 22: The ledger: every statement of the account, its kind and its verdict.

Statement	Kind	Verdict	Where
The frame			
Free fall is universal; the frame weighs all energy	derived	holds	Sec. 4, 21
$\gamma = 1$ for the metric; the information scalar barely couples	derived	holds	Sec. 4, 21
The frame is classical after the first record	derived, by construction	—	Lemma 2
A classical mediator creates no entanglement	proven	—	Lemma 3
DMP can host the account (compatibility)∗	derived	—	Prop. 3

Statement	Kind	Verdict	Where
No graviton as a quantum of the frame	derived	not testable by counting	Sec. 4
The frame decoheres matter and diffuses	derived	live	Sec. 5, 16
Record and recompute is measurement and feedback*	identification	—	Prop. 4
The balance rule: the trade-off at equality	hypothesis	live	Sec. 5.4, 16
Bridge A: the horizon fixes the remnant's scale*	hypothesis	refuted	Sec. 11, 22
Bridge B: the Diósi–Penrose rate	hypothesis	live; not yet sensitive	Sec. 5.4, 21
The experiments			
Gravity entangles no two masses (Hypothesis 1′)	prediction	live	Sec. 6, 17
An entanglement test with two spheres decides between classical and quantum gravity*	—	refuted if the Diósi–Penrose rate is real	Sec. 17
The LOCC inequality holds	prediction	live	Sec. 6
The original spin witness, as printed in its arXiv version*	—	refuted	Sec. 17
Simulation			
A recorded frame keeps masses in product states and reproduces Newton's force	proven	—	Sec. 16
Hosted configurations return their known answers	proven	validates the host	Sec. 19
Holographic minimal cuts for contiguous regions and open bulk legs*	proven	refuted for every region	Sec. 18
Existence witnesses detect faults that leave the sector, not faults within it*	proven	—	Sec. 18
Black holes and holography			
A horizon is a containing boundary; its counts are M, J, Q	reading	holds as a reading	Sec. 7
Spinning holes give up energy outside the horizon	derived	holds	Sec. 7, 21
The Hawking part of the radiation of real holes	—	untested	Sec. 7
Information returns from black holes	—	open	Sec. 7
The area is a number (one fixed-area sector)	derived	—	Sec. 8
The tree is our space	—	refuted as literal; holds as routing	Sec. 8
2+1 gravity is an exception*	—	no; holds as a reading	Sec. 9
Cosmology			
Inflation with its oscillon decay is one start, the minting	hypothesis	holds as a reading	Sec. 4.5
Quantum geometry only within the start	derived	—	Sec. 4.5
Causality begins with the strong split	hypothesis	holds as a reading; unification untested	Sec. 10.1, 24
Primordial tensor modes, standard spectrum*	—	holds as a reading	Sec. 10
Plateau inflation, borrowed*	—	conditional; inherited tension	Sec. 10, 21
The balanced state as the de Sitter future	reading	holds as a reading; inherited tensions	Sec. 10, 21
Dark matter			
Remnants of interactions are trapped in the frame	hypothesis	live, through the contained remnant	Sec. 11
The stochastic remnant is the dark matter*	prediction	refuted	Sec. 22, 23
The contained remnant: primordial black holes*	prediction	live; untested here	Sec. 23
The radiated remnant or the known neutrinos are the dark matter	—	refuted	Sec. 11, 21

Statement	Kind	Verdict	Where
The information scalar and the Standard Model			
α does not grow*	derived	untested	Sec. 12.1, 21
Only α varies: $\mu = m_p/m_e$ varies far less	prediction	live; the window fixed in advance failed*	Sec. 12.1, 24
α is born near 1/128 at the electroweak split	reading	holds	Sec. 12.4, 24
The Higgs vacuum grants rest mass; the couplings follow mass	prediction	holds for the measured couplings	Sec. 12, 24
Neutrinos are Dirac; no neutrinoless double-beta decay (with $B - L$ as a count)*	prediction	live	Sec. 12, 24
Elementary spin stops at 1	reading	holds as a reading	Sec. 12, 24
No unification within the Standard Model's running	reading	holds as a reading	Sec. 10.1, 24
The Higgs field is the gravitational field	—	refuted	Sec. 12
Higgs inflation drives the start	—	open	Sec. 12
Hadrons			
Hadrons are trees of colour-conjugate halves (H8)	hypothesis	holds as a reading	Sec. 12.6
Compact-tree masses, from a standard colour-spin model	prediction	refuted for most	Sec. 25
The doubly bottom tetraquark is bound (the same model)	prediction	live	Sec. 25
A pentaquark is a minting: no open-flavour pentaquark (H9)	hypothesis	holds as registered; no statistical weight	Sec. 25
Hidden-bottom pentaquarks exist near 11.1 GeV (H9)	prediction	live	Sec. 25
Every exotic hadron carries a pair of one flavour	prediction	refuted	Sec. 25
Exotic doublets are split by $\alpha M/2$	prediction	refuted	Sec. 25
Readings			
The readings of Table 2	reading	holds as a reading	Sec. 13

29 Where we expect it to break

1. *The hypotheses.* Everything physical rests on the steps of Table 1, above all on Hypothesis 1′, which places the field that pulls laboratory masses in the frame. Another faithful reading changes the predictions; the dissent of Section 6.5 is one.
2. *Experiment.* A positive test of gravitationally mediated entanglement, a violation of the LOCC inequality, or decoherence and diffusion both measured below the trade-off (8) would each show that gravity is not GUID's classical frame, unless classical gravity can entangle after all [81]. A null entanglement test is not decisive (Section 17).
3. *Constants.* The axioms derive none of G, $\hbar$, Λ, the entropy per area or the strength of the channel. The bridges fix the strength only by adding a principle. If a superposition of about 10^{-14} kg over more than its own size keeps its coherence well beyond $\hbar R/GM^2$, Bridge B is refuted.
4. *Dark matter.* The contained remnant now carries the dark matter alone. If evaporation and microlensing close the asteroid-mass window, or if halos of primordial black holes cannot form the observed structures, the trapped-remnant hypothesis loses its last route.
5. *The start.* GUID borrows its inflation model; if the scalar tilt settles where plateau models need many more e-folds, another model replaces it, and GUID's reading of the start does not depend on which. If Axiom 1 forbids a coupling to curvature in every frame, plateau inflation and GUID exclude each other.

6. *Black holes.* A derivation of the Page curve that needs superposed geometries after the start would contradict Lemma 2.
7. *Space.* A literal tree is not our space; the routing reading avoids the objection but has not been shown to give a Lorentzian 3+1 manifold in any limit.
8. *The frame's place.* Hypothesis $1'$ keeps the classical prediction only because the frame is a DMP structure shaped by the counts. Read as lying outside DMP's structures, it would leave DMP silent on whether gravity is quantum.
9. *Unification.* A proton decay found would show that a unification existed, as Hypothesis $6'$ allows; continued silence cannot refute one that lies before time.
10. *Neutrinos and hadrons.* A confirmed neutrinoless double-beta decay would refute the Dirac statement; a confirmed open-flavour pentaquark would refute H9.
11. *The simulations and checks.* Classical hosting is exponential in the number of entangled qubits; the coherent field is simulated in full only for small systems; every data confrontation depends on the likelihood and error model stated in its section; and the checks of Part I use round inputs that move numbers by tens of per cent, not the verdicts.

30 Conclusion

Read as physics, the three axioms of the Dark Matter Protocol give a definite account of gravity and of what it contains. Gravity is a classical frame, the tetrad of general relativity, shaped by the counts of energy and momentum, weighing all energy alike, and classical because the axioms make each of its values a record that independent witnesses hold in full. Consistency with quantum mechanics then makes it decohere matter and diffuse, and its own bridge hypotheses fix the rate at Diósi and Penrose's. The axioms host either reading of the field that pulls laboratory masses; the classical one is GUID's choice, and an experiment can refute it.

Tested, the account holds wherever general relativity, ΛCDM and the Standard Model hold, inherits their open tensions, and loses some of its ideas: Bridge A, the stochastic dark-matter remnant, a numerical window it fixed in advance, and the masses a standard model gives its compact hadron trees. What remains is a classical frame over quantum content, primordial black holes from the start as its candidate dark matter, Dirac neutrinos, a pentaquark rule of its own with hidden-bottom pentaquarks to find, and one sharp number, the Diósi–Penrose rate. Every GUID-specific statement the data can already reach has failed; the rest wait for experiments that can. The simulations showed that, if that rate is real, the experiment usually named as decisive cannot separate classical from quantum gravity; the first experiment to decide is whether a single mesoscopic superposition decoheres at that rate. We claim that the construction exists, reproduces and agrees with the data named. Whether it is right is for the experiments, and for the reader, to break.

Acknowledgements

The first author remembers the late Peter Higgs, and thanks the many theorists and experimentalists whose work this account requires and builds on, and the SPARC team for making their rotation curves public. The review passes that tested earlier forms of this work, one of which gave the dissenting reading of Section 6.5, and the two referee reports on a draft of this version are part of its record.

Contributions and use of AI

Steven Hrovat conceived DMP, the reading of trust as existence, the two-field structure, and every hypothesis tested here, asked for the tests of Parts II and III, chose which ideas to keep

after each round of tests, and takes responsibility for the axiom system and the implications of its theorems. Sirius (Claude, Anthropic) made the derivations, related the account to the literature, designed the tests, wrote `docs/guid-v6/checks.py` and `docs/simulation/bmv_witness.py`, and edited this text. The other simulation and data scripts of Parts II and III and the first drafts of their sections were written by AI agents working under Sirius's coordination, each with its own checks, and three further agents reviewed drafts of the paper independently; the holographic-code assembly, the Lean development and the review passes were written by AI systems under the first author's direction. Every numerical claim is a line of a public script; every reference was checked against a publisher, arXiv or collaboration record, or is marked as not re-checked, as listed in `docs/guid-v6/README.md`.

Licence and availability

This paper and its source are released under the Apache License, Version 2.0. DMP: `https://github.com/StevenHrovat/dmp` (version 0.9.9 at commit `4400cce`). Gravity: `https://github.com/StevenHrovat/gravity` (this paper, its checks and `reproduce.py` in `docs/guid-v6/`; the simulations and data confrontations in `docs/simulation/`; the holographic-code assembly in `docs/qg-verdict/`; the Lean development in `evidence/`). The SPARC rotation curves are not redistributed; Appendix A names the files and their SHA-256 digests.

A Reproducibility

Every number of this paper is printed by one of the scripts in Table 23. Each script prints OK or FAIL for each of its checks and exits with code 0 only if all pass. The one exception is the data-confrontation script, which exits with code 1 on purpose because three scored confrontations fail at 95% (Section 21). `python3 docs/guid-v6/reproduce.py` runs all of them and prints the table again from the scripts' own output.

All scripts are deterministic. They fix their random seeds, pin the linear-algebra libraries to one thread, and read no clock except to report their own runtime. Two runs of each gave identical output files apart from that runtime field. The runtimes in the table are for one core of a shared machine and are not results. The scripts need Python 3 with numpy and scipy; `checks.py` also needs sympy, `holographic_code.py` needs stim 1.16, and the scripts that draw figures need matplotlib. None of them needs a network connection.

Table 23: The scripts behind the paper. The first two are `docs/guid-v6/checks.py` and `docs/qg-verdict/assembly.py`; the others are in `docs/simulation/`.

Script	Sections	Checks	Runtime	Labels
`checks.py`	Part I	125 of 125	seconds	[Q1]–[Q20]
`assembly.py`	8	18 of 18	seconds	[A1]–[A5]
`hosted_benchmarks.py`	19	53 of 53	8 min	[hosted:B1]–[hosted:B4]
`holographic_code.py`	18	77 of 77	8 min	[holo:H0]–[holo:H6]
`gravity_simulated.py`	16, 17	110 of 110	3 min	[gravsim:G1]–[gravsim:G4]
`bmv_witness.py`	17	34 of 34	1 min	[witness:B1]–[witness:B4]
`sparc_rotation.py`	22	39 of 39	1 min	[sparc:S0]–[sparc:S3]
`remnant_history.py`	23	39 of 39	2.5 min	[remnant:R1]–[remnant:R6]
`predictions_vs_data.py`	21	6 of 6; 28 of 31 scored	4 s	[data:D1.1]–[data:D9.6]

Script	Sections	Checks	Runtime	Labels
`standard_model.py`	24	32 of 32	5 s	[sm:SM1]–[sm:SM5]
`exotic_hadrons.py`	25	80 of 80	1 min	[exotics:A]–[exotics:H]
`tally.py`	26	6 of 6	1 s	[tally:total]

Data. The rotation-curve analyses read the SPARC tables of Lelli, McGaugh and Schombert [251]. Their licence is not stated, so they are not redistributed with this paper. Both scripts check the files by SHA-256 before use and stop if a digest differs:

- `MassModels_Lelli2016c.mrt`: `9108994b12cc401b94a1768beca61c53ec354779385c9c9cc571049f3043244c`;
- `SPARC_Lelli2016c.mrt`: `5aa0501f6b0d881fa579030e315e7b5b6ef561a5bd3a07472f9929c7e5728243`.

Pass their folder with `-data <folder>` or the environment variable `SPARC_DATA_DIR`. All other data enter the scripts as published values, each with its source in the script and in Sections 21 to 25.

Results files. Each simulation script writes its numbers to a JSON file and a readable account to a Markdown file of the same name, both in `docs/simulation/results/`; its figures go to `docs/simulation/figures/`. The JSON files are the reference for the labels of the form [script:label] in the text; a label that names a whole analysis step (such as [sparc:S0] or [remnant:R6]) points to the step in the script and its results note.

B The checks of Part I

`python3 docs/guid-v6/checks.py` (numpy, scipy, sympy) prints 125 checks, all of which pass.

- Q1: Planck units, the existence energy, Unruh, Hawking, Bekenstein–Hawking, Jacobson's entropy density, and the size of the quantum term of (1).
- Q2: the lapse.
- Q3: no-cloning and broadcasting.
- Q4: classical against coherent mediation, 300 random couplings each.
- Q5: the Bose et al. geometry.
- Q6: the MICROSCOPE bound on the information scalar, at the one-standard-deviation level and at 95%, and the Higgs-share test.
- Q7: Brans–Dicke and Cassini.
- Q8: polarisations and Weyl components in D dimensions.
- Q9: the single-graviton threshold.
- Q10: Kerr horizon, ergosphere, Christodoulou, efficiencies.
- Q11: Hawking radiation of real holes.
- Q12: the Hubble mass at its Schwarzschild threshold.
- Q13: flatness, the de Sitter floor and the onset of acceleration.
- Q14: the tensor spectrum, the BICEP/Keck bound and Starobinsky's r.
- Q15: what every trapped remnant must satisfy, and why the known neutrinos do not (with the DESI DR2 mass bound).
- Q16: α as self-interaction strength, the electron's lengths, gravity against electromagnetism, and the running of α.
- Q17: the speed of gravity.
- Q18: remnants trapped in the field: the stochastic remnant's rotation curve and acceleration scale, the contained remnant's Hawking temperatures and evaporation, and the radiated remnant's energy density.

- Q19: the two bridges, Bridge A against the Tully–Fisher normalisation, Bridge B's windows and Diósi–Penrose rates, and the granular term at both smearing bounds.
- Q20: the radiation-era clock and the time of the electroweak crossover, the W and Z masses from the Higgs vacuum, α at its birth and today, and the proton lifetime for a unification at the start.

`python3 docs/qg-verdict/assembly.py` (numpy) prints the 18 checks of the holographic-code assembly. A1 to A5 in this paper are its checks [1] to [5]; its check [6] is the special case of [Q4] with CNOT and CZ couplings.

C Changes from version 5.1

This paper replaces version 5.1 and is complete without it. For readers of the earlier version, this appendix lists what the tests of Parts II and III and three independent reviews of drafts of this paper corrected or sharpened. Each change is already applied in the main text, and each names its evidence.

Bounds and data.

1. *The bound on the information scalar.* Version 5.1 gave $|\lambda| < 2\times10^{-5}$ from MICROSCOPE; that value corresponds to $|\eta| < 2.7 \times 10^{-15}$, a one-standard-deviation bound. The 95% bound is $|\lambda| < 3.1 \times 10^{-5}$ [Q6], [data:D1.1].
2. *The clock bound.* The newest clock comparison, $(1.8 \pm 2.5) \times 10^{-19}$ per year [204], includes and supersedes the value $(1.0 \pm 1.1) \times 10^{-18}$ per year cited before [data:D1.3].
3. *The sign of the drift of α.* With the action as written, without a potential, "zero or negative" is derived, but the drift is below 10^{-23} per year; a measurable drift needs a potential, and then the sign is not predicted [data:D1.12].
4. *The smearing length of Bridge B.* "At least about 10^{-10} m" becomes $R_0 \geq 4\times10^{-10}$ m, the summary of current bounds; Donadi et al.'s own 95% bound is 0.54×10^{-10} m [data:D6.5], [data:D6.6].
5. *Starobinsky inflation.* "Fits the scalar data" holds for Planck 2018; against the Atacama Cosmology Telescope's sixth release combined with Planck, lensing and baryon acoustic oscillations it needs $N \geq 63$ e-folds [data:D4.3].
6. *The neutrino bound.* DESI's second release bounds the mass sum below 0.064 eV, so the known neutrinos are below 0.6% of the dark matter, not 0.7% [Q15], [data:D9.3].
7. *The late universe.* DESI's evolving dark energy stands at 2.8 to 3.8 standard deviations after the recalibration of the DES supernovae (2.8 to 4.2 before it), and the inherited tensions in the Hubble constant (4.9σ) and in curvature (2.1σ) are now named [data:D3.3], [data:D3.4], [data:D3.7].
8. *Gravitational waves.* The speed of gravity is quoted as the published interval, $-3 \times 10^{-15} \leq \Delta c/c \leq 7 \times 10^{-16}$, and the bound $m_g \leq 2.42 \times 10^{-23}\,\mathrm{eV}/c^2$ is added [data:D2.11], [data:D2.12].

The decisive experiments.

9. *What a null entanglement experiment shows.* "A Bose–Marletto–Vedral experiment sees decoherence and no entanglement" is right as GUID's prediction but does not discriminate: with Diósi–Penrose decoherence a quantised field entangles no two spheres either. The decoherence measurement decides Bridge B first; only in its absence does the entanglement test separate classical from quantum gravity [gravsim:G2].
10. *What a positive result shows.* It excludes every classical mediator; it implies decoherence below $\omega/2$ only if the noise is local and uncorrelated (Section 17).
11. *The dissenting reading* of Section 6.5 predicts entanglement only as long as the decoherence stays below $\omega/2$.

12. *The spin witness.* The witness of the original proposal, as printed in its arXiv version, is reached by product states, most likely a misprint; the form quoted in later work [26] is valid but detects only entangling phases above $\pi/2$ in this geometry. A two-setting witness that detects every entangling phase is given in Section 17 [witness:B1], [witness:B2].

Dark matter.

13. *Bridge A* is refuted by data as well as by the estimate: it lies above 98% of the SPARC points [sparc:S2b].
14. *The stochastic remnant.* Version 5.1 said that it "already gives the shape and scale of galactic rotation curves in a first calculation". With one universal constant it fits worse than the radial acceleration relation by $\Delta\mathrm{BIC} = +5066$, the low-acceleration excess is not constant (slope 0.44, which a mock analysis shows the estimator does not produce from a constant), and the mass scaling of Oppenheim and Russo is disfavoured [sparc:S2a], [sparc:S2c], [sparc:S2d]. Its accumulated form is MOND's interpolating function and fails in the solar system [remnant:R1a].
15. *The dark-matter verdict.* Version 5.1 kept the stochastic and the contained remnant both live. The stochastic remnant is refuted as the dark matter in every form computed; with halos present the data leave its square-root form 1 to 5% and a constant 8 to 31% of the dark acceleration; the two remnants compete rather than combine; and primordial black holes from the start, the contained remnant, are the reading that survives, which the rotation curves cannot tell apart from particle dark matter [remnant:R3], [remnant:R4].
16. *The acceleration scale.* The closeness of $c^2\sqrt{\Lambda/3}/2\pi$ to Milgrom's a_0 is Milgrom's known coincidence; among nine natural scales a random one comes as close with a probability of 10 to 20% [remnant:R1b].

Statements of Part I sharpened.

17. *What "existence" means.* Hypotheses 1, 2 and $1'$ are reconciled: Hypothesis $1'$, adopted, keeps "existence" for the rest mass the Higgs vacuum grants and makes gravity a frame shaped by the counts $T_{\mu\nu}$ of everything (Section 3).
18. *Free fall and light bending.* The proof of Proposition 1 now states its geodesic step. $\gamma = 1$ is claimed for the metric; the scalar shifts the inferred γ by less than 10^{-14}, and the exclusion of $F(\phi)R$ depends on the conformal frame.
19. *The mediator lemma.* Lemma 3 assumes that the mediator is dephased after each interaction, which the records of Lemma 2 supply; a diagonal marginal alone is not enough.
20. *Hosting and the channel.* Proposition 3 is a statement of compatibility, not a derivation of the dynamics, and Proposition 4 is an identification. Hosting on a classical computer costs of order 2^n for n qubits of generic content [hosted:B4].
21. *The trade-off.* Its normalisation is written out (9), with the map to Oppenheim et al.'s, and the strength D_2 of Bridge B is defined.
22. *The holographic assembly.* The minimal-cut formula holds for contiguous regions and, for every region, when the bulk legs are open; the tile orientation is stated; existence witnesses detect faults that leave the existence sector and pass faults within it [holo:H2], [holo:H5].
23. *2+1 dimensions.* The verdict is *holds as a reading*, not *derived*: a classical 2+1 frame with classical holonomies would also be consistent.

Nothing in the axioms or in the hypotheses' content changed. The checks of Part I grew from 123 to 125.

D Hosting benchmarks in detail

This appendix gives the models, methods and results behind Table 10 of Section 19. Headline values are not repeated here; the table is their home.

D.1 Conventions

Witnesses. Witness T and witness I share no code path with each other or with the host. Where the host counts set bits with a lookup table, witness T uses bit-parallel arithmetic. Where witness T reads a determinant from a pivoted LU factorisation, witness I uses a Householder QR factorisation. The gate passes only when both witnesses pass.

Controls. Each benchmark carries four controls.

- $[H, \Sigma] = 0$ is checked on a small, complete Hilbert space.
- An injected fault that leaves the existence sector must be refused by both witnesses.
- Two orthogonal states of the sector, that is two different contents, must receive the same verdict.
- In B4, a fault that changes only the content must pass both witnesses.

Uncertainties. Exact identities (closed forms, commutators) are checked to floating-point accuracy. An extrapolated number carries, as 1σ, the largest change over a family of alternative fits (windows and functional forms) fixed in the code. In B3 a second fit uses these spreads as weights; the uncertainty it propagates is combined in quadrature. A comparison passes at $|\text{deviation}| \le 2\sigma$.

D.2 B1: the Heisenberg chain

Model. $H = \sum_{i=1}^{N} \mathbf{S}_i \cdot \mathbf{S}_{i+1}$ acts on spins 1/2 on a ring, for even N from 8 to 24. The syndrome $\Sigma = (S^z_{\text{tot}})^2$ selects the sector $S^z_{\text{tot}} = 0$, with 2 704 156 states at $N = 24$. The ground-state energy per site tends to $e_0 = \frac{1}{4} - \ln 2$ [334, 335].

Method. Lanczos iteration gives the two lowest levels. At every N the ground energy is checked against the Bethe ansatz, the exact solution of this chain. With $M = N/2$ rapidities u_j and quantum numbers $I_j = -\frac{M-1}{2}, \ldots, \frac{M-1}{2}$, the Bethe equations and the energy are

$$N \arctan 2u_j = \pi I_j + \sum_k \arctan(u_j - u_k), \qquad E_0 = \frac{N}{4} - \sum_j \frac{2}{1 + 4u_j^2}.$$

A Newton solution agrees with Lanczos to 1.3×10^{-15} per site. For a critical chain, conformal invariance gives $E_0(N)/N = e_0 - \pi v c/(6N^2) + \ldots$ [336]. Here $c = 1$ and $v = \pi/2$, and the form is corrected by powers of $1/\ln N$ [337]. We fit $a + bN^{-2} + dN^{-4}$ on $N \ge 12$. The nine alternatives are the windows $N \ge 8, 10, 14, 16$, and all five windows with an added N^{-6} term.

Results. All nine alternatives lie above the exact value. The logarithmic corrections predict this, since no power series in $1/N^2$ absorbs them. The triplet gap $\Delta_N = E_1 - E_0$ closes:

- $N\Delta_N$ grows from 4.18 at $N = 8$ to 4.39 at $N = 24$;
- the local exponent $-\mathrm{d}\ln\Delta_N/\mathrm{d}\ln N$ rises from 0.946 to 0.968;
- a fit $\Delta_\infty + \beta_1/N + \beta_2/N^2$ on $N \ge 12$ gives $\Delta_\infty = 0.0024$, against $\Delta_{24} = 0.183$.

The chain is gapless, and $N\Delta_N$ approaches $\pi^2/2$ only logarithmically.

Witnesses. T counts the up spins of every register index that carries amplitude. I evaluates $\langle (S^z_{\text{tot}})^2 \rangle$ site by site. $[H, S^z_{\text{tot}}] = 0$ holds exactly on the full 2^{10}-dimensional space. The $N = 24$ ground state and the orthogonal triplet pass. A single spin flip gives $\langle (S^z_{\text{tot}})^2 \rangle = 1$ and is refused by both.

D.3 B2: the transverse-field Ising ring at $L = 4000$

Model. $H = -\sum_i Z_i Z_{i+1} - g\sum_i X_i$ acts on a ring of $L = 4000$ spins, at $g = 1$ (critical), 0.5 and 1.5. The Jordan–Wigner transformation maps the spins to fermions. With the Majorana operators $a_j = (\prod_{l<j} X_l)Z_j$ and $b_j = (\prod_{l<j} X_l)Y_j$, one has $X_j = ia_jb_j$ and $Z_jZ_{j+1} = ib_ja_{j+1}$. The bond that closes the ring carries the parity $P = \prod_i X_i$. For $P = +1$ it becomes an antiperiodic fermion bond [338, 339], and that sector is the existence sector. There, and only there, $H = i\sum_{jk} A_{jk}a_jb_k$, with $A_{jj} = -g$, $A_{j+1,j} = 1$, $A_{1L} = -1$ and all other entries zero. At $L = 10$ this reproduces the lowest even-parity eigenvalue of the full spin Hamiltonian to 1.8×10^{-14}.

Method. The singular-value decomposition $A = U\,\mathrm{diag}(s_m)V^{\mathsf{T}}$ gives the ground energy $E_0 = -\sum_m s_m$. The ground state is Gaussian, with correlation matrix $G_{jk} = \langle ia_jb_k\rangle = -(UV^{\mathsf{T}})_{jk}$. A block of l consecutive spins has entropy

$$S(l) = \sum_m h\left(\tfrac{1+\nu_m}{2}\right), \qquad h(p) = -p\ln p - (1-p)\ln(1-p),$$

where ν_m are the singular values of G restricted to the block [340, 341]. Each coupling costs one SVD of a 4000×4000 matrix, about 27 s on one core.

Results. $E(m)$ and $K(k)$ denote the complete elliptic integrals of the second kind (parameter m) and of the first kind (modulus k).

(a) The ground energy equals the finite-L closed form $-\sum_k \sqrt{1 + g^2 - 2g\cos k}$, with $k = 2\pi(n + \frac{1}{2})/L$, at all three couplings. At $g = 0.5$ and 1.5, E_0/L equals the thermodynamic limit $e_\infty = -\frac{2}{\pi}(1+g)E(4g/(1+g)^2)$ to 4×10^{-16} [339].

(b) At $g = 1$, $E_0/L - e_\infty = -3.27 \times 10^{-8}$. Read as the Casimir term $-\pi vc/(6L^2)$ with $v = 2$, it gives the first value of c in Table 10. Its uncertainty is floating-point error, and the exact expansion predicts $\frac{1}{2} + 7\pi^2/(480L^2) = 0.5000000090$.

(c) At $g = 1$ we fit the Calabrese–Cardy form for a ring [342], $S(l) = \frac{c}{3}\ln[\frac{L}{\pi}\sin\frac{\pi l}{L}] + c_1'$, on $100 \le l \le 2000$. This gives the second value of c, with $c_1' = 0.478558$. The uncertainty is the spread over windows that start at 50, 200 or 400 or end at 1000.

(d) At $g = 0.5$ and 1.5 the entropy saturates, $S(L/2) - S(L/8) = 2 \times 10^{-12}$: an area law. The saturated values match the corner-transfer-matrix prediction. The corner transfer matrix is Baxter's transfer matrix for a quadrant of the two-dimensional lattice; its spectrum gives the entanglement levels of a half-infinite chain. These are $(2j+1)\epsilon$ for $g > 1$ and $2j\epsilon$ for $g < 1$, with $\epsilon = \pi K(k')/K(k)$ and $k = \min(g, 1/g)$ [342, Eqs. 4.30–4.34]. A block has two cuts, so $S = 2S_{\mathrm{half}}$ for $g > 1$. For $g < 1$ the even-parity ground state is a cat state whose $\ln 2$ the two cuts share, so $S = 2S_{\mathrm{half}} - \ln 2$. We derived this before the comparison.

Witnesses. The syndrome is the parity, $\langle P\rangle = \det G$, together with purity ($G$ orthogonal). T reads $\det G$ from a pivoted LU factorisation: parity $+1$, with $|\ln|\det G|| < 10^{-13}$. I reads it from a Householder QR, using the reflector count and the signs of the diagonal of R, and finds $\max|GG^{\mathsf{T}} - \mathbb{1}| \le 7 \times 10^{-15}$. A spin flip Z_1 reverses the first row of G; both witnesses read -1 and refuse.

D.4 B3: the Schwinger model

Model. Quantum electrodynamics in 1+1 dimensions, with one fermion of mass m and charge g, confines. At $m = 0$ it is a free boson of mass $g/\sqrt{\pi}$, the vector particle [192]. We use Kogut–Susskind staggered fermions [343]: one fermion component per site, the two components of the continuum spinor sitting on alternate sites. The lattice has N sites, open boundaries and zero background field. Gauss's law reads $E_n - E_{n-1} = Q_n$, with $Q_n = \frac{1}{2}(\sigma_n^z + (-1)^n)$ the charge on

site n and E_n the integer electric flux on the link from site n to $n+1$. It fixes $E_n = \sum_{k \le n} Q_k$ and leaves the spin Hamiltonian [344, Eq. 2.6]

$$W = \frac{2H}{ag^2} = x \sum_{n=0}^{N-2} (\sigma_n^+ \sigma_{n+1}^- + \text{h.c.}) + \sqrt{x}\, \frac{m_{\text{lat}}}{g} \sum_{n=0}^{N-1} (1 + (-1)^n \sigma_n^z) + \sum_{n=0}^{N-2} E_n^2, \tag{59}$$

with lattice spacing a and $x = (ag)^{-2}$. The vector mass is $M/g = (W_1 - W_0)/(2\sqrt{x})$, from the two lowest levels, taken to $N \to \infty$ and then to $x \to \infty$. We run two lattice masses: the naive $m_{\text{lat}} = 0$, and $m_{\text{lat}} = -g^2 a/8$. The discrete chiral symmetry of the lattice model identifies the second as the massless point [345]. The shift vanishes as $a \to 0$, so both lattices have the same continuum limit.

Existence and witnesses. The existence sector is fixed by Gauss's law at every site, with $E_{-1} = 0$ (no background) and $E_{N-1} = 0$ (no field beyond the last site). Together these imply total charge zero. The host keeps an integer link register $E_0, \ldots, E_{N-2}$ for every basis configuration.

- T checks $\mathcal{G}_n \equiv E_n - E_{n-1} - Q_n = 0$ at every site of every configuration carrying amplitude. I evaluates $\langle \sum_n \mathcal{G}_n^2 + Q_{\text{tot}}^2 \rangle$.
- The ground and vector states at $N = 22$ pass. A charge without its field (one spin flipped) and a flux line without charges (one link raised) give the value 2 and are refused by both.
- The elimination itself is checked with explicit integer links at $N = 6$ (1 075 648 states; $|E_n| \le 3$ is exact there). $[H, \mathcal{G}_n] = 0$ for all n, and the 20 states with all $\mathcal{G}_n = 0$ reproduce the spectrum of (59) to 2.5×10^{-14}.

Method. Lanczos iteration runs for $N = 8, \ldots, 22$ (705 432 states at $N = 22$) at $x = 0.5, 1, 1.5, 2, 3, 4$.

- *$N \to \infty$ at each x.* We fit $M_N/g = w + \alpha N^{-2} + \beta N^{-3}$ on $N \ge 12$. The N^{-2} term is the kinetic energy $\pi^2/(2M\ell^2)$ of the vector particle in a box of length ℓ; the N^{-3} term absorbs a boundary shift of ℓ. The alternatives are $N \ge 14$, $N \ge 16$, $\alpha N^{-2} + \beta N^{-4}$, and αN^{-2} alone on the last four sizes.
- *Which x enter.* Only x with $N_{\max}/\sqrt{x} \ge 10$, a box at least ten units of $1/g$ long. All six qualify.
- *$x \to \infty$.* We fit $w(x)$ with a quadratic in $y = 1/\sqrt{x} = ag$, weighted by the inverse squared N-extrapolation errors. The alternatives are linear on $x \ge 1.5$, quadratic on $x \ge 1$, cubic, and, for the shifted lattice only, quadratic without the linear term.

Table 24: Schwinger model at $m = 0$: vector mass M/g at each $x = 1/(ag)^2$, extrapolated to $N \to \infty$ (fit-family spread in units of the last digit), and at $N = 22$ (S) [hosted:B3]. Exact value: $1/\sqrt{\pi} = 0.5641896$.

		shifted, $m_{\text{lat}} = -g^2 a/8$		naive, $m_{\text{lat}} = 0$	
x	ag	$N \to \infty$	$N = 22$	$N \to \infty$	$N = 22$
0.5	1.41	0.5835(6)	0.5892	0.8955(4)	0.8986
1	1.00	0.5648(15)	0.5784	0.7848(10)	0.7939
1.5	0.82	0.5620(24)	0.5836	0.7422(18)	0.7580
2	0.71	0.5616(35)	0.5910	0.7180(28)	0.7405
3	0.58	0.5620(60)	0.6067	0.6900(49)	0.7259
4	0.50	0.5627(87)	0.6222	0.6740(69)	0.7228

Results. Table 24 gives the values at each x.

- On the shifted lattice the continuum uncertainty combines a propagated N-extrapolation uncertainty of 0.016 with a continuum fit-family spread of 0.035. On the naive lattice the two parts are 0.012 and 0.028.

- The per-x errors are fit-family spreads, strongly correlated between x. The main continuum fits have $\chi^2 = 0.04$ (shifted) and 0.18 (naive) for 3 degrees of freedom. With $\chi^2/\mathrm{dof} \ll 1$, the deviations in units of σ are consistency checks, not significances.
- The continuum alternatives on the shifted lattice give 0.562 (linear, $x \geq 1.5$), 0.578 (quadratic, $x \geq 1$), 0.570 (cubic) and 0.550 (no linear term).
- The finite-size form passes an out-of-sample test. At $x = 4$ the fit on $N \leq 22$ predicts the $N = 24$ gap as 0.613467; the computed value is 0.613388.

Precision limits.

- Matrix product states (MPS) give 0.56421(9) [344]. The density-matrix renormalisation group (DMRG) gives 0.5642(2) [346], as tabulated in [344]. Their uncertainties are 200 to 400 times smaller than ours.
- The MPS study uses x from 5 to 600 with $N \geq 20\sqrt{x}$ [344]. Exact diagonalisation stops at $N = 22$, hence at $x \leq 4$.
- The weighted continuum fit is dominated by the coarsest lattice, $x = 0.5$. Its N-extrapolation is the most precise, but a quadratic in ag describes its lattice artefacts worst. This pulls the central value up and sets the 7% error.
- The shifted values at $1.5 \leq x \leq 4$ lie within 0.5% of $1/\sqrt{\pi}$ [345]. They are not continuum values, and we claim nothing from them.
- We did not attempt the massive model, whose binding energies are known to 5×10^{-5} [344], or the chiral condensate. A continuum error of several per cent cannot test either.

D.5 B4: Shor's period finding

Circuit. To factor N we use a counting register of $t = 2n$ qubits and a work register of $n = \lceil \log_2 N \rceil$ qubits, held as an exact state vector [193]. The run has three steps.

1. The counting register starts in uniform superposition, and the work register in $|1\rangle$.
2. Controlled U^{2^j} follow, with $U|y\rangle = |ay \bmod N\rangle$ for $y < N$ and $U|y\rangle = |y\rangle$ otherwise.
3. The inverse quantum Fourier transform is applied as the exact discrete Fourier transform. It agrees with the explicit gate circuit to 1.8×10^{-16}.

U permutes $\{y < N\}$, so the existence sector (unit norm, work register $y < N$) is invariant. An outcome k gives the period r as the first continued-fraction convergent denominator $q < N$ of $k/2^t$ with $a^q \equiv 1$. It gives a factor $\gcd(a^{r/2} - 1, N)$ when r is even and $a^{r/2} \not\equiv -1 \pmod N$; this is one run, with no repetition. We host $N = 15, 21, 33, 35, 39, 51, 55, 57$ with every base a coprime to N, and $N = 143$ (24 qubits) with $a = 2, 3, 5, 7$: 192 states in all.

Results. The simulated distribution is compared with Shor's analytic distribution [193],

$$P(k) = Q^{-2} \sum_{x_0 < r} \Big| \sum_{j < m(x_0)} e^{-2\pi i j r k / Q} \Big|^2, \qquad Q = 2^t, \quad m(x_0) = \lfloor (Q - 1 - x_0)/r \rfloor + 1.$$

Table 25 lists the register sizes and the success probabilities. The textbook failure appears where it should: for $N = 33$ and 57 with $a = 2$ the period is found but $2^{r/2} \equiv -1$, so no factor follows.

Witnesses. T requires every amplitude with $y \geq N$ to be exactly zero and the norm to be one. I evaluates $\langle \Pi_{y \geq N} \rangle = 0$, with the projector applied as an operator. All 192 states pass.

- A faulty multiplier that exchanges $|1\rangle$ and $|2^n - 1\rangle$ moves amplitude out of the sector. Both witnesses refuse it in every case.
- A phase Z on one work qubit changes the content but not the existence, and both witnesses pass it.

Table 25: Shor's period finding on an exact state vector (S) [hosted:B4]. The simulated register has $3n$ qubits; Beauregard's circuit needs $2n + 3$ [194]. Exact single-run probabilities for $a = 2$, and the mean over all hosted bases. P(factor) can exceed P(period) because a recovered multiple of r can also yield a factor.

N	factors	$3n$	$2n+3$	bases	r $(a = 2)$	P(period)	P(factor)	mean P(factor)
15	3×5	12	11	7	4	0.500	0.500	0.429
21	3×7	15	13	11	6	0.331	0.331	0.211
33	3×11	18	15	19	10	0.398	0.000	0.220
35	5×7	18	15	23	12	0.331	0.331	0.303
39	3×13	18	15	23	12	0.331	0.331	0.303
51	3×17	18	15	31	8	0.500	0.500	0.484
55	5×11	18	15	39	20	0.393	0.393	0.320
57	3×19	18	15	35	18	0.329	0.000	0.180
143	11×13	24	19	4	60	0.265	0.265	0.232

D.6 The cost map

One random layer consists of n Haar-random (uniformly distributed) single-qubit unitaries and $n - 1$ controlled-Z gates. They act on a dense state vector of 2^n amplitudes, 16 bytes each, on one core. The measured peak memory is the state plus a gate buffer of at most 16 MiB (Table 26). The layer time divided by n grows as $2^{1.068n}$ for $n = 18$ to 26. The excess over 2^n comes from cache and memory bandwidth. The rate is 6.9×10^{-9} s per qubit and amplitude for one layer, with the $n - 1$ controlled-Z gates folded in. The extrapolations of Section 19 use this rate.

Table 26: Cost of a dense state-vector simulation, one random layer on one core (S) [hosted:B4].

n	state	measured peak	seconds per layer
10	16 KiB	54 KiB	0.0011
14	256 KiB	770 KiB	0.0024
18	4 MiB	10 MiB	0.032
22	64 MiB	80 MiB	0.63
24	256 MiB	272 MiB	3.7
26	1 GiB	1.02 GiB	15.5

D.7 Choices made after seeing the data

(i) In B1 the fit family and the 2σ rule were fixed after a prototype had shown the extrapolated value. Under a 1σ rule B1 would fail (1.3σ).

(ii) In B2 the entropy-fit windows were fixed after a prototype had shown $c = 0.500000$ to six digits.

(iii) In B3 the x grid and $N_{\max} = 22$ were set from the cost of exploratory runs. The fit rules were written before the first continuum fit (on $N \leq 18$) and were kept, with one exception. The alternative without a linear term was then dropped for the naive lattice, whose mass shift is an $O(a)$ effect [345]. Keeping it widens the naive uncertainty from 0.031 to 0.075 (recomputed from Table 24); the comparison passes either way.

(iv) In B4 the faulty multiplier replaced an X on the top work qubit as the fault. For $N = 15$, $a = 2$, that X maps the orbit $\{1, 2, 4, 8\}$ to $\{9, 10, 12, 0\}$, inside the existence sector.

D.8 Caveats

- These are classical emulations of hosting. A classical host carries generic content only at a cost of order 2^n.

- The modular multiplications of B4 are exact permutation gates, not elementary-gate circuits. This changes gate counts but not states. The registers use the textbook $t = 2n$, not Beauregard's layout.
- B1 stops at $N = 24$, the memory limit of a sector code without translation symmetry.
- Wall times are single-core on a shared machine and are reported, not checked. All other numbers are deterministic.

E The holographic code in detail

This appendix holds the construction, the methods, the proofs and the remaining results behind Section 18.

E.1 Construction and method

The tile. A tile is the six-leg perfect tensor of the five-qubit code [96, 97]:

$$T_{b\,q_1\ldots q_5} = \langle q_1 \ldots q_5 | E | b \rangle / \sqrt{2}.$$

Here E is the encoding isometry of $[[5, 1, 3]]$, whose stabilisers are $XZZXI$ and its cyclic shifts, with $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. The leg b is the logical leg. Every split of the six legs into three and three is unitary [99].

The tree. The root has its bulk leg b and five legs to children. Every other tile has a bulk leg, a parent leg and four child legs. The leaves' legs form the boundary, $n = 5 \cdot 4^D$ qubits in ring order, carried by $1 + 5(4^D - 1)/3$ tiles. For $D = 1, \ldots, 5$ this gives $n = 20$, 80, 320, 1280 and 5120 boundary qubits on 6, 26, 106, 426 and 1706 tiles. With one reference per bulk qubit, setting (b) has 6826 qubits at $D = 5$.

Orientations. The children sit on $q_2, \ldots, q_5$ in both orientations.

- *Orientation A*, the main construction, is the 25-qubit assembly of Section 8: the parent edge sits on b and the bulk on q_1.
- *Orientation H*, the HaPPY convention, puts the bulk on b and the parent edge on q_1.

Both make every tile a perfect tensor. They differ only in which legs meet across an edge.

Exact computation. Every tile map is a Clifford isometry, $V|v\rangle = U|v, 0, \ldots, 0\rangle$. We find U from the dense tensor and check it against that tensor. The generators are pushed through the tree as signed Pauli strings [196]. In all 20 networks (five depths, two settings, two orientations) they are N independent generators on N qubits. Each one, sign included, is a $+1$ stabiliser of an independent composition of the same Cliffords by stim [347] [holo:H1]. Entropies, in bits, follow exactly from the generator matrix G over GF(2) [190, 348]:

$$S(A) = \operatorname{rank}(G|_A) - |A|. \tag{60}$$

At $D = 1$ these entropies agree with a dense contraction of the tensors to 2×10^{-15}. On a 3×3 toric code they match the dense state vector exactly [holo:H1], [holo:H6].

Recovery. Bulk qubit t is recoverable from region A when $I(R_t : A) = S(R_t) + S(A) - S(R_t A) = 2$ bits. For a stabiliser state,

$$I(R_t : A) = \dim\big(V \cap \langle X_{R_t}, Z_{R_t} \rangle\big),$$

where V is the reference part of the stabilisers supported on $A \cup R$. One elimination, with columns ordered as complement, references, A, gives V for all bulk qubits at once.

Cuts, wedges and the residual region.

- The minimal cut $\mathrm{mc}(A)$ assigns every tile to the side of A or of its complement and counts the legs crossed, with unit capacity per leg. Dynamic programming on the tree finds the cheapest assignment.
- In setting (b) a bulk leg is crossed when its tile lies on A's side, because its reference belongs to the complement.
- The greedy algorithm of Ref. [99] absorbs a tile into A's wedge when at least 3 of its 5 non-bulk legs lie in the region. We run it to its fixed point.
- The *residual region* consists of the tiles absorbed neither from A nor from its boundary complement.

E.2 The exceptions to the minimal-cut formula

Section 18 gives the exceptions in setting (a) and their cause. Here we characterise them over all setting-(a) cases [holo:H2]:

- Every exception has a non-empty residual region, so the two greedy cuts never meet.
- The deficit never exceeded $\lfloor r/2 \rfloor$ for r residual tiles. We observed this bound; we did not prove it.
- Degenerate minimal cuts are not the explanation: 95 735 exceptions have a unique minimal cut.
- The argument of Ref. [99] for connected regions rests on the greedy cuts meeting. On a tree with fixed bulk they need not meet, and then the minimal cut is only an upper bound.

One interval is confirmed as an exception in orientation H by dense contraction alone [holo:H1].

E.3 Entanglement-wedge reconstruction

The pairs tested. These are all 2^{20} regions, 6 bulk qubits and 2 orientations at $D = 1$, plus intervals and random subsets of 11 sizes at $D = 2, \ldots, 5$ [holo:H3].

The predictions compared. In every pair, four predictions agree:

- the exact recoverability;
- the greedy wedge;
- the recursion of Proposition 6;
- the min-cut prediction $I_{\mathrm{cut}} = 1 + \mathrm{mc}(A) - \mathrm{mc}(A \cup R_t)$.

I_{cut} never takes the value 1. Wedge reconstruction [102, 103] therefore holds in this code.

Two regimes. Table 27 shows them.

- An interval holds the bulk geometrically. A deep qubit is in the wedge when its subtree is, and the root needs more than half the ring.
- For a random subset, the root and the upper levels hold nothing until about 77 % of the boundary survives, which is $1 - p_c$. Then they hold nearly everything. Qubits near the leaves are recovered in proportion to their local coverage.

E.4 The erasure recursion

Definitions. Let A be the boundary qubits that survive erasure. The edge above a tile is *available from below* when its qubit can be recovered from the part of A below the tile. It is *available from above* when its qubit can be recovered from the part of A outside the tile's subtree. A surviving boundary leg counts as available.

Proposition 6 (exact erasure recursion, setting (b); *proven*)**.**

- *Seen from either side, every edge is either available or decoupled; that is, its channel to that part of A is constant.*

Table 27: Fraction of bulk qubits recoverable at $D = 5$ (orientation A, setting b), by depth (0 is the root) and by the fraction $|A|/n$ of the boundary held. Each entry averages 6 interval starts or 6 random subsets. [holo:H3]

	interval, $\|A\|/n =$				random subset, $\|A\|/n =$			
depth	0.3	0.5	0.6	0.7	0.6	0.7	0.8	0.9
0	0.00	0.17	0.67	1.00	0.00	0.00	1.00	1.00
1	0.27	0.43	0.53	0.77	0.00	0.03	1.00	1.00
3	0.29	0.50	0.60	0.70	0.07	0.44	0.99	1.00
5	0.30	0.50	0.60	0.70	0.52	0.75	0.96	1.00

- *An edge is available from below if and only if at least 3 of the 4 legs below its tile are available.*
- *An edge is available from above if and only if at least 3 of the parent's other 4 non-bulk legs are available.*
- *Bulk qubit t has $I(R_t : A) = 2$ if at least 3 of its tile's 5 non-bulk legs are available, and $I(R_t : A) = 0$ otherwise. For the root, this means 3 of its 5 children.*

Proof. Two properties of a perfect tensor are used. Any three legs are a unitary image of the other three, and any three legs are maximally mixed.

- Suppose at least 3 legs below a tile are available. Decoding them recovers the remaining legs, of which there are at most three: the edge, the bulk leg and at most one child.
- Suppose at most 2 legs are available. Then the edge and those legs form three maximally mixed legs, so $I = 1 + 2 - 3 = 0$.
- By induction on the height, an unavailable subtree is a constant channel. After local decoding, an available subtree is equivalent to its edge qubit.

The same argument runs from above, and with the bulk leg as the target. □

The probabilities. The availabilities in Proposition 6 depend on disjoint sets of boundary qubits, so they are independent. This gives Eq. (34), with the binomial step $a_h = P(\mathrm{Bin}(4, a_{h-1}) \geq 3)$. Analogous closed forms hold at every depth. At a^* the map has slope $f'(a^*) = 12a^{*2}(1-a^*) = 1.643 > 1$. Its fixed point is therefore unstable, and the transition window narrows as 1.643^{-D} [holo:H4].

Table 28: Probability that the root is recoverable after erasure at rate p: the Monte Carlo estimate, with the recursion (34) in brackets. There are 4000, 2000, 1500, 1000 and 400 samples per p for $D = 1, \ldots, 5$. The row 0.2324 is p_c. [holo:H4]

p	$D = 1$	$D = 2$	$D = 3$	$D = 4$	$D = 5$
0.22	0.938 (0.932)	0.946 (0.942)	0.958 (0.956)	0.976 (0.973)	0.990 (0.988)
0.2324	0.913 (0.914)	0.909 (0.914)	0.907 (0.914)	0.905 (0.914)	0.912 (0.914)
0.25	0.886 (0.884)	0.854 (0.860)	0.821 (0.815)	0.704 (0.722)	0.517 (0.529)
0.28	0.818 (0.819)	0.713 (0.730)	0.549 (0.546)	0.247 (0.236)	0.020 (0.017)

Checks of the recursion.

- At $D = 1$ the recursion equals, to 3×10^{-13}, the exact recovery polynomial built from all 2^{20} erasure patterns.
- The Monte Carlo samples of Section 18 cover orientation A at $D = 1, \ldots, 5$, at 11 values of p from 0.05 to 0.5. A further check in orientation H at $D = 5$ agrees in the same way.
- The Monte Carlo fractions match the probabilities at every point of D, p and depth, with largest $|z| = 2.55$.

- The Monte Carlo curves for $D = 4$ and $D = 5$ cross at $p = 0.2331$, within the noise of 400 samples [holo:H4].

E.5 The witnesses in detail

Contents. The six bulk contents of Section 18 are five random products of cardinal states and one with 400 bulk Bell pairs.

The two witnesses. Witness I reads the basis $g_1, g_1g_2, g_2g_3, \ldots$ of the stabiliser group, not the generators themselves, and decodes the result. In the noisy runs, both witnesses also match the commutation pattern of the error with the generators, which is computed separately.

Reading the content. After the content of the 86 read qubits is read, the other 1620 bulk qubits keep fidelity 1 [holo:H5].

One tile. We enumerated every Pauli error on the five qubits of a tile. The syndrome never depends on the content, and lookup correction restores every single-qubit error. By weight $0, \ldots, 5$, the correctable errors number $1, 15, 0, 60, 135, 45$. Under depolarising noise of rate p, the logical failure probability is therefore exactly

$$P_{\text{fail}} = 1 - \left[(1-p)^5 + 5p(1-p)^4 + \tfrac{20}{9}p^3(1-p)^2 + \tfrac{5}{3}p^4(1-p) + \tfrac{5}{27}p^5\right] = 10p^2 + O(p^3). \quad (61)$$

- This lies below the closed form $1-(1-p)^5-5p(1-p)^4$ for a distance-3 code that corrects only weight-1 errors. The two give 0.0795 against 0.0815 at $p = 0.1$, and 0.249 against 0.263 at $p = 0.2$.
- The difference comes from errors of weight 3 to 5 that equal a correctable error times a stabiliser.
- The pseudo-threshold, where $P_{\text{fail}} = p$, is $p = 0.1376$.
- For Haar-random contents the syndrome distribution is independent of the content to machine precision. Reading the content leaves fidelity exactly $\frac{1}{2}$ [holo:H5].

E.6 The classical centre

Section 8 closes on Harlow's formula [22], $S(\rho_R) = \operatorname{Tr}(\rho\,\mathcal{L}_R) + S_{\text{bulk}}(\rho, \mathcal{M}_R)$. In it the area operator $\mathcal{L}_R$ lies in the centre of the reconstructable algebra $\mathcal{M}_R$. In setting (b) we can read the formula off directly, in three steps.

- The cut function is submodular, so minimal cuts are closed under union. The largest A-side among them is therefore itself a minimal cut.
- $I_{\text{cut}}(t) = 2$ holds exactly when tile t lies on A's side of some minimal cut. So this largest side is the set of tiles with $I_{\text{cut}} = 2$. Because $I_{\text{cut}} = I$ in every pair, it is the wedge W_A.
- The cut crosses one bulk leg per tile of W_A and $|\gamma_A|$ non-bulk legs. Hence
$$S(A) = |\gamma_A| + |W_A| \qquad \text{(setting b)}. \quad (62)$$

An empty residual region. Every bulk qubit then lies either in W_A or in the wedge of the complement. Operators on A commute with operators on the complement, so $\mathcal{M}_A$ is exactly the algebra of the qubits in W_A. Equation (62) is then Harlow's formula, with $S_{\text{bulk}} = |W_A|$ and $\operatorname{Tr}(\rho\,\mathcal{L}_A) = |\gamma_A|$, which is a number. On a fixed tree the area can only be a number. The assembly realises one fixed-area sector, as Lemma 2 requires, and it cannot represent a superposition of areas. That *holds* by construction. The superposition of geometries stays *untested*.

Setting (a). With an empty residual region and a pure bulk, Harlow's formula gives $S = |\gamma_A|$, and we find $S = \text{mc}$ in every such case.

A non-empty residual region. A residual tile is a bulk qubit that neither side can reconstruct alone. Complementary recovery of whole bulk qubits then fails, and Harlow's theorem in this form does not apply. The deficits have the tensor-network cause given in Section 18. Whether a non-trivial centre of the reconstructable algebra also describes them is *untested*: $I(R_t : A)$ takes only the values 0 and 2, and the script constructs no centre.

E.7 A control: the toric code

We built the ground state of the toric code [349] on $L \times L$ tori for $L = 8$, 16, 24, 32, 40 and 45, which reaches 4050 qubits. The generators are the stars, the plaquettes and two non-contractible Z loops; their GF(2) rank is $2L^2$. The regions are sets of edges in disks or annuli about a generic centre [holo:H6]. We applied two constructions.

- *Kitaev and Preskill* [350], $-\gamma = S_A + S_B + S_C - S_{AB} - S_{BC} - S_{CA} + S_{ABC}$, on 33 disks of radius 2 to 18. Each disk is cut into three sectors.
- *Levin and Wen* [351], $-2\gamma = (S_1 - S_2) - (S_3 - S_4)$, on 20 annuli, from radii 2–4.5 to 10–19.5. Region 1 is the annulus, regions 2 and 3 have one gap each, and region 4 has both gaps.

Every case gives $\gamma = 1$ bit $= \ln 2$, the value for $\mathbb{Z}_2$ topological order, whatever the region size and L (P). The same machinery that finds the deficits of Table 9 reproduces this universal constant exactly.

E.8 Choices and caveats

Choices made after seeing the data.

- Orientation H was added after the exhaustive run at $D = 1$ showed interval exceptions in the HaPPY convention.
- A first characterisation, that exceptions need a degenerate minimal cut, failed on 95 735 regions. The residual region and the observed $\lfloor r/2 \rfloor$ bound replaced it.
- The first witness run at scale failed: shortening the pushed generators had mixed reference factors into the code stabilisers. Shortening now uses only reference-free generators, and no entropy changes.
- The Monte Carlo samples at $D = 5$ were raised from 160 to 400.
- Two floating-point tolerances were widened tenfold, to 10^{-11} and 10^{-13}, after rounding exceeded them.
- A counting error in the script had added the setting-(b) sets at $D = 1$ to the setting-(a) total. It is fixed. The checks and every other number are unchanged.
- The region sets were sized for runtime.

Caveats.

- This is kinematics on a fixed background.
- The states are stabiliser states. Non-stabiliser contents were tested on one tile only.
- The noise is Pauli or erasure noise, and syndrome extraction is noiseless.
- The $\lfloor r/2 \rfloor$ bound is observed in every setting-(a) case, not proved, and the regions at $D \geq 2$ are samples.
- The recursion is exact only in setting (b).
- A tree is not our space (Section 8).

F The exotic-hadron model in detail

This appendix gives the model, the fit, the full spectra, the threshold catalogue and the details of the H9 tests behind Section 25. Headline values are not repeated here; Table 19 and Table 20 are their home.

F.1 The colour-spin Hamiltonian in the H8 trees

Colour and spin. In Eq. (58), $\boldsymbol{\lambda}_i \cdot \boldsymbol{\lambda}_j = \sum_{a=1}^{8} \lambda_i^a \lambda_j^a$ is the colour product of the Gell-Mann matrices, with $-\lambda^{a*}$ for an antiquark, and $\boldsymbol{\sigma}$ are the Pauli matrices. The state is the H8 tree. Every internal node below the root couples its two branches to a colour (anti)triplet, with quadratic Casimir 4/3; the root couples to a singlet; every node carries a spin. The colour factors that follow are $\langle \boldsymbol{\lambda}_i \cdot \boldsymbol{\lambda}_j \rangle = -16/3$ for a quark and antiquark in a singlet and $-8/3$ for a pair in a diquark or a baryon. A quark and an antiquark in the two halves of $[qq][\bar{q}\bar{q}]$ have $-4/3$, the two diquarks of $([qq][qq])\bar{q}$ have $-2/3$ per pair, and its antiquark has $-4/3$ with each quark [exotics:E]. Within a fixed colour tree, node spins that give the same total J are mixed and the Hamiltonian is diagonalised. For neutral states with hidden flavour, the eigenvectors are sorted by C parity, the exchange of the two halves. Two identical s, c or b quarks in a diquark have spin 1 (Pauli), and a light $[ud]$ pair has isospin equal to its spin. The S-wave pentaquark multiplets have the multiplicities of Maiani, Polosa and Riquer [321].

The trees span the colour space. The products $\mathbf{3}\otimes\mathbf{3} = \bar{\mathbf{3}}\oplus\mathbf{6}$, $\bar{\mathbf{3}}\otimes\bar{\mathbf{3}} = \mathbf{3}\oplus\bar{\mathbf{6}}$ and $\mathbf{3}\otimes\bar{\mathbf{3}} = \mathbf{1}\oplus\mathbf{8}$ each contain the conjugate triplet or the singlet. So any quarks and antiquarks that can form a singlet can be coupled pairwise until two conjugate halves meet at the root (P). For a tetraquark the two branches of H8 exhaust the colour space. The singlets of $qq\bar{q}\bar{q}$ form a two-dimensional space, and the compact tree $[q_1q_2]_{\bar{3}}[\bar{q}_3\bar{q}_4]_3$ and the molecular tree $(q_1\bar{q}_3)_1(q_2\bar{q}_4)_1$ are two vectors of it with $|\langle \text{compact}|\text{molecular}\rangle|^2 = 1/3$ [exotics:E]. Every tetraquark colour state is therefore a superposition of the two trees, and H8 cannot fail on a single state. No numerology is used: the five quarks of a pentaquark are not connected with the five parts of the minting, since the axioms supply no map between them.

Couplings. Mesons and baryons have separate effective masses m_f^m and m_f^b ($f = q, s, c, b$, $q = u, d$). The coefficient of $\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$ is $v_{ij}^m = 2b\,k_{ij}/(m_i^m m_j^m)$ in a meson and $v_{ij}^b = a\,k_{ij}/(m_i^b m_j^b)$ in a baryon, so that

$$C_{ij} = \tfrac{3}{16}\, v_{ij}^m \quad \text{(quark–antiquark pair)}, \qquad C_{ij} = \tfrac{3}{8}\, v_{ij}^b \quad \text{(quark–quark pair)} \tag{63}$$

returns v^m and v^b for the colour factors $-16/3$ and $-8/3$. The factor k_{ij} is 1 except for (c, s) and (b, s), where it is KR's enhancement $k_s = a_{cs}/a$. KR apply it in baryons; we share it with mesons, where KR found the same enhancement of the $D_s^* - D_s$ splitting. Heavy–heavy pairs take $v_{c\bar{c}}^m = [M(J/\psi) - M(\eta_c)]/4$ and $v_{b\bar{b}}^m = [M(\Upsilon) - M(\eta_b)]/4$. $v_{b\bar{c}}^m$ is interpolated between them with $|\psi(0)|^2 \propto \mu^p$ in the reduced mass μ [165, Sec. VII], and a heavy–heavy quark pair has the same C. The binding terms are

$$B_{ij} = B_{ij}^{(1)}\, \frac{\langle \boldsymbol{\lambda}_i \cdot \boldsymbol{\lambda}_j \rangle}{-16/3}, \tag{64}$$

with colour-singlet values $B^{(1)}$ for $(c, \bar{s})$, $(b, \bar{s})$, $(c, \bar{c})$, $(b, \bar{b})$ and $(b, \bar{c})$ fitted to mesons, and none for pairs with a u or d quark. In a baryon or diquark this is KR's rule $B(QQ) = B(Q\bar{Q})/2$. For a pair in different halves of a multiquark it gives $B^{(1)}/4$: KR's stated assumption, a potential that factorises into a colour factor and one spatial factor, applied to every pair.

F.2 Variants of the mass convention

Two variants were fixed with the model and are reported for every state, never scored. *P-in* drops the binding between constituents of different halves. *J* replaces the two mass sets by one universal set and adds a constant S per string junction [166, 171]: none for a meson, one for a baryon, two for a tetraquark and three for a pentaquark, the internal nodes below the root. Variant J fits ordinary hadrons as well as the primary model ($\sigma_{\text{model}} = 8.2\,\text{MeV}$, $S = 160.6\,\text{MeV}$). It shifts the tetraquarks by 90 to 113 MeV and the lowest pentaquarks by 204 to 210 MeV, ten to

Table 29: Fitted parameters (1σ from the covariance scaled by σ_{model}) and KR's values [165], all in MeV; the rows b and a give $b/(310\,\text{MeV})^2$ and $a/(363\,\text{MeV})^2$ [exotics:F].

parameter	fit	KR	parameter	fit	KR
m_q^m	309.6 ± 2.1	310	m_q^b	364.0 ± 1.4	363
m_s^m	481.8 ± 3.1	483	m_s^b	536.0 ± 1.8	538
m_c^m	1658.1 ± 5.2	1663.3	m_c^b	1715.9 ± 4.4	1710.5
m_b^m	5001.6 ± 5.4	5003.8	m_b^b	5047.8 ± 4.5	5043.5
b	79.3 ± 1.5	80	a	51.1 ± 1.0	50
k_s	1.79 ± 0.12	1.65 (baryons)	$B^{(1)}_{c\bar{s}}$	-66.1 ± 6.7	-69.9
$B^{(1)}_{b\bar{s}}$	-80.4 ± 7.0	-83.6	$B^{(1)}_{c\bar{c}}$	-247.5 ± 12.3	-258.0
$B^{(1)}_{b\bar{b}}$	-558.4 ± 12.6	-562.8	$B^{(1)}_{b\bar{c}}$	-334.6 ± 12.4	-341.5
$v^m_{c\bar{c}}$	28.2 ± 2.9	28.3	$v^m_{b\bar{b}}$	15.4 ± 2.9	15.6

twenty-five times σ_{model}. The ordinary hadrons cannot choose between the conventions, and this choice dominates every compact mass. Of the nine scored confrontations, the primary model holds 4, P-in 2 and J 2 [exotics:A].

F.3 The fit and the hosting check

Inputs. The 42 hadrons are $\pi, \rho, \omega, K, K^*, \phi$, $D^{(*)}, D_s^{(*)}, B^{(*)}, B_s^{(*)}, \eta_c, J/\psi, \eta_b, \Upsilon(1S), B_c$, the light octet and decuplet baryons, and the singly charmed and bottom baryons $\Lambda_Q, \Sigma_Q^{(*)}, \Xi_Q, \Xi'_Q, \Xi^*_Q, \Omega_Q^{(*)}$ (no Ω_b^*). The masses are the values tabulated by KR [165] or current values of the Particle Data Group [132], read state by state; the Ξ'_b uses LHCb's mass difference [352]. Isospin averages are taken, since the model has no isospin breaking; η and η' are left out because of flavour mixing, as by KR.

Uncertainty. $\sigma_{\text{model}} = \sqrt{\text{RSS}/(N-p)}$ with $N = 42$ and $p = 18$. A prediction carries $\sigma^2_{\text{pred}} = \sigma^2_{\text{model}} + g^{\mathsf{T}}\Sigma g$, with g the gradient of the predicted mass, $\Sigma = \sigma^2_{\text{model}}(\mathsf{D}^{\mathsf{T}}\mathsf{D})^{-1}$ the parameter covariance and D the Jacobian of the fitted masses. The residuals have root-mean-square 5.5 MeV for mesons and 6.6 MeV for baryons, and the largest is -16.3 MeV, for the Σ (Fig. 24). Fitting with each hadron left out in turn and predicting it gives 11.2 MeV over 37 hadrons; the five that fix a parameter exactly are skipped. Our parameters (Table 29) differ from KR's mainly in m_c^b and $B^{(1)}_{c\bar{c}}$, because KR fix each parameter on one or two hadrons and we fit all at once with current charmonium masses [exotics:F].

Hosting check. With KR's published parameters the code reproduces KR's tables to their quoted precision: 0.15 MeV for the heavy hadrons, and the rounding to 1 MeV for the light ones. It gives Ξ_{cc} at 3626.7 MeV, and $T(bb\bar{u}\bar{d}) = 10389.3$, $T(cc\bar{u}\bar{d}) = 3882.15$ and $T(bc\bar{u}\bar{d}) = 7133.75$ MeV against KR's 10389.4, 3882.2 and 7133.7 MeV [166]. It also reproduces the fully charmed 0^{++} estimate of Karliner, Nussinov and Rosner, 6191.5 MeV [171], from their formula [exotics:H].

F.4 The full spectra

Table 30 adds to Table 19 the same trees with KR's own parameters, the two variants, the states that are not scored and the isospin partners; Fig. 25 shows the pentaquark and four-charm levels [exotics:A].

Doubly heavy states. T_{cc}^+ is LHCb's pole 360 ± 40 keV below the $D^{*+}D^0$ threshold of 3875.10 MeV, with $J^P = 1^+$ and isospin zero [168, 306]. For T_{bb}^-, lattice QCD finds binding

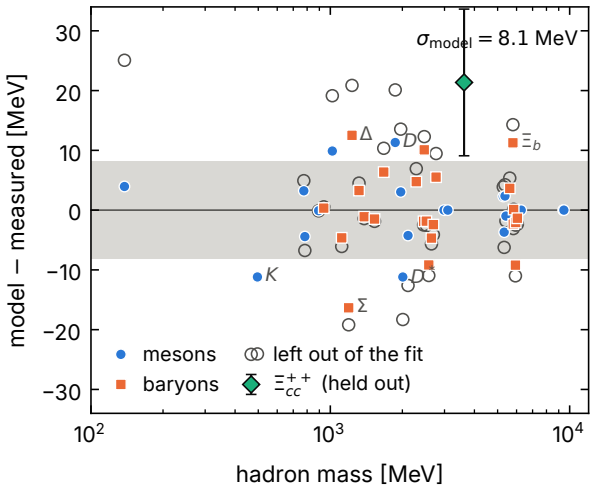


Figure 24: Fit of the colour-spin model to 42 ordinary ground-state hadrons: model minus measured mass against mass, mesons (circles) and baryons (squares); open circles are the residuals when the hadron is left out of the fit; the band is $\pm\sigma_{\text{model}} = \pm 8.1$ MeV. The diamond is the held-out Ξ_{cc}^{++} with $\pm\sigma_{\text{pred}}$. Data: KR's tables [165] and the Particle Data Group [132]. What to look for: no trend with mass, and the held-out baryon within 2σ [exotics:F].

Table 30: All compact trees: the model (KR's convention, our fit) with σ_{pred}, the same tree with KR's parameters, the variants P-in and J, the datum, the pull, and model mass minus the lowest S-wave two-hadron threshold. Data as in Table 19, $X(3872)$ [353] and $Z_c(3900)$ [354]. Masses in MeV [exotics:A].

state	tree, $J^{P(C)}$	model	KR par.	P-in	J	data	z	$M - T$
T_{cc}^{+}	$[cc]_1[\bar{u}\bar{d}]_0$, 1^+	3897 ± 12	3882	3897	3995	3874.74	+1.86	+22
$X_0(2900)$	$[ud]_0[\bar{c}\bar{s}]_0$, 0^+	2755 ± 10	2754	2755	2861	2866	−9.28	+392
$T_{c\bar{s}0}(2900)$	$[cq][\bar{s}\bar{q}]$, 0^+	2809 ± 9	2812	2825	2916	2908	−4.06	+701
$X(6900)$	$[cc]_1[\bar{c}\bar{c}]_1$, 2^{++}	6425 ± 29	6383	6672	6515	6905	−15.2	+231
$P_c(4312)$	lowest $([cq][qq])\bar{c}$	4277 ± 11	4264	4339	4485	4311.9	−3.06	+354
$P_c(4440)^{\dagger}$	$([cq][qq]_1)\bar{c}$, $\frac{1}{2}^-$	4428 ± 11	4446	4396	4485	4440.3	−0.98	+506
$P_c(4457)^{\dagger}$	$([cq][qq]_1)\bar{c}$, $\frac{1}{2}^-$	4462 ± 11	4456	4490	4485	4457.3	+0.38	+539
$P_{\psi s}(4338)$	$([cs][ud]_0)\bar{c}$, $\frac{1}{2}^-$	4392 ± 11	4382	4470	4602	4338.2	+4.81	+292
$P_{cs}(4459)^{\dagger}$	$([cs][ud]_0)\bar{c}$, $\frac{3}{2}^-$	4460 ± 11	4442	4470	4602	4458.8	+0.13	+361
not scored								
$X(6900)$	$[cc]_1[\bar{c}\bar{c}]_1$, 0^{++}	6340 ± 26	6298	6588	6430	6905	−19.1	+372
$X(3872)$	$[cq][\bar{c}\bar{q}]$, 1^{++}	4108 ± 11	4093	4170	4212	3871.70	+21.0	+237
$Z_c(3900)$	$[cq][\bar{c}\bar{q}]$, 1^{+-}	3985 ± 11	3969	4047	4087	3899	+6.72	+749
live								
T_{bb}^{-}	$[bb]_1[\bar{u}\bar{d}]_0$, 1^+	10399 ± 12	10389	10399	10512	–	–	−205
T_{bc}	$[bc]_0[\bar{u}\bar{d}]_0$, 0^+	7146 ± 11	7134	7146	7251	–	–	+2
T_{bc}	$[bc]_1[\bar{u}\bar{d}]_0$, 1^+	7180 ± 12	7168	7180	7285	–	–	−9
P_{cc}	$([cc]_1[ud]_0)\bar{q}$, $\frac{1}{2}^-$	4232 ± 12	4215	4232	4436	–	–	+471
$cc\bar{c}\bar{c}$	$[cc]_1[\bar{c}\bar{c}]_1$, 1^{+-}	6369 ± 27	6326	6616	6459	–	–	+288
T_{cc}, $I = 1$	$[cc]_1[\bar{q}\bar{q}]_1$, $0^+, 1^+, 2^+$	4042, 4071, 4131				–	–	+312, +200, +117
T_{bb}, $I = 1$	$[bb]_1[\bar{q}\bar{q}]_1$, $0^+, 1^+, 2^+$	10583, 10593, 10613				–	–	+25, −11, −37

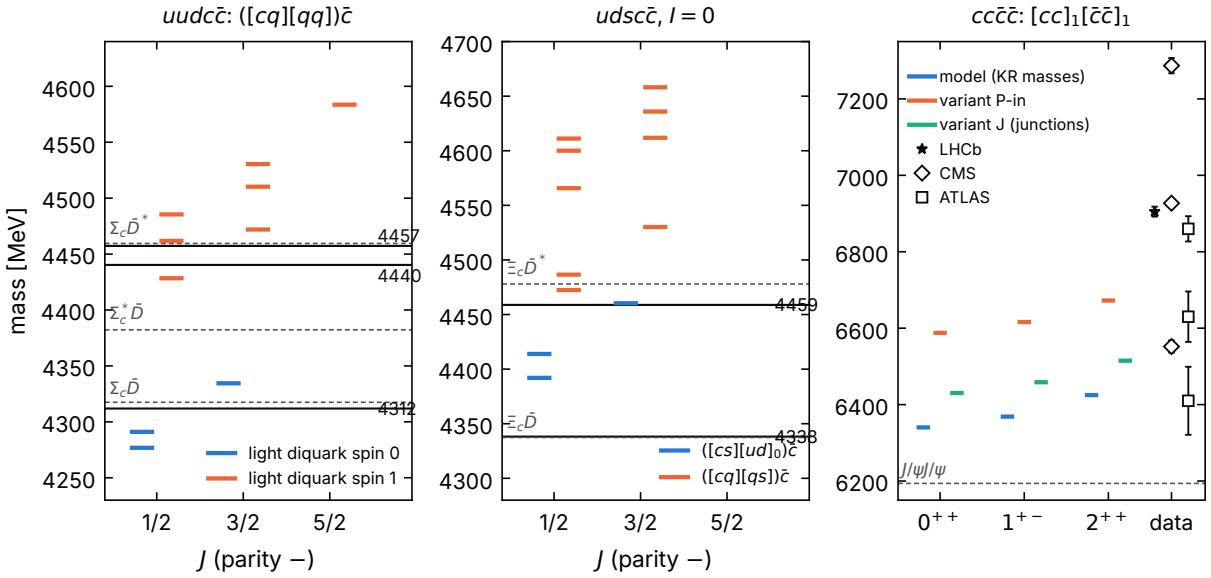


Figure 25: Levels of the compact trees (model, KR's convention, our fit). Left: hidden-charm pentaquarks $([cq][qq])\bar{c}$ with $I = 1/2$ by J (all of negative parity), coloured by the spin of the light diquark, against LHCb's $P_c(4312)$, $P_c(4440)$, $P_c(4457)$ [169] (solid) and the $\Sigma_c\bar{D}^{(*)}$, $\Sigma_c^*\bar{D}$ thresholds (dashed). Centre: $udsc\bar{c}$ with $I = 0$ in the two trees $([cs][ud]_0)\bar{c}$ and $([cq][qs])\bar{c}$ against $P_{\psi s}(4338)$ [310] and $P_{cs}(4459)$ [311], with the $\Xi_c\bar{D}^{(*)}$ thresholds. Right: $[cc]_1[\bar{c}\bar{c}]_1$ for the model and both variants against the di-J/ψ structures of LHCb (model I) [309], CMS (no interference) [356] and ATLAS (model A) [357]. What to look for: the lowest model levels sit 35 to 55 MeV from the lowest observed pentaquarks, the levels are dense enough above 4.4 GeV to match any peak, and the S-wave four-charm levels lie far below 6.9 GeV [exotics:A].

energies relative to $B\bar{B}^*$ of 189 ± 10 [312], 128 ± 26 [313], 83 ± 22 [314], 112 ± 13 [315] and 116^{+36}_{-30} MeV [316]; against our 205 MeV the pulls run from -1.0 to -5.1. These are calculations and enter no verdict. For the 1^+ $bc\bar{u}\bar{d}$ state, lattice QCD finds binding of 15 to 61 MeV [317] or $\Delta E = -43^{+15}_{-25}$ MeV [318], pulls $+1.1$ and $+1.8$ against our -9 MeV. Alexandrou et al. find shallow bound or virtual states a few MeV or less below both $B^{(*)}\bar{D}$ thresholds [355].

Open charm and strangeness. $X_0(2900)$ is a spin-0 $\bar{c}d\bar{s}u$ state in $B^+ \to D^+D^-K^+$ [307], and $T_{c\bar{s}0}(2900)^{++,0}$ an isospin triplet with $J^P = 0^+$ in $D_s^+\pi^\pm$ [308]. The spin-1 $X_1(2900)$ has negative parity, needs an orbital excitation and lies outside the S-wave model.

Four charm quarks. The S-wave levels lie 146 to 231 MeV above $J/\psi J/\psi$. LHCb's $X(6900)$ is $6905 \pm 11 \pm 7$ MeV without interference with the continuum [309]. CMS's no-interference fit finds structures at 6552 ± 16, 6927 ± 10 and 7287^{+21}_{-19} MeV [356]; ATLAS's model A finds $6.41^{+0.11}_{-0.09}$, $6.63^{+0.09}_{-0.05}$ and $6.86^{+0.03}_{-0.04}$ GeV [357]; LHCb sees a broad enhancement from 6.2 to 6.8 GeV. Only the lowest ATLAS structure lies near the S-wave levels; this was noticed after the comparison and is not a test.

Pentaquarks. The $([cq][qq])\bar{c}$ multiplet has 10 levels with isospin 1/2, from 4277 to 4584 MeV. The lowest has a spin-0 light diquark and $J = 1/2$. Variant P-in, without the $c\bar{c}$ binding across the halves, gives $z = +2.0$ for $P_c(4312)$. For $udsc\bar{c}$ with isospin 0 we compute the two trees $([cs][ud]_0)\bar{c}$ and $([cq][qs])\bar{c}$; the lowest $J = 1/2$ level belongs to the first. $P_{cs}(4459)$ lies 1.5 MeV from a $J = 3/2$ level.

Table 31: Threshold proximity: mass (MeV), $J^{P(C)}$ (? unknown), nearest allowed S-wave threshold, $M - T$ (MeV) and p_i of Eq. (65) for $W = 150\,\text{MeV}$ [exotics:B].

state	M	$J^{P(C)}$	threshold	$M-T$	p_i	state	M	$J^{P(C)}$	threshold	$M-T$	p_i
$\chi_{c1}(3872)$	3871.7	1^{++}	$D^0\bar{D}^{*0}$	+0.0	0.0001	$\psi(4230)$	4222.0	1^{--}	$\omega\chi_{c0}$	+24.6	0.59
$Z_c(3900)$	3899.0	1^+	$\psi(3770)\pi$	−14.3	0.39	$Z_b(10610)$	10607.2	?	$B\bar{B}^*$	+2.8	0.075
$Z_c(4020)$	4022.9	?	$D^*\bar{D}^*$	+5.8	0.11	$Z_b(10650)$	10652.2	?	$B^*\bar{B}^*$	+2.7	0.055
$Z_c(4430)$	4475.0	1^+	$D_1\bar{D}_1$	−373.3	0.72	T_{cc}^+	3874.7	1^+	D^0D^{*+}	−0.4	0.0072
$Z_{cs}(3985)$	3982.5	?	$\bar{D}^0D_s^{*+}$	+5.5	0.34	$X_0(2900)$	2866.0	0^+	$D^{*-}K^{*+}$	−36.1	0.24
$Z_{cs}(4000)$	4003.0	1^+	$J/\psi K^{*+}$	+14.2	0.33	$X_1(2900)$	2904.0	1^-	$\bar{D}_1^0K^0$	−15.7	0.11
$Z_{cs}(4220)$	4216.0	$1^?$	$\psi(2S)K^+$	+36.2	0.89	$T_{c\bar{s}0}(2900)$	2908.0	0^+	$D^{*+}K^{*+}$	+5.9	0.039
$\chi_{c1}(4140)$	4118.0	1^{++}	$J/\psi\phi$	+1.6	0.022	$P_c(4312)$	4311.9	?	$\Sigma_c^+\bar{D}^0$	−5.6	0.25
$\chi_{c1}(4274)$	4294.0	1^{++}	$\omega\psi(2S)$	−174.8	0.99	$P_c(4440)$	4440.3	?	$p\chi_{c1}$	−8.6	0.55
$\chi_{c0}(4500)$	4474.0	0^{++}	$\omega\psi(2S)$	+5.2	0.10	$P_c(4457)$	4457.3	?	$\Lambda_c(2595)\bar{D}^0$	+0.2	0.022
$\chi_{c0}(4700)$	4694.0	0^{++}	$\phi\psi(2S)$	−11.6	0.42	$P_{\psi s}(4338)$	4338.2	$\frac{1}{2}^-$	$\Xi_c^+D^-$	+0.8	0.063
$X(4685)$	4684.0	1^{++}	$\phi\psi(2S)$	−21.6	0.49	$X(6900)$	6905.0	$?^{?+}$	$\chi_{c0}\chi_{c1}$	−20.4	0.86
$X(4630)$	4626.0	1^{-+}	$D_s^*\bar{D}_{s1}(2536)$	−21.3	0.79	$X(3960)$	3956.0	0^{++}	$\eta'\eta_c$	+14.1	0.33

F.5 The threshold catalogue

Catalogue. The 26 states, each from its primary publication, are $\chi_{c1}(3872)$ [353, 358], $Z_c(3900)$ [354, 359], $Z_c(4020)$ [360], $Z_c(4430)$ [361], $Z_{cs}(3985)$ [362], $Z_{cs}(4000)$, $Z_{cs}(4220)$, $\chi_{c1}(4140)$, $\chi_{c1}(4274)$, $\chi_{c0}(4500)$, $\chi_{c0}(4700)$, $X(4685)$, $X(4630)$ [363, 364], $X(3960)$ [365], $\psi(4230)$ [366], $Z_b(10610)$, $Z_b(10650)$ [367], T_{cc}^+ [306], $X_0(2900)$, $X_1(2900)$ [307], $T_{c\bar{s}0}(2900)$ [308], $P_c(4312)$, $P_c(4440)$, $P_c(4457)$ [169], $P_{\psi s}(4338)$ [310] and $X(6900)$ [309].

Thresholds. We take every pair of hadrons narrower than 60 MeV, from 83 hadrons and their antiparticles [132]: light, open-charm and open-bottom mesons including the narrow P-wave ones, charmonia and bottomonia below open flavour, and light, charmed and bottom baryons. A pair counts if its quark flavours add up to the state's and it can form the state in an S wave. That requires J in $|j_1 - j_2|, \ldots, j_1 + j_2$ and parity P_1P_2. For neutral states with hidden flavour it also requires $C = C_1C_2$ for two self-conjugate hadrons, or $(-1)^S$ for a particle and its own antiparticle. Unknown quantum numbers impose nothing; isospin and G parity are not imposed.

Null and statistic. Each mass is placed uniformly in $[M - W, M + W]$ with $W = 150\,\text{MeV}$, and

$$p_i = \frac{1}{2W}\Big|\,[M_i - W, M_i + W] \cap \bigcup_k [T_k - d_i, T_k + d_i]\Big| \tag{65}$$

is the probability that the nearest allowed threshold T_k lies at most as far as the observed d_i. It is computed exactly and agrees with 20000 random placements per state. The p_i are combined by Fisher's method, $X^2 = -2\sum_i \ln p_i \sim \chi^2_{52}$; $X^2 = 101.1$. The windows contain 5.5 allowed thresholds on average; this look-elsewhere count is built into each p_i [exotics:B].

Robustness. Within 5 MeV of an allowed threshold lie 7 states, where 3.8 are expected. Restricted to the 17 states with measured J^P, $p = 1.8 \times 10^{-4}$. Perturbing the 45 threshold masses we could not re-read by 1 MeV (3 MeV) moves p within 7.0×10^{-6} to 7.5×10^{-5} (5.0×10^{-7} to 1.4×10^{-4}). Twelve of the 26 states lie below their nearest threshold and fourteen above.

F.6 The tests of H9 in detail

I1. $P_c(4380)$ was reported above 9 standard deviations in 2015 [329]; its verification awaits an amplitude analysis [169]. $P_c(4337)$ is evidence at 3.1 to 3.7 standard deviations [330], and $P_{cs}(4459)$ at 3.1 [311]. In $\Lambda_c^+D_s^-$, the production ratios of $P_{\psi s}(4338)$ and $P_{cs}(4459)$ are below 0.12 and 0.20 at 95% CL [331]. The hidden row of the 2×2 table holds 4 established and 3

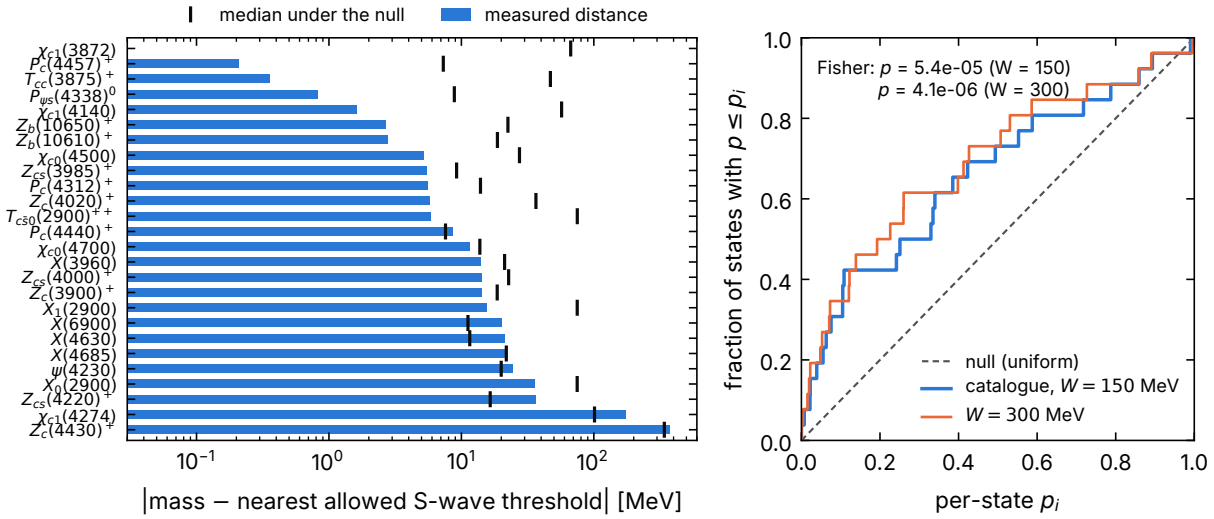


Figure 26: Threshold proximity of the 26 exotic hadrons of Table 31. Left: distance to the nearest allowed S-wave two-hadron threshold (bars, logarithmic) and the median distance under the null of uniform placement within ±150 MeV (ticks). Right: empirical distribution of the per-state p_i against the uniform null, for $W = 150$ and 300 MeV, with Fisher's combined p. What to look for: the excess of small p_i comes from a handful of states, two of them within 1 MeV of threshold [exotics:B].

Table 32: Hidden-bottom pentaquarks as molecules: the P_c and $P_{\psi s}$ binding carried over unchanged, or scaled with the reduced mass. Masses in MeV [exotics:I2].

partner of	bottom threshold	T	binding	M (same binding)	M (scaled)
$P_c(4312)$	$\Sigma_b\bar{B}$	11093.7	+8.8	11084.9	11070.8
$P_c(4440)$	$\Sigma_b\bar{B}^*$	11139.0	+21.7	11117.4	11084.5
$P_c(4457)$	$\Sigma_b\bar{B}^*$	11139.0	+4.7	11134.4	11127.3
$P_{\psi s}(4338)$	$\Xi_b\bar{B}$	11072.1	−1.9	11074.0	11077.0

unconfirmed claims, the open row 0 and 3; Fisher's exact p is 0.17 one-sided and 0.20 two-sided [exotics:I1].

I2. Binding energies are taken relative to isospin-averaged thresholds: $P_c(4312)$ lies 8.8 MeV below $\Sigma_c\bar{D}$, $P_c(4440)$ and $P_c(4457)$ lie 21.7 and 4.7 MeV below $\Sigma_c\bar{D}^*$, and $P_{\psi s}(4338)$ lies 1.9 MeV above $\Xi_c\bar{D}$. The threshold masses are KR's averages [165] and the Particle Data Group's B^* [132]. In the variant, the binding scales with the reduced mass of the pair (a ratio of 2.5 to 2.6), since a heavier pair binds more deeply in the same potential. The compact multiplet $([bq][qq])\bar{b}$ spans 10883 to 11141 MeV (variant J: lowest 11100 MeV), and the lowest $uds b\bar{b}$ level with $J = 1/2$ lies at 10993 ± 11 MeV [exotics:I2].

I4: eligibility. Eligibility was fixed from quark content and J^P only, and the list was written into the script before any splitting was computed. A pair is two exotic states with the same quark content (isospin partners counted once), each reported as an observation, with equal or unknown parity and spins that differ by at most one unit or are unknown. The 25 eligible pairs are: three among $Z_c(3900)$, $Z_c(4020)$, $Z_c(4430)$; three among the Z_{cs}; fifteen among the six $c\bar{c}s\bar{s}$ states of positive parity ($X(3960)$, $\chi_{c1}(4140)$, $\chi_{c1}(4274)$, $\chi_{c0}(4500)$, $X(4685)$, $\chi_{c0}(4700)$); $Z_b(10610)/Z_b(10650)$; $P_c(4312)$ with each of $P_c(4440)$ and $P_c(4457)$; and the CMS pair at 6552 and 6927 MeV [356]. Not eligible are $X(4630)$, $X(3872)/\psi(4230)$ and $X_0/X_1(2900)$, whose parities differ. The two-peak fit of $P_{cs}(4459)$ is excluded because LHCb can neither confirm nor refute it [311], and $X(7100)$, $P_c(4337)$ and $P_c(4380)$ because they are not observations.

Table 33: The splitting rule $\Delta = \alpha M/2$: measured splitting, the rule, its pull, and the colour-spin model's splitting of the corresponding pair of S-wave levels with its pull. Shown: the pairs that pass and the pairs with a model baseline; the other 18 eligible pairs fail by 4.8 to 38 standard deviations (Fig. 27). Below the line, two pairs that are not eligible. Masses in MeV [exotics:I4].

pair	Δ	$\alpha M/2$	pull	model Δ	pull
$Z_c(3900)/Z_c(4020)$	123.9 ± 6.7	14.5	+16.3	131.6 ± 8.9	−0.7
$Z_b(10610)/Z_b(10650)$	45.0 ± 2.5	38.8	+2.5	143.8 ± 8.9	−10.7
$Z_{cs}(3985)/Z_{cs}(4000)$	20.5 ± 11.4	14.6	+0.5	63.9 ± 8.3	−3.1
$X(4685)/\chi_{c0}(4700)$	10.0 ± 18.0	17.1	−0.4	–	–
$\chi_{c0}(4500)/\chi_{c0}(4700)$	220.0 ± 10.8	16.7	+18.7	32.5 ± 8.4	+13.7
$X(3960)/\chi_{c0}(4500)$	518.0 ± 12.0	15.4	+42.0	32.5 ± 8.4	+33.2
$X(3960)/\chi_{c0}(4700)$	738.0 ± 15.0	15.8	+48.1	32.5 ± 8.4	+41.0
$P_c(4440)/P_c(4457)$, suggested the rule	17.0	16.2	–	–	–
$P_{cs}(4459)$, unresolved two-peak fit	12.9 ± 4.6	16.3	–	–	–

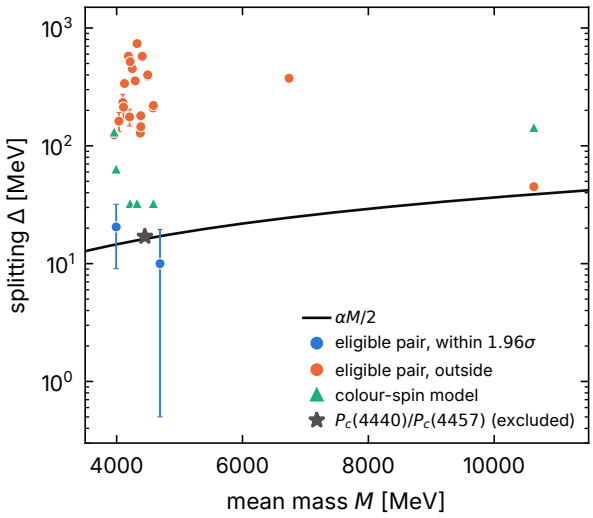


Figure 27: Splittings of the 25 eligible pairs of exotic hadrons against their mean mass (logarithmic scale; lower error bars clipped at the axis), with the rule $\alpha M/2$ (line), the colour-spin model's splitting where it has a corresponding pair of S-wave levels (triangles), and the pair that suggested the rule (star, not counted). Data: Table 31 and CMS [356]. What to look for: the eligible pairs scatter over two orders of magnitude, and the two within 1.96σ of the line have errors as large as the prediction [exotics:I4].

I4: test. A pair passes if $|\Delta - \alpha M/2| < 1.96\,\sigma_\Delta$, with $\alpha = 1/137.035999177$ [300]. Under a null in which Δ is uniform between 0 and 300 MeV, a pair passes by chance with the probability that this interval covers; the sum over the 25 pairs, 4.2, is the look-elsewhere count, and the Poisson probability of two or more passes is 0.92. As the QCD baseline we take the colour-spin model's splitting of two S-wave levels with the same quantum numbers, where the model has such a pair (Table 33, Fig. 27) [exotics:I4].

F.7 Caveats in full

- KR's mass convention dominates the compact masses (Appendix F.2).
- σ_{model} measures ordinary hadrons. Carrying the model to four and five quarks adds uncertainty it does not contain: the size of a diquark, binding across the halves, and colour configurations outside the tree, such as $\mathbf{6} \otimes \bar{\mathbf{6}}$, whose mixing would lower the lowest levels.
- Only S-wave ground states are computed, so a mismatch refutes an S-wave assignment, not every compact reading.

- The threshold test inherits the selection of its catalogue, and its look-elsewhere count depends on the threshold list.
- Lattice results for T_{bb} scatter by 100 MeV.
- The server of the Particle Data Group was not reachable. Its current values were read through a search of its listings, and 45 of the 83 threshold hadrons were not re-read; perturbing them moves no conclusion.

G Readings in detail

Section 13 lists the readings that add no prediction a measurement could tell apart from the established account. This appendix gives each in full.

G.1 The thermodynamics of the frame

G.1.1 The existence of a boundary is its temperature

GUID defines the existence a boundary asserts as the product of the local strength of gravity g and Planck's constant h, divided by the speed of light:

$$E_{\mathrm{ex}} \equiv \frac{g\,h}{c}. \tag{66}$$

Proposition 7 (existence and the Unruh temperature)**.** *E_{ex} is the only energy that an acceleration, Planck's constant and the speed of light can form, and*

$$E_{ex} = 4\pi^2\, k_B T_U, \qquad k_B T_U = \frac{\hbar g}{2\pi c}, \tag{67}$$

where T_U is the temperature an observer of proper acceleration g assigns to the vacuum (Unruh [20]). At a horizon of surface gravity κ it is $4\pi^2$ times the Hawking temperature $k_B T_H = \hbar\kappa/2\pi c$; for a Schwarzschild hole $\kappa = c^4/4GM$.

Proof. With dimensions $[g] = LT^{-2}$, $[h] = ML^2T^{-1}$, $[c] = LT^{-1}$, the monomial $g^a h^b c^k$ has the dimension of energy, ML^2T^{-2}, only for $(a, b, k) = (1, 1, -1)$ [Q1]. With $h = 2\pi\hbar$, $E_{\mathrm{ex}}/k_B T_U = 4\pi^2$. □

The uniqueness means the form of (66) is not itself evidence; quantum field theory supplies the factor and the meaning. At standard gravity $T_U = 4.0 \times 10^{-20}\,\mathrm{K}$; at the horizon of a solar-mass hole $T_H = 6.2 \times 10^{-8}\,\mathrm{K}$ [Q1]. *Verdict: holds as a reading* of the Unruh and Hawking temperatures.

G.1.2 Time: the lapse

In the GUID reading a clock asserts its existence at a rate that falls where the frame is strong. In general relativity that rate is the lapse, $N = \sqrt{1 + 2\Phi/c^2}$ in a weak field and $N = \sqrt{1 - 2GM/rc^2}$ outside a mass, which vanishes at the horizon [Q2]. It reproduces the redshift over the 22.5 m Pound–Rebka tower, 2.46×10^{-15} [368], and the rate difference of two optical clocks 33 cm apart, 3.6×10^{-17} [369]. *Verdict: holds*; it adds nothing to general relativity.

G.1.3 Einstein's equation as an equation of state

Jacobson derived Einstein's equation from thermodynamics [21], and GUID's two thermal inputs are exactly his. Take any point p and any local Rindler horizon through it, generated by a null vector field k^a with affine parameter λ, vanishing expansion and shear at p, and approximate boost Killing field $\chi^a = -\kappa\lambda k^a$. Assume:

1. the horizon has the Unruh temperature $T = \hbar\kappa/2\pi$ (units $c = k_B = 1$), Proposition 7;
2. its entropy is proportional to its area, $S = sA$, the boundary's floor (Hypothesis 5);
3. the Clausius relation $\delta Q = T\,\mathrm{d}S$ holds for the energy flux across it.

The heat flux is $\delta Q = \int T_{ab}\chi^a \mathrm{d}\Sigma^b = -\kappa \int \lambda\, T_{ab}k^a k^b \,\mathrm{d}\lambda\,\mathrm{d}\mathcal{A}$. The area change follows from the Raychaudhuri equation, $\mathrm{d}\theta/\mathrm{d}\lambda = -R_{ab}k^a k^b$ to first order, as $\delta A = -\int \lambda R_{ab}k^a k^b \,\mathrm{d}\lambda\,\mathrm{d}\mathcal{A}$. Equating $\delta Q = (\hbar\kappa/2\pi)s\,\delta A$ for every p and every null k^a gives $R_{ab} + f g_{ab} = (2\pi/\hbar s)T_{ab}$ for some function f; the contracted Bianchi identity and $\nabla^a T_{ab} = 0$ fix $f = -R/2 + \Lambda$, so

$$G_{ab} + \Lambda g_{ab} = \frac{2\pi}{\hbar s}\,T_{ab}, \qquad G = \frac{1}{4\hbar s} \quad (c = 1). \tag{68}$$

Newton's constant is set by the entropy per area, $s = 1/4L_P^2$ [Q1], and the cosmological constant enters as an integration constant. Jacobson's own conclusion fits GUID's classical frame: on this view, "it may be no more appropriate to quantize the Einstein equation than it would be to quantize the wave equation for sound in air" [21]. Two limits: DMP supplies neither the value of s nor a manifold, so the derivation is borrowed, not produced; and the entanglement-equilibrium version [370] needs a state, not a schedule, which the axioms do not define. *Verdict: holds as a reading*; the constant is open.

G.1.4 Information radiation and the floor

DMP's T31 says that a containing boundary emits only thermal radiation, carrying "entropy and no information", and that an outside reader learns of the inside only its size and its staleness. With Hypothesis 5 that is the semiclassical horizon: a black hole of mass M radiates at T_H and an outside observer knows its mass, spin and charge (P). Erasing a bit costs at least $k_B T \ln 2$ of heat (Landauer [371]); in the GUID reading entropy is the inefficiency of converting between existence and information, which is Landauer's principle in other words. *Verdict: holds as a reading.*

G.2 The containment split gives Christodoulou's mass formula

DMP splits a contained boundary into what is inside and the counts every outside reader sees (D). Give the contained part a mass M_{irr} that sets a Schwarzschild boundary of radius $2GM_{\mathrm{irr}}/c^2$; let the spin count J be carried at that boundary as a momentum $p = Jc^2/(2GM_{\mathrm{irr}})$ (an identification, (H)); and combine the two as energy and momentum combine in special relativity. In units $G = c = 1$,

$$M^2 = M_{\mathrm{irr}}^2 + \left(\frac{J}{2M_{\mathrm{irr}}}\right)^2. \tag{69}$$

This is Christodoulou's mass formula for the Kerr black hole [230], with $M_{\mathrm{irr}} = \sqrt{A/16\pi}$ (P); solved for M_{irr} it returns Kerr's horizon area for every spin $a = J/M^2 \in [0, 1]$, has no real solution above $a = 1$ (the spin bound), and gives the extractable share $1 - M_{\mathrm{irr}}/M = 1 - 1/\sqrt{2} \approx 29\%$ at maximal spin [Q10]. The derivation uses DMP's split, two physical inputs and one identification, and gives no number general relativity does not. *Verdict: holds as a rederivation.*

G.3 The universe at its own Schwarzschild threshold

For a flat Friedmann universe with Hubble rate H, the mass inside the Hubble radius, $M_H = \frac{4}{3}\pi(c/H)^3\rho_c$ with $\rho_c = 3H^2/8\pi G$, has a Schwarzschild radius equal to the Hubble radius,

$$\frac{2GM_H}{c^2} = \frac{c}{H}, \tag{70}$$

exactly and at every epoch [Q12]. This is the coincidence behind Pathria's "universe as a black hole" [372]. GUID reads the early universe as a Kugelblitz, a horizon formed by energy alone,

that could not contain its own existence; the reading holds in the sense of (70) and at every epoch, not only the earliest, and the universe is not a black hole in the sense of general relativity, having no outside. *Verdict: holds as a reading.*

G.4 Inflation as the possibility space of existence

In GUID, inflation is the space that existence makes for itself wherever existence is possible. Physically it is a period of nearly exponential expansion, $a \propto e^{Ht}$, which dilutes a spatial curvature $|\Omega - 1|$ by e^{-2N} over N e-folds (Guth [117]) (P). Afterwards the curvature share grows again, as a^2 while radiation dominates and as a while matter does; from a reheating temperature of 10^{15} GeV to today it grows by about 5×10^{51}, so an order-one initial curvature needs about 62 e-folds to meet Planck's bound $|\Omega_k| \lesssim 5 \times 10^{-3}$ [116] [Q13]. The newest combination, DESI's second data release with the microwave background, finds $10^3\,\Omega_K = 2.3 \pm 1.1$, a 2.1σ departure from flatness that inflation and ΛCDM share and that DESI does not claim as a detection (Section 21, [data:D3.7]). *Verdict: holds as a reading*; it supplies the motive, not the number.

G.5 The VIBE coefficient

GUID names α the *VIBE coefficient*, after the Variable Influence Balancing Equation (2), and reads it as the strength with which existence interacts with itself before it must re-ground itself. In quantum electrodynamics α is the strength with which a charge interacts with the electromagnetic field and so, through it, with itself; each emission and reabsorption of a virtual photon costs one power of α. The cleanest measure is the electron's anomalous magnetic moment, whose first term $a_e = \alpha/2\pi$ (Schwinger [373]) gives the measured value [297] to 0.15% [Q16]. The same constant is the ratio of the electron's three lengths, $r_e/\bar{\lambda}_C = \bar{\lambda}_C/a_0 = \alpha$ [Q16]. *Verdict: holds as a reading*; the VIBE coefficient is a name for α, not a new constant.

References

[1] M. H. Goroff, A. Sagnotti, "The ultraviolet behavior of Einstein gravity", *Nucl. Phys. B* 266, 709 (1986).

[2] G. 't Hooft, M. Veltman, "One-loop divergencies in the theory of gravitation", *Ann. Inst. Henri Poincaré A* 20, 69 (1974).

[3] J. F. Donoghue, "General relativity as an effective field theory: the leading quantum corrections", *Phys. Rev. D* 50, 3874 (1994).

[4] N. E. J. Bjerrum-Bohr, J. F. Donoghue, B. R. Holstein, "Quantum gravitational corrections to the nonrelativistic scattering potential of two masses", *Phys. Rev. D* 67, 084033 (2003).

[5] S. W. Hawking, "Particle creation by black holes", *Commun. Math. Phys.* 43, 199 (1975).

[6] J. D. Bekenstein, "Black holes and entropy", *Phys. Rev. D* 7, 2333 (1973).

[7] D. N. Page, "Information in black hole radiation", *Phys. Rev. Lett.* 71, 3743 (1993).

[8] G. Penington, "Entanglement wedge reconstruction and the information paradox", *JHEP* 09 (2020) 002.

[9] A. Almheiri, N. Engelhardt, D. Marolf, H. Maxfield, "The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole", *JHEP* 12 (2019) 063.

[10] K. Eppley, E. Hannah, "The necessity of quantizing the gravitational field", *Found. Phys.* 7, 51 (1977).

[11] D. N. Page, C. D. Geilker, "Indirect evidence for quantum gravity", *Phys. Rev. Lett.* 47, 979 (1981).

[12] J. Oppenheim, "A postquantum theory of classical gravity?", *Phys. Rev. X* 13, 041040 (2023).

[13] J. Oppenheim, C. Sparaciari, B. Šoda, Z. Weller-Davies, "Gravitationally induced decoherence vs space-time diffusion: testing the quantum nature of gravity", *Nat. Commun.* 14, 7910 (2023), arXiv:2203.01982.

[14] S. Bose, A. Mazumdar, G. W. Morley, H. Ulbricht, M. Toroš, M. Paternostro, A. A. Geraci, P. F. Barker, M. S. Kim, G. Milburn, "Spin entanglement witness for quantum gravity", *Phys. Rev. Lett.* 119, 240401 (2017).

[15] C. Marletto, V. Vedral, "Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity", *Phys. Rev. Lett.* 119, 240402 (2017).

[16] T. D. Galley, F. Giacomini, J. H. Selby, "A no-go theorem on the nature of the gravitational field beyond quantum theory", *Quantum* 6, 779 (2022).

[17] S. Hrovat, *The Dark Matter Protocol*, version 0.9.9, `https://github.com/StevenHrovat/dmp`, commit `4400cce` (2026).

[18] D. Kafri, J. M. Taylor, G. J. Milburn, "A classical channel model for gravitational decoherence", *New J. Phys.* 16, 065020 (2014).

[19] A. Tilloy, L. Diósi, "Sourcing semiclassical gravity from spontaneously localized quantum matter", *Phys. Rev. D* 93, 024026 (2016).

[20] W. G. Unruh, "Notes on black-hole evaporation", *Phys. Rev. D* 14, 870 (1976).

[21] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state", *Phys. Rev. Lett.* 75, 1260 (1995).

[22] D. Harlow, "The Ryu–Takayanagi formula from quantum error correction", *Commun. Math. Phys.* 354, 865 (2017).

[23] M. Gorton, A. M. Green, "How open is the asteroid-mass primordial black hole window?", *SciPost Phys.* 17, 032 (2024).

[24] W. H. Zurek, "Quantum Darwinism", *Nat. Phys.* 5, 181 (2009).

[25] F. G. S. L. Brandão, M. Piani, P. Horodecki, "Generic emergence of classical features in quantum Darwinism", *Nat. Commun.* 6, 7908 (2015).

[26] H. Chevalier, A. J. Paige, M. S. Kim, "Witnessing the non-classical nature of gravity in the presence of unknown interactions", *Phys. Rev. A* 102, 022428 (2020), arXiv:2005.13922.

[27] A. Tilloy, L. Diósi, "Principle of least decoherence for Newtonian semiclassical gravity", *Phys. Rev. D* 96, 104045 (2017), arXiv:1706.01856.

[28] L. de Moura, S. Ullrich, "The Lean 4 theorem prover and programming language", in *CADE 28*, LNCS 12699, 625 (2021).

[29] D. E. Denning, "A lattice model of secure information flow", *Commun. ACM* 19, 236 (1976).

[30] D. Malkhi, M. Reiter, "Byzantine quorum systems", *Distrib. Comput.* 11, 203 (1998).

[31] P. O'Hearn, J. Reynolds, H. Yang, "Local reasoning about programs that alter data structures", in *CSL 2001*, LNCS 2142, 1 (2001).

[32] J. A. Goguen, J. Meseguer, "Security policies and security models", in *IEEE Symposium on Security and Privacy*, 11 (1982).

[33] J. H. Saltzer, M. D. Schroeder, "The protection of information in computer systems", *Proc. IEEE* 63, 1278 (1975).

[34] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984).

[35] R. Aldrovandi, J. G. Pereira, *Teleparallel Gravity: An Introduction* (Springer, Dordrecht, 2013).

[36] T. Jacobson, D. Mattingly, "Gravity with a dynamical preferred frame", *Phys. Rev. D* 64, 024028 (2001).

[37] B. P. Abbott et al. (LIGO, Virgo, Fermi GBM, INTEGRAL), "Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A", *Astrophys. J. Lett.* 848, L13 (2017).

[38] P. Touboul et al. (MICROSCOPE Collaboration), "MICROSCOPE mission: final results of the test of the equivalence principle", *Phys. Rev. Lett.* 129, 121102 (2022).

[39] C. M. Will, "The confrontation between general relativity and experiment", *Living Rev. Relativ.* 17, 4 (2014).

[40] W. Israel, "Event horizons in static vacuum space-times", *Phys. Rev.* 164, 1776 (1967).

[41] R. P. Kerr, "Gravitational field of a spinning mass as an example of algebraically special metrics", *Phys. Rev. Lett.* 11, 237 (1963).

[42] S. W. Hawking, "Gravitational radiation from colliding black holes", *Phys. Rev. Lett.* 26, 1344 (1971).

[43] J. M. Bardeen, B. Carter, S. W. Hawking, "The four laws of black hole mechanics", *Commun. Math. Phys.* 31, 161 (1973).

[44] J. D. Bekenstein, "Fine-structure constant: Is it really a constant?", *Phys. Rev. D* 25, 1527 (1982).

[45] H. B. Sandvik, J. D. Barrow, J. Magueijo, "A simple cosmology with a varying fine structure constant", *Phys. Rev. Lett.* 88, 031302 (2002).

[46] Y.-B. Yang, J. Liang, Y.-J. Bi, Y. Chen, T. Draper, K.-F. Liu, Z. Liu, "Proton mass decomposition from the QCD energy momentum tensor", *Phys. Rev. Lett.* 121, 212001 (2018).

[47] G. Dvali, M. Zaldarriaga, "Changing α with time: implications for fifth-force-type experiments and quintessence", *Phys. Rev. Lett.* 88, 091303 (2002).

[48] S. Schlamminger, K.-Y. Choi, T. A. Wagner, J. H. Gundlach, E. G. Adelberger, "Test of the equivalence principle using a rotating torsion balance", *Phys. Rev. Lett.* 100, 041101 (2008), arXiv:0712.0607.

[49] C. Brans, R. H. Dicke, "Mach's principle and a relativistic theory of gravitation", *Phys. Rev.* 124, 925 (1961).

[50] B. Bertotti, L. Iess, P. Tortora, "A test of general relativity using radio links with the Cassini spacecraft", *Nature* 425, 374 (2003).

[51] M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed. (Cambridge University Press, Cambridge, 2010).

[52] R. Horodecki, P. Horodecki, M. Horodecki, K. Horodecki, "Quantum entanglement", *Rev. Mod. Phys.* 81, 865 (2009), arXiv:quant-ph/0702225.

[53] G. Vidal, R. F. Werner, "Computable measure of entanglement", *Phys. Rev. A* 65, 032314 (2002).

[54] A. Peres, "Separability criterion for density matrices", *Phys. Rev. Lett.* 77, 1413 (1996), arXiv:quant-ph/9604005.

[55] M. Horodecki, P. Horodecki, R. Horodecki, "Separability of mixed states: necessary and sufficient conditions", *Phys. Lett. A* 223, 1 (1996), arXiv:quant-ph/9605038.

[56] W. K. Wootters, "Entanglement of formation of an arbitrary state of two qubits", *Phys. Rev. Lett.* 80, 2245 (1998), arXiv:quant-ph/9709029.

[57] O. Gühne, G. Tóth, "Entanglement detection", *Phys. Rep.* 474, 1 (2009), arXiv:0811.2803.

[58] W. K. Wootters, W. H. Zurek, "A single quantum cannot be cloned", *Nature* 299, 802 (1982).

[59] D. Dieks, "Communication by EPR devices", *Phys. Lett. A* 92, 271 (1982).

[60] H. Barnum, C. M. Caves, C. A. Fuchs, R. Jozsa, B. Schumacher, "Noncommuting mixed states cannot be broadcast", *Phys. Rev. Lett.* 76, 2818 (1996).

[61] M. A. Amin, R. Easther, H. Finkel, R. Flauger, M. P. Hertzberg, "Oscillons after inflation", *Phys. Rev. Lett.* 108, 241302 (2012).

[62] L. Kofman, A. Linde, A. A. Starobinsky, "Reheating after inflation", *Phys. Rev. Lett.* 73, 3195 (1994).

[63] J. J. Halliwell, "Decoherence in quantum cosmology", *Phys. Rev. D* 39, 2912 (1989).

[64] C. Kiefer, "Continuous measurement of mini-superspace variables by higher multipoles", *Class. Quantum Grav.* 4, 1369 (1987).

[65] J. B. Hartle, S. W. Hawking, "Wave function of the Universe", *Phys. Rev. D* 28, 2960 (1983).

[66] D. Deutsch, "Quantum theory, the Church–Turing principle and the universal quantum computer", *Proc. R. Soc. Lond. A* 400, 97 (1985).

[67] S. Weinberg, "Photons and gravitons in S-matrix theory: derivation of charge conservation and equality of gravitational and inertial mass", *Phys. Rev.* 135, B1049 (1964).

[68] S. Deser, "Self-interaction and gauge invariance", *Gen. Relativ. Gravit.* 1, 9 (1970).

[69] F. Dyson, "Is a graviton detectable?", *Int. J. Mod. Phys. A* 28, 1330041 (2013).

[70] T. Rothman, S. Boughn, "Can gravitons be detected?", *Found. Phys.* 36, 1801 (2006).

[71] C. A. Mead, "Possible connection between gravitation and fundamental length", *Phys. Rev.* 135, B849 (1964).

[72] G. Tobar, S. K. Manikandan, T. Beitel, I. Pikovski, "Detecting single gravitons with quantum sensing", *Nat. Commun.* 15, 7229 (2024).

[73] D. Carney, V. Domcke, N. L. Rodd, "Graviton detection and the quantization of gravity", *Phys. Rev. D* 109, 044009 (2024).

[74] C. Møller, "The energy-momentum complex in general relativity and related problems", in *Les Théories Relativistes de la Gravitation*, ed. A. Lichnerowicz, M.-A. Tonnelat (CNRS, Paris, 1962).

[75] L. Rosenfeld, "On quantization of fields", *Nucl. Phys.* 40, 353 (1963).

[76] L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics", *Phys. Lett. A* 120, 377 (1987).

[77] R. Penrose, "On gravity's role in quantum state reduction", *Gen. Relativ. Gravit.* 28, 581 (1996).

[78] A. Bassi, K. Lochan, S. Satin, T. P. Singh, H. Ulbricht, "Models of wave-function collapse, underlying theories, and experimental tests", *Rev. Mod. Phys.* 85, 471 (2013), arXiv:1204.4325.

[79] S. Donadi, K. Piscicchia, C. Curceanu, L. Diósi, M. Laubenstein, A. Bassi, "Underground test of gravity-related wave function collapse", *Nat. Phys.* 17, 74 (2021).

[80] L. Figurato, M. Dirindin, J. L. Gaona-Reyes, M. Carlesso, A. Bassi, S. Donadi, "On the effectiveness of the collapse in the Diósi–Penrose model", *New J. Phys.* 26, 113004 (2024).

[81] J. Aziz, R. Howl, "Classical theories of gravity produce entanglement", *Nature* 646, 813 (2025).

[82] C. Marletto, J. Oppenheim, V. Vedral, E. Wilson, "Classical gravity cannot mediate entanglement", arXiv:2511.07348 (2025).

[83] L. Diósi, "No, classical gravity does not entangle quantized matter fields", arXiv:2511.00852 (2025).

[84] M. D. Schneider, N. Huggett, N. Linnemann, "A demonstration that classical gravity does not produce entanglement", arXiv:2511.19242 (2025).

[85] L. Gundhi, A. Infantino, A. Bassi, "Can classical theories of gravity produce entanglement?", arXiv:2604.19696 (2026).

[86] M. Christodoulou, A. Di Biagio, M. Aspelmeyer, Č. Brukner, C. Rovelli, R. Howl, "Locally mediated entanglement in linearized quantum gravity", *Phys. Rev. Lett.* 130, 100202 (2023).

[87] L. Lami, J. S. Pedernales, M. B. Plenio, "Testing the quantumness of gravity without entanglement", *Phys. Rev. X* 14, 021022 (2024).

[88] T. Westphal, H. Hepach, J. Pfaff, M. Aspelmeyer, "Measurement of gravitational coupling between millimetre-sized masses", *Nature* 591, 225 (2021).

[89] R. Colella, A. W. Overhauser, S. A. Werner, "Observation of gravitationally induced quantum interference", *Phys. Rev. Lett.* 34, 1472 (1975).

[90] V. V. Nesvizhevsky et al., "Quantum states of neutrons in the Earth's gravitational field", *Nature* 415, 297 (2002).

[91] D. Carney, P. C. E. Stamp, J. M. Taylor, "Tabletop experiments for quantum gravity: a user's manual", *Class. Quantum Grav.* 36, 034001 (2019).

[92] R. Penrose, "Gravitational collapse: the role of general relativity", *Riv. Nuovo Cimento* 1, 252 (1969).

[93] R. D. Blandford, R. L. Znajek, "Electromagnetic extraction of energy from Kerr black holes", *Mon. Not. R. Astron. Soc.* 179, 433 (1977).

[94] J. M. Bardeen, W. H. Press, S. A. Teukolsky, "Rotating black holes: locally nonrotating frames, energy extraction, and scalar synchrotron radiation", *Astrophys. J.* 178, 347 (1972).

[95] D. Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, California Institute of Technology (1997), arXiv:quant-ph/9705052.

[96] R. Laflamme, C. Miquel, J. P. Paz, W. H. Zurek, "Perfect quantum error correcting code", *Phys. Rev. Lett.* 77, 198 (1996).

[97] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, W. K. Wootters, "Mixed-state entanglement and quantum error correction", *Phys. Rev. A* 54, 3824 (1996).

[98] R. Cleve, D. Gottesman, H.-K. Lo, "How to share a quantum secret", *Phys. Rev. Lett.* 83, 648 (1999).

[99] F. Pastawski, B. Yoshida, D. Harlow, J. Preskill, "Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence", *JHEP* 06 (2015) 149.

[100] J. M. Maldacena, "The large N limit of superconformal field theories and supergravity", *Adv. Theor. Math. Phys.* 2, 231 (1998).

[101] S. Ryu, T. Takayanagi, "Holographic derivation of entanglement entropy from AdS/CFT", *Phys. Rev. Lett.* 96, 181602 (2006).

[102] A. Almheiri, X. Dong, D. Harlow, "Bulk locality and quantum error correction in AdS/CFT", *JHEP* 04 (2015) 163.

[103] X. Dong, D. Harlow, A. C. Wall, "Reconstruction of bulk operators within the entanglement wedge in gauge-gravity duality", *Phys. Rev. Lett.* 117, 021601 (2016).

[104] X. Dong, D. Harlow, D. Marolf, "Flat entanglement spectra in fixed-area states of quantum gravity", *JHEP* 10 (2019) 240.

[105] M. Heydeman, M. Marcolli, I. Saberi, B. Stoica, "Tensor networks, p-adic fields, and algebraic curves: arithmetic and the AdS_3/CFT_2 correspondence", *Adv. Theor. Math. Phys.* 22, 93 (2018).

[106] S. S. Gubser, J. Knaute, S. Parikh, A. Samberg, P. Witaszczyk, "p-adic AdS/CFT", *Commun. Math. Phys.* 352, 1019 (2017).

[107] Y. Rabinovich, R. Raz, "Lower bounds on the distortion of embedding finite metric spaces in graphs", *Discrete Comput. Geom.* 19, 79 (1998).

[108] M. Gell-Mann, J. B. Hartle, "Quantum mechanics in the light of quantum cosmology", in *Complexity, Entropy and the Physics of Information*, ed. W. H. Zurek (Addison-Wesley, Redwood City, 1990), 425.

[109] S. Deser, R. Jackiw, G. 't Hooft, "Three-dimensional Einstein gravity: dynamics of flat space", *Ann. Phys.* 152, 220 (1984).

[110] E. Witten, "2+1 dimensional gravity as an exactly soluble system", *Nucl. Phys. B* 311, 46 (1988).

[111] J. D. Brown, M. Henneaux, "Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity", *Commun. Math. Phys.* 104, 207 (1986).

[112] A. Maloney, E. Witten, "Quantum gravity partition functions in three dimensions", *JHEP* 02 (2010) 029.

[113] J. E. Nelson, T. Regge, "Homotopy groups and (2+1)-dimensional quantum gravity", *Nucl. Phys. B* 328, 190 (1989).

[114] G. 't Hooft, "Non-perturbative 2 particle scattering amplitudes in 2+1 dimensional quantum gravity", *Commun. Math. Phys.* 117, 685 (1988).

[115] B. P. Abbott et al. (LIGO Scientific and Virgo Collaborations), "Observation of gravitational waves from a binary black hole merger", *Phys. Rev. Lett.* 116, 061102 (2016).

[116] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters", *Astron. Astrophys.* 641, A6 (2020).

[117] A. H. Guth, "Inflationary universe: a possible solution to the horizon and flatness problems", *Phys. Rev. D* 23, 347 (1981).

[118] Planck Collaboration, "Planck 2018 results. X. Constraints on inflation", *Astron. Astrophys.* 641, A10 (2020), arXiv:1807.06211.

[119] DESI Collaboration, "DESI DR2 results II: measurements of baryon acoustic oscillations and cosmological constraints", *Phys. Rev. D* 112, 083515 (2025).

[120] M. Chevallier, D. Polarski, "Accelerating universes with scaling dark matter", *Int. J. Mod. Phys. D* 10, 213 (2001), arXiv:gr-qc/0009008.

[121] E. V. Linder, "Exploring the expansion history of the universe", *Phys. Rev. Lett.* 90, 091301 (2003), arXiv:astro-ph/0208512.

[122] T. Louis et al. (Atacama Cosmology Telescope), "The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters", *JCAP* (accepted, 2025), arXiv:2503.14452.

[123] A. G. Riess et al., *Astrophys. J. Lett.* (2022), doi:10.3847/2041-8213/ac5c5b, arXiv:2112.04510.

[124] W. L. Freedman et al., "Status report on the Chicago-Carnegie Hubble Program (CCHP): measurement of the Hubble constant using the Hubble and James Webb Space Telescopes", *Astrophys. J.* 985, 203 (2025), arXiv:2408.06153.

[125] U. Amaldi, W. de Boer, H. Fürstenau, "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP", *Phys. Lett. B* 260, 447 (1991).

[126] J. P. Preskill, "Cosmological production of superheavy magnetic monopoles", *Phys. Rev. Lett.* 43, 1365 (1979).

[127] M. Ambrosio et al. (MACRO Collaboration), "Final results of magnetic monopole searches with the MACRO experiment", *Eur. Phys. J. C* 25, 511 (2002).

[128] A. Takenaka et al. (Super-Kamiokande Collaboration), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in Super-Kamiokande I-IV", *Phys. Rev. D* 102, 112011 (2020).

[129] H. Georgi, S. L. Glashow, "Unity of all elementary-particle forces", *Phys. Rev. Lett.* 32, 438 (1974).

[130] K. Abe et al. (Hyper-Kamiokande Proto-Collaboration), "Hyper-Kamiokande design report", arXiv:1805.04163 (2018).

[131] M. D'Onofrio, K. Rummukainen, "Standard model cross-over on the lattice", *Phys. Rev. D* 93, 025003 (2016).

[132] S. Navas et al. (Particle Data Group), "Review of particle physics", *Phys. Rev. D* 110, 030001 (2024), and its online listings.

[133] A. A. Starobinsky, "Spectrum of relict gravitational radiation and the early state of the universe", *JETP Lett.* 30, 682 (1979).

[134] L. M. Krauss, F. Wilczek, "Using cosmology to establish the quantization of gravity", *Phys. Rev. D* 89, 047501 (2014).

[135] P. A. R. Ade et al. (BICEP/Keck), "Improved constraints on primordial gravitational waves using Planck, WMAP, and BICEP/Keck observations through the 2018 observing season", *Phys. Rev. Lett.* 127, 151301 (2021).

[136] M. Tristram et al., "Improved limits on the tensor-to-scalar ratio using BICEP and Planck data", *Phys. Rev. D* 105, 083524 (2022).

[137] A. A. Starobinsky, "A new type of isotropic cosmological models without singularity", *Phys. Lett. B* 91, 99 (1980).

[138] F. L. Bezrukov, M. Shaposhnikov, "The Standard Model Higgs boson as the inflaton", *Phys. Lett. B* 659, 703 (2008).

[139] E. Allys et al. (LiteBIRD Collaboration), "Probing cosmic inflation with the LiteBIRD cosmic microwave background polarization survey", *Prog. Theor. Exp. Phys.* 2023, 042F01 (2023).

[140] D. Polarski, A. A. Starobinsky, "Semiclassicality and decoherence of cosmological perturbations", *Class. Quantum Grav.* 13, 377 (1996).

[141] D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, D. Zaritsky, "A direct empirical proof of the existence of dark matter", *Astrophys. J. Lett.* 648, L109 (2006).

[142] S. Tremaine, J. E. Gunn, "Dynamical role of light neutral leptons in cosmology", *Phys. Rev. Lett.* 42, 407 (1979).

[143] T. Piffl et al., "The RAVE survey: the Galactic escape speed and the mass of the Milky Way", *Astron. Astrophys.* 562, A91 (2014).

[144] J. Oppenheim, A. Russo, "Anomalous contribution to galactic rotation curves due to stochastic spacetime", arXiv:2402.19459 (2024).

[145] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis", *Astrophys. J.* 270, 365 (1983).

[146] M. P. Hertzberg, A. Loeb, "Critical analysis of replacing dark matter and dark energy with a model of stochastic spacetime", *J. Cosmol. Astropart. Phys.* 09 (2024) 046, arXiv:2404.13037.

[147] F. W. Hehl, B. Mashhoon, "Nonlocal gravity simulates dark matter", arXiv:0812.1059 (2008).

[148] E. Cotner, A. Kusenko, V. Takhistov, "Primordial black holes from inflaton fragmentation into oscillons", *Phys. Rev. D* 98, 083513 (2018).

[149] G. Agazie et al. (NANOGrav Collaboration), "The NANOGrav 15 yr data set: evidence for a gravitational-wave background", *Astrophys. J. Lett.* 951, L8 (2023).

[150] S. L. Glashow, "Partial-symmetries of weak interactions", *Nucl. Phys.* 22, 579 (1961).

[151] S. Weinberg, "A model of leptons", *Phys. Rev. Lett.* 19, 1264 (1967).

[152] A. Salam, "Weak and electromagnetic interactions", in *Elementary Particle Theory*, ed. N. Svartholm (Almqvist and Wiksell, Stockholm, 1968), 367.

[153] P. W. Higgs, "Broken symmetries and the masses of gauge bosons", *Phys. Rev. Lett.* 13, 508 (1964).

[154] F. Englert, R. Brout, "Broken symmetry and the mass of gauge vector mesons", *Phys. Rev. Lett.* 13, 321 (1964).

[155] A. David et al. (LHC Higgs Cross Section Working Group), "LHC HXSWG interim recommendations to explore the coupling structure of a Higgs-like particle", arXiv:1209.0040 (2012).

[156] J. Erler, A. Freitas, "Electroweak model and constraints on new physics", in Particle Data Group, *Review of Particle Physics*, 2017 edition.

[157] D. Buttazzo, G. Degrassi, P. P. Giardino, G. F. Giudice, F. Sala, A. Salvio, A. Strumia, "Investigating the near-criticality of the Higgs boson", *JHEP* 12 (2013) 089, arXiv:1307.3536.

[158] S. Weinberg, *Phys. Rev. Lett.* 43, 1566 (1979).

[159] KamLAND-Zen Collaboration, "Search for Majorana neutrinos with the complete KamLAND-Zen dataset", *Phys. Rev. Lett.* 135, 262501 (2025), arXiv:2406.11438.

[160] LEGEND Collaboration, "First results on the search for lepton number violating neutrinoless double beta decay with the LEGEND-200 experiment", arXiv:2505.10440 (2025), doi:10.1103/25tk-nctn.

[161] R. L. Jaffe, "Exotica", *Phys. Rep.* 409, 1 (2005), arXiv:hep-ph/0409065.

[162] L. Maiani, F. Piccinini, A. D. Polosa, V. Riquer, "Diquark-antidiquarks with hidden or open charm and the nature of $X(3872)$", *Phys. Rev. D* 71, 014028 (2005), arXiv:hep-ph/0412098.

[163] F.-K. Guo, C. Hanhart, U.-G. Meißner, Q. Wang, Q. Zhao, B.-S. Zou, "Hadronic molecules", *Rev. Mod. Phys.* 90, 015004 (2018), arXiv:1705.00141.

[164] A. De Rújula, H. Georgi, S. L. Glashow, *Phys. Rev. D* 12, 147 (1975).

[165] M. Karliner, J. L. Rosner, "Baryons with two heavy quarks: masses, production, decays, and detection", *Phys. Rev. D* 90, 094007 (2014), arXiv:1408.5877.

[166] M. Karliner, J. L. Rosner, "Discovery of doubly-charmed Ξ_{cc} baryon implies a stable $bb\bar{u}\bar{d}$ tetraquark", *Phys. Rev. Lett.* 119, 202001 (2017), arXiv:1707.07666.

[167] LHCb Collaboration, "Exotic hadron naming convention", arXiv:2206.15233 (2022).

[168] LHCb Collaboration, "Observation of an exotic narrow doubly charmed tetraquark", *Nature Phys.* (2022), doi:10.1038/s41567-022-01614-y, arXiv:2109.01038.

[169] LHCb Collaboration, "Observation of a narrow pentaquark state, $P_c(4312)^+$, and of two-peak structure of the $P_c(4450)^+$", *Phys. Rev. Lett.* 122, 222001 (2019), arXiv:1904.03947.

[170] G. C. Rossi, G. Veneziano, "A possible description of baryon dynamics in dual and gauge theories", *Nucl. Phys. B* 123, 507 (1977).

[171] M. Karliner, J. L. Rosner, S. Nussinov, "$QQ\bar{Q}\bar{Q}$ states: masses, production, and decays", *Phys. Rev. D* 95, 034011 (2017), arXiv:1611.00348.

[172] A. Ashtekar, "New variables for classical and quantum gravity", *Phys. Rev. Lett.* 57, 2244 (1986).

[173] C. Rovelli, L. Smolin, "Discreteness of area and volume in quantum gravity", *Nucl. Phys. B* 442, 593 (1995).

[174] S. Weinberg, "Ultraviolet divergences in quantum theories of gravitation", in *General Relativity: An Einstein Centenary Survey*, ed. S. W. Hawking, W. Israel (Cambridge University Press, 1979), 790.

[175] M. Reuter, "Nonperturbative evolution equation for quantum gravity", *Phys. Rev. D* 57, 971 (1998).

[176] J. Ambjørn, A. Görlich, J. Jurkiewicz, R. Loll, "Planckian birth of the quantum de Sitter universe", arXiv:0712.2485 (2007).

[177] L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, "Space-time as a causal set", *Phys. Rev. Lett.* 59, 521 (1987).

[178] M. Van Raamsdonk, "Building up spacetime with quantum entanglement", *Gen. Relativ. Gravit.* 42, 2323 (2010).

[179] J. Maldacena, L. Susskind, "Cool horizons for entangled black holes", *Fortschr. Phys.* 61, 781 (2013).

[180] M. J. Bremner, R. Jozsa, D. J. Shepherd, "Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy", *Proc. R. Soc. A* 467, 459 (2011), arXiv:1005.1407.

[181] H. M. Wiseman, G. J. Milburn, "Quantum theory of optical feedback via homodyne detection", *Phys. Rev. Lett.* 70, 548 (1993).

[182] D. N. Page, "Average entropy of a subsystem", *Phys. Rev. Lett.* 71, 1291 (1993), arXiv:gr-qc/9305007.

[183] C. E. Porter, R. G. Thomas, "Fluctuations of nuclear reaction widths", *Phys. Rev.* 104, 483 (1956).

[184] M. J. Bremner, A. Montanaro, D. J. Shepherd, "Average-case complexity versus approximate simulation of commuting quantum computations", *Phys. Rev. Lett.* 117, 080501 (2016), arXiv:1504.07999.

[185] R. Simon, "Peres-Horodecki separability criterion for continuous variable systems", *Phys. Rev. Lett.* 84, 2726 (2000), arXiv:quant-ph/9909044.

[186] R. Howl, R. Penrose, I. Fuentes, "Exploring the unification of quantum theory and general relativity with a Bose–Einstein condensate", *New J. Phys.* 21, 043047 (2019), arXiv:1812.04630.

[187] J. I. de Vicente, "Separability criteria based on the Bloch representation of density matrices", *Quantum Inf. Comput.* 7, 624 (2007), arXiv:quant-ph/0607195.

[188] D. Trillo, M. Navascués, "The Diósi–Penrose model of classical gravity predicts gravitationally induced entanglement", *Phys. Rev. D* 111, L121101 (2025), arXiv:2411.02287.

[189] T. Feng, V. Vedral, C. Marletto, "Collapse-based models for gravity do not violate the entanglement-based witness of non-classicality", arXiv:2503.19774 (2025).

[190] D. Fattal, T. S. Cubitt, Y. Yamamoto, S. Bravyi, I. L. Chuang, "Entanglement in the stabilizer formalism", arXiv:quant-ph/0406168 (2004).

[191] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, "Capacities of quantum erasure channels", *Phys. Rev. Lett.* 78, 3217 (1997).

[192] J. Schwinger, "Gauge invariance and mass. II", *Phys. Rev.* 128, 2425 (1962).

[193] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer", *SIAM J. Comput.* 26, 1484 (1997).

[194] S. Beauregard, "Circuit for Shor's algorithm using 2n+3 qubits", *Quantum Inf. Comput.* 3, 175 (2003).

[195] D. Gottesman, "The Heisenberg representation of quantum computers", arXiv:quant-ph/9807006 (1998).

[196] S. Aaronson, D. Gottesman, "Improved simulation of stabilizer circuits", *Phys. Rev. A* 70, 052328 (2004).

[197] G. Schwarz, "Estimating the dimension of a model", *Ann. Statist.* 6, 461 (1978).

[198] R. E. Kass, A. E. Raftery, "Bayes factors", *J. Am. Stat. Assoc.* 90, 773 (1995).

[199] B. Efron, "Bootstrap methods: another look at the jackknife", *Ann. Statist.* 7, 1 (1979).

[200] S. Holm, "A simple sequentially rejective multiple test procedure", *Scand. J. Statist.* 6, 65 (1979).

[201] R. A. Fisher, *Statistical Methods for Research Workers*, 4th ed. (Oliver and Boyd, Edinburgh, 1932).

[202] E. Gross, O. Vitells, "Trial factors for the look elsewhere effect in high energy physics", *Eur. Phys. J. C* 70, 525 (2010), arXiv:1005.1891.

[203] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, D. T. Mellor, "The preregistration revolution", *Proc. Natl. Acad. Sci. USA* 115, 2600 (2018).

[204] M. Filzinger, S. Dörscher, R. Lange, J. Klose, M. Steinel, E. Benkler, E. Peik, C. Lisdat, N. Huntemann, "Improved limits on the coupling of ultralight bosonic dark matter to photons from optical atomic clock comparisons", *Phys. Rev. Lett.* 130, 253001 (2023), arXiv:2301.03433.

[205] E. D. Davis, L. Hamdan, "Reappraisal of the limit on the variation in α implied by the Oklo natural fission reactors", *Phys. Rev. C* 92, 014319 (2015), arXiv:1503.06011.

[206] M. T. Murphy et al., "Fundamental physics with ESPRESSO: precise limit on variations in the fine-structure constant towards the bright quasar HE 0515−4414", *Astron. Astrophys.* 658, A123 (2022).

[207] M. R. Wilczynska et al., "Four direct measurements of the fine-structure constant 13 billion years ago", *Sci. Adv.* (2020), arXiv:2003.07627.

[208] L. Hart, J. Chluba, *Mon. Not. R. Astron. Soc.* (2018), doi:10.1093/mnras/stx2783, arXiv:1705.03925.

[209] P. P. Avelino et al., "Early-universe constraints on a time-varying fine structure constant", *Phys. Rev. D* 64, 103505 (2001), arXiv:astro-ph/0102144.

[210] A. Fienga, O. Minazzoli, "Testing theories of gravity with planetary ephemerides", *Living Rev. Relativ.* 27, 1 (2024), arXiv:2303.01821.

[211] A. Genova et al., "Solar system expansion and strong equivalence principle as seen by the NASA MESSENGER mission", *Nat. Commun.* 9, 289 (2018).

[212] L. Biskupek, J. Müller, J.-M. Torre, "Benefit of new high-precision LLR data for the determination of relativistic parameters", *Universe* 7, 34 (2021), arXiv:2012.12032.

[213] G. Voisin et al., "An improved test of the strong equivalence principle with the pulsar in a triple star system", *Astron. Astrophys.* 638, A24 (2020), arXiv:2005.01388.

[214] P. Delva et al., "A gravitational redshift test using eccentric Galileo satellites", *Phys. Rev. Lett.* 121, 231101 (2018), arXiv:1812.03711.

[215] LIGO Scientific, Virgo and KAGRA Collaborations, "Tests of general relativity with GWTC-3", *Phys. Rev. D* 112, 084080 (2025), arXiv:2112.06861 (version 3).

[216] F. Shankar, D. H. Weinberg, J. Miralda-Escudé, "Self-consistent models of the AGN and black hole populations: duty cycles, accretion rates, and the mean radiative efficiency", *Astrophys. J.* 690, 20 (2009), arXiv:0710.4488.

[217] S. W. Davis, A. Laor, "The radiative efficiency of accretion flows in individual AGN", *Astrophys. J.* 728, 98 (2011), arXiv:1012.3213.

[218] M. Isi, W. M. Farr, M. Giesler, M. A. Scheel, S. A. Teukolsky, "Testing the black-hole area law with GW150914", *Phys. Rev. Lett.* 127, 011103 (2021), arXiv:2012.04486.

[219] LIGO Scientific, Virgo and KAGRA Collaborations, "GW250114: testing Hawking's area law and the Kerr nature of black holes", doi:10.1103/kw5g-d732 (2025), arXiv:2509.08054.

[220] Y. Y. Fein et al., "Quantum superposition of molecules beyond 25 kDa", *Nat. Phys.* 15, 1242 (2019).

[221] S. Pedalino et al., "Probing quantum mechanics using nanoparticle Schrödinger cats", arXiv:2507.21211 (2025).

[222] T. Kovachy et al., "Quantum superposition at the half-metre scale", *Nature* 528, 530 (2015).

[223] V. Xu et al., "Probing gravity by holding atoms for 20 seconds", *Science* 366, 745 (2019), arXiv:1907.03054.

[224] C. S. Reynolds, "Observational constraints on black hole spin", *Annu. Rev. Astron. Astrophys.* (2021), doi:10.1146/annurev-astro-112420-035022, arXiv:2011.08948.

[225] ATLAS Collaboration, *Phys. Rev. Lett.* 131, 251802 (2023), arXiv:2308.04775.

[226] K. Kajantie, M. Laine, K. Rummukainen, M. Shaposhnikov, "Is there a hot electroweak phase transition at $m_H \gtrsim m_W$?", *Phys. Rev. Lett.* 77, 2887 (1996), arXiv:hep-ph/9605288.

[227] KATRIN Collaboration, "Direct neutrino-mass measurement based on 259 days of KATRIN data", *Science* 388, 180 (2025), arXiv:2406.13516.

[228] T. Damour, F. Dyson, *Nucl. Phys. B* 480, 37 (1996), arXiv:hep-ph/9606486.

[229] R. Lange et al., "Improved limits for violations of local position invariance from atomic clock comparisons", *Phys. Rev. Lett.* 126, 011102 (2021).

[230] D. Christodoulou, "Reversible and irreversible transformations in black-hole physics", *Phys. Rev. Lett.* 25, 1596 (1970).

[231] A. Marconi et al., "Local supermassive black holes, relics of active galactic nuclei and the X-ray background", *Mon. Not. R. Astron. Soc.* 351, 169 (2004), arXiv:astro-ph/0311619.

[232] Q. Yu, S. Tremaine, "Observational constraints on growth of massive black holes", *Mon. Not. R. Astron. Soc.* 335, 965 (2002), arXiv:astro-ph/0203082.

[233] M. Moresco et al., "A 6% measurement of the Hubble parameter at $z \sim 0.45$: direct evidence of the epoch of cosmic re-acceleration", *JCAP* 05 (2016) 014, arXiv:1601.01701.

[234] O. Farooq, F. Madiyar, S. Crandall, B. Ratra, "Hubble parameter measurement constraints on the redshift of the deceleration-acceleration transition, dynamical dark energy, and space curvature", *Astrophys. J.* 835, 26 (2017), arXiv:1607.03537.

[235] J. F. Jesus, R. F. L. Holanda, S. H. Pereira, "Model independent constraints on transition redshift", *JCAP* 05 (2018) 073, arXiv:1712.01075.

[236] B. Popovic et al. (DES Collaboration), *Mon. Not. R. Astron. Soc.* (2026), doi:10.1093/mnras/stag632, arXiv:2511.07517.

[237] MuLan Collaboration, "Detailed report of the MuLan measurement of the positive muon lifetime and determination of the Fermi constant", *Phys. Rev. D* 87, 052003 (2013), arXiv:1211.0960.

[238] LHC-TeV MW Working Group, "Compatibility and combination of world W-boson mass measurements", *Eur. Phys. J. C* 84, 451 (2024), arXiv:2308.09417.

[239] H. S. Poh et al., "Approaching Tsirelson's bound in a photon pair experiment", *Phys. Rev. Lett.* 115, 180408 (2015), arXiv:1506.01865.

[240] B. Hensen et al., "Experimental loophole-free violation of a Bell inequality using entangled electron spins separated by 1.3 km", *Nature* 526, 682 (2015), arXiv:1508.05949.

[241] C. J. A. P. Martins, "Primordial nucleosynthesis with varying fundamental constants: degeneracies with cosmological parameters", *Astron. Astrophys.* 646, A47 (2021), arXiv:2012.10505.

[242] S. S. McGaugh, F. Lelli, J. M. Schombert, "The radial acceleration relation in rotationally supported galaxies", *Phys. Rev. Lett.* 117, 201101 (2016), arXiv:1609.05917.

[243] F. Lelli, S. S. McGaugh, J. M. Schombert, "The small scatter of the baryonic Tully-Fisher relation", *Astrophys. J. Lett.* 816, L14 (2016), arXiv:1512.04543.

[244] F. Lelli, S. S. McGaugh, J. M. Schombert, M. S. Pawlowski, "One law to rule them all: the radial acceleration relation of galaxies", *Astrophys. J.* 836, 152 (2017), arXiv:1610.08981.

[245] B. Famaey, S. S. McGaugh, "Modified Newtonian dynamics (MOND): observational phenomenology and relativistic extensions", *Living Rev. Relativ.* 15, 10 (2012), arXiv:1112.3960.

[246] J. F. Navarro, C. S. Frenk, S. D. M. White, "The structure of cold dark matter halos", *Astrophys. J.* 462, 563 (1996), arXiv:astro-ph/9508025.

[247] A. A. Dutton, A. V. Macciò, "Cold dark matter haloes in the Planck era: evolution of structural parameters for Einasto and NFW profiles", *Mon. Not. R. Astron. Soc.* (2014), doi:10.1093/mnras/stu742, arXiv:1402.7073.

[248] A. Burkert, "The structure of dark matter haloes in dwarf galaxies", *Astrophys. J. Lett.* 447, L25 (1995), arXiv:astro-ph/9504041.

[249] W. J. G. de Blok, "The core-cusp problem", *Adv. Astron.* 2010, 789293 (2010), arXiv:0910.3538.

[250] B. P. Moster, T. Naab, S. D. M. White, "Galactic star formation and accretion histories from matching galaxies to dark matter haloes", *Mon. Not. R. Astron. Soc.* 428, 3121 (2013), arXiv:1205.5807.

[251] F. Lelli, S. S. McGaugh, J. M. Schombert, "SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves", *Astron. J.* 152, 157 (2016), arXiv:1606.09251.

[252] M. Cappellari et al., "The ATLAS3D project – XV. Benchmark for early-type galaxies scaling relations from 260 dynamical models", *Mon. Not. R. Astron. Soc.* 432, 1709 (2013), arXiv:1208.3522.

[253] E. V. Pitjeva, N. P. Pitjev, "Relativistic effects and dark matter in the Solar system from observations of planets and spacecraft", *Mon. Not. R. Astron. Soc.* 432, 3431 (2013), arXiv:1306.3043.

[254] P. Li, F. Lelli, S. S. McGaugh, J. M. Schombert, "Fitting the radial acceleration relation to individual SPARC galaxies", *Astron. Astrophys.* 615, A3 (2018), arXiv:1803.00022.

[255] D. C. Rodrigues, V. Marra, A. del Popolo, Z. Davari, "Absence of a fundamental acceleration scale in galaxies", *Nat. Astron.* 2, 668 (2018), arXiv:1806.06803.

[256] B. Carr, K. Kohri, Y. Sendouda, J. Yokoyama, "Constraints on primordial black holes", *Rep. Prog. Phys.* (2021), doi:10.1088/1361-6633/ac1e31, arXiv:2002.12778.

[257] M. Milgrom, "High-redshift rotation curves and MOND", arXiv:1703.06110 (2017).

[258] H. Übler et al., "The evolution of the Tully–Fisher relation between $z \sim 2.3$ and $z \sim 0.9$ with KMOS^{3D}", *Astrophys. J.* 842, 121 (2017), arXiv:1703.04321.

[259] B. I. Ciocan et al., "MUSE-DARK III: the evolution of the radial acceleration relation at intermediate redshifts", *Astron. Astrophys.* 709, L16 (2026), arXiv:2604.22613.

[260] A. Nestor Shachar et al., "RC100: rotation curves of 100 massive star-forming galaxies at $z = 0.6$–2.5 reveal little dark matter on galactic scales", *Astrophys. J.* 944, 78 (2023), arXiv:2209.12199.

[261] R. H. Sanders, "Clusters of galaxies with modified Newtonian dynamics", *Mon. Not. R. Astron. Soc.* 342, 901 (2003), arXiv:astro-ph/0212293.

[262] ATLAS Collaboration, "Search for squarks and gluinos in final states with jets and missing transverse momentum using 139 fb^{-1} of $\sqrt{s} = 13$ TeV pp collision data with the ATLAS detector", *JHEP* 02 (2021) 143, arXiv:2010.14293.

[263] ATLAS Collaboration, "Search for resonances decaying into photon pairs in 139 fb^{-1} of pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Phys. Lett. B* 822, 136651 (2021), arXiv:2102.13405.

[264] CMS Collaboration, "Search for pair production of excited top quarks in the lepton+jets final state", *Phys. Lett. B* 778, 349 (2018), arXiv:1711.10949.

[265] ATLAS Collaboration, "Study of the spin and parity of the Higgs boson in diboson decays with the ATLAS detector", *Eur. Phys. J. C* 75, 476 (2015), arXiv:1506.05669.

[266] ATLAS Collaboration, "A detailed map of Higgs boson interactions by the ATLAS experiment ten years after the discovery", *Nature* 607, 52 (2022), arXiv:2207.00092.

[267] CMS Collaboration, "A portrait of the Higgs boson by the CMS experiment ten years after the discovery", *Nature* 607, 60 (2022), arXiv:2207.00043.

[268] S. Antusch, K. Hinze, S. Saad, "Updated running quark and lepton parameters at various scales", arXiv:2510.01312 (2025).

[269] ATLAS Collaboration, "Measurements of WH and ZH production with Higgs boson decays into bottom quarks and direct constraints on the charm Yukawa coupling in 13 TeV pp collisions with the ATLAS detector", *JHEP* 04 (2025) 075, arXiv:2410.19611.

[270] CMS Collaboration, "Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s} = 13$ TeV", *Phys. Lett. B* 846, 137783 (2023), arXiv:2208.00265.

[271] E. Ma, "Neutrino mass: mechanisms and models", arXiv:0905.0221 (2009).

[272] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz, "NuFit-6.0: updated global analysis of three-flavor neutrino oscillations", arXiv:2410.05380 (2024).

[273] LEGEND Collaboration, "LEGEND-1000 preconceptual design report", arXiv:2107.11462 (2021).

[274] nEXO Collaboration, "nEXO: neutrinoless double beta decay search beyond 10^{28} year half-life sensitivity", *J. Phys. G* 49, 015104 (2022), arXiv:2106.16243.

[275] W. Elbers et al. (DESI Collaboration), "Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape", arXiv:2503.14744 (2025), accepted in *Phys. Rev. D*.

[276] E. Calabrese et al. (Atacama Cosmology Telescope), "The Atacama Cosmology Telescope: DR6 constraints on extended cosmological models", *JCAP* 11 (2025) 063, arXiv:2503.14454.

[277] X. Luo, W. Rodejohann, X.-J. Xu, "Dirac neutrinos and $N_{\rm eff}$", *JCAP* 06 (2020) 058, arXiv:2005.01629.

[278] J. J. Bennett, G. Buldgen, P. F. de Salas, M. Drewes, S. Gariazzo, S. Pastor, Y. Y. Y. Wong, "Towards a precision calculation of $N_{\rm eff}$ in the Standard Model II: neutrino decoupling in the presence of flavour oscillations and finite-temperature QED", *JCAP* 04 (2021) 073, arXiv:2012.02726.

[279] J. de Blas, M. Pierini, L. Reina, L. Silvestrini, "Impact of the recent measurements of the top-quark and W-boson masses on electroweak precision fits", *Phys. Rev. Lett.* 129, 271801 (2022), arXiv:2204.04204.

[280] ATLAS Collaboration, "Measurement of the W-boson mass and width with the ATLAS detector using proton-proton collisions at $\sqrt{s} = 7\,\text{TeV}$", *Eur. Phys. J. C* 84, 1309 (2024), arXiv:2403.15085.

[281] CMS Collaboration, "High-precision measurement of the W boson mass with the CMS experiment", *Nature* 652, 321 (2026), arXiv:2412.13872.

[282] LHCb Collaboration, "Measurement of the W boson mass", *JHEP* 01 (2022) 036, arXiv:2109.01113.

[283] CDF Collaboration, *Science* 376, 170 (2022); the value as quoted in Ref. [279].

[284] M. Steinhauser, "Leptonic contribution to the effective electromagnetic coupling constant up to three loops", *Phys. Lett. B* 429, 158 (1998), arXiv:hep-ph/9803313.

[285] M. Davier, A. Hoecker, B. Malaescu, Z. Zhang, "A new evaluation of the hadronic vacuum polarisation contributions to the muon anomalous magnetic moment and to $\alpha(m_Z^2)$", *Eur. Phys. J. C* 80, 241 (2020), arXiv:1908.00921.

[286] J. Gasser, H. Leutwyler, *Phys. Rep.* 87, 77 (1982).

[287] S. Stern, "Dynamical dark energy and variation of fundamental 'constants"', dissertation, University of Heidelberg (2008), arXiv:0812.3377.

[288] J. Gasser, H. Leutwyler, A. Rusetsky, "On the mass difference between proton and neutron", *Phys. Lett. B* 814, 136087 (2021), arXiv:2003.13612.

[289] Sz. Borsanyi, Z. Fodor, C. Hoelbling, L. Lellouch, K. K. Szabo, C. Torrero, L. Varnhorst, "Ab-initio calculation of the proton and the neutron's scalar couplings for new physics searches", arXiv:2007.03319 (2020).

[290] M. Hoferichter, J. Ruiz de Elvira, B. Kubis, U.-G. Meißner, "High-precision determination of the pion-nucleon σ-term from Roy-Steiner equations", *Phys. Rev. Lett.* 115, 092301 (2015), arXiv:1506.04142.

[291] M. C. Ferreira, M. D. Julião, C. J. A. P. Martins, A. M. R. V. L. Monteiro, "Probing unification scenarios with atomic clocks", *Phys. Rev. D* 86, 125025 (2012), arXiv:1212.4164.

[292] X. Calmet, H. Fritzsch, "The cosmological evolution of the nucleon mass and the electroweak coupling constants", *Eur. Phys. J. C* 24, 639 (2002), arXiv:hep-ph/0112110.

[293] N. Kanekar et al., "Constraints on changes in the proton-electron mass ratio using methanol lines", *Mon. Not. R. Astron. Soc.* 448, L104 (2015), arXiv:1412.7757.

[294] W. Ubachs, J. Bagdonaite, E. J. Salumbides, M. T. Murphy, L. Kaper, "Search for a drifting proton-electron mass ratio from H_2", *Rev. Mod. Phys.* 88, 021003 (2016), arXiv:1511.04476.

[295] R. H. Parker, C. Yu, W. Zhong, B. Estey, H. Müller, "Measurement of the fine-structure constant as a test of the Standard Model", *Science* 360, 191 (2018), arXiv:1812.04130.

[296] L. Morel, Z. Yao, P. Cladé, S. Guellati-Khélifa, "Determination of the fine-structure constant with an accuracy of 81 parts per trillion", *Nature* 588, 61 (2020).

[297] X. Fan, T. G. Myers, B. A. D. Sukra, G. Gabrielse, "Measurement of the electron magnetic moment", *Phys. Rev. Lett.* 130, 071801 (2023).

[298] Muon $g-2$ Collaboration, "Measurement of the positive muon anomalous magnetic moment to 127 ppb", arXiv:2506.03069 (2025).

[299] R. Aliberti et al., "The anomalous magnetic moment of the muon in the Standard Model: an update", *Phys. Rept.* 1143, 1 (2025), arXiv:2505.21476.

[300] CODATA Task Group on Fundamental Constants, "CODATA recommended values of the fundamental physical constants: 2022", arXiv:2409.03787.

[301] S. P. Martin, M. T. Vaughn, "Two-loop renormalization group equations for soft supersymmetry-breaking couplings", *Phys. Rev. D* 50, 2282 (1994), arXiv:hep-ph/9311340.

[302] S. P. Martin, "A supersymmetry primer", arXiv:hep-ph/9709356 (version 7, 2016).

[303] D. d'Enterria et al., "The strong coupling constant: state of the art and the decade ahead", *J. Phys. G* 51, 090501 (2024), arXiv:2203.08271.

[304] J. Erler, "Calculation of the QED coupling $\hat{\alpha}(M_Z)$ in the modified minimal-subtraction scheme", *Phys. Rev. D* 59, 054008 (1999), arXiv:hep-ph/9803453.

[305] LHCb Collaboration, "Precision measurement of the Ξ_{cc}^{++} mass", *JHEP* 02 (2020) 049, arXiv:1911.08594.

[306] LHCb Collaboration, "Study of the doubly charmed tetraquark T_{cc}^{+}", *Nature Commun.* 13, 3351 (2022), arXiv:2109.01056.

[307] LHCb Collaboration, "Amplitude analysis of the $B^+ \to D^+D^-K^+$ decay", *Phys. Rev. D* 102, 112003 (2020), arXiv:2009.00026.

[308] LHCb Collaboration, "First observation of a doubly charged tetraquark and its neutral partner", *Phys. Rev. Lett.* 131, 041902 (2023), arXiv:2212.02716.

[309] LHCb Collaboration, "Observation of structure in the J/ψ-pair mass spectrum", *Sci. Bull.* 65, 1983 (2020), arXiv:2006.16957.

[310] LHCb Collaboration, "Observation of a $J/\psi\Lambda$ resonance consistent with a strange pentaquark candidate in $B^- \to J/\psi\Lambda\bar{p}$ decays", *Phys. Rev. Lett.* 131, 031901 (2023), arXiv:2210.10346.

[311] LHCb Collaboration, "Evidence of a $J/\psi\Lambda$ structure and observation of excited Ξ^- states in the $\Xi_b^- \to J/\psi\Lambda K^-$ decay", *Sci. Bull.* 66, 1278 (2021), arXiv:2012.10380.

[312] A. Francis, R. J. Hudspith, R. Lewis, K. Maltman, "Lattice prediction for deeply bound doubly heavy tetraquarks", *Phys. Rev. Lett.* 118, 142001 (2017), arXiv:1607.05214.

[313] L. Leskovec, S. Meinel, M. Pflaumer, M. Wagner, "Lattice QCD investigation of a doubly-bottom $\bar{b}\bar{b}ud$ tetraquark with quantum numbers $I(J^P) = 0(1^+)$", *Phys. Rev. D* 100, 014503 (2019), arXiv:1904.04197.

[314] T. Aoki, S. Aoki, T. Inoue, "Lattice study on a tetra-quark state T_{bb} in the HAL QCD method", arXiv:2306.03565 (2023).

[315] R. J. Hudspith, D. Mohler, "Exotic tetraquark states with two $\bar{b}$-quarks and $J^P = 0^+$ and 1^+ B_s states in a nonperturbatively-tuned lattice NRQCD setup", *Phys. Rev. D* 107, 114510 (2023), arXiv:2303.17295.

[316] B. S. Tripathy, N. Mathur, M. Padmanath, "$bb\bar{u}\bar{d}$ and $bs\bar{u}\bar{d}$ tetraquarks from lattice QCD using two-meson and diquark-antidiquark variational basis", *Phys. Rev. D* 111, 114504 (2025), arXiv:2503.09760.

[317] A. Francis, R. J. Hudspith, R. Lewis, K. Maltman, "Evidence for charm-bottom tetraquarks and the mass dependence of heavy-light tetraquark states from lattice QCD", *Phys. Rev. D* 99, 054505 (2019), arXiv:1810.10550.

[318] M. Padmanath, A. Radhakrishnan, N. Mathur, "Bound isoscalar axial-vector $bc\bar{u}\bar{d}$ tetraquark T_{bc} from lattice QCD", *Phys. Rev. Lett.* 132, 201902 (2024), arXiv:2307.14128.

[319] BaBar Collaboration, "Search for a charged partner of the $X(3872)$ in the B meson decay $B \to X^-K$, $X^- \to J/\psi\pi^-\pi^0$", *Phys. Rev. D* 71, 031501 (2005), arXiv:hep-ex/0412051.

[320] Belle Collaboration, "Bounds on the width, mass difference and other properties of $X(3872) \to \pi^+\pi^- J/\psi$ decays", *Phys. Rev. D* 84, 052004 (2011), arXiv:1107.0163.

[321] L. Maiani, A. D. Polosa, V. Riquer, "The new pentaquarks in the diquark model", *Phys. Lett. B* 749, 289 (2015), arXiv:1507.04980.

[322] T. Nakano et al. (LEPS Collaboration), "Evidence for narrow $S = +1$ baryon resonance in photoproduction from neutron", *Phys. Rev. Lett.* 91, 012002 (2003), arXiv:hep-ex/0301020.

[323] B. McKinnon, K. Hicks et al. (CLAS), "Search for the Θ^+ pentaquark in the reaction $\gamma d \to pK^-K^+n$", *Phys. Rev. Lett.* 96, 212001 (2006), arXiv:hep-ex/0603028.

[324] NA49 Collaboration, "Observation of an exotic $S = -2$, $Q = -2$ baryon resonance in proton-proton collisions at the CERN SPS", *Phys. Rev. Lett.* 92, 042003 (2004), arXiv:hep-ex/0310014.

[325] H1 Collaboration, "Evidence for a narrow anti-charmed baryon state", *Phys. Lett. B* 588, 17 (2004), arXiv:hep-ex/0403017.

[326] K. T. Knoepfle, M. Zavertyaev, T. Zivko for the HERA-B Collaboration, "Search for Θ^+ and $\Xi^{--}_{3/2}$ pentaquarks in HERA-B", *J. Phys. G* 30, S1363 (2004), arXiv:hep-ex/0403020.

[327] ZEUS Collaboration, "Search for a narrow charmed baryonic state decaying to $D^{*\pm}p^{\mp}$ in ep collisions at HERA", *Eur. Phys. J. C* 38, 29 (2004), arXiv:hep-ex/0409033.

[328] LHCb Collaboration, "Search for prompt production of pentaquarks in charm hadron final states", *Phys. Rev. D* 110, 032001 (2024), arXiv:2404.07131.

[329] LHCb Collaboration, "Observation of $J/\psi p$ resonances consistent with pentaquark states in $\Lambda_b^0 \to J/\psi K^- p$ decays", *Phys. Rev. Lett.* 115, 072001 (2015), arXiv:1507.03414.

[330] LHCb Collaboration, "Evidence for a new structure in the $J/\psi p$ and $J/\psi \bar{p}$ systems in $B_s^0 \to J/\psi p\bar{p}$ decays", *Phys. Rev. Lett.* 128, 062001 (2022), arXiv:2108.04720.

[331] LHCb Collaboration, "First observation of the $\Lambda_b^0 \to \Lambda_c^+ D_s^- K^+ K^-$ decay and search for pentaquarks in the $\Lambda_c^+ D_s^-$ system", *Phys. Rev. D* 112, 052013 (2025), arXiv:2507.10713.

[332] Belle Collaboration, "Search for $\Lambda_c^+ \to \phi p \pi^0$ and branching fraction measurement of $\Lambda_c^+ \to K^-\pi^+ p\pi^0$", *Phys. Rev. D* 96, 051102 (2017), arXiv:1707.00089.

[333] X. Cao, F.-K. Guo, Y.-T. Liang, J.-J. Wu, J.-J. Xie, Y.-P. Xie, Z. Yang, B.-S. Zou, "Photoproduction of hidden-bottom pentaquark and related topics", *Phys. Rev. D* 101, 074010 (2020), arXiv:1912.12054.

[334] H. Bethe, "Zur Theorie der Metalle. I. Eigenwerte und Eigenfunktionen der linearen Atomkette", *Z. Phys.* 71, 205 (1931).

[335] L. Hulthén, "Über das Austauschproblem eines Kristalles", *Ark. Mat. Astron. Fys.* 26A, No. 11 (1938).

[336] H. W. J. Blöte, J. L. Cardy, M. P. Nightingale, "Conformal invariance, the central charge, and universal finite-size amplitudes at criticality", *Phys. Rev. Lett.* 56, 742 (1986).

[337] I. Affleck, D. Gepner, H. J. Schulz, T. Ziman, "Critical behaviour of spin-s Heisenberg antiferromagnetic chains: analytic and numerical results", *J. Phys. A* 22, 511 (1989).

[338] E. Lieb, T. Schultz, D. Mattis, "Two soluble models of an antiferromagnetic chain", *Ann. Phys.* 16, 407 (1961).

[339] P. Pfeuty, "The one-dimensional Ising model with a transverse field", *Ann. Phys.* 57, 79 (1970).

[340] I. Peschel, "Calculation of reduced density matrices from correlation functions", *J. Phys. A* 36, L205 (2003).

[341] G. Vidal, J. I. Latorre, E. Rico, A. Kitaev, "Entanglement in quantum critical phenomena", *Phys. Rev. Lett.* 90, 227902 (2003).

[342] P. Calabrese, J. Cardy, "Entanglement entropy and quantum field theory", *J. Stat. Mech.* (2004) P06002.

[343] J. Kogut, L. Susskind, "Hamiltonian formulation of Wilson's lattice gauge theories", *Phys. Rev. D* 11, 395 (1975).

[344] M. C. Bañuls, K. Cichy, J. I. Cirac, K. Jansen, "The mass spectrum of the Schwinger model with matrix product states", *JHEP* 11 (2013) 158.

[345] R. Dempsey, I. R. Klebanov, S. S. Pufu, B. Zan, "Discrete chiral symmetry and mass shift in the lattice Hamiltonian approach to the Schwinger model", *Phys. Rev. Research* 4, 043133 (2022).

[346] T. Byrnes, P. Sriganesh, R. J. Bursill, C. J. Hamer, "Density matrix renormalization group approach to the massive Schwinger model", *Phys. Rev. D* 66, 013002 (2002).

[347] C. Gidney, "Stim: a fast stabilizer circuit simulator", *Quantum* 5, 497 (2021).

[348] A. Hamma, R. Ionicioiu, P. Zanardi, "Bipartite entanglement and entropic boundary law in lattice spin systems", *Phys. Rev. A* 71, 022315 (2005).

[349] A. Yu. Kitaev, "Fault-tolerant quantum computation by anyons", *Ann. Phys.* 303, 2 (2003).

[350] A. Kitaev, J. Preskill, "Topological entanglement entropy", *Phys. Rev. Lett.* 96, 110404 (2006).

[351] M. Levin, X.-G. Wen, "Detecting topological order in a ground state wave function", *Phys. Rev. Lett.* 96, 110405 (2006).

[352] LHCb Collaboration, "Observation of two new Ξ_b^- baryon resonances", *Phys. Rev. Lett.* 114, 062004 (2015), arXiv:1411.4849.

[353] LHCb Collaboration, "Study of the lineshape of the $\chi_{c1}(3872)$ state", *Phys. Rev. D* 102, 092005 (2020), arXiv:2005.13419.

[354] BESIII Collaboration, "Observation of a charged charmoniumlike structure in $e^+e^- \to \pi^+\pi^- J/\psi$ at $\sqrt{s} = 4.26\,\text{GeV}$", *Phys. Rev. Lett.* 110, 252001 (2013), arXiv:1303.5949.

[355] C. Alexandrou, J. Finkenrath, T. Leontiou, S. Meinel, M. Pflaumer, M. Wagner, "Shallow bound states and hints for broad resonances with quark content $\bar{b}\bar{c}ud$ in B-$\bar{D}$ and B^*-$\bar{D}$ scattering from lattice QCD", *Phys. Rev. Lett.* 132, 151902 (2024), arXiv:2312.02925.

[356] CMS Collaboration, "New structures in the $J/\psi J/\psi$ mass spectrum in proton-proton collisions at $\sqrt{s} = 13\,\text{TeV}$", *Phys. Rev. Lett.* 132, 111901 (2024), arXiv:2306.07164.

[357] ATLAS Collaboration, "Observation of an excess of di-charmonium events in the four-muon final state with the ATLAS detector", *Phys. Rev. Lett.* 131, 151902 (2023), arXiv:2304.08962.

[358] LHCb Collaboration, "Determination of the $X(3872)$ meson quantum numbers", *Phys. Rev. Lett.* 110, 222001 (2013), arXiv:1302.6269.

[359] BESIII Collaboration, "Determination of spin and parity of the $Z_c(3900)$", *Phys. Rev. Lett.* 119, 072001 (2017), arXiv:1706.04100.

[360] BESIII Collaboration, "Observation of a charged charmoniumlike structure $Z_c(4020)$ and search for the $Z_c(3900)$ in $e^+e^- \to \pi^+\pi^- h_c$", *Phys. Rev. Lett.* 111, 242001 (2013), arXiv:1309.1896.

[361] LHCb Collaboration, "Observation of the resonant character of the $Z(4430)^-$ state", *Phys. Rev. Lett.* 112, 222002 (2014), arXiv:1404.1903.

[362] BESIII Collaboration, "Observation of a near-threshold structure in the K^+ recoil-mass spectra in $e^+e^- \to K^+(D_s^- D^{*0} + D_s^{*-} D^0)$", *Phys. Rev. Lett.* 126, 102001 (2021), arXiv:2011.07855.

[363] LHCb Collaboration, "Observation of new resonances decaying to $J/\psi K^+$ and $J/\psi\phi$", *Phys. Rev. Lett.* 127, 082001 (2021), arXiv:2103.01803.

[364] LHCb Collaboration, "Observation of $J/\psi\phi$ structures consistent with exotic states from amplitude analysis of $B^+ \to J/\psi\phi K^+$ decays", *Phys. Rev. Lett.* 118, 022003 (2017), arXiv:1606.07895.

[365] LHCb Collaboration, "Observation of a resonant structure near the $D_s^+ D_s^-$ threshold in the $B^+ \to D_s^+ D_s^- K^+$ decay", *Phys. Rev. Lett.* 131, 071901 (2023), arXiv:2210.15153.

[366] BESIII Collaboration, "Precise measurement of the $e^+e^- \to \pi^+\pi^- J/\psi$ cross section at center-of-mass energies from 3.77 to 4.60 GeV", *Phys. Rev. Lett.* 118, 092001 (2017), arXiv:1611.01317.

[367] Belle Collaboration, "Observation of two charged bottomonium-like resonances in $\Upsilon(5S)$ decays", *Phys. Rev. Lett.* 108, 122001 (2012), arXiv:1110.2251.

[368] R. V. Pound, G. A. Rebka, Jr., "Apparent weight of photons", *Phys. Rev. Lett.* 4, 337 (1960).

[369] C. W. Chou, D. B. Hume, T. Rosenband, D. J. Wineland, "Optical clocks and relativity", *Science* 329, 1630 (2010).

[370] T. Jacobson, "Entanglement equilibrium and the Einstein equation", *Phys. Rev. Lett.* 116, 201101 (2016).

[371] R. Landauer, "Irreversibility and heat generation in the computing process", *IBM J. Res. Dev.* 5, 183 (1961).

[372] R. K. Pathria, "The universe as a black hole", *Nature* 240, 298 (1972).

[373] J. Schwinger, "On quantum-electrodynamics and the magnetic moment of the electron", *Phys. Rev.* 73, 416 (1948).